claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4.jsonl`
- SHA-256: `7a76c66087150e05dbf3372e87910bf7df62261bad38e5c86bd4d46ac42bc0a4`
- Source modified: `2026-06-13T16:41:29+00:00`
- Imported at: `2026-07-05T16:48:19+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `da34752f-c664-4262-9602-f5d4cfcc6db4`

Transcript

1. user (2026-06-12T09:02:24.604Z)

Hello. Is it feasible to add an option in one of the linked repositories to be able to re-encode a video to a lower resolution specifically the one which Roora would be comfortable with. We likely have documented that their annotated videos are almost always 0.5+ seconds delayed. So I am wondering if we can process the videos before giving it to them so that their tools which operate at lower resolution or a frame by frame can be reused so that their rallies are much more precisely annotated. So essentially, the questions are, whether it is possible and if yes, should we even add that in the repos or just make a single tool for it which I can run on every video before handing them for annotation.

2. assistant (2026-06-12T09:02:51.110Z)

I'll research this across the three repos before answering ? specifically: where the Roora ~0.5s delay is documented, what video-prep/ffmpeg tooling already exists, and how the vendor handoff + annotation ingestion flow works (which determines where a re-encode step should live).

3. user (2026-06-12T09:03:26.846Z)

Workflow launched in background. Task ID: wr6mnel6c
Summary: Research Roora annotation delay + video re-encode feasibility across 3 repos
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_cffc9fdf-8f2
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\roora-reencode-feasibility-wf_cffc9fdf-8f2.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\roora-reencode-feasibility-wf_cffc9fdf-8f2.js"} to iterate without resending the script.)
Run ID: wf_cffc9fdf-8f2
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\roora-reencode-feasibility-wf_cffc9fdf-8f2.js", resumeFromRunId: "wf_cffc9fdf-8f2"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

4. assistant (2026-06-12T09:03:33.129Z)

While that runs, let me refresh the vendor-pilot context from memory.

5. user (2026-06-12T09:03:34.097Z)

<system-reminder>This memory is 6 days old. Memories are point-in-time observations, not live

state ? claims about code behavior or file:line citations may be outdated. Verify against current code before asserting as fact.</system-reminder>
1 ---
2 name: golden-set-vendoring
3 description: Plan to generate golden-set rally labels via third-party vendors + an extensible regression-testing flow
4 metadata:
5   node_type: memory
6   type: project
7   originSessionId: aa329f09-28b7-4acc-b062-98ec7e905abc
8 ---
9
10 How we produce trusted **golden-set ground truth** (hand-verified rally `[start,end)` labels) and use it to regression-test the production tool. Full design + vendor shortlist is version-controlled in `docs/GOLDEN_SET_VENDORING.md` (and ADR-006); this file holds the durable decisions.
11
12 **Keystone insight:** a vendor's label CSV is structurally identical to a pipeline prediction, so the existing `backend/eval/metrics.py` (`interval_iou`?`match_intervals`?`score`) grades all three of: product-vs-GT, vendor-vs-answer-key, and annotator-vs-annotator (IAA). No new scoring code ? only a new *source* of intervals.
13
14 **Decisions locked with the user (2026-05-30):**
15 - **Licensed/owned video only** ? vendors never see user uploads (this is the monetization **privacy lever**, see [[monetization-plan]]). Start corpus = the user's own growing match collection + ShuttleSet.
16 - **Rally `[start,end)` depth** (cheapest; extensible later to shot-counts/(x,y)).
17 - **Tiered turnaround:** slow human tier (authoritative) + fast automated tier (CI / "surprise regression"). ?? **Oracle rule:** the fast-tier auto-labeler MUST be a different model lineage than production, else shared blind spots ? false green.
18 - **Vendor bake-off**, favouring **Bangalore/India-local** (user is Bangalore-based). *(Superseded 2026-06-02 ? started as a single-vendor **Roora** pilot; see Status below.)*
19
20 **Architecture (partly built ? 2026-06-02):** the `AnnotationProvider` registry + `FileProvider` (GS-1), the `regress(video)` runner (GS-2 ? `backend/eval/regression.py` + `run_regression.py`), and the `AutomatedProvider` oracle scaffold (GS-4) have **landed**. The vendor-dependent **`HumanVendorProvider` (GS-3)** is the remaining slice, parked until the Roora pilot confirms quality.
21
22 **Vendor leads (verified India presence + video; pricing NOT verified ? all quote-only):** primary = **iMerit** (Bengaluru office, sports domain, enterprise QA) and **Taskmonk** (Bengaluru, platform+service, likely cheaper/agile); heavyweight third = **TELUS/Playment** (Bangalore-founded CV). Excluded: **Karya** (Bengaluru but speech/text only, no video). Tier-2 leads + confidence labels in the doc. (PR #12.)
23
24 **Status (2026-06-02):**
25 - **Vendor selected for the initial pilot: Roora** (Roora ML Pvt Ltd, HSR Layout, Bengaluru; `info@rooragrp.com`) ? site/capability (image/video/text annotation; sports industry) and location verified; **quality TBD via the pilot**. Run as a **single-vendor pilot first**, NOT a multi-vendor bake-off; iMerit/Taskmonk/TELUS are the fallback if it misses the quality bar.
26 - **Annotation guideline authored** (Step 0): collector repo `docs/annotation/` ? `ROORA_BRIEF.md` (rally CSV schema: `rally_number,start_time,end_time,shots_count,ending_reason`, mm:ss?seconds), `ANNOTATION_GUIDE.md` (illustrated; three golden rules = ignore dead time / include EVERY rally incl. service faults & lets / shot = one racket-paddle-bat contact; `ending_reason` = forced/unforced-error taxonomy (`winner | forced_error | unforced_error | service_fault | let | other`, shared across net-separated sports; superseded observational `into_net`/`out` 2026-06-06); **singles & doubles** for badminton & pickleball), + intake-manifest & NDA templates.
27 - **Pilot scope = 3 sports** (badminton, pickleball, table tennis), scaling to ~12?15 videos/sport. The **collector now ingests all three** via `import-local` (collector ADR-001 keeps explicit per-sport code over a registry; collector test coverage 87%).
28 - **Still pending:** the internal **gold key** (5?10 expert-labeled clips) and **GS-3 `HumanVendorProvider`**.

29

6. user (2026-06-12T09:03:34.815Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-467 of 1379 total (62639 tokens, cap 25000). Call Read with offset=468 limit=467 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1	---
2	name: current-state
3	description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
4	metadata:
5	  node_type: memory
6	  type: project
7	  originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8	---
9
10	# Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12	**Checkpoint:** 2026-06-11 (UI-alpha **PR-2 #99 MERGED + docs-sync #100 MERGED** ? owner authorized merge w/ due
13	diligence; master `020ff61`). **LIVE E2E AT MERGE (server up on :8123, real artifacts):** 86MB smoke video uploaded in
14	3 sequential parts ? reassembled to **MD5 byte-identity** with the source; negatives verified (out-of-order 409,
15	bad-ext/traversal 400, unknown-id 404; download traversal blocked at BOTH layers ? handler 400 on backslash form,
16	router 404 on %2F form); **compile of the real confirmed rally ? 3.9MB reel streamed via GET /api/highlights** (the
17	fixed segment_ids path works browser-equivalent). #100 = doc due diligence: NEXT_STEPS (PR-2 merged + sweep recorded),
18	UI plan PR-2 checked off w/ E2E evidence, **BACKLOG shlex item closed** (residual tenancy note kept). The CUJ runbook
19	(PR2_HANDOFF §"Tomorrow-morning") is now fully unblocked for the owner. ?? NEXT: owner CUJ; PR-3 identity (remember:
20	`allowed_video_root` must contain `upload_dir`). Original handoff context below:
21	(superseded) **Checkpoint:** 2026-06-11 (UI-alpha PR-2 executed ? #99 was awaiting review ? per the Cowork handoff
22	`scratch/PR2_HANDOFF.md`; protocol = Cowork designs, Claude Code runs mechanics, **Avi reviews/merges ? do NOT
23	auto-merge #99**). #88 confirmed CLOSED ? PR88_FIX skipped. Cowork's uncommitted B2+B3 implementation verified to
24	layer cleanly on post-sweep master (AI-7 CODE_MAP entries + pydantic fixes intact) ? Step 0 run here (Cowork sandbox
25	down): suite **388 green** (15 new `tests/test_upload_download.py`), mypy clean; 3 trivial ruff B904s in the new
26	endpoints fixed per the handoff's trivial-fix rule (exception chaining only, no design changes). Branch
27	`feat/upload-download`, exactly the 8 handoff files committed ? **including the two long-preserved doc edits**
28	(DEFERRED_HARDENING_PLAN #16-banner + UI_ALPHA_EXECUTION_PLAN working-agreement rewrite), which are now COMMITTED
29	(tree finally clean of them). CI 4/4 green on #99. ?? NEXT: owner reviews/merges #99 ? then the tomorrow-morning CUJ
30	runbook in the handoff (record ? localhost:8000 ? path-paste ? index ? compile ? ? download; gemini default,
31	rotated key in .env, ~$0.05/run). PR-3 = per-user dirs + allowed_video_root?upload_dir (hosted-mode note).
32
33	**Checkpoint:** 2026-06-11 (8?9 QUALITY SWEEP ? COMPLETE ? all approved AIs landed; master `31069cb`). Final stretch:
34	**AI-7 hybrid-twin collapse** = shared `segmenters/trajectory_hybrid.py` engine (tracknet/wasb now ~40-line subclasses;
```

one `_shot_count_from`; `_wsl_bash` ? `detectors/base.py::WslCommandMixin`; DetectorRunner ABC now declares the parse/windowing contract; mypy quarantine ratcheted ?2; 13 equivalence tests; CODE_MAP 2.1/2.2 updated). ? PROCESS SLIP (disclosed): AI-7 commit `2c241a8` went DIRECTLY to master (post-#96 script left checkout on master) ? content was pre-approved + locally verified; master CI green post-hoc; chose NOT to force-rewind (concurrent sessions live).
- **QUALITY CHECKPOINT (owner-mandated) PASSED:** offline LOVO reproduced **EXACTLY 0.626** (Mahadevpura Gen-0 0.744) post-sweep ? featurizer/eval edits numerically inert. **Live smoke = wasb_hybrid's FIRST-EVER verified E2E run** (3-min Mahadevpura slice): P1 healthcheck?P2 trajectory (9 candidates, correct video_id keying)?P3 Gemini (503 capacity wave; 6-attempt backoff worked; 1 segment confirmed)?P4 add_play_segment + candidate link (`wasb-hybrid-gemini-flash-latest`). 1/9 confirms = Gemini weather, NOT code.
- ? Smoke found 2 REAL pre-existing live-path bugs (masked: trajectories were always hand-scripted, tests mock WSL) ? **#97 MERGED**: (1) shlex.quote strangled `~` expansion in staging (the BACKLOG "shlex-quote breaks ~" item ? CLOSED; fix = `wsl_tilde_quote`, quotes only after `~/`); (2) relative `output/wasb` resolved against WSL cwd ? 8-min GPU run wrote CSV inside WSL invisibly (fix = abspath before /mnt translation). Regression tests pin both command shapes.
- Merged this sweep: **#90** hygiene · **#91** pydantic model_dump · **#92** RUFF lane (+194 fixes; DELETED never-compiled scripts/generate_khelacharya_flyer.py ? line 39 was a literal AI-placeholder sentence) · **#93** MYPY lane (7-module quarantine w/ ratchet rule) · **#94** SQLite conn-leak fix (340?4 ResourceWarnings) · **#95** coverage floor 70 (CI-measured 72%) · **#96** config unknown-key warning · **#97** WSL tilde/abspath · (**#98** Node-24 actions bump = owner-run chip session). Suite **373 green**; CI = 4 lanes (smoke/lint/mypy/full+cov-floor).
- NEW session gotchas: `Select-Object -Last N` on a background pipe loses EVERYTHING if the process is killed ? pipe long background runs through `Out-File` instead; **sibling sessions flip branches in THIS shared checkout** (the node24 chip did, mid-session) ? re-verify `git branch --show-current` before every commit; idle/suspend KILLS background runs silently (smoke run #3 died mid-Gemini, no notification, stale report on disk).
- ?? NEXT: unchanged owner-gated tracks (UI-alpha PR-2 upload/download per `docs/UI_ALPHA_EXECUTION_PLAN.md`; corpus growth; ship-gate target). Deferred-with-intent from the sweep: ruff I/UP rule groups; yolo_hybrid P3/P4 fold-in; pyproject packaging (AI-8 ? owner deselected); extra='forbid' flip (awaits gemini_* alias retirement).

Checkpoint: 2026-06-11 (CI Node-24 action bump ? **#98 MERGED**, master `7717e15`). GitHub forces actions onto Node 24 from 2026-06-16 (Node 20 removed from runners 2026-09-16); ci.yml pinned checkout@v4 + setup-python@v5 (Node 20). Verified latest majors via `gh api releases/latest`: **checkout v6** (v6.0.3 ? NOT v5 as the run-log warning suggested) + **setup-python v6** (v6.2.0). Bumped both in all 4 lanes; all green on the PR (lint 11s / mypy 29s / smoke 15s / full+cov 1m27s), squash-merged on verified rollup. PR #97 (fix/wsl-tilde-quoting) was open + untouched throughout; working tree restored to it (PR-2 doc edits intact).

Checkpoint: 2026-06-11 (8?9 QUALITY SWEEP IN PROGRESS ? owner approved 8 AIs via question UI; AI-8 packaging NOT selected). **MERGED on green CI: #90** (hygiene: .coveragerc repointed post-#40 +

TEST_RUN_SUMMARY ? docs/archives), **#91** (pydantic .dict()?model_dump, 7 sites, warnings 15?0), **#92** (? RUFF LANE: ruff.toml E/F/W/B py38 + fix 194 violations + `lint` CI job; **DELETED scripts/generate_khelacharya_flyer.py ? never compiled, line 39 was a literal AI-placeholder sentence**, invisible because nothing swept scripts/), **#93** (MYPY LANE: mypy.ini lenient green baseline, explicit 7-module quarantine w/ ratchet rule ? main.py+5 segmenters+registry; fixed 18 annotation-level errors), **#94** (? DB FIX surfaced by coverage: `with sqlite3.connect()` = transaction scope NOT close ? all 24 Database call sites leaked connections; new `_connect()` CM; ResourceWarnings 340?4), **#95** (COVERAGE FLOOR: full-suite lane runs --cov, **--cov-fail-under=70** vs CI-measured 72% ? ratchet-only), **#96** (config unknown-key warning: dotted-path [CONFIG WARNING] for typo'd/dropped keys at all nesting levels; repo config.json verified clean; extra='forbid' still deferred). Suite 349?**351 green**. CI = 4 lanes (smoke/lint/mypy/full+cov). ?? NEXT: **AI-7 hybrid-twin collapse** (shared trajectory-hybrid base, fix shot_count drift, dedupe _wsl_bash, equivalence tests) + **post-AI-7 quality-eval checkpoint** (owner: verify ~0.66-class evals stick; GPU free, Gemini key ROTATED, live calls OK). Deferred-with-intent this sweep: ruff I/UP rule groups; yolo_hybrid P3/P4 fold-in (noted as follow-up).

Checkpoint: 2026-06-11 (PR #88 ? **#89 MERGED** ? KhelSutra UI-alpha docs bundle on master `34cf26c`). #88 conflicted (branch cut from `feat/async-job-model` pre-squash of #87) ? per `scratch/PR88_FIX.md`, recreated as `docs/khelsutra-plan-2` off post-#87 master keeping 6 paths (UI_PROPOSAL, HOSTING_PLAN, UI_ALPHA_EXECUTION_PLAN, mockups/, DECISIONS **ADR-009**, NEXT_STEPS sync); #88 closed + branch deleted. CI green-gated (statusCheckRollup), squash-merged.
- **Dropped from #89, preserved for PR-2 as UNCOMMITTED working-tree edits on master:** (1) `docs/DEFERRED_HARDENING_PLAN.md` "#16 DONE ? MERGED (PR #87)" status hunk (also saved as `scratch/PR2_deferred_hardening_status.patch`); (2) `docs/UI_ALPHA_EXECUTION_PLAN.md` working-agreement rewrite (Cowork designs + leaves `scratch/PRn_HANDOFF.md`; **Claude Code executes all git/PR mechanics** ? ratified 2026-06-11). Don't `checkout --` these away; they commit with PR-2.
- ?? Next UI-alpha slice = **PR-2 (upload/download)** per `docs/UI_ALPHA_EXECUTION_PLAN.md`.
- ? **SLATE CLEANED (2026-06-11):** indexer = NO open PRs, master==origin (`34cf26c`), **all 23 stale local branches pruned** (every one mapped to a MERGED PR #62?#87 before `-D`); only `master` remains. Collector = NO open PRs, on master. Intentional carryovers (NOT mess): the 2 uncommitted PR-2 doc edits (above), untracked `scratch/` handoffs, untracked `artifacts/rally_tal_gen0/0.1.0/` (Gen-0 bundle, ADR-008). ?? Collector has a stray uncommitted `.gitignore` edit (adds `artifacts/` ignore w/ ADR-008 comment) ? mega-session leftover, owner to commit (micro-PR) or fold into the next collector PR.

Checkpoint: 2026-06-10 (GEMINI BATTERY COMPLETE ? full coverage; owner's billing was fine, postpay active; ALL failures were burst/capacity artifacts). Merged this stretch: **#80** (chunker re-encode/downscale, from the chip session ? reviewed+merged), **#81** (rescued the chip branch's orphaned run_signals test via cherry-pick), **#83** (provider: 6 generate attempts exponential backoff+jitter ? saved chunks live), **#84**

(tracked
105   `eval_baselines/gemini_wrapper_2026-06-10.json`). Suite 336 green.
106   - **FINAL QUALITY TABLE (IoU .5 vs human golden labels):**
107   - **Mahadevpura** (clean-ish, 31 rallies): heuristic 0.657 · Gen-0 0.744 · **two-stage WASB+Gemini refine 0.83**
108   (P .81 R .84, 32 preds) · **full-video wrapper 0.93** (P 1.0 R .87).
109   - **Kushagra** (multi-court, 32 rallies, FULL 7/7 coverage + audio): heuristic 0.157 · Gen-0 0.561 ·
110   **wrapper 0.66** (P .69 R .62, ±1.2s; FPs = background-game pickups + one 53s dead-time merge ? exactly the
111   contrast-metric confound).
112   - **STRATEGY (refined):** clean footage commoditized (0.93 cheap); **multi-court drops the wrapper to 0.66 = the
113   owned model's ship target** (Gen-0 is 0.10 behind at n=6 ? within corpus-growth reach). Two-stage refine quality
114   (0.83) < full wrapper on short videos; its value = COST on long academy tapes (only candidate clips uploaded).
115   Wrapper reliability is genuinely flaky (503 waves all session) ? provider-fallback architecture is load-bearing.
116   - Artifacts: `output/mahadevpura_gemini_refined.csv`, `output/MahadevpuraSingles_highlights.mp4` (demo reel),
117   Kushagra preds under the ORIGINAL corpus MD5 now. Total Gemini spend today: ~$1-2.
118   - PENDING/owner: rotate the chat-pasted key; ship-gate F1@tIoU decision (multi-court target now has a number: 0.66);
119   C13 interviews + camera pilot; repo rename.
120
121   **Checkpoint:** 2026-06-10 (GEMINI BASELINE LANDED ? owner's "gradual ramp-up" theory CORRECT). New key in gitignored
122   `.env` (rotate later ? pasted in chat). The "credits depleted" 429 was a BURST-THROTTLE artifact: text ping + serial
123   upload worked ? **PR #79 merged**: gemini segmenter now honors `indexing.ai_max_workers` (was hardcoded 5; config ships
124   1=serial ? at 5 workers the same video 429'd, at 1 it worked).
125   - ? **WRAPPER BASELINE (gemini-flash-latest, ~$0.2-0.5 total): Mahadevpura F1 0.93** (P 1.00 R 0.87, mIoU 0.80, ±0.7s;
126   3 of 4 misses = short rallies Gemini FOUND but our min_segment_duration=3.0 rejected) vs Gen-0 learned 0.744 vs
127   heuristic 0.657. **Clean footage = commoditized.**
128   - ?? **MOAT TEST (Kushagra multi-court, windowed n=12 GT): wrapper F1 0.56** (P .54 R .58 mIoU .69) ? Gen-0 learned
129   0.561, both >> heuristic 0.157. **The multi-court confound hurts Gemini too ? the wedge survives on the target
130   footage class.** Caveats: only chunks 2+4 answered (5/7 lost to GENUINE gemini-flash-latest capacity 503s ? retry
131   off-peak for full coverage); visual-only (audio stripped in downscale); 1280p re-encode (4K chunks were 1.3GB >
132   500MB upload cap ? the audit's silent-oversize-skip fired live; fix = chip task_44e72c41 ? see PR #80 checkpoint
133   below). Kushagra preds keyed under downscaled-file MD5 (`output/kushagra_1280.mp4`).
134   - ? **DEMO ARTIFACT: `output/MahadevpuraSingles_highlights.mp4`** ? real 27-rally reel compiled by VideoCompiler
135   (the once-broken module) from Gemini segments. The academy-pitch object exists.
136   - **STRATEGY READ:** product path = Gemini for clean footage TODAY (cheap, F1 0.93) + owned model's job = multi-court
137   (0.56 is the bar to beat there, not 0.93); 503 flakiness = real reliability cost of the wrapper (provider-fallback
138   story strengthens). PENDING: gemini_refine two-stage (503-gated, run off-peak), full Kushagra re-run, PR #80 merge.
139
140   **Checkpoint:** 2026-06-10 (4K gemini oversize fix ? **MERGED: #80 + #81 + #82; master `f01387e`, CI rollup SUCCESS**).
141   Root cause of the live "0 segments / success" on KushagraYashAviFirstVideo.MP4 (4K): gemini segmenter stream-copies 90s
142   chunks ? all 7 at 832?1299MB > provider `MAX_UPLOAD_MB=500` ? every chunk silently skipped. Fix (3 layers): (1) chunks

RE-ENCODED + downscaled to new declared knob `indexing.ai_chunk_scale_width` (default 1280 = gemini_refine's `--clip-scale-width`; 0 = legacy stream-copy; also fixes keyframe drift); (2) skip is LOUD ? provider `metrics["oversize_skipped"]` + segmenter aggregation (chunked + single-file) ? `logger.error` + NEW thread-safe `backend/reporting/run_signals` channel (incr/pop by video_id; popped unconditionally in `emit_indexer_report`); (3) report flags it ? `ai_handoff.details.oversize_clips_skipped` + spec.yaml ERROR rule `oversize_clip_skipped`. Suite 325?**336 green** locally (CI runs fewer: reporting tests importorskip obsreport, absent there BY DESIGN ? `tests/test_run_signals.py` deliberately has NO importorskip = the CI-visible half of the channel). **Merge dance (process note):** owner merged #80 mid-session (auto-deleting the remote branch) WHILE the follow-up test commit was being pushed ? push re-created the branch orphaned; owner rescued it as #81, 39s before my cherry-picked #82 ? **#82's squash `f01387e` = EMPTY commit** (harmless dup in history). Lesson: re-check `gh pr view` state before pushing more commits to a PR branch ? owner + sibling sessions work concurrently (this checkpoint file too: another session prepended the GEMINI-BASELINE checkpoint mid-edit). ? **OWNER/SIBLING WIP IN TREE (uncommitted, do NOT clobber):** `backend/providers/gemini.py` ? generate retries 3?6 w/ exponential backoff + jitter (~3?96s) for the 503 capacity waves. ?? commit/PR the 503-backoff WIP · machine-side 4K re-run (chunks should be ~tens of MB) · task-17 battery still pending Gemini billing top-up.

Checkpoint: 2026-06-10 (M0 GREENLIT SEQUENCE EXECUTED ? session wrap). Owner ratified ADR-008 ("split driven by productionization, not isolation") + greenlit M0.4+M0.1. **All 5 steps shipped, every PR merged on verified-green CI:**

- **#73 ADR-008** (DECISIONS.md): training in-repo behind an executable wall; artifact-bundle boundary; ship-gate product-side; pre-committed split trigger ? `sports-temporal-models` at Gen-2/second-consumer.
- **#74 canonical GT loader** `backend/eval/gt_loader.py::load_gt_intervals` ? closes the two-loader fork (accepts BOTH `start,end` and `start_time,end_time`); legacy loaders delegate; round-trip contract test; LOVO 0.626 re-verified.
- **#75+#76 featurizer promotion + import wall**: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1, fps semantics explicit) imported by train AND serve; `training/` skeleton (5 wall rules, requirements-train.txt); smoke tests: runtime no-training-in-backend.main, static sweep, featurizer purity (torch/cv2 blocked; importorskips numpy on the minimal CI lane ? #76 hotfixed a red lane after #75 merged early; merges now gated on verified SUCCESS rollup).
- **collector #19 ? n=6 OWNED CORPUS IN THE SoR**: all 6 golden matches via `import-local --normalize` (262 rallies, 0 dropped, Proprietary, explicit fps); commercial export now 6 real matches + demo entry.
- **#77 Gen-0 harness** `python -m training.gen0.harness --manifest ? --floor eval_baselines/heuristic_lovo_n6.json --version 0.1.0`: train?honest LOVO?ship-gate?artifact bundle (gitignored weights.json + manifest.json w/ schema pin, calibration, lineage gt_hashes, attestations incl. C3, gate verdict). **First real bundle built: LOVO 0.626, floor PASSED (0.480), paired sign test 5/6 wins p=0.109 = honestly NOT significant, ship=False** (blockers: target undefined, C3, significance). Committed floor baseline `eval_baselines/heuristic_lovo_n6.json`. Suite **325 green**.
- ? **Gemini baseline STILL BLOCKED after owner provided a NEW key (2026-06-10 later):**

key written to gitignored `.env`, validated via `models.list` (55 models incl. `gemini-flash-latest`; **config migrated off deprecating gemini-2.5-flash ? `gemini-flash-latest` alias, PR #78 merged**; gemini_refine already defaulted to it). But inference = 429 RESOURCE_EXHAUSTED "prepayment credits depleted" ? **the AI-Studio PROJECT behind the key has no credits** (key auth ? billing). Owner: add credits at https://ai.studio/projects (+ consider rotating the chat-pasted key). Planned battery on unblock (in task 17): wrapper Mahadevpura ? score; wrapper Kushagra (multi-court MOAT test); two-stage gemini_refine; compile a real highlight reel (academy-demo artifact). Pipeline handled both failed runs gracefully (validators pass, reports emitted, artifacts cleaned); shared-genai-client thread-safety finding observed live.
- **C13 field plan (owner executing):** recruit interviewer; camera idea RATIFIED w/ 3 constraints (consent-at-capture/ adults-only for training; demo via Gemini-or-human-polish, NOT gated on owned model; interviews?cameras sequencing).
- ?? NEXT: owner tops up Gemini billing (wrapper baseline) · owner sets the ship-gate F1@tIoU target · grow corpus (camera pilot videos ? import-local ? re-run harness) · repo rename · M0.1 event-index contract in collector.

Checkpoint: 2026-06-10 (Roora PB+TT pilot review ? REPORT DELIVERED to owner; owner emails Roora himself) · ? Owner cross-checked the rally tables in VLC ? **CONFIRMED: every boundary off ?0.5s (up to ~2s)** vs the ±0.3s bar (Brief §4). **Root cause (my forensics): all 24 boundaries sit EXACTLY on the 30-frame sampling grid** (~0.5s at 59.94fps) ? first/last box never on the true contact/dead frame ? structurally cannot meet ±0.3s. Also decoded their bogus `start/end time` attrs = **frame÷900 (decimal minutes @ assumed 15fps, ~4× off wall-clock)**. Report findings: boundary accuracy (CRITICAL), skip-rallies-in-progress-at-file-start/end convention (PB rally#1 starts at frame 0 ? shouldn't have been labeled), dead-time+warmup reminders for Phase-B, schema cut to exactly `rally_number,shot_count,ending_reason,last_shot_outcome,note` (drop their time attrs + scores), field quality (PB outcome='0' off-vocab; TT 5/6 'other' + missing required notes; TT#5 intra-rally other|winner). Per owner: NO redelivery demand in the report (he words the email himself; thread = Jamshad@roora on avidullu@live.com Outlook). **Artifacts:** PDF `Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Pickleball and Table Tennis (2026-06-10).pdf` + md record at `Annotation Setup\Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md` (generator: `%TEMP%\roora\make_review_pdf.py`). ? Local guideline docs are **DRAFT v2** ? owner mentioned "v7"; no v7 found on disk (possibly in the email thread; report cites v2). Spawned chip `task_c55f0be6`: collector `rally_cvat.py` ATTR_MAP drops `shot_count` (present in Roora XML, empty in our CSVs) ? OUR bug, kept out of the vendor report. ?? owner sends email; Roora's badminton long-video + redelivery decisions pending owner.

Checkpoint: 2026-06-10 (REPO-TOPOLOGY decision panel + C13 plan ? AWAITING OWNER RATIFICATION). Owner leaning M0.4+M0.1 greenlight; asked: model in-repo vs separate repo consumed as a segmenter (owner prefers isolation). Ran a 4-agent judge panel (`wf_471a9cdd-d1d`; full output `?\tasks\wnksv545g.output`). **Verdict (my rec): "artifact boundary NOW, repo split at Gen-2"** ? M0.4 lands as a top-level `training/` tree in the indexer w/ executable isolation (import-smoke asserts backend.main loads no training module; own requirements-train.txt + 3rd CI lane), featurizer promoted to `backend/features/rally_seq.py` (single source, train==serve), artifact bundle = gitignored weights (private Releases) + committed manifest (feature-schema ver, normalization, calibration, corpus ref+gt_hash, consent/license attestation incl. C3), ship-gate stays product-side (regression.py extension); **pre-committed split trigger** to a named repo (`sports-temporal-models`) when Gen-2/cloud-training or a 2nd artifact consumer arrives. Decisive evidence: Gen-0's input = the indexer's OWN featurization (mid-pipeline seam ? WASB/Gemini video-boundary); indexer not pip-installable ? split today = copy-paste = the observed drift class; obsreport "precedent" never actually exercised (pin is a commented line, repo unpublished, CI skips). **FIRST contract to freeze: golden-label/corpus contract** ? indexer carries TWO incompatible GT loaders (annotations.py `start,end` vs calibrate_wasb `start_time, end_time` ? fed to 7 modules) vs collector export `start,end`; consolidate + round-trip test BEFORE any training epoch. Panel extras: n=6 corpus's durable home = loose `Annotation Setup\` folder (neither repo!) ? M0.1 first deliverable = hash-addressed in collector; training fixtures synthetic-only until consent ledger exists; run the Gemini-wrapper baseline before/with M0.4. **C13 field plan:** owner recruiting an interviewer; my rec = bake in the camera idea (paid recordings = consented owned multi-court corpus + demo), demo served via Gemini path or human-polish (owned model honest 0.626 ? >0.8 yet; the comparison doubles as the wrapper baseline). ?? awaiting owner ratification of topology + M0.4/M0.1 scope; then: ADR + GT-loader consolidation + featurizer promotion + harness scaffold.

Checkpoint: 2026-06-10 (CORPUS RECLASS EXECUTED + NEXT_STEPS refreshed ? session wrap). Owner approved + I ran the corpus correction: `reclassify-licenses --apply` (23 ShuttleSet broadcast rows ? annotation_license=MIT, video=Unknown, is_commercial=False; **zero annotations deleted**, 1,917 rallies intact) + `curate export` (commercial manifest now owned-data-only: 1 video/2 segments; stale ShuttleSet CSVs removed from the gitignored eval_export; eval-only re-export available via `--include-staging --include-noncommercial`). **Merged: collector #17** (DB + .gitignore backup rule); pre-change backup at `data/sports_metadata.backup-2026-06-10.db` (gitignored). Quality signals UNAFFECTED (n=6 corpus never flows through the collector DB; learned 0.626 re-verified earlier). **Indexer #72 merged:** NEXT_STEPS.md refreshed (sweep banner, 0.626 vs 0.480, M0.3 partially-done with corrected distill_local pointer, NEW prioritized PMF-validation item, durable artifact paths). **Owner is now reading NEXT_STEPS to chalk out next steps** ? open owner decisions stand: greenlight M0.x (rec M0.4+M0.1), confirm collector=SoR, ship-gate F1@tIoU, repo rename, C13 interviews / Gemini-wrapper baseline. Both repos clean on master, 0 open PRs.

Checkpoint: 2026-06-10 (HIGH-RISK REMEDIATION COMPLETE ? owner approved "everything A?K, open+merge PRs after CI green, re-run quality evals, GPU available"). **ALL 11 PRs MERGED + CI green; both repos clean on master.**
- **Indexer (badminton-highlight-indexer):** #63 compiler-syntax fix+CI (prior), **#64** detector keying by video_id,
  #65 re-ingest destroy-after-confirm (add_video UPSERT + watermark/prune), **#66** nested-config wasb typed,
  #67 eval gate + metric footguns (iou=0, midrank AUC, fps-normalized speed, baseline?`eval_baselines/` exit-3),

#68 API path validation, **#69** reliability (timeouts 1800s, SQLite WAL+busy_timeout, chunk-step guard, CORS),
#70 test coverage (TestClient endpoints + compiler + WasbRunner), **#71** docs truth-sync. Suite 256?**308 green**.
- **Collector (sports-data-collector):** **#14** licensing gate (video vs annotation license; `reclassify-licenses`
found 23 conflated ShuttleSet rows) + **added collector CI**, **#15** import-local golden-label guard (+`--force`,
`--normalize`, duplicate-rally guard), **#16** validate-delivery fails on empty/corrupt/zip CVAT. Suite 124?**136 green**.
- ? **QUALITY RE-VALIDATION (owner asked): metric fixes STRENGTHENED the conclusion.** After midrank-AUC +
fps-normalized speed + occlusion-gap guard, **learned LOVO 0.583 ? 0.626** (per-video AUC unchanged 0.83?0.92 =
never tie-inflated; fps-fix mainly helped the 30fps short fold 0.30?0.56). Heuristic LOVO **0.480** unchanged (scored
at IoU 0.5; my iou>0 guard is a no-op there). **Learned beats heuristic by +0.146** (was +0.10).
- **Eval artifacts RESCUED from `%TEMP%` ? durable `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\`**
(+ `\roora\`); manifest `output/human_lovo_manifest.json` repointed there + reverified 0.626.
- **Deferred WITH INTENT (in BACKLOG): repo-rename; Gen-2 cost benchmark; ship-gate F1@tIoU target; packaging
(pyproject); app-factory/lifespan + lazy torch-cv2 imports (with async slice #16); WSL temp/frame/cache + Gemini
remote-file GC (retention policy); shlex-quote WSL config values (breaks `~` expansion ? non-trivial); deeper
validator tests + 3-hybrid contract test (partial coverage shipped).**
- ? **Process note:** during PR-F I briefly `git checkout --`'d the owner's uncommitted `data/sports_metadata.db`
change (200704b), then RECOVERED it from dangling stash `d3c2e77` and restored it to the working tree (no loss).
Going forward stash-and-pop that DB for collector PRs; never `checkout --` it. The corpus-DB reclassify (`curate
reclassify-licenses --apply` + `curate export`) is left for the OWNER to run deliberately (binary SoR data).
- Plan doc (reference): `scratch/HIGH_RISK_REMEDIATION_PLAN.md` (untracked). Audit JSON: `?\tasks\wf38bovts.output`
(batch 1) + `?\tasks\wa4d2980l.output` (full 8 auditors).

Checkpoint: 2026-06-10 (compiler SyntaxError FIX ? **PR #63 MERGED**, squash `master 13d4573`; owner approved merge-on-green) ·
? **master UNBROKEN again ? fixed the ? SyntaxError from 7fbbeb0** (= the audit's fix chip `task_2b7a19d4` work):
restored the f-string quotes on the 7 mangled lines of `backend/pipeline/stitcher/compiler.py`
(100/104/156/160/181/191/213; message texts verified against the pre-conversion file at `7fbbeb0^`; line 191 =
plain string, no placeholders) + fixed line 168 `f"\nSUMMARY]` ? `f"\n[SUMMARY]`. **NEW regression guard
`tests/test_import_smoke.py`**: (a) py_compile sweep over every file in backend/ (imports nothing ? runs on
cv2-free machines), (b) subprocess `import backend.main` with missing cv2/torch/numpy stubbed. Negative check:
the broken compiler.py fails the sweep (`SyntaxError: invalid decimal literal`, line 100).
? **CI NOW EXISTS ? the audit's no-CI finding is CLOSED** (`.github/workflows/ci.yml`, runs on every PR + master
push): job 1 **import-smoke** = smoke tests on a GPU-stack-free runner (minimal dep set verified empirically in a
fresh venv: pytest/fastapi/uvicorn/pydantic/python-dotenv/**google-genai** ? needed at import time by
backend.providers); job 2 **full-suite** = requirements.txt with **CPU-only torch wheels**

(`--index-url download.pytorch.org/whl/cpu`) + libgl for opencv ? `python run_all_tests.py` (obsreport absent ? reporting tests skip by design). **Both jobs GREEN on the first run** (27s / 1m29s). Verified post-merge: `import backend.main` OK on master, suite 256 green (254+2). ?? Audit's **coverage-inversion** finding (wasb_hybrid/validators untested) remains open ? CI only guards the import/syntax class + whatever the suite covers.

Checkpoint: 2026-06-10 (DEEP AUDIT session ? ultracode 18-agent audit `wf_2a64bc88-2a8` over indexer+collector: engineering/testing/reliability/docs-consistency/PMF; 3 of 8 auditors completed+verified, 5 rate-limited).
- ? **CRITICAL ? master is BROKEN since 7fbbeb0 (2026-06-07 print?logger sweep):** `backend/pipeline/stitcher/compiler.py` has **7 unquoted f-string lines (100/104/156/160/181/191/213)** ? SyntaxError ? `import backend.main` fails ? FastAPI server + `/api/compile` dead. Triple-verified + reproduced first-hand (`py_compile` exit 1). The **254-green suite never noticed** (zero tests import backend.main / compiler / wasb_hybrid / motion / any concrete validator; **no CI exists at all**). Spawned fix chip `task_2b7a19d4` (fix + import/compileall smoke test). Coverage is INVERTED vs production risk (tracknet twin 8 tests vs live wasb_hybrid 0; TrackNetRunner 21 vs WasbRunner 1). Suite itself healthy where it exists (254 green / 19.9s / faithful mocks / low flake risk); "fully mocked runs anywhere" is false on cv2-free machines (bare `import cv2` at collection time, only obsreport importorskips).
- **Doc drift (verified, one-directional ? newest decisions not back-propagated):** (1) licensing "Apache-2.0 only" still in CLAUDE.md:33 / PLATFORM:269,301 / COMMERCIALIZATION C14 vs ratified **MIT/Apache/BSD** (a literal CLAUDE.md reader rejects the plan's own MIT ActionFormer/ASFormer heads); (2) **Gemma-4 still "approved"** in those same docs vs AMBER/bench-only in the plan; (3) **M0.3 purge pointer WRONG** ? `training.label_candidates` is a pure GT-agnostic fn; the live Gemini-silver training violation = `backend/eval/distill_local.py` (+ calibrate_local thresholds) + gitignored `output/local_rally_model.json`; (4) **C3 escape hatch broken** ? "train-own weights (Phase-4 roadmap)" cites a phase that contains no such plan and ADR-002 says corpus labels are temporal-only (CAN'T train a detector) ? own-weights is an unscoped new workstream (coordinate labels or detector swap); (5) OWNED_MODEL_IMPLEMENTATION_PLAN + NEXT_STEPS are linked from NO entry point (CLAUDE.md "read first" / docs/README index) and NEXT_STEPS is branch-only; (6) README "yolo_hybrid by default" vs config.json `gemini` vs CODE_MAP "wasb = LIVE default"; (7) ROADMAP still says audio "gated on golden labels" (trigger now satisfied!) vs plan "parked, not in reserve"; (8) Roora multisport CSVs live ONLY in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ? headline TT-0.909/pickleball-0.308 evidence is one cleanup from loss; (9) TT-on-WASB idea silently walks into the C3 ship-gate ("multiplies per sport"); (10) ship-gate F1@tIoU still undefined while 0.44/0.54/0.480 floors circulate; repo-rename "ASAP" tracked in no tracker (BACKLOG lacks it).
- **PMF (web-grounded, sources in audit output):** pricing/cost benchmarks VERIFIED (SwingVision $179.99/yr ~20k paying subs rev $2.75M +12% YoY; PB Vision $99/yr + **$8/hr-of-video partner API**; Gemini COGS ~$0.30?0.60/hr ? fat margin); but **"India badminton whitespace" FALSIFIED ? Bangalore is the most contested badminton-AI city**: Machaxi ($1.5M Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School), Game Theory (Gopichand-backed smart courts w/ auto-highlights), Stupa ($3.3M, BWF-approved, TT?badminton) ? NONE in our

2026-06-07 competitive table.
313 Surviving wedge (narrow, real): **software-only rally-precise indexing of EXISTING messy multi-court BYO footage**
314 (market solves it only with per-court hardware) sold per-hour as engine/API, possibly THROUGH those players
315 (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) + **Gemini-wrapper baseline vs the
316 n=6 LOVO corpus** (if wrapper ? learned 0.583, owned-model is premature) + 1 paid academy pilot.
317 - **AUDIT NOW COMPLETE (8/8 auditors, resumed `wa4d2980l` after Max-20x refresh; 40 agents).** New high/critical findings
318 beyond the first batch:
319 - ? **CRITICAL (collector) ? LICENSING GUARDRAIL VIOLATION:** `src/cli/ingest.py:12` defaults `--license MIT`; the
320 ShuttleSet **annotation** license is stamped onto **YouTube broadcast video** (e.g. shuttleset_21 = a YONEX Thailand
321 Open broadcast, `is_commercial=true, license_type=MIT`). **24 videos / 1915 segments currently exported as
322 "commercially-safe" on unsound grounds** ? directly violates the "training data must be commercial-clean" guardrail.
323 - **HIGH (indexer, all verified):** (a) **nested config sections silently drop unknown keys** ? only `AppConfig` has
324 `extra='allow'`, so the ENTIRE `indexing.wasb` block (weights/conda_env/velocity_threshold) + `yolo_prefetch_workers`
325 + `debug_duration` are unconfigurable via config.json; meanwhile `yolo_parallel_chunks/yolo_chunk_workers` are read
326 nowhere. (b) **detector artifacts keyed by file BASENAME not video_id** ? two `match.mp4` files reuse stale frames ?
327 a trajectory + candidate_segments computed from the WRONG video (silent training-data poisoning). (c) **re-ingest is
328 destroy-before-confirm** ? `add_video('processed'/'failed')` INSERT-OR-REPLACE cascade-deletes prior good segments
329 BEFORE the new run; a failed/empty re-run leaves zero. (d) **stitching padding config not wired** ? API reels always
330 end_padding=1.0 despite config 0.5. (e) **`/api/process` arbitrary server-file read + `/api/compile` path-traversal/
331 absolute write** via output_filename. (f) **run_regression "CI gate" not in CI + silently passes** on missing baseline;
332 baseline omits iou/detector/GT-hash. (g) **metrics bug**: `match_intervals` at iou_threshold=0.0 matches
333 non-overlapping intervals (P=R=F1=1.0 for garbage); **rally_seq_proto AUC has no tie handling** (order-dependent;
334 all-zero dead-time rows guarantee ties) + speed features fps-inconsistent within one vector.
335 - **HIGH (collector):** silent DELETE-then-INSERT replace of golden labels (no warn/backup, filename-derived video_id);
336 validate-delivery **PASSES on a zero-rally CVAT export**; not yet SoR (no provenance/maturity/labeled_window cols,
337 import-local hardcodes CURATED); `int(float())` rally_number crashes mid-import leaving a committed video row w/
338 rolled-back annotations (normalize_annotations guard is opt-in, NOT wired into import-local).
339 - **MEDIUM batch:** WSL `timeout_sec=0` = hang-forever (both runners + provider PROCESSING poll); `uvicorn backend.main:app`
340 leaves globals None (init only in main()); SQLite no WAL/no busy_timeout + swallow-on-lock; chunk_overlap?duration =
341 infinite loop; WSL temp/frame/cache + Gemini remote-file leaks; CORS `*`+credentials; sport-profile relevance config
342 dead under typed config (bites first non-badminton sport ? the current branch!).
343 - **VERDICT NUANCE:** verifier DOWNGRADED 2 "high"?not-real (Result-contract gap + DB-swallow are DOCUMENTED CODE_MAP
344 gotchas/deliberate-incremental, medium at most; DB "no busy_timeout" claim factually wrong ? sqlite3 default
345 timeout=5s). Eng-pipeline/eval/collector strengths confirmed: registry seams, DetectorRunner ABC, WASB crash-cache,

346   honest fit-on-self labeling, leakage-free LOVO, frame-true CVAT conversion all genuinely good.
347   - ? **SPOOF-CHECK PASSED ? quality iterations are REAL (re-ran the evals first-hand on this box, pure numpy/scipy):**
348   all 6 videos + 6 human label CSVs + 6 trajectories exist; traj row-counts match docs exactly. **`rally_seq_proto` LOVO
349   reproduced: mean 0.583** (AUC 0.877/0.851/0.916/0.832/0.730/0.902 vs doc 0.879/0.853/0.915/0.833/0.726/0.902).
350   **`calibrate_local` heuristic LOVO reproduced: 0.480 ± 0.242** (per-fold Adarsh 0.751/Kushagra 0.157/doubles 0.737/
351   gBaaddy 0.309/testlarge 0.267/mahadevpura 0.657 ? exact). **Contrast metric recomputed independently from raw CSVs:**
352   3.7/1.4/3.7/1.9/1.4/2.2× vs doc 3.9/1.4/3.6/1.9/1.4/2.2. Opus 4.8 did NOT spoof. ? one nit: gBaaddy CSV has 48 rallies
353   (doc says 47). Caveat: trajectories were WASB-GPU-generated (can't regenerate here) but sizes/frame-counts/visibility all sane.
354   - **HIGH-RISK REMEDIATION PLAN written for owner approval:** `scratch/HIGH_RISK_REMEDIATION_PLAN.md` (12 PRs, sequenced;
355   PR-1 compiler fix already DONE via #63). Awaiting owner scope + push/PR decision before execution. Full audit JSON:
356   `%LOCALAPPDATA%\Temp\claude\...\tasks\wa4d2980l.output` (8 auditors) + `wf38bovts.output` (first batch).
357   Full audit JSON: `%LOCALAPPDATA%\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\tasks\wf38bovts.output`.
358
359   **Checkpoint:** 2026-06-09 (Roora MULTISPORT recon ? for FRESH next session) · ?? **Pickleball + Tabletennis golden labels
360   arrived; INGESTION = SOLVED, but the "merged model" experiment is BLOCKED on per-sport trajectory detectors (validates the
361   roadmap blocker).** Owner left a recon-only ask (deliberately did NOT drain context / spin heavy jobs ? owner wants me fresh).
362   - **Deliverables:** `Annotation Setup/6 June 2026/{pickleball,tabletennis}/*.zip` (3 CVAT export FORMATS each: COCO
363   `instances_default.json`, **CVAT-XML `annotations.xml`** ? use this, Pascal-VOC). Videos: `?/6 June 2026/Annotation Videos/
364   Video 2 - Pickleball.mp4` (4K, 59.94fps, 97s, 5737fr) + `Video 3 - Tabletennis.mp4` (4K, 59.94fps, 58s, 3490fr). Short clips.
365   - **What Roora annotated:** label class "rally" as **per-frame BOUNDING BOXES on sampled frames (~every 30fr)** ? box geometry
366   is just a CARRIER; the boxes carry the FULL rally schema as CVAT **attributes** (rally_number, start/end time, shot_count,
367   score_before/after(A-B), ending_reason, last_shot_outcome [+note for TT]). 456/513 attr instances. Boxed frames cluster into
368   rallies (runs of ~30-spaced frames = a rally; big gaps = dead time).
369   - **INGESTION = YES, cleanly:** the collector's `src/parsers/cvat/rally_cvat.py` (`curate convert-cvat`) is PURPOSE-BUILT for
370   exactly this ? ignores box geometry, **recomputes rally interval from the frame range** (`sec=frame/fps` at true fps, ignoring
371   vendor start/end times which are unreliable), groups boxes by rally_number, maps attrs?canonical. Minor: verify the attr-name
372   mapping handles spaces/"(A-B)" (`ATTR_MAP`). Collector already has pickleball + tabletennis schemas.
373   - ? **"MERGED LEARNT MODEL" across sports is BLOCKED:** `rally_seq_proto` features come from a WASB **shuttle** trajectory =
374   BADMINTON-ONLY; pickleball (wiffle ball) + TT (tiny fast ball on a table) are out-of-distribution ? no clean trajectory yet.
375   This EMPIRICALLY validates the roadmap's binding multisport blocker (clean per-sport MIT/Apache ball detectors not identified).
376   - **PROGRESS (owner asked to complete 1/2/3 before badminton):**
377   - ? **(1) convert-cvat DONE ? INGESTION PROVEN.** `python -m src.cli.curate convert-cvat --input <annotations.xml> --out
378   <csv> --fps 59.94` ? **pickleball 6 rallies + tabletennis 6 rallies**, sane intervals, attrs mapped. CSVs at
379   `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv` (+ extracted XMLs `%TEMP%\roora\{pb,tt}\annotations.xml`). ? Roora data

380 nits (non-blocking): pickleball `last_shot_outcome='0'` (off-vocab in/n_a/net/out); TT `ending_reason` mostly 'other' with the
381 real outcome in `notes` ("out of table"/"net") ? ending_reason taxonomy mapping could improve. Both clips SHORT (97s/58s).
382 - ? **(2) WASB-transfer test DONE (`%TEMP%\{pickleball,tabletennis}_traj.csv`; log `~/clips/multisport_run.log`):** WASB
383 (badminton shuttle detector) detects SOMETHING on both ? **pickleball 28% / tabletennis 39% visibility** (NOT near-zero;
384 comparable to badminton 40?67%). So it's not blind.
385 - ? **(3) DONE ? SURPRISE RESULT (directional, n=6 each, TINY so noisy):** `eval_partial_labels` on (WASB-transfer traj + Roora
386 6-rally CSV, fps 59.94, fw 1920): **TABLE TENNIS best F1 0.909** (tuned+gate2: P1.0/R0.833, 5/6 matched, boundary err 0.49s ?
387 **WASB transfers to TT remarkably well!**); **PICKLEBALL best F1 0.308** (poor ? wiffle ball too out-of-distribution).
388 ? **NUANCES the roadmap blocker: per-sport detector transfer is SPORT-DEPENDENT ? TT may be servable with WASB short-term;
389 pickleball needs a proper detector.** ? BIG caveat: 6 rallies each = could be partly luck ? owner sending more videos to
390 confirm. A "merged" prototype LOVO across badminton+pickleball+TT is the natural next experiment once n grows per sport.
391 Roora LABELS ingest cleanly + are corpus-gold regardless. **Great honest surprise to greet owner with.**
392
393 **Checkpoint:** 2026-06-09 (PR #61 MERGED ? roadmap SET, session wrap) · ? **Owner agreed + merged #61 (`master 4d04a73`).
394 ALL CLEAN: master, 0 uncommitted, 0 open PRs, branches pruned.** The owned-model **ROADMAP is now set + agreed** ?
395 authoritative doc = **`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`** (full verified decisions in the "(PR review)" checkpoint
396 just below: ms-swift + Qwen3-VL, TAL-head-first, conversational-QA seam-now-model-later, multisport one-head/base, Python
397 everywhere, MIT/Apache/BSD ratified, "envision now build later"). [[session-handoff]] rewritten to this phase.
398 - ?? **NEXT SESSION (awaiting owner greenlight ? NO build until then):** greenlight a Phase-0 milestone (rec **M0.1
399 corpus+event-index** ? **M0.4 Gen-0 TAL harness**, both $0/local); confirm **collector = system-of-record**; define the
400 **ship-gate F1@tIoU target** (floor = beat heuristic LOVO 0.480); **rename repo off "badminton"**; M0.3 purge Gemini
401 training labels + WASB C3 (own-weights/clean swap). Corpus = n=6 (`output/human_lovo_manifest.json`); learned prototype
402 is the Gen-0 seed. Committed next PLATFORM slice stays the async job model (single-tenant). Owner flagged context ~85% ? this is the wrap checkpoint.
403
404 **Checkpoint:** 2026-06-09 (PR review) · ? **OWNER REVIEWED PR #61 ? REFRAMED ROADMAP (important checkpoint).** Owner left
405 ~12 inline comments (mostly state-updates + recommendations wanted). **Key roadmap shifts to internalize:**
406 - **Immediate goal REFRAMED:** not "on-device model" but a **TRAINING FRAMEWORK + HARNESS to produce a VERY HIGH
407 precision/recall model that extracts HIGHLIGHTS + RALLIES from a clip.** Quality first. On-device = LATER (compression
408 for power users to save upload cost), definitely on roadmap but not now.
409 - **MULTISPORT FROM THE START** ? all net-separated racquet sports ? eventually all sports; a SINGLE sports-agnostic model.
410 Collector + indexer already sport-generic by design. ? **ACTION: rename the repo off "badminton" ASAP** (owner emphatic).
411 - **CONVERSATIONAL VIDEO-QA = north-star feature:** upload video ? ask contextual questions ("longest rally", "make a clip
412 of all service faults", "rally >20 shots"). Needs per-video STATE/bookkeeping ? design the framework for it. Owner asks:
413 embark now or punt + revisit after quality? (My lean: design the bookkeeping/event-store layer now, punt the heavy
414 conversational model ? pending the strategy workflow.)

415   - **LICENSING GUARDRAIL (owner: "never corrupted"):** EVERY dependency ? code, data, AND weights ? must be usable in
416   PRIVATE PROPRIETARY COMMERCIAL software. **MIT / Apache-2.0 / BSD only.** Ratify (was "Apache-2.0 only"). Enforce religiously.
417   - Owner wants a **coherent recommendation: local Gemma 4 vs ms-swift starting point** (multisport, single, conversational,
418   commercial). NB: ms-swift = training FRAMEWORK, Gemma 4 = BASE model ? orthogonal; clarify in the reply.
419   - Owner: **upgrade WASB C3 (own-weights/provenance) to a first-class action item now.** M0.2 (grow n) = practically DONE
420   (n=6); owner will add 1 more varied multi-court-club video soon ? Phase-0 ready to start.
421   - **LANGUAGE QUESTION (owner callout, no preference):** is Python right for the upcoming work, or brainstorm alternatives?
422   - ? **STRATEGY WORKFLOW done (`wf_f9aa1d1c-c83`, 10 agents, verified+critiqued) ? VERIFIED DECISIONS (now in the rewritten
423   `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`, the authoritative roadmap; pushed to PR #61 + a consolidated PR comment reply):**
424   - **Framework = ms-swift (Apache-2.0)** (only forge w/ native video+audio+timestamp-grounding+video-GRPO); unsloth = 8GB-local
425   SFT/compression tool (avoid its AGPL Studio). **Base = Qwen3-VL (Apache-2.0, timestamp emission)** primary; **Gemma-4 STAYS
426   AMBER/bench-only** (critic corrected the workflow's "clear AMBER" ? posture unconfirmed + 60s/no-timestamp dispute);
427   InternVL3 (MIT) fallback. ms-swift?Gemma4 (framework vs base ? clarified for owner).
428   - **START WITH THE TAL HEAD, NOT THE VLM** (VLMs miss strict temporal boundaries; boundary accuracy = the product; head
429   de-risked AUC 0.83?0.92, $0 local, doubles as teacher+baseline). Head now ? VLM later (two-stage: VLM proposes, head refines).
430   - **Conversational-QA: build the SEAM now (per-video event store + structured text-to-SQL + ffmpeg clip-cut), PUNT the NL
431   model** until quality clears the bar. Queries ("longest rally"/"service faults"/">20 shots") = structured-numeric ? SQL, not
432   a VLM. Reserve fault/event taxonomy in M0.1. ? real DB delta (new cols + a `highlights` table; play_segments lacks them today).
433   - **Multisport: one shared head + one base; scale DATA + per-sport clean trajectory detectors (the binding blocker ? none
434   identified for tennis/TT/pickleball/padel yet); WASB C3 multiplies. Single-model temporal generalization = OPEN HYPOTHESIS
435   (RacketVision evidence REFUTED = ball-tracking not temporal). Repo rename off "badminton" = action item.**
436   - **Language: Python everywhere** (training Python-locked; platform I/O-bound, no hot path ? reject Go/Rust split); on-device =
437   deferred EXPORTED runtime shell (ONNX/CoreML/GGUF), doesn't dictate dev language. Constraint NOW: **train export-clean ops**.
438   - **Honest gaps:** cost-to-Gen-2 UNQUANTIFIED (biggest risk); ship-gate "very high P/R" target undefined (floor = beat heuristic
439   LOVO 0.480 ? set explicit F1@tIoU); per-sport detectors TBD; base PRETRAINING provenance unauditable ? owner risk-acceptance;
440   **Gemini-silver TRAINING labels = LIVE violation ? hard-blocker purge M0.3.**
441   - **Owner principle recorded: "envision now, build later"** ? design every deferred seam (event store, export boundary, sport-
442   agnostic head) up front; punt the feature, never the seam.
443   - ?? **NEXT:** owner reviews the reframed plan + decisions (greenlight a Phase-0 milestone ? rec M0.1 corpus+event-index then M0.4
444   harness; confirm collector=system-of-record). Then start the greenlit milestone. Repo rename. Define the ship-gate F1@tIoU target.
445
446   **Checkpoint:** 2026-06-09 · ? **n=6: MahadevpuraSingles added while owner reviews PR #61.** Owner moved ALL golden videos
447   + CSVs to `Annotation Setup/Collect/Golden Labelled/` (manifest label paths updated; trajectories unaffected in `%TEMP%`).
448   New: **MahadevpuraSingles** (singles, 1080p, 29.97fps, 31 rallies; traj `%TEMP%\mahadevpura_traj.csv`, script
449   `run_wasb_mahadevpura.sh`). **Borderline multi-court (contrast 2.2×)** ? heuristic best

0.667, learned held-out **AUC
450   0.902/F1 0.712** (clean generalization to an unseen match). **n=6 LOVO: heuristic 0.480
± 0.242 vs learned 0.583** (learned
451   holds ~+0.10 since n=3). Contrast metric now predicts heuristic F1 MONOTONICALLY across
5 matches (Mahadevpura's 2.2×?0.67
452   lands on the curve). **Decoupling headline: learned held-out AUC 0.83?0.92 regardless of
contrast; heuristic swings 0.16?0.75
453   with it.** Committed to PR #61 (n=6 doc update). ? GOTCHA: don't combine inner `&` with
run_in_background (orphans the job);
454   calibrate_local buffers output to end + is slow (~17?20min). "Ingested" = added to EVAL
corpus (manifest); collector
455   import-local (system-of-record) still = M0.1, plan-only pending greenlight. ?? owner
reviewing PR #61 (4 plan decisions pending).
456
457   **Checkpoint:** 2026-06-08 (later+4, SESSION WRAP) · ? **n=5 corpus + MULTI-COURT root
cause + heuristic discovery + SINGLE PR #61 for offline review.**
458   Owner tagged gBaaddy (47, 1080p, multi-court) + TestLargeVideo (6, short) ? corpus
n=5 (`output/human_lovo_manifest.json`).
459   - **HEURISTIC DISCOVERY (user asked "discover more heuristics like audio"):** tested 2
label-free trajectory pre-filters
460   for multi-court (measured on all 5): foreground spatial-density gate = **net ?0.04**
(helps multi-court +0.03, hurts clean
461   ?0.12); track-continuity/jump filter = **no-op**. ? simple single-cue trajectory
heuristics CAN'T isolate the foreground
462   court. **AUDIO-foreground idea PROTOTYPED (no more videos needed) ? WEAK ?:** on the 2
multi-court videos + clean
463   control, audio onset/RMS energy AUC only **0.53?0.65** for in-rally vs dead-time; HF
bands (8?22kHz @48kHz) raise
464   CONTRAST (Kushagra 1.1?1.7×, gBaaddy 1.3?1.9×, confirming distance-attenuation
physics) but AUC stays ~0.6. Fails:
465   dead-time isn't quiet ? foreground court loud during dead-time too
(footwork/talk/pickup) + gym reverb ? audio loudness
466   ? court occupancy, not rally (same wall as E1 ~2.2×). **Audio alone ? multi-court
gate.** Remaining heuristic lever =
467   real court-region detection (player-cluster/court-line, ROADMAP #2); robust path = the
learned visual model (AUC 0.85).

7. user (2026-06-12T09:04:08.050Z)

No files found

8. user (2026-06-12T09:04:12.674Z)

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9. user (2026-06-12T09:04:19.012Z)

```
Name           Mode
----           ----
__pycache__    d----
.git           d--h-
.github        d----
.pytest_cache  d----
.sixth         d----
config         d----
data           d----
docs           d----
src            d----
tests          d----
.coverage      -a---
.gitattributes -a---
.gitignore     -a---
cleanup.py     -a---
pyproject.toml -a---
```

```
README.md      -a---
```

10. user (2026-06-12T09:04:26.313Z)

```
FullName
--------
C:\Users\avidu\Projects\sports-data-collector\docs\DECISIONS.md
C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md
C:\Users\avidu\Projects\sports-data-collector\docs\MANUAL_INGESTION_POSITIONING.md
C:\Users\avidu\Projects\sports-data-collector\docs\PIPELINE_ROLE.md
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ANNOTATION_GUIDE.md
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\INTAKE_MANIFEST_TEMPLATE.csv
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\NDA_TEMPLATE.md
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md
```

11. user (2026-06-12T09:04:32.196Z)

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# Roora Annotation Brief ? Rally Golden Set

**Project:** racket-sport rally segmentation golden / regression dataset
**Sports in scope:** Badminton, Pickleball, Table Tennis (each in **singles _and_ doubles**)
**Companion doc:** [`ANNOTATION_GUIDE.md`](./ANNOTATION_GUIDE.md)

> This brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV that
> imports directly via `python -m src.cli.curate import-local`.

## 0. What we need in one sentence
For each provided match video, produce **one CSV file** listing **every rally** with its
start time, end time, number of shots, and how it ended ? **excluding all dead time**.

## 1. Deliverable format (canonical)
One CSV per video (UTF-8), named exactly `<video_id>.csv` (we supply the `video_id`).
One row per rally, chronological, header row required.

| column | meaning |
|---|---|
| `rally_number` | integer, 1-based, chronological. No gaps, no repeats. |
| `start_mmss` | rater-entered time as `m:ss.mmm` (e.g. `1:12.40`) ? serve contact. |
| `end_mmss` | rater-entered time as `m:ss.mmm` ? ball/shuttle dead. |
| `start_time` | decimal **seconds** (e.g. `72.40`). **Auto-filled** from `start_mmss`. |
| `end_time` | decimal **seconds**. **Auto-filled** from `end_mmss`. |
| `shots_count` | integer. Total racket/paddle/bat contacts (serve = shot 1). |
| `ending_reason` | one of: `winner` \| `forced_error` \| `unforced_error` \| `service_fault` \| `let` \| `other` (**why** it ended) |
| `last_shot_outcome` | one of: `in` \| `net` \| `out` \| `n_a` (**where** the last shot landed; `n_a` for service_fault/let) |
| `score_before` | optional, `"A-B"`. Blank if not confidently readable. |
| `score_after` | optional, `"A-B"`. |
| `notes` | optional free text. **Required** when `ending_reason = other`. |

**Times (mm:ss ? decimal seconds).** Raters type only `start_mmss`/`end_mmss`. The template
Google Sheet fills `start_time`/`end_time` with this formula (shown for `start_time`, row 2):

```
=IF(B2="","", IFERROR(VALUE(LEFT(B2,FIND(":",B2)-1))*60 + VALUE(MID(B2,FIND(":",B2)+1,20)), VALUE(B2)))
```

On **Download as CSV**, the computed decimal seconds are exported. (If your tool already
shows decimal seconds, type those into `start_time`/`end_time` and leave the mm:ss
~~~~

columns blank.)

Example (badminton):
```csv
rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot_outcome,score_before,score_after,notes
1,0:12.50,0:25.00,12.50,25.00,9,winner,in,0-0,1-0,
2,0:45.00,0:47.30,45.00,47.30,1,service_fault,n_a,1-0,1-1,serve into net
3,1:01.20,1:18.85,61.20,78.85,14,unforced_error,out,1-1,2-1,final clear sailed long (no pressure)
```

Critical format rules
- `start_time`/`end_time` that reach us **must be decimal seconds** (the formula guarantees this).
- Times are relative to the **start of the file we provide**. Do **not** trim, clip, or
  re-encode the video ? our pipeline identifies a video by its file checksum.
- `end_time` > `start_time` for every row.
- No overlaps: `end_time[N] <= start_time[N+1]`.
- Everything between one rally's `end_time` and the next rally's `start_time` is **dead time** and gets **no row**.

2. The three non-negotiable rules (full detail + pictures in the Guide)
1. **Ignore all dead time.** Only label active play.
2. **Include every rally ? exclude none.** Every served point, including **service faults**,
   1?2 shot rallies, and **lets/replays**. A missed rally is the worst error in this project.
   Exception: warm-up / knock-up is NOT a rally ? pre-match and between-games warm-up looks
   like rallying but isn't scored play, so **don't label it**. Tell it apart by: no real serve,
   cooperative (not competitive) hitting, the score not progressing, often several shuttles/balls
   in use (full how-to in the Guide, Part 2). When unsure if the match has started, flag it.
3. **A shot = one contact** with the racket/paddle/bat. Serve = shot 1. Bounces and net-clips are not shots.

3. Scope & phasing
Phase A ? Pilot (this engagement): 3 videos (1 badminton + 1 pickleball + 1 table tennis),
each `< 3 min` with ~10?12 rallies. This is the **paid calibration round**; we review jointly and
lock the Guide before scale-up.

Phase B ? Scale (after a successful pilot): ~12?15 videos **per sport** (~36?45 total), each
~30 min. Same format, rules, and template.

> **Doubles are included** across the program. Rally rules are identical to singles; the
> badminton- and pickleball-specific doubles notes are in the Guide.

4. Quality bar (acceptance criteria)
Automated validation (no overlaps, positive durations, schema-valid) runs on every file, then a
~20% human spot-check. A file is returned for rework (in scope) if it misses:
- **Completeness:** every rally present; zero missed rallies; zero dead-time rows.
- **Boundaries:** `start_time`/`end_time` within **±0.3 s** of true serve-contact / landing.
- **shots_count:** exact for rallies ? 15 shots; ±1 allowed for longer/faster rallies.
- **ending_reason:** one of the six allowed values, per the Guide's decision tree.
- **Format:** decimal seconds present, valid columns, non-overlapping, chronological.

5. Per-sport quick reference (full version + examples in the Guide)

```
**Badminton (singles & doubles)** ? start: racket?shuttle serve contact · end: shuttle floors / fault · shot: each racket?shuttle contact.
**Pickleball (singles & doubles)** ? start: paddle?ball serve contact · end: double bounce / out / net / fault · shot: each paddle?ball contact (floor bounces are **not** shots; two-bounce rule is normal play).
**Table tennis** ? start: racket?ball serve strike (not the toss) · end: failed legal return · shot: each racket?ball contact (table bounces are **not** shots). Net-cord serve = `let`.

## 6. ending_reason values (decision tree in the Guide)
This vocabulary attributes **why** the point ended; it is shared across all
net-separated sports (badminton, pickleball, table tennis, tennis). Record **where**
the last shot physically landed in the separate **`last_shot_outcome`** column
(`in` \| `net` \| `out` \| `n_a`) ? e.g. a forced error that flew long is
`ending_reason=forced_error, last_shot_outcome=out`.

| value | meaning |
|---|---|
| `winner` | last shot landed **in** and the opponent made no legal contact (a clean winner). |
| `forced_error` | the losing player erred (netted / out / missed) **under pressure** ? stretched, wrong-footed, or rushed by the opponent's strong shot. |
| `unforced_error` | the losing player erred on a **routine, unpressured** ball they had time and position to make. |
| `service_fault` | the serve itself failed / was illegal. |
| `let` | the point was replayed. |
| `other` | anything else / cannot tell. **Must** add a note. |

## 7. Recommended capture method
Use any tool that shows current time (mm:ss.mmm or seconds, or frame + known fps) and can
**step one frame** forward/back: a frame-steppable player (VLC frame-step) or a timeline tool
(Label Studio / CVAT). The deliverable is the CSV in the schema above.
`seconds = frame_number / fps` (we provide fps per video).

## 8. Logistics & contacts
- **TODO(avidullu): paste the Google Drive folder link** where source videos are shared with Roora.
- Per-video `fps` + `video_id`: provided in the intake manifest ? template [`INTAKE_MANIFEST_TEMPLATE.csv`](./INTAKE_MANIFEST_TEMPLATE.csv). **TODO(avidullu): fill the manifest (one row per pilot video) and share its link.**
- **TODO(avidullu): paste the Drive folder link** where Roora uploads completed `<video_id>.csv` files.
- Blank annotation template Sheet (mm:ss ? seconds formula from §1 pre-filled): **TODO(avidullu): create it and paste the link.**
- **TODO(avidullu): add the escalation channel** (email / WhatsApp group) for edge-case questions.
- NDA before any footage is shared ? template [`NDA_TEMPLATE.md`](./NDA_TEMPLATE.md). **TODO(avidullu): have counsel review, fill the blanks, and have Roora sign.**
- Roora contact on file: `info@rooragrp.com` (HSR Layout, Bangalore).

## 9. Pricing request (please quote)
1. Price **per rally** and **per minute** of footage.
2. **Pilot** price (3 short videos, ~30?36 rallies) as a paid calibration round.
3. Indicative **scale** price for ~36?45 longer videos.
4. Turnaround for the pilot and throughput (rallies/day) at scale.
5. QA process and rework policy.
6. Whether annotators have racket-sport familiarity (helpful, not required).
7. Confirmation you can deliver the CSV schema above.
```

12. user (2026-06-12T09:05:04.958Z)

```
backend\providers\gemini.py:48:                         f"Re-encode/downscale chunks
```

```
(indexing.ai_chunk_scale_width) or chunk smaller.")
backend\reporting\spec.yaml:55:      indexing.ai_chunk_scale_width at its default (1280) so AI
chunks are
backend\reporting\spec.yaml:56:      re-encoded/downscaled before upload (0 disables re-encoding
and reintroduces
backend\config\models.py:83:    # Width (px) the gemini segmenter re-encodes chunks to before
upload. Stream-copied
backend\config\models.py:89:    ai_chunk_scale_width: int = 1280
backend\utils\video.py:21:                        output_path: str, reencode: bool = False,
backend\utils\video.py:22:                        scale_width: Optional[int] = None) -> bool:
backend\utils\video.py:26:        reencode: If True, re-encodes with libx264 (slower but frame-
accurate).
backend\utils\video.py:29:        scale_width: If set (requires reencode), downscale to this
width (height auto,
backend\utils\video.py:39:    if reencode:
backend\utils\video.py:40:        if scale_width:
backend\utils\video.py:41:            cmd += ["-vf", f"scale={int(scale_width)}:-2"]
backend\utils\video.py:42:        cmd += ["-c:v", "libx264", "-preset", "ultrafast", "-crf",
"23", "-c:a", "aac"]
backend\api\static\index.html:5:    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
backend\tools\rally_slicer.py:142:            "-c:v", "libx264", "-preset", "superfast", "-crf",
"22",
backend\eval\gemini_refine.py:73:                  clip_scale_width: Optional[int] = 1280) ->
Tuple[int, Interval, List[Interval]]:
backend\eval\gemini_refine.py:83:    if not extract_video_chunk(video, cs, ce - cs, clip,
reencode=True, scale_width=clip_scale_width):
backend\eval\gemini_refine.py:104:        min_dur: float = 2.0, limit: int = 0,
clip_scale_width: Optional[int] = 1280) -> dict:
backend\eval\gemini_refine.py:123:                          tmpdir, i, clip_scale_width)
backend\eval\gemini_refine.py:150:                       out_csv: str =
"output/gemini_fullvideo.csv", clip_scale_width: Optional[int] = 1280,
backend\eval\gemini_refine.py:171:        if not extract_video_chunk(video, cstart, cend -
cstart, clip, reencode=True, scale_width=clip_scale_width):
backend\eval\gemini_refine.py:184:    print(f"[gemini_full] {len(starts)} chunk(s) of {video}
(<= {chunk_sec}s, width {clip_scale_width}) "
backend\eval\gemini_refine.py:219:    scale = args.clip_scale_width or None
backend\eval\gemini_refine.py:222:                           clip_scale_width=scale,
min_dur=args.min_dur, max_workers=args.max_workers)
backend\eval\gemini_refine.py:229:        min_dur=args.min_dur, limit=args.limit,
clip_scale_width=scale)
backend\pipeline\stitcher\compiler.py:144:                    "-c:v", "libx264",
backend\pipeline\segmenters\gemini.py:117:        ai_scale_width =
int(idx_cfg.get("ai_chunk_scale_width", 1280) or 0)
backend\pipeline\segmenters\gemini.py:148:                    reencode=ai_scale_width > 0,
backend\pipeline\segmenters\gemini.py:149:                    scale_width=ai_scale_width if
ai_scale_width > 0 else None,
backend\pipeline\segmenters\gemini.py:217:                    f"Set
indexing.ai_chunk_scale_width (default 1280) so chunks are re-encoded/"
backend\pipeline\segmenters\trajectory_hybrid.py:206:            if not
extract_video_chunk(video_path, w_start, w_dur, w_path, reencode=True):
backend\pipeline\segmenters\yolo_hybrid.py:491:            success =
extract_video_chunk(video_path, w_start, w_dur, w_path, reencode=True)
```

13. user (2026-06-12T09:05:26.210Z)

```
No matches found
```

14. user (2026-06-12T09:05:27.583Z)

```
C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py:64:
video_id=source,
C:\Users\avidu\Projects\sports-data-collector\src\crawlers\badminton_crawler.py:103:
video_id = f"shuttleset_{match_id}"
C:\Users\avidu\Projects\sports-data-collector\src\crawlers\badminton_crawler.py:195:
```

```
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\ingest.py:39:
video_id=args.video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:63:    def _guard_import(self,
sport_str, video_id, annotations, force=False):
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:114:        video_id =
args.video_id or match_name.lower().replace(" ", "_")
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:170:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:216:    def
_parse_local_annotations(self, sport: str, video_id: str, filepath: str) -> List[Any]:
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:234:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:246:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:257:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:270:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:282:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:301:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:313:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:324:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:337:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:349:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:437:            videos = [v for
v in videos if v.video_id == args.video_id]
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py:555:        indexer keys videos
by *file MD5*, not by our `video_id`, so it must run
C:\Users\avidu\Projects\sports-data-collector\src\parsers\badminton\coachai.py:47:
video_id = kwargs.get("video_id") or match_name.lower().replace(" ", "_")
C:\Users\avidu\Projects\sports-data-collector\src\parsers\badminton\coachai.py:135:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\parsers\badminton\coachai.py:147:
video_id=video_id,
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:227:            video_id =
rallies[0].video_id
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:228:
cursor.execute("DELETE FROM badminton_rallies WHERE video_id = ?", (video_id,))
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:255:            video_id =
points[0].video_id
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:256:
cursor.execute("DELETE FROM tennis_points WHERE video_id = ?", (video_id,))
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:282:            video_id =
deliveries[0].video_id
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:283:
cursor.execute("DELETE FROM cricket_deliveries WHERE video_id = ?", (video_id,))
C:\Users\avidu\Projects\sports-data-collector\src\core\db.py:311:            video_id =
rallies[0].video_id

[Showing results with pagination = limit: 30]
```

15. user (2026-06-12T09:11:01.126Z)

<retrieval_status>timeout</retrieval_status>

<task_id>wr6mnel6c</task_id>

<task_type>local_workflow</task_type>

<status>running</status>

16. user (2026-06-12T09:14:06.773Z)

<retrieval_status>success</retrieval_status>

<task_id>wr6mnel6c</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>
[Truncated. Full output: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\tasks\wr6mnel6c.output]

eo keyed by human `video_id` (supplied in intake manifest)\n- Rally keyed by 1-based integer `rally_number` (chronological, no gaps)\n- Database schema in `/src/core/db.py` lines 76-88 (badminton_rallies table):\n ```sql\n PRIMARY KEY (video_id, rally_number)\n FOREIGN KEY (video_id) REFERENCES videos(video_id)\n ```\n\n**Timestamp units & precision:**\n- **Canonical format:** Decimal **seconds** (floats)\n- **Precision:** ±0.3 s tolerance on boundaries (ROORA_BRIEF.md §4, line 83; ANNOTATION_GUIDE.md Part 3)\n- **Formula source:** mm:ss.mmm ? seconds conversion via Google Sheet formula (shown in ROORA_BRIEF.md §1, lines 35-36)\n\n**Ingestion pipeline:**\n1. **CSV parsing** (`/src/cli/curate.py` lines 216-362):\n   - Reads canonical CSV with headers: `rally_number, start_time, end_time, ...` (lowercase-folded)\n   - Uses `safe_float_required()` to parse start_time/end_time as decimal seconds (required fields; raises if missing)\n   - Falls back to mm:ss parsing if decimal is missing: `/src/core/parsing.py` lines 32-49 (`parse_time_string`)\n\n2. **CVAT alternative** (if vendor uses CVAT instead of CSV):\n   - Converts CVAT XML box export ? canonical CSV via `/src/parsers/cvat/rally_cvat.py`\n   - **Critical:** Recomputes times from frame ranges: `seconds = frame / fps` (lines 266-267)\n   - **Ignores vendor's time attributes** ? a CVAT job that assumes 30 fps yields systematically wrong times (lines 23-24)\n\n3. **Schema validation** (`/src/core/validator.py` lines 29-57):\n   - `end_time > start_time` (strictly greater)\n   - No overlaps: `end_time[N] <= start_time[N+1]`\n   - Chronological by rally_number\n\n4. **Annotation objects** created as `BadmintonRally` (Pydantic model, `/src/core/schemas/badminton.py` lines 1-21):\n   - Fields: `video_id, rally_number, start_time, end_time, shots_count, ending_reason, last_shot_outcome, score_before, score_after`\n   - Validation: start_time ? 0, end_time > start_time\n\n**Critical join behavior:**\n- Annotations are keyed to the **original video file** via `video_id`\n- The collector DB stores the video's `local_path` (from import command, e.g. `--video-path match.mp4`)\n- Indexer repo retrieves the video's MD5 and local_path from the manifest (`/data/eval_export/manifest.json`)\n- **No timestamp offset handling during ingestion** ? times are taken as-is from CSV (must be pre-aligned to the vendor's received video)\n\n---\n\n### 3. TIMESTAMP NORMALIZATION & OFFSET HANDLING\n\n**Normalization (before import):** `/src/cli/normalize_annotations.py`\n- Converts mm:ss / HH:MM:SS to decimal seconds\n- Drops rows with invalid/missing times or negative start_time\n- Detects & renumbers fractional `rally_number` (e.g., 7.5 ? collides on PK) or duplicate IDs\n- Handles pipe-delimited `ending_reason` ('forced_error|other' ? 'forced_error')\n- Output columns: `rally_number, start_time, end_time, ending_reason, sport, shots_count, original_rally_number, notes`\n\n**During CVAT conversion:** `/src/parsers/cvat/rally_cvat.py` lines 366-394\n- Explicit fps resolution order: user-supplied `--fps` > ffprobe of `--video-path` > error (never defaults to 30)\n- Recomputes frame range ? seconds: `start_time = frame_min / fps`, `end_time = frame_max / fps`\n- **No configurable offset correction** ? this is a TODO if the vendor's encoded video has a leading padding\n\n**No per-batch offset correction** currently implemented in the code. If Roora receives a re-encoded copy with leading silence/padding, the returned timestamps will be offset relative to the original. **This is documented as a constraint: do not re-encode.**\n\n---\n\n### 4. GOLDEN-SET BATCHES (GS-1..GS-4) ? WHAT WAS SENT & QUALITY NOTES\n\n**Pilot status (GS-1 candidate):** \n- **File:** `/data/validation/pilot_badminton_01_report.md`\n- **Scope:** 1 badminton video (shorter clip, ~2?3 min estimated)\n- **What was sent:** Unknown; intake manifest still has TODOs (INTAKE_MANIFEST_TEMPLATE.csv lines 2-4)\n- **What came back:** 24 rallies in CVAT XML export\n- **Format received:** CVAT export (image or track mode) with box attributes (rally_number, shots_count, ending_reason, etc.)\n\n**Roora pilot delivery quality issues (from pilot_badminton_01_report.md):**\n- **Blocker (1):** Rally 1 overlaps Rally 2 by 18.52 s (ends 21.02 s but Rally 2 starts at 2.50 s) ? invalid chronology, suggests overlapping/duplicate box annotations in CVAT\n- **Medium (3):** \n - Rally 2: 2 shots over 30.0 s (0.07/s) ? implausibly

low shot density, likely inflated span from stray CVAT box\n - Rally 7: 2 shots over 16.0 s (0.12/s) ? same issue\n - Rally 16: 2 shots over 17.5 s (0.11/s) ? same issue\n- **Root cause:** CVAT boxes not properly trimmed or sampled; coarse sampling (30-frame step at 59.94 fps ? 0.50 s per frame) mentioned in issues.txt\n- **Gate:** BLOCKED pending rework\n\n**GS-2, GS-3, GS-4:** Not yet delivered (Phase 0 guideline complete, vendor bake-off deferred; Roora pilot in progress).\n\n---\n\n### 5. CONCRETE FILE PATHS & SCHEMAS\n\n**Annotation spec:**\n- `/sports-data-collector/docs/annotation/ROORA_BRIEF.md` ? vendor-facing spec, rally CSV schema, phasing\n- `/sports-data-collector/docs/annotation/ANNOTATION_GUIDE.md` ? illustrated guide, per-sport rules, edge cases\n- `/sports-data-collector/docs/annotation/INTAKE_MANIFEST_TEMPLATE.csv` ? per-video fps/video_id template (to be filled)\n- `/sports-data-collector/docs/annotation/NDA_TEMPLATE.md` ? vendor contract template\n\n**Ingestion code:**\n- `/sports-data-collector/src/cli/curate.py` lines 97-214 (`import_local` command) ? CSV/JSON parsing, validation, DB insert\n- `/sports-data-collector/src/cli/curate.py` lines 216-362 (`_parse_local_annotations`) ? CSV DictReader + BadmintonRally object construction\n- `/sports-data-collector/src/core/parsing.py` lines 32-49 ? mm:ss/HH:MM:SS to seconds conversion\n- `/sports-data-collector/src/core/db.py` lines 76-88 ? badminton_rallies table schema with (video_id, rally_number) PK\n- `/sports-data-collector/src/core/validator.py` lines 29-57 ? chronological + no-overlap checks\n\n**CVAT conversion (vendor alternative):**\n- `/sports-data-collector/src/parsers/cvat/rally_cvat.py` lines 366-394 (`convert` function) ? XML?records, fps resolution, issue detection\n- `/sports-data-collector/src/parsers/cvat/rally_cvat.py` lines 283-363 (`detect_issues`) ? overlap, zero_duration, shot_density, possible_missed_rally checks\n- `/sports-data-collector/data/roora_badminton_issues.txt` ? pilot delivery issue report (coarse sampling, blocker overlap)\n\n**Pilot delivery:**\n- `/sports-data-collector/data/validation/pilot_badminton_01_report.md` ? validation report, 24 rallies, 1 blocker, 3 medium issues\n\n**Indexer bridge (golden-set architecture):**\n- `/badminton-highlight-indexer/docs/GOLDEN_SET_VENDORING.md` ? vendor selection process, security controls, bake-off spec\n- `/badminton-highlight-indexer/docs/GOLDEN_SET_IMPLEMENTATION.md` ? GS-1..GS-5 PR breakdown, AnnotationProvider registry\n- `/badminton-highlight-indexer/docs/GOLDEN_SET_PHASE1_VIDEOS.md` ? round-1 corpus spec (resolution/fps per sport, diversity matrix)\n\n**Example manifest (output side):**\n- `/sports-data-collector/data/eval_export/manifest.json` ? `local_path` + video_id mappings for indexer ingestion\n",
        "keyFiles": [
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\annotation\\ROORA_BRIEF.md",
            "why": "Vendor-facing specification: rally CSV schema, timestamps (mm:ss?decimal seconds), format rules, non-recode constraint, intake manifest, phasing (3-video pilot)",
            "lines": "1-132"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\annotation\\ANNOTATION_GUIDE.md",
            "why": "Illustrated supplement: per-sport rules (badminton/pickleball/table-tennis, singles & doubles), edge cases, warm-up exclusion, boundary tolerance (±0.3s)",
            "lines": "1-227"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py",
            "why": "import_local command: CSV parsing (DictReader, case-folded headers), safe float parsing, validation, DB insert, (video_id, rally_number) join key",
            "lines": "97-214, 216-362"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\core\\parsing.py",
            "why": "Timestamp parsing: mm:ss/HH:MM:SS?decimal seconds conversion, safe_float_required (raises on missing, critical field)",
            "lines": "32-49"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\core\\db.py",
            "why": "Database schema: badminton_rallies table, PRIMARY KEY (video_id, rally_number), columns (start_time, end_time as REAL)",
            "lines": "76-88"

```
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\src\\core\\validator.py",
            "why": "Validation: no-overlap check (end[N] <= start[N+1]), end_time > start_time,
chronological by rally_number",
            "lines": "29-57"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\src\\parsers\\cvat\\rally_cvat.py",
            "why": "CVAT?CSV conversion: fps resolution (explicit > ffprobe > error),
frame?seconds recompute (seconds = frame/fps), issue detection (overlap, shot_density,
missed_rally)",
            "lines": "1-35, 366-394, 283-363"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\data\\validation\\pilot_badminton_01_report.md",
            "why": "Pilot delivery: 24 rallies, 1 blocker (overlap Rally 1?2), 3 medium
(shot_density), coarse CVAT sampling issue (30 frames ~0.5s step)",
            "lines": "1-16"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\docs\\annotation\\INTAKE_MANIFEST_TEMPLATE.csv",
            "why": "Per-video metadata sent to vendor: video_id, fps, duration, expected_rallies
(template, unfilled)",
            "lines": "1-5"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\src\\cli\\normalize_annotations.py",
            "why": "Pre-import normalization: mm:ss?seconds, drop invalid rows, renumber
fractional/duplicate rally_numbers, output lean schema",
            "lines": "1-50, 77-189"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\PIPELINE_ROLE.md",
            "why": "Bridge to indexer: video identity (collector uses video_id, indexer uses
MD5), manifest carries local_path, re-encode changes MD5 (no timestamp offset handling)",
            "lines": "40-45"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\GOLDEN_SET_VENDORING.md",
            "why": "Vendor selection & security: bake-off process, Roora selected for initial
pilot (verified Bangalore location), DPA controls, no user uploads",
            "lines": "1-65, 130-165"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\GOLDEN_SET_IMPLEMENTATION.md",
            "why": "Architecture: AnnotationProvider registry (FileProvider,
HumanVendorProvider, AutomatedProvider), GS-1..GS-5 PR breakdown, GS-3 parked on Roora pilot",
            "lines": "1-68"
          },
          {
            "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\GOLDEN_SET_PHASE1_VIDEOS.md",
            "why": "Corpus spec: resolution/fps per sport (badminton 30 ok/50-60 better,
pickleball 30-60, table tennis 60 min), static-primary profile, diversity matrix",
            "lines": "38-67"
          }
        ],
```

"quotes": [
"C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\annotation\\ROORA_BRIEF.md:52 ? 'Times are relative to the **start of the file we provide**. Do **not** trim, clip, or re-encode the video ? our pipeline identifies a video by its file checksum.'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\annotation\\ROORA_BRIEF.md:83-86 ? '**Boundaries:** `start_time`/`end_time` within **±0.3 s** of true serve-contact / landing. ... **Format:** decimal seconds present, valid columns, non-overlapping, chronological.'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\annotation\\ROORA_BRIEF.md:51-54 ? '`start_time`/`end_time` that reach us **must be decimal seconds** (the formula guarantees this). ... No overlaps: `end_time[N] <= start_time[N+1]`.'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py:302-303 ? 'rally_number=_safe_int(row.get(\"rally_number\"), idx + 1), start_time=_safe_float(row.get(\"start_time\")), end_time=_safe_float(row.get(\"end_time\")), ...'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\core\\db.py:86 ? 'PRIMARY KEY (video_id, rally_number), FOREIGN KEY (video_id) REFERENCES videos(video_id) ON DELETE CASCADE'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\parsers\\cvat\\rally_cvat.py:23-24 ? 'A CVAT job that samples every `step` frames and assumes 30 fps yields systematically wrong times ... The frame index is reliable; the vendor's seconds are not.'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\parsers\\cvat\\rally_cvat.py:266-267 ? 'start_time=fmin / fps, end_time=fmax / fps, ...'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\data\\validation\\pilot_badminton_01_report.md:8-9 ? '[BLOCKER] - (overlap) Rally 1 (ends 21.02s) overlaps Rally 2 (starts 2.50s) by 18.52s ? rallies must be disjoint.'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\data\\roora_badminton_issues.txt:2-3 ? 'fps used : 59.940 ... sampling step : 30 frames (~0.50s) ... [BLOCKER] - (overlap) Rally 1 (ends 21.02s) overlaps Rally 2 (starts 2.50s) by 18.52s'",
"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\GOLDEN_SET_VENDORING.md:152 ? 'Roora (Roora ML Pvt Ltd) | **Bengaluru** ? HSR Layout (verified, `rooragrp.com`) | ... **Selected for the initial paid pilot (2026-06-02)** ? existence/capability/location verified; **quality TBD via the pilot**.'",
"C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\PIPELINE_ROLE.md:42-45 ? 'The indexer keys videos by **file MD5**, this repo by a human `video_id`. A re-encode/re-download changes the bytes, hence the MD5. So the manifest carries the exact `local_path`, and any downstream processing must run on **that** file.'",
"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\GOLDEN_SET_PHASE1_VIDEOS.md:44-46 ? 'Badminton | 1080p (720p floor) | 30 ok, **50/60 better** | Shuttle decelerates fast; 60 sharpens smashes. Pickleball | 1080p | **30?60** | Ball slower than TT; 30 likely fine, 60 safer. Table tennis | 1080p | **60 min (120 ideal)**'"
],
"gaps": "\n1. **Pilot metadata unfilled:** The INTAKE_MANIFEST_TEMPLATE.csv still has all-TODO placeholders ? the actual fps, video_id, filenames, and Google Drive links used for the pilot are not documented in the code/docs (TODOs in ROORA_BRIEF.md §8).\n\n2. **Re-encoded video join handling:** No explicit timestamp offset correction is implemented in the ingestion code. If the vendor receives a re-encoded copy with leading padding/silence, the returned timestamps will drift relative to the original video. The spec says 'do not re-encode,' but there is no automated correction if this rule is broken.\n\n3. **Roora tooling specifics:** The brief recommends frame-steppable tools (VLC, Label Studio, CVAT) but does not confirm what Roora actually uses. CVAT is mentioned as an acceptable deliverable format, but whether Roora's annotators use CVAT, Google Sheets, or a custom tool is not documented.\n\n4. **GS-2, GS-3, GS-4 actual deliveries:** Only the GS-1 pilot (pilot_badminton_01) has been attempted; no pickleball or table-tennis pilot data is documented. The phase-1 scale (12?15 videos per sport) has not been sent.\n\n5. **Vendor resolution/fps constraints:** No maximum or minimum frame rate or resolution was communicated to Roora in the documented brief. The GOLDEN_SET_PHASE1_VIDEOS.md spec (badminton 50/60 fps, table tennis 60 min) applies to our corpus, not necessarily Roora's constraints or capabilities.\n\n6. **Post-delivery metadata:** No documentation of how Roora's tooling/environment (monitor refresh rate, playback frame rate, any drift between wall-clock time and video frame time) might affect timestamp precision.\n\n7. **GS-1 pilot source file

details:** The pilot_badminton_01_report.md does not record what resolution/fps the source video was, or what format Roora received it in (MP4/MOV/etc.).\n"
      }
    ],
    "verification": {
      "confirmed": [
        "Offset magnitudes verbatim: 'Every rally's start/end in both videos is off by 0.5 s or more ? up to ~2 s in several ? against the agreed tolerance of ±0.3 s' (C:\\Users\\avidu\\Projects\\Annotation Setup\\Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md, §1 line 11; §2 line 15: 'A full frame-by-frame re-check of all 12 rallies shows every start/end boundary off by 0.5 s or more, several by up to ~2 s').",
        "Root cause verbatim (§2 line 17): 'All 24 rally boundaries sit exactly on the 30-frame sampling grid ? at 59.94 fps that grid is ~0.5 s apart, so a boundary placed only on grid frames cannot meet ±0.3 s.' Impact quotes also verbatim (line 19): '1?8.5 s' rallies, '0.5?2 s boundary error corrupts 10?100% of a rally's span', 'the 1.0 s table-tennis rally #2 can be erased entirely', and 'at our standard evaluation gate (overlap ? 50%), a typical 4.5 s rally labeled 2 s late fails to match even a perfectly correct detection'.",
        "Badminton review quote verbatim: 'After a full frame-by-frame review, the rally timings are generally good' (Roora Pilot Review ? Badminton (2026-06-06).md, §1, lines 13-14), and the pickleball/TT review explicitly calls the boundary problem 'a regression' (§1 line 11).",
        "Tolerance spec confirmed in both vendor docs: ROORA_BRIEF.md line 83 ('Boundaries: start_time/end_time within ±0.3 s of true serve-contact / landing', §4) and ANNOTATION_GUIDE.md line 96 ('Boundary tolerance: ±0.3 s (~±9 frames at 30 fps)', Part 3 ? which also mandates exact serve-contact/dead frames, lines 83-95).",
        "Join key confirmed: collector keys annotations by a HUMAN video_id string + rally_number, not MD5. curate.py:113-114: video_id = args.video_id or match_name.lower().replace(' ', '_') (match_name defaults to the video filename stem); db.py:86-87: PRIMARY KEY (video_id, rally_number), FOREIGN KEY (video_id) REFERENCES videos(video_id) ON DELETE CASCADE. The indexer separately keys by MD5 of the whole file (badminton-highlight-indexer\\backend\\utils\\hashing.py:23-33, compute_video_id). PIPELINE_ROLE.md lines 42-45 quote verified verbatim ('The indexer keys videos by file MD5, this repo by a human video_id... the manifest carries the exact local_path').",
        "Timestamp unit/precision confirmed: canonical decimal seconds as floats (SQLite REAL columns, db.py:79-80); CSV import parses start_time/end_time with safe_float_required (aliased as _safe_float, curate.py:11, used at lines 303-304); CVAT path recomputes start_time=fmin/fps, end_time=fmax/fps (rally_cvat.py:266-267) and deliberately ignores vendor time attributes (rally_cvat.py:17-24); fps resolution order explicit --fps > ffprobe(video_path) > hard error, never a guessed 30 (rally_cvat.py:366-385).",
        "metrics.py boundary-error quote verbatim at exactly lines 124-125 (mean abs start/end error over matched pairs); Score has mean_start_error/mean_end_error fields (lines 97-98).",
        "Badminton pilot validation artifacts confirmed: data\\validation\\pilot_badminton_01_report.md ? 24 rallies, gate BLOCKED, 1 blocker (Rally 1 ends 21.02s overlaps Rally 2 starting 2.50s by 18.52s), 3 medium shot_density (rallies 2/7/16); data\\roora_badminton_issues.txt ? fps 59.940, 'sampling step : 30 frames (~0.50s)', plus a [HIGH] coarse_sampling line.",
        "No CFR or keyframe-dense re-encode helper exists anywhere: grep across badminton-highlight-indexer\\backend found no -r, -g, force_key_frames, vsync, or fps-filter in any ffmpeg invocation; sports-data-collector\\src contains zero ffmpeg encode calls (yt-dlp download + ffprobe probe_resolution_fps only, downloader.py:112-140); rally-annotator has none. Encoding params as reported: compiler.py libx264/superfast/crf22/aac128k then concat -c copy (lines 225-234, 304-311); rally_slicer.py same (lines 142-143); extract_video_chunk libx264/ultrafast/crf23 + optional scale_width (video.py:39-44); hit_detector audio-only -vn -ac 1 -ar <sr> pcm_s16le (lines 34-35).",
        "ROORA_BRIEF.md no-re-encode rule verbatim (lines 52-53): 'Times are relative to the start of the file we provide. Do not trim, clip, or re-encode the video ? our pipeline identifies a video by its file checksum.' No vendor-prep re-encode step exists anywhere in the collector flow (INGESTION_PIPELINE.md four commands: convert-cvat ? validate-delivery ? import-local ? export).",
        "Roora selection confirmed: GOLDEN_SET_VENDORING.md line 152 ('Selected for the initial paid pilot (2026-06-02)', Bengaluru HSR Layout, rooragrp.com, 'quality TBD via the pilot') and line 164 (single-vendor pilot instead of the iMerit/Taskmonk bake-off). Contact info@rooragrp.com per ROORA_BRIEF.md line 122. INTAKE_MANIFEST_TEMPLATE.csv still all TODO(avidullu) placeholders; ROORA_BRIEF.md §8 logistics TODOs unfilled.",
        "Validator no-overlap/chronology checks confirmed (validator.py:43-55: start>=end error, r.start_time < prev.end_time overlap error); normalize_annotations.py drops bad rows, renumbers

fractional/duplicate rally_numbers, converts mm:ss, folds pipe-delimited ending_reason (file docstring + OUT_FIELDS, lines 1-42)."
      ],
      "corrected": [
        "Direction overstated (delay-evidence): the review never states the offsets were LATE ('delay'). It says boundaries are 'off by 0.5 s or more' ? magnitude only, direction uncharacterized. The 'labeled 2 s late' phrase is a hypothetical in the impact illustration, not a measurement. Calling it 'the annotation vendor timing delay' / 'systematic delay' asserts a direction the source does not.",
        "Wrong location (delay-evidence): the two review docs are NOT in 'the badminton-highlight-indexer project's Annotation Setup folder'. C:\\Users\\avidu\\Projects\\Annotation Setup\\ is a standalone sibling folder under Projects, outside all three repos.",
        "Badminton pilot was NOT 'only minor boundary and completeness issues' (delay-evidence): the badminton delivery was gate-BLOCKED (pilot_badminton_01_report.md) ? an 18.52s rally overlap blocker, a box starting ~27s into dead time (rally 2: delivered 0:02?0:32 vs true 0:29?0:32), a box bridging two rallies, and 2 missed rallies (review §3.1?3.2). 'Timings generally good' is the doc's diplomatic summary; the boundary defects were severe, just different in kind (stray/merged boxes vs grid snapping).",
        "'Regression from the badminton pilot' needs nuance (delay-evidence): the same 30-frame sampling grid was already present AND flagged in the badminton delivery ? roora_badminton_issues.txt [HIGH]: 'Frames sampled every 30 (~0.50s at 59.94fps); rally boundaries carry +/-0.50s uncertainty, exceeding the +/-0.3s spec.' That coarse_sampling check was subsequently REMOVED from detect_issues (rally_cvat.py:288-294 now deliberately does not flag sampling density). The grid risk was known before the pickleball/TT round.",
        "No mm:ss fallback in the import path (handoff-flow): claim 'Falls back to mm:ss parsing if decimal is missing (parse_time_string, parsing.py:32-49)' is wrong for import-local. curate.py:11 aliases safe_float_required as _safe_float ? float('1:12.40') raises ValueError and ABORTS the whole import (normalize_annotations.py docstring documents exactly this landmine). parse_time_string is only used by the CoachAI parser. mm:ss handling exists only via the opt-in --normalize pre-step or the CVAT converter. (INGESTION_PIPELINE.md's FAQ 'mm:ss is also accepted everywhere' overstates the code too.)",
        "Manifest carries no MD5 (handoff-flow): claim 'Indexer repo retrieves the video's MD5 and local_path from the manifest' is wrong ? data\\eval_export\\manifest.json fields are video_id (human), csv, local_path, fps, resolution, license, etc. No MD5 field. The indexer resolves local_path (backend\\eval\\batch.py::resolve_video_path) and computes the MD5 itself by hashing that file (backend\\utils\\hashing.py).",
        "GS-1..GS-5 are NOT vendor annotation batches (handoff-flow): per GOLDEN_SET_IMPLEMENTATION.md they are indexer PR slices ? GS-1 AnnotationProvider registry + FileProvider, GS-2 regression runner, GS-3 HumanVendorProvider (parked on the Roora pilot), GS-4 automated oracle stub, GS-5 triggers. GS-1/GS-2/GS-4 LANDED as code (doc line 8); nothing called 'GS-2/GS-3/GS-4' was ever 'sent to' or 'not yet delivered by' Roora. The whole 'GOLDEN-SET BATCHES ? WHAT WAS SENT' section conflates PR slices with the 3-video pilot (video_ids pilot_badminton_01/pilot_pickleball_01/pilot_tabletennis_01).",
        "Pilot delivery status stale (handoff-flow): all THREE pilot videos have been delivered ? pickleball and table tennis came back and were reviewed 2026-06-10 (the second review doc). The agent reported only the badminton delivery and missed the 2026-06-10 round entirely, including its boundary-accuracy findings.",
        "Downscale-for-API IS used in the indexing flow (ffmpeg-tooling): claims 'extract_video_chunk(..., scale_width=...) is not used in the indexing flow' and 'No preprocessing/proxy generation... no low-res copies for inference anywhere' are wrong. The production Gemini segmenter downscales every uploaded chunk via ai_chunk_scale_width (default 1280) ? gemini.py:117 and 146-150 (reencode=ai_scale_width > 0, scale_width=ai_scale_width); backend\\eval\\gemini_refine.py:71-83 re-encodes + downscales candidate clips (clip_scale_width default 1280, comment: 'Downscale the uploaded clip (Gemini analyses at low res) -> small, fast, under the 500MB upload cap even for 4K/5K sources'). What remains true: no CFR conversion and no keyframe-interval control exist anywhere.",
        "Stitcher padding self-contradiction unresolved (ffmpeg-tooling): the report gives 0.5/0.5 (config.json:72-74) in one place and 0.5/1.0 (compiler) in another without noticing they conflict. Reality: VideoCompiler.__init__ defaults are begin=0.5, end=1.0 (compiler.py:10), and main.py:508 and :633 construct VideoCompiler(output_dir, watermark=...) WITHOUT passing the config paddings ? so the LIVE stitcher pads 0.5s/1.0s and the config.json stitching.begin_padding/end_padding (0.5/0.5) are only read by gemini.py:406-408 for a log-line duration estimate. The config padding is effectively not wired to the compiler.",
        "Schema presented as current is superseded (handoff-flow): the brief's CSV schema (with start/end mmss, score_before/score_after) is no longer the operative vendor contract. The

2026-06-10 review §5 drops the score fields and the vendor time attributes ('decimal minutes computed at an assumed 15 fps, ~4x off') and fixes the CVAT label schema going forward to exactly: rally_number, shot_count, ending_reason, last_shot_outcome, note."
      ],
      "missed": [
        "An offset/slack-absorption mechanism ALREADY EXISTS and is documented in the QA-gate code ? directly contradicting the 'no offset handling' framing and tempering the 0.5s-corruption narrative: rally_cvat.py::detect_issues docstring (lines 288-294): 'we deliberately do NOT flag the sampling density (step/fps). A boundary slack of ~0.5s is acceptable for this pipeline because the downstream highlight compiler pads every clip on output (begin_padding=0.5s, end_padding=1.0s), which absorbs the sampling uncertainty.' Plus the badminton review §2: 'Minor boundary slack is absorbed by padding downstream.' The earlier issues.txt shows the [HIGH] coarse_sampling check existed and was later removed. So the codebase's own gate was deliberately relaxed on EXACTLY the defect the 2026-06-10 review calls 'the product itself' ? any remediation recommendation must reconcile these two opposing positions (golden-set training labels need ±0.3s; highlight-stitching consumption tolerates ~0.5s).",
        "Roora's actual tool/workflow constraint is documented and supersedes the CSV brief: badminton review §2 ? 'Keep working in CVAT; we ingest your bounding-box export as-is. No need to produce a CSV' and 'We recompute each rally's time from the box frame range at the true fps (seconds = frame / fps), so the CVAT time attributes don't matter.' Operative deliverable = CVAT XML; ALL timing derives from frame indices / fps. Critical consequence for any vendor-prep re-encode recommendation: what must be preserved is the FRAME<->TIME mapping (fps and frame count) of the file the vendor annotates, not its bytes. A prep copy at a different fps (e.g., 59.94 VFR -> 30 CFR) silently changes the meaning of every CVAT frame index (convert-cvat must be run with the prep copy's fps), and Roora's 30-frame sampling grid becomes 1.0s at 30fps ? twice as coarse. The MD5 'do not re-encode' rule is the lesser constraint on the collector side, since annotations join by human video_id, not MD5.",
        "Vendor wrote garbage time attributes that the pipeline already neutralizes: pickleball/TT review §5 ? the delivered CVAT 'start time/end time' values were decimal MINUTES computed at an assumed 15 fps (~4x off real time at 59.94 fps). rally_cvat.py preserves them only as vendor_start_time/vendor_end_time on each record (lines 277-278) and never uses them. This is the concrete instance behind the converter's design rationale (lines 17-24).",
        "New boundary convention from the 2026-06-10 review §3 that affects offset analysis: pickleball rally #1 starts at frame 0:00.000 because the file begins MID-RALLY ? its start is unknowable, not 'offset'. Going-forward rule: skip rallies already in progress at file start/end, never extrapolate. Any boundary-error measurement over the pilot data should exclude that rally.",
        "Filename-derived video_id collisions are a known, guarded risk in the collector: curate.py:83-95 (_guard_import) ? comment 'filename-derived video_ids collide easily'; an import to an existing video_id with annotations hard-exits unless --force, which REPLACES the segments. Relevant to any vendor-prep flow that creates renamed/derived copies of the same source video.",
        "Live stitcher padding bug nobody flagged: config.json's stitching.begin_padding/end_padding are silently ignored by the actual highlight compiler (main.py:508,633 construct VideoCompiler without them; live values are the constructor defaults 0.5s/1.0s). The CVAT QA gate's slack-absorption rationale (rally_cvat.py:293) cites the 0.5/1.0 defaults, so changing config.json paddings would change nothing in output and the cross-repo rationale would silently diverge.",
        "Phase B is imminent in the review docs and changes scale of the problem: ~30-minute videos, ~12-15 per sport (ROORA_BRIEF §3; pickleball/TT review §4 re-emphasizes dead-time/warm-up rules for ~30-min footage; §8: conventions to be folded into a Guide revision, and Roora asked to confirm which Brief/Guide version they worked from). Recommendations should target the Phase-B contract revision, not just retro-fixing the 3 pilot videos."
      ]
    }
  }
}
</output>

17. user (2026-06-12T09:15:42.687Z)

Noted (position 1, task_id: task_d8df38bc). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_d8df38bc "Wire stitching padding config into VideoCompiler". Continue your current work.

18. user (2026-06-12T09:16:06.635Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

19. user (2026-06-12T09:16:17.630Z)

```
---
name: current-state
description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
metadata:
  node_type: memory
  type: project
  originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
---

# Current state ? updated continuously (see [[working-agreement-checkpoints]])

**Checkpoint:** 2026-06-11 (UI-alpha **PR-2 #99 MERGED + docs-sync #100 MERGED** ? owner authorized merge w/ due
diligence; master `020ff61`). **LIVE E2E AT MERGE (server up on :8123, real artifacts):** 86MB smoke video uploaded in
3 sequential parts ? reassembled to **MD5 byte-identity** with the source; negatives verified (out-of-order 409,
bad-ext/traversal 400, unknown-id 404; download traversal blocked at BOTH layers ? handler 400 on backslash form,
```

20. user (2026-06-12T09:16:33.720Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

21. assistant (2026-06-12T09:17:06.768Z)

Yes, it's feasible ? and worth doing ? but the research corrected one important premise: the 0.5s problem isn't a "delay," and re-encoding alone won't fix it. My recommendation on your second question: put it in the **collector** as a small `curate` subcommand, not a standalone script. Details below.

What the evidence actually says

The documented finding (in `Annotation Setup\Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md`) is that every boundary is **off by ?0.5s, up to ~2s** ? magnitude only; no direction ("late") was ever measured. The root cause is precisely diagnosed there: **all 24 boundaries sit exactly on a 30-frame sampling grid** (~0.5s at 59.94fps). Roora works in CVAT and only places boxes every 30th frame, so the first/last box of a rally structurally cannot land on the true contact/dead frame. Notably, the same grid was already flagged `[HIGH]` in the badminton round (`data/roora_badminton_issues.txt` in the collector) ? it's their standing workflow, not a one-off regression.

So your hypothesis is half right: the heavy 4K/59.94 source is very likely *why* they sample every 30 frames (a 4K CVAT task at frame-step 1 is painful), and a light proxy removes that excuse. But the binding fix is the convention change already drafted in the review ? **first/last box on the exact contact/dead frame, frame-step to find it** ? and the proxy is what makes that convention cheap for them to actually follow.

Why a re-encoded proxy is safe to hand them

The pipeline turns out to be proxy-friendly by construction:

- The operative contract is CVAT XML, and [rally_cvat.py](C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py) **recomputes all times as `frame / fps`**,

deliberately ignoring their (garbage) time attributes. So what must be preserved is the *frame?time mapping*, not the bytes.
- The collector joins annotations by the **human-assigned `video_id`** (PK `(video_id, rally_number)`), not MD5. The brief's "do not re-encode" rule (ROORA_BRIEF.md §1) binds *the vendor*; nothing breaks if *we* hand them a proxy, as long as the intake manifest keeps pointing downstream processing at the original file (exactly what `PIPELINE_ROLE.md` already prescribes for the indexer's MD5 keying).
- The ffmpeg primitive already exists: [video.py:21-42](backend/utils/video.py) (`extract_video_chunk(reencode=True, scale_width=?)`) is what the production Gemini path uses to downscale 4K chunks to 1280px.

The non-negotiable design constraints for the proxy:

1. **Same fps, no trimming, no leading padding** ? then seconds map 1:1 to the original and golden CSVs apply to the original directly. **Never reduce fps**: if their 30-frame habit persists at 30fps, the grid becomes 1.0s ? twice as bad ? and every frame index changes meaning.
2. **Verify, don't assume**: assert proxy frame count == original frame count and durations match (also catches VFR sources, where frame/fps math is already shaky).
3. **Keep audio** ? racket/paddle contact sound is the annotator's best boundary cue.
4. Sensible profile: 4K?**1080p** (720p risks losing the TT ball), CRF ~18?20, **dense keyframes** (`-g 15`) so scrubbing/back-stepping is snappy, audio copied.
5. **Record provenance**: original MD5 + proxy MD5 + video_id + fps in the intake manifest. We've already been bitten once by the re-encoded-copy keying confusion (Kushagra preds keyed under the downscaled file's MD5).

Repo vs. standalone tool

Collector, as `curate prep-annotation-proxy` (one small PR). The reasoning:

- The collector already owns the entire vendor flow (`docs/annotation/`, intake manifest, `convert-cvat` ? `validate-delivery` ? `import-local`) and already shells out to ffprobe ? no new dependency, and the proxy step slots in as command #0 of that pipeline.
- The ffmpeg one-liner is the trivial 20%; the **parity check + manifest provenance is the actual value**, and that's exactly what a loose script never does (compare: the unfilled intake-manifest TODOs, the `%TEMP%` artifact-loss audit finding).
- **Phase B makes this routine, not one-off**: ~36?45 videos of ~30 minutes each. A 30-min 4K/59.94 file is >100k frames ? Roora genuinely cannot annotate that at frame-step 1 without a proxy, so this tool becomes mandatory at scale anyway.

For the immediate next handoff, running the equivalent ffmpeg command by hand is fine as a stopgap ? but I'd codify it before Phase B starts.

Two companion items that the research surfaced, for the same decision:

- **Pair the tool with a contract line** in the Phase-B brief revision: "annotate at frame step 1; boundary boxes on the exact contact/dead frame; we supply annotation-optimized proxies."
- **Re-add a grid-snap QA check to `validate-delivery`**: the old `coarse_sampling` check was deliberately removed from `detect_issues` on the rationale that the highlight compiler's 0.5/1.0s padding absorbs the slack ? which is true for highlight *consumption* but flatly conflicts with the ±0.3s bar golden *training labels* need. A cheap check (flag when most boundaries share a common frame stride) would have auto-caught this delivery.

Per the workflow conventions I haven't written any code ? say the word and I'll draft the collector PR (subcommand + parity verification + manifest emission + tests). Separately, the verifier re-confirmed an old audit finding still live (config stitching paddings never reach `VideoCompiler` ? `main.py:508/633`); I've flagged that as a chip so it doesn't get lost.

22. user (2026-06-12T10:00:17.908Z)

Amazing. I think I agree with your recommendation, and as usual lets document this choice (specially the rationale for picking collector and the concrete reason that Roora can't scale at our need so this is almost necessary)

23. assistant (2026-06-12T10:00:53.020Z)

Agreed ? let me document this as a collector ADR (that's where the decision lands) and open a small docs PR. First, checking the collector's DECISIONS.md format and current git state (memory says there's a stray owner-owned `.gitignore` edit in that tree I must not clobber).

24. user (2026-06-12T10:00:53.284Z)

````
# Decision records

Lightweight ADRs for this repo. (The cross-repo training-pipeline ADRs live in
the indexer repo's `docs/DECISIONS.md`; this file is for collector-local choices.)

---

## ADR-001 ? Keep explicit per-sport code instead of a sport registry (for now)

**Status:** Accepted · **Date:** 2026-06-02

### Context
Supporting a sport currently requires parallel edits across several sites:
- `SEGMENT_TABLES` + a `CREATE TABLE` + `insert_*` / `get_*` in `src/core/db.py`
- a `validate_*` method in `src/core/validator.py`
- `import-local` parse/validate/insert branches and interactive review/play in `src/cli/curate.py`
- a schema in `src/core/schemas/`

A code review flagged this as copy-paste duplication and proposed a registry/plugin
mechanism (e.g. `SPORT_REGISTRY: dict[SportType, SportSpec]` driving generic
insert/get/validate/dispatch) to make adding a sport a single entry.

### Decision
**We are keeping the explicit per-sport code and NOT introducing a registry
abstraction at this time.**

### Rationale
- **It's well-tested and mechanical.** Coverage is ~87%; adding a sport is a known,
  ~30-minute "mirror an existing sport" task (checklist below), not a research problem.
- **Schemas are heterogeneous.** Rally sports (badminton, pickleball, table tennis)
  share `rally_number / shots_count / ending_reason`, but tennis (`set_score`,
  `game_score`, `server`, `ending_type`) and cricket (`over`, `ball`, `runs`,
  `wicket`, `extras`) differ enough that a single generic table/row mapper would leak
  special-cases ? trading visible duplication for hidden complexity.
- **Explicit code is greppable.** Contributors can follow one sport end-to-end without
  learning a registry indirection. Premature abstraction is harder to undo than dup.
- **Low, slow growth.** Five sports today; new ones arrive rarely.

### Consequences
- Adding a sport touches ~5 sites (see checklist). We accept that boilerplate.
- If the cost rises, ADR-001 should be revisited and superseded.

### Revisit triggers (when to build the registry)
- Sport count grows beyond ~8, **or**
- We add a 4th+ rally-identical sport and the dispatch churn starts causing bugs, **or**
- A new cross-cutting feature would otherwise require editing every per-sport branch.

### Future design sketch (when revisited)
```
@dataclass
class SportSpec:
    schema_class: type
    table_name: str
    pk_columns: tuple[str, ...]
    columns: tuple[str, ...]
    validate_fn: Callable

SPORT_REGISTRY: dict[SportType, SportSpec] = { ... }
````
````

~~~~
# Generic: insert_segments(sport, items) / get_segments(sport, video_id) /
#          validate(sport, items) / import-local dispatch driven by the registry.
```

---

## Checklist: adding a new sport today (under ADR-001)

1. `src/core/schemas/<sport>.py` ? define the segment model (mirror the closest existing sport) with a `start_time < end_time` validator.
2. `src/core/db.py` ? add to `SEGMENT_TABLES`, add a `CREATE TABLE`, `insert_<sport>()`, `get_<sport>()`, and a cascade `DELETE` in `delete_video`.
3. `src/core/validator.py` ? add `validate_<sport>()` (duration + no-overlap).
4. `src/cli/curate.py` ? add branches in `import_local` (validate + insert), `_parse_local_annotations` (JSON **and** CSV), and the interactive review/play.
5. `config/config.yaml` ? add the sport to `supported_sports` (and confirm the `SportType` enum value in `schemas/base.py`).
6. `tests/` ? mirror the existing db/validator/cli tests for the new sport.

~~~~

25. user (2026-06-12T10:00:55.620Z)

master...origin/master

```
 M .gitignore
9e6f204 feat(corpus): ingest the n=6 owned golden corpus (262 rallies, Proprietary/commercial) (#19)
db5e1c2 fix(cvat): fold attribute names so shot_count (singular) maps to shots_count (#18)
e4dd8f1 chore(corpus): reclassify 23 ShuttleSet broadcast videos as non-commercial (owner-approved) (#17)
21b7b5d fix(cvat): fail the delivery gate on empty/corrupt/zipped CVAT instead of passing (#16)
4390a9d fix(import-local): protect golden labels + guard the fractional-rally_number landmine (#15)
```

26. user (2026-06-12T10:01:51.488Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\DECISIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)
```

27. user (2026-06-12T10:01:59.342Z)

```
Switched to a new branch 'docs/adr-002-annotation-proxy'
 M .gitignore
M  docs/DECISIONS.md
```

28. user (2026-06-12T10:02:11.975Z)

```
[docs/adr-002-annotation-proxy 265392c] docs: ADR-002 - vendor annotation proxies generated in-repo (curate prep-annotation-proxy)
 1 file changed, 80 insertions(+)
remote:
remote: Create a pull request for 'docs/adr-002-annotation-proxy' on GitHub by visiting:
remote:      https://github.com/avidullu/sports-data-collector/pull/new/docs/adr-002-annotation-proxy
remote:
branch 'docs/adr-002-annotation-proxy' set up to track 'origin/docs/adr-002-annotation-proxy'.
To https://github.com/avidullu/sports-data-collector.git
 * [new branch]      docs/adr-002-annotation-proxy -> docs/adr-002-annotation-proxy
```

29. user (2026-06-12T10:02:30.246Z)

```
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/21
```

30. user (2026-06-12T10:02:37.463Z)

```
M       .gitignore
Your branch is up to date with 'origin/master'.
Switched to branch 'master'
```

master...origin/master

```
 M .gitignore
```

31. user (2026-06-12T10:02:50.898Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

32. user (2026-06-12T10:02:57.662Z)

```
1       ---
2       name: current-state
3       description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4       metadata:
5         node_type: memory
6         type: project
7         originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8       ---
9
10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      **Checkpoint:** 2026-06-12 (stitching-padding wiring fix ? **indexer #102 + collector
#20 OPEN, CI green, awaiting
13      owner review**). Audit finding (d) (2026-06-10) fixed: config.json
`stitching.{begin,end}_padding` was silently
14      ignored by the live compiler ? both `backend/main.py` VideoCompiler sites
(`/api/compile` + CLI trim) passed only
15      `watermark=`, so reels always used ctor defaults 0.5/**1.0** vs configured 0.5/0.5. Fix
threads the typed
16      StitchingConfig paddings into both sites (+ `StitchingConfig` re-export from
backend.config); regression test drives
17      the REAL compiler through /api/compile (mocked ffmpeg) asserting a config override
(2.0/3.0) reaches `-ss/-to`
18      (8.0/17.0 for a 10?14s segment ? unwired defaults would give 9.5/15.0); CODE_MAP truth-
synced (ctor-vs-config
19      distinction + recipe row now cites the main.py wiring). ? **Behavior change on merge:
live API reels end padding
20      1.0?0.5s** (the config value). Companion **collector #20** = docstring-only:
`rally_cvat.py::detect_issues` cited the
21      hardcoded 0.5/1.0 + a stale module path; now cites config-driven 0.5/0.5 (rationale
still holds; ? cross-checks the
22      sibling Roora-proxy thread below ? boundary-slack rationale now assumes 0.5s end
padding, not 1.0s). Verified: indexer
23      suite **401 green** + 4/4 CI lanes; collector **137 green** + CI. Both repos back on
master; owner's uncommitted
24      collector `.gitignore` edit untouched. NOTE: untracked `arial.ttf` at indexer root (not
mine ? left alone, likely a
25      watermark-font experiment). ?? owner reviews/merges #102 + collector #20.
26
27      **Checkpoint:** 2026-06-12 (Roora ANNOTATION-PROXY assessment delivered ? AWAITING OWNER
DECISION, no code).
28      Owner asked: re-encode videos to a Roora-friendly profile before handoff? Researched via
workflow `wr6mnel6c`
29      (3 sweeps + adversarial verify; full JSON
`%LOCALAPPDATA%\Temp\claude\?\tasks\wr6mnel6c.output`). **Facts:** root
30      cause of the ?0.5s boundaries = 30-frame CVAT sampling grid (NOT playback delay;
direction never measured ? docs say
```

33. user (2026-06-12T10:03:19.162Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

34. assistant (2026-06-12T10:03:38.817Z)

Documented and shipped for review: **[sports-data-collector#21](https://github.com/avidullu/sports-data-collector/pull/21)** ? a docs-only PR adding **ADR-002** to the collector's `docs/DECISIONS.md` (matching ADR-001's format).

The ADR captures exactly the two things you called out, plus the rest of the decision record:

- **Why the collector:** it owns the vendor flow end-to-end, so proxy prep is simply command #0 of the existing `convert-cvat` ? `validate-delivery` ? `import-local` pipeline; the ffmpeg call is the trivial 20% while the durable value ? frame-parity verification and manifest provenance with both MD5s ? is precisely what one-off scripts historically skip (cited precedents: the unfilled intake-manifest TODOs, the `%TEMP%` artifact losses, the downscaled-MD5 keying incident). Standalone-script and indexer homes are recorded as rejected alternatives, with the standalone explicitly allowed as a stopgap for an immediate redelivery.
- **The necessity argument:** Phase B is ~12?15 videos per sport at ~30 minutes each; a 30-minute 4K/59.94 file is >100k frames, so frame-step-1 annotation on originals is not realistic for Roora ? the pilot already showed their coping mechanism (the 30-frame grid) structurally violates the ±0.3s bar. The ADR frames the proxy as a **near-precondition for Phase B, not an optimization**.
- Also recorded: the full proxy contract (same fps ? never reduce ? no trimming, parity check, 1080p/CRF 18?20, dense keyframes, audio kept), the follow-ups (Phase-B brief convention line + re-adding a grid-snap QA check to `validate-delivery`), and revisit triggers.

Housekeeping: the collector checkout is back on master with your uncommitted `.gitignore` edit untouched, and the memory checkpoint is updated. One coordination note: the padding-fix chip session you started has opened [collector#20](https://github.com/avidullu/sports-data-collector/pull/20), which edits the `rally_cvat.py` docstring rationale (0.5/1.0 ? 0.5/0.5 padding) ? no file overlap with #21, and both texts stay accurate regardless of merge order.

Implementation of `curate prep-annotation-proxy` itself remains gated on your greenlight, per the ADR and the repo convention.

35. user (2026-06-12T10:14:07.510Z)

Please go ahead and start the work! Appreciate the research oriented strategy here! One note that the watermark work has been merged, so please ensure your local is in sync with remote before proceeding.

36. assistant (2026-06-12T10:14:29.163Z)

Starting now. First, syncing both repos with remote and checking the PR states (#20/#21/#102), then I'll implement against the ADR-002 contract.

37. user (2026-06-12T10:14:33.802Z)

From https://github.com/avidullu/sports-data-collector
 - [deleted]   (none)   -> origin/feat/golden-corpus-n6

master...origin/master

 M .gitignore
9e6f204 feat(corpus): ingest the n=6 owned golden corpus (262 rallies, Proprietary/commercial) (#19)
db5e1c2 fix(cvat): fold attribute names so shot_count (singular) maps to shots_count (#18)
e4dd8f1 chore(corpus): reclassify 23 ShuttleSet broadcast videos as non-commercial (owner-approved) (#17)
21b7b5d fix(cvat): fail the delivery gate on empty/corrupt/zipped CVAT instead of passing (#16)
4390a9d fix(import-local): protect golden labels + guard the fractional-rally_number landmine (#15)

38. user (2026-06-12T10:14:36.633Z)

```
From https://github.com/avidullu/badminton-highlight-indexer
 - [deleted]         (none)     -> origin/chore/repo-hygiene
 - [deleted]         (none)     -> origin/ci/coverage-floor
 - [deleted]         (none)     -> origin/ci/mypy-lane
 - [deleted]         (none)     -> origin/ci/node24-action-bump
 - [deleted]         (none)     -> origin/docs/pr2-merge-status-sync
 - [deleted]         (none)     -> origin/feat/config-unknown-key-warning
 - [deleted]         (none)     -> origin/feat/upload-download
 - [deleted]         (none)     -> origin/fix/db-connection-close
 - [deleted]         (none)     -> origin/fix/pydantic-v2-model-dump
 - [deleted]         (none)     -> origin/fix/wsl-tilde-quoting
```

master...origin/master

```
?? arial.ttf
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
ffdfde0 Merge pull request #103 from avidullu:claude/watermark-bundled-font
112e75d feat(stitcher): bundle Apache-2.0 default watermark font + log fallback reason
82e479f Merge pull request #102 from avidullu:fix/stitching-padding-config
```

39. user (2026-06-12T10:14:54.134Z)

```
Already up to date.
9e6f204 feat(corpus): ingest the n=6 owned golden corpus (262 rallies, Proprietary/commercial)
(#19)
db5e1c2 fix(cvat): fold attribute names so shot_count (singular) maps to shots_count (#18)
e4dd8f1 chore(corpus): reclassify 23 ShuttleSet broadcast videos as non-commercial (owner-
approved) (#17)

number state  title
------ -----  -----
    21 OPEN   docs: ADR-002 ? vendor annotation proxies generated in-repo (curate prep-
annotation-proxy)
    20 OPEN   docs(cvat): update detect_issues padding rationale to config-driven values
    19 MERGED Corpus: ingest the n=6 owned golden corpus (262 rallies, commercial-clean)
    18 MERGED fix(cvat): fold attribute names so shot_count (singular) maps to shots_count
    17 MERGED Corpus: reclassify 23 ShuttleSet broadcast videos as non-commercial (owner-
approved)
```

40. user (2026-06-12T10:14:54.816Z)

```
The user doesn't want to proceed with this tool use. The tool use was rejected (eg. if it was a
file edit, the new_string was NOT written to the file). STOP what you are doing and wait for the
user to tell you how to proceed.
```

41. user (2026-06-12T10:14:54.817Z)

```
[Request interrupted by user for tool use]
```

42. user (2026-06-12T10:27:22.482Z)

```
Sorry I interrupted your work so that I could merge some PRs. Can you restart and feel free to
merge the PRs as well to complete the task. All the collector and highlighter PRs are now merged
so you must have a clean slate
```

43. assistant (2026-06-12T10:27:31.736Z)

```
No problem ? restarting with a sync of both repos and verifying the clean slate.
```

44. user (2026-06-12T10:27:39.807Z)

```
From https://github.com/avidullu/sports-data-collector
 - [deleted]         (none)     -> origin/docs/adr-002-annotation-proxy
 - [deleted]         (none)     -> origin/docs/compiler-padding-rationale
   9e6f204..4fb2d6f  master     -> origin/master
Updating 9e6f204..4fb2d6f
Fast-forward
 docs/DECISIONS.md              | 80 ++++++++++++++++++++++++++++++++++++++++++
 src/parsers/cvat/rally_cvat.py |  3 +-
 2 files changed, 82 insertions(+), 1 deletion(-)
```

master...origin/master

```
 M .gitignore
4fb2d6f docs: ADR-002 - vendor annotation proxies generated in-repo (curate prep-annotation-proxy) (#21)
a2d08f8 docs(cvat): update detect_issues padding rationale to config-driven values (#20)
9e6f204 feat(corpus): ingest the n=6 owned golden corpus (262 rallies, Proprietary/commercial) (#19)
db5e1c2 fix(cvat): fold attribute names so shot_count (singular) maps to shots_count (#18)
```

45. user (2026-06-12T10:27:44.621Z)

```
Already up to date.
ffdfde0 Merge pull request #103 from avidullu:claude/watermark-bundled-font
112e75d feat(stitcher): bundle Apache-2.0 default watermark font + log fallback reason
82e479f Merge pull request #102 from avidullu:fix/stitching-padding-config
30e4899 fix(api): thread config stitching paddings into live VideoCompiler sites
```

46. user (2026-06-12T10:28:23.128Z)

```
import os
import sys
import time
import json
import csv
import argparse
from typing import List, Dict, Any, Tuple
from tabulate import tabulate

from src.core.config import load_config
from src.core.parsing import safe_int as _safe_int, safe_float_required as _safe_float
from src.core.db import SportsDatabase
from src.core.schemas.base import VideoMetadata, SportType, LicenseType, CurationStatus
from src.core.schemas.badminton import BadmintonRally
from src.core.schemas.tennis import TennisPoint
from src.core.schemas.cricket import CricketDelivery
from src.core.schemas.pickleball import PickleballRally
from src.core.schemas.tabletennis import TableTennisRally
from src.core.validator import DatasetValidator
from src.core.downloader import download_best_video, probe_resolution_fps, height_to_label


class SportsDataCLI:
    def __init__(self, config_path: str = "config/config.yaml"):
        self.config_path = config_path
        self.config = self._load_config()

        # Resolve DB path
        db_path = self.config.get("database_path", "data/sports_metadata.db")
        self.db = SportsDatabase(db_path)
        self.validator = DatasetValidator(config_path)

    def _load_config(self) -> Dict[str, Any]:
```

```
        if not os.path.exists(self.config_path):
            print(f"Warning: Configuration file not found at {self.config_path}. Using defaults.")
        return load_config(self.config_path)

    def status(self):
        """Displays status summary of all videos in the database."""
        # Query total count by status via the public DB read API.
        rows = self.db.get_status_counts()  # list of (sport, status, cnt)

        if not rows:
            print("\nDatabase is currently empty. Run crawlers or import local data.\n")
            return

        data = [[sport, status, cnt] for sport, status, cnt in rows]
        print("\n=== Dataset Ingestion Status ===")
        print(tabulate(data, headers=["Sport", "Status", "Count"], tablefmt="grid"))
        print()

    @staticmethod
    def _annotation_key(sport_str, a):
        """The per-sport primary-key value of one annotation (for duplicate detection)."""
        if sport_str in ("badminton", "pickleball", "tabletennis"):
            return getattr(a, "rally_number", None)
        if sport_str == "tennis":
            return getattr(a, "point_number", None)
        if sport_str == "cricket":
            return (getattr(a, "over", None), getattr(a, "ball", None))
        return None

    def _guard_import(self, sport_str, video_id, annotations, force=False):
        """Pre-insert safety checks (run BEFORE any DB write, so a rejection is atomic)."""
        # (1) Duplicate primary-key guard. int(float()) truncation (e.g. 7 and 7.5 both
        # become 7) would otherwise raise IntegrityError mid-insert, leaving a committed
        # video row with rolled-back annotations. Catch it before touching the DB.
        keys = [self._annotation_key(sport_str, a) for a in annotations]
        keys = [k for k in keys if k is not None]
        seen, dupes = set(), set()
        for k in keys:
            if k in seen:
                dupes.add(k)
            seen.add(k)
        if dupes:
            print(f"Error: duplicate rally/point identifier(s) {sorted(dupes)} in the annotations "
                  f"(often a fractional id like 7.5 truncating to 7).")
            print("  -> re-run with --normalize, or normalize first: "
                  "python -m src.cli.normalize_annotations --in <csv> --out clean.csv")
            sys.exit(1)

        # (2) Golden-label protection ? refuse to silently clobber an existing video's
        # annotations (filename-derived video_ids collide easily).
        if self.db.get_video(video_id):
            try:
                n = len(self.db.get_segment_intervals(video_id, SportType(sport_str)))
            except Exception:
                n = 0
            if n > 0 and not force:
                print(f"Error: video_id '{video_id}' already has {n} annotated segment(s); "
                      f"importing would REPLACE them.")
                print("  -> pass --force to overwrite, or use a distinct --video-id to keep both.")
```

```
                sys.exit(1)
            if n > 0:
                print(f"Warning: --force given ? replacing {n} existing segment(s) for video_id '{video_id}'.")

    def import_local(self, args):
        """Manually registers a local video and annotation file into the database."""
        sport_str = args.sport.lower()
        if sport_str not in self.config.get("supported_sports", ["badminton"]):
            print(f"Error: Sport '{sport_str}' is not supported. Supported: {self.config.get('supported_sports')}")
            sys.exit(1)

        # Validate video path
        if not os.path.exists(args.video_path):
            print(f"Warning: Local video path '{args.video_path}' does not exist on disk, but continuing registration.")

        # Validate annotation file
        if not os.path.exists(args.annotations_path):
            print(f"Error: Annotations file '{args.annotations_path}' not found.")
            sys.exit(1)

        match_name = args.match_name or os.path.basename(args.video_path).split('.')[0]
        video_id = args.video_id or match_name.lower().replace(" ", "_")

        # Parse license
        try:
            license_type = LicenseType(args.license)
        except ValueError:
            print(f"Warning: Unknown license '{args.license}'. Defaulting to Unknown (non-commercial).")
            license_type = LicenseType.UNKNOWN

        is_commercial = self.validator.check_license_commercial(license_type)

        # Optional: normalize a raw CSV annotator export before parsing (drops unusable
        # rows, renumbers fractional/duplicate rally ids) so the landmines below can't fire.
        annotations_path = args.annotations_path
        if getattr(args, "normalize", False) and annotations_path.lower().endswith(".csv"):
            from src.cli.normalize_annotations import normalize_file
            normalized_path = annotations_path + ".normalized.csv"
            rep = normalize_file(annotations_path, normalized_path)
            print("=== normalize-annotations ===")
            print(rep.summary())
            annotations_path = normalized_path

        # 1. Parse Annotations
        annotations = []
        try:
            annotations = self._parse_local_annotations(sport_str, video_id, annotations_path)
        except Exception as e:
            print(f"Error parsing annotations: {e}")
            sys.exit(1)

        # 2. Run Validation
        is_valid = True
        errors = []
        if sport_str == "badminton":
            is_valid, errors = self.validator.validate_badminton_rallies(annotations)
        elif sport_str == "tennis":
            is_valid, errors = self.validator.validate_tennis_points(annotations)
```

```
            elif sport_str == "cricket":
                is_valid, errors = self.validator.validate_cricket_deliveries(annotations)
            elif sport_str == "pickleball":
                is_valid, errors = self.validator.validate_pickleball_rallies(annotations)
            elif sport_str == "tabletennis":
                is_valid, errors = self.validator.validate_tabletennis_rallies(annotations)

            if not is_valid:
                print(f"Error: Annotations failed validation!")
                for err in errors:
                    print(f"  - {err}")
                sys.exit(1)

            # 2b. Pre-insert safety ? checked BEFORE any DB write, so a rejection never
            # leaves a half-updated DB (a committed video row with rolled-back annotations).
            self._guard_import(sport_str, video_id, annotations, force=getattr(args, "force", False))

            # 3. Create Video Metadata
            video_meta = VideoMetadata(
                video_id=video_id,
                sport=SportType(sport_str),
                video_url=None,
                local_path=args.video_path,
                license_type=license_type,
                license_url=None,
                is_commercial=is_commercial,
                status=CurationStatus.CURATED,  # Custom local imports bypass staging by default
                match_name=match_name,
                fps=args.fps,
                resolution=args.resolution
            )

            # 4. Insert into Database
            self.db.insert_video(video_meta)

            count = 0
            if sport_str == "badminton":
                count = self.db.insert_badminton_rallies(annotations)
            elif sport_str == "tennis":
                count = self.db.insert_tennis_points(annotations)
            elif sport_str == "cricket":
                count = self.db.insert_cricket_deliveries(annotations)
            elif sport_str == "pickleball":
                count = self.db.insert_pickleball_rallies(annotations)
            elif sport_str == "tabletennis":
                count = self.db.insert_tabletennis_rallies(annotations)

            print(f"\nSuccessfully imported local video '{match_name}':")
            print(f"  - Video ID: {video_id}")
            print(f"  - Sport: {sport_str}")
            print(f"  - Commercial Use Allowed: {is_commercial} (License: {license_type.value})")
            print(f"  - Registered {count} annotated segments.\n")

            # Per-video observability report (next to the video). Never raises.
            from src.reporting import emit_import_report
            src_fmt = os.path.splitext(args.annotations_path)[1].lstrip(".").lower() or "unknown"
            report_paths = emit_import_report(
                video_meta_obj=video_meta, annotations=annotations, is_valid=True,
                errors=[], count=count, sport=sport_str, source_format=src_fmt,
                fallback_dir=os.path.dirname(os.path.abspath(args.annotations_path)),
            )
```

```python
        if report_paths:
            print(f"  - Observability report: {report_paths.get('technical_md')}")
            print(f"  - Customer summary    : {report_paths.get('customer_md')}\n")

    def _parse_local_annotations(self, sport: str, video_id: str, filepath: str) -> List[Any]:
        """Parses simple JSON or CSV annotations files."""
        parsed_items = []
        ext = os.path.splitext(filepath)[1].lower()

        if ext == '.json':
            with open(filepath, 'r', encoding='utf-8') as f:
                data = json.load(f)

            if not isinstance(data, list):
                raise ValueError("JSON file must contain a list of objects.")

            # Uses the same safe casts as the CSV path: a missing/blank required
            # time raises a clear ValueError (caught by the caller) instead of a
            # bare KeyError, and optional ints degrade to sensible defaults.
            for idx, obj in enumerate(data):
                if sport == "badminton":
                    parsed_items.append(BadmintonRally(
                        video_id=video_id,
                        rally_number=_safe_int(obj.get("rally_number"), idx + 1),
                        start_time=_safe_float(obj.get("start_time")),
                        end_time=_safe_float(obj.get("end_time")),
                        score_before=obj.get("score_before"),
                        score_after=obj.get("score_after"),
                        shots_count=_safe_int(obj.get("shots_count"), None),
                        ending_reason=obj.get("ending_reason"),
                        last_shot_outcome=obj.get("last_shot_outcome")
                    ))
                elif sport == "tennis":
                    parsed_items.append(TennisPoint(
                        video_id=video_id,
                        point_number=_safe_int(obj.get("point_number"), idx + 1),
                        start_time=_safe_float(obj.get("start_time")),
                        end_time=_safe_float(obj.get("end_time")),
                        set_score=obj.get("set_score"),
                        game_score=obj.get("game_score"),
                        server=obj.get("server"),
                        ending_type=obj.get("ending_type")
                    ))
                elif sport == "cricket":
                    parsed_items.append(CricketDelivery(
                        video_id=video_id,
                        over=_safe_int(obj.get("over"), 0),
                        ball=_safe_int(obj.get("ball"), 0),
                        start_time=_safe_float(obj.get("start_time")),
                        end_time=_safe_float(obj.get("end_time")),
                        batsman=obj.get("batsman"),
                        bowler=obj.get("bowler"),
                        runs=_safe_int(obj.get("runs"), None),
                        wicket=bool(obj.get("wicket", False)),
                        extras=_safe_int(obj.get("extras"), 0)
                    ))
                elif sport == "pickleball":
                    parsed_items.append(PickleballRally(
                        video_id=video_id,
                        rally_number=_safe_int(obj.get("rally_number"), idx + 1),
                        start_time=_safe_float(obj.get("start_time")),
                        end_time=_safe_float(obj.get("end_time")),
                        score_before=obj.get("score_before"),
                        score_after=obj.get("score_after"),
```

```
                shots_count=_safe_int(obj.get("shots_count"), None),
                ending_reason=obj.get("ending_reason"),
                last_shot_outcome=obj.get("last_shot_outcome")
            ))
        elif sport == "tabletennis":
            parsed_items.append(TableTennisRally(
                video_id=video_id,
                rally_number=_safe_int(obj.get("rally_number"), idx + 1),
                start_time=_safe_float(obj.get("start_time")),
                end_time=_safe_float(obj.get("end_time")),
                score_before=obj.get("score_before"),
                score_after=obj.get("score_after"),
                shots_count=_safe_int(obj.get("shots_count"), None),
                ending_reason=obj.get("ending_reason"),
                last_shot_outcome=obj.get("last_shot_outcome")
            ))
elif ext == '.csv':
    with open(filepath, 'r', encoding='utf-8') as f:
        reader = csv.DictReader(f)
        # Normalize headers
        reader.fieldnames = [n.lower().strip() for n in reader.fieldnames] if reader.fieldnames else []

        for idx, row in enumerate(reader):
            if sport == "badminton":
                parsed_items.append(BadmintonRally(
                    video_id=video_id,
                    rally_number=_safe_int(row.get("rally_number"), idx + 1),
                    start_time=_safe_float(row.get("start_time")),
                    end_time=_safe_float(row.get("end_time")),
                    score_before=row.get("score_before"),
                    score_after=row.get("score_after"),
                    shots_count=_safe_int(row.get("shots_count"), None),
                    ending_reason=row.get("ending_reason"),
                    last_shot_outcome=row.get("last_shot_outcome")
                ))
            elif sport == "tennis":
                parsed_items.append(TennisPoint(
                    video_id=video_id,
                    point_number=_safe_int(row.get("point_number"), idx + 1),
                    start_time=_safe_float(row.get("start_time")),
                    end_time=_safe_float(row.get("end_time")),
                    set_score=row.get("set_score"),
                    game_score=row.get("game_score"),
                    server=row.get("server"),
                    ending_type=row.get("ending_type")
                ))
            elif sport == "cricket":
                parsed_items.append(CricketDelivery(
                    video_id=video_id,
                    over=_safe_int(row.get("over"), 0),
                    ball=_safe_int(row.get("ball"), 0),
                    start_time=_safe_float(row.get("start_time")),
                    end_time=_safe_float(row.get("end_time")),
                    batsman=row.get("batsman"),
                    bowler=row.get("bowler"),
                    runs=_safe_int(row.get("runs"), None),
                    wicket=(row.get("wicket") or "").lower() in ["true", "1", "yes"],
                    extras=_safe_int(row.get("extras"), 0)
                ))
            elif sport == "pickleball":
                parsed_items.append(PickleballRally(
                    video_id=video_id,
                    rally_number=_safe_int(row.get("rally_number"), idx + 1),
```

```
                            start_time=_safe_float(row.get("start_time")),
                            end_time=_safe_float(row.get("end_time")),
                            score_before=row.get("score_before"),
                            score_after=row.get("score_after"),
                            shots_count=_safe_int(row.get("shots_count"), None),
                            ending_reason=row.get("ending_reason"),
                            last_shot_outcome=row.get("last_shot_outcome")
                        ))
                    elif sport == "tabletennis":
                        parsed_items.append(TableTennisRally(
                            video_id=video_id,
                            rally_number=_safe_int(row.get("rally_number"), idx + 1),
                            start_time=_safe_float(row.get("start_time")),
                            end_time=_safe_float(row.get("end_time")),
                            score_before=row.get("score_before"),
                            score_after=row.get("score_after"),
                            shots_count=_safe_int(row.get("shots_count"), None),
                            ending_reason=row.get("ending_reason"),
                            last_shot_outcome=row.get("last_shot_outcome")
                        ))
        else:
            raise ValueError("Unsupported annotations file extension. Use .json or .csv.")

        return parsed_items

    def _fetch_one_with_retry(self, url, dest_path, args, max_height, max_size_mb):
        """Download one video at best resolution, retrying transient failures.

        A below-min-height result almost always means YouTube rate-limited the
        request and served a low fallback (not that the video lacks HD), so we
        retry with a growing backoff. Returns (DownloadResult, None) on success,
        or (None, failure_reason) once retries are exhausted. A below-min file is
        discarded each attempt so it never lands in the DB.
        """
        attempts = max(1, args.retries + 1)
        reason = "unknown error"
        for i in range(attempts):
            result = download_best_video(
                url, dest_path,
                max_height=max_height, max_size_mb=max_size_mb,
                cookies_from_browser=args.cookies_from_browser,
                cookies_file=args.cookies,
                player_client=args.player_client,
            )
            if result.success and (result.height is None or result.height >= args.min_height):
                return result, None

            if not result.success:
                reason = result.error or "unknown error"
            else:
                reason = (f"got {result.resolution_label} (< min-height {args.min_height}p) ? "
                          f"YouTube served a low format (rate-limited?)")
                try:
                    os.remove(result.path)
                except OSError:
                    pass

            if i < attempts - 1:
                backoff = args.retry_backoff * (i + 1)
                print(f"  attempt {i+1}/{attempts} failed: {reason}\n  retrying in {backoff:.0f}s...")
                time.sleep(backoff)
```

```
            return None, reason

        def fetch(self, args):
            """(Re)download videos at best available resolution, updating the DB in place.

            This is the in-place upgrade path: rows, rally annotations, and curation
            status are all preserved ? only the video file plus its `local_path`,
            `resolution`, and `fps` are refreshed. By default it skips videos that
            already have a local file at or above `--min-height`, so re-runs are cheap
            and resumable; `--force` re-fetches everything.
            """
            storage = self.config.get("video_storage", {})
            local_dir = storage.get("local_dir", "./data/videos")
            max_download_size_mb = storage.get("max_download_size_mb")
            max_height = args.max_height if args.max_height is not None else storage.get("max_height")
            os.makedirs(local_dir, exist_ok=True)

            commercial_only = not args.include_noncommercial
            if args.status == "all":
                statuses = [CurationStatus.CURATED, CurationStatus.STAGING]
            else:
                statuses = [CurationStatus(args.status)]

            supported = self.config.get("supported_sports", ["badminton"])
            sport_strs = [args.sport] if args.sport else supported

            # Collect target videos.
            videos = []
            for sport_str in sport_strs:
                try:
                    sport = SportType(sport_str.lower())
                except ValueError:
                    print(f"Warning: skipping unknown sport '{sport_str}'.")
                    continue
                for status in statuses:
                    videos.extend(self.db.get_videos_by_status(status, sport))
            if args.video_id:
                videos = [v for v in videos if v.video_id == args.video_id]

            if not videos:
                print("\nNo matching videos to fetch.\n")
                return

            results = []  # (video_id, action, detail)
            for video in videos:
                if commercial_only and not video.is_commercial:
                    results.append((video.video_id, "skip", "non-commercial"))
                    continue
                if not video.video_url or "watch?v=example" in video.video_url:
                    # Either no URL, or the parser's placeholder (the ShuttleSet catalog
                    # had no URL for this match) ? nothing to download.
                    results.append((video.video_id, "skip", "no real source URL"))
                    continue

                dest_path = os.path.join(local_dir, f"{video.video_id}.mp4")

                # Skip if we already have a good-enough local file (unless --force).
                if not args.force and os.path.exists(dest_path):
                    _, height, _ = probe_resolution_fps(dest_path)
                    if height is not None and height >= args.min_height:
                        results.append((video.video_id, "have", f"{height_to_label(height)} already"))
                        continue

```

```
                print(f"Fetching {video.video_id} ({video.match_name})...")
                result, fail_reason = self._fetch_one_with_retry(
                    video.video_url, dest_path, args, max_height, max_download_size_mb)
                if fail_reason is not None:
                    results.append((video.video_id, "FAIL", fail_reason))
                    if args.sleep:
                        time.sleep(args.sleep)
                    continue

                # Update only the file-derived fields; preserve everything else (UPSERT
                # mutates the row in place, leaving rallies intact).
                video.local_path = result.path
                if result.resolution_label:
                    video.resolution = result.resolution_label
                if result.fps:
                    video.fps = result.fps
                self.db.insert_video(video)

                detail = (f"{result.resolution_label or '?'}"
                          f"{f' {result.fps:g}fps' if result.fps else ''}, {result.size_mb:.0f} MB")
                results.append((video.video_id, "fetched", detail))
                if args.sleep:
                    time.sleep(args.sleep)

            print("\n=== Fetch summary ===")
            print(tabulate([[vid, action, detail] for vid, action, detail in results],
                           headers=["Video ID", "Action", "Detail"], tablefmt="grid"))
            fetched = sum(1 for _, a, _ in results if a == "fetched")
            failed = [r for r in results if r[1] == "FAIL"]
            print(f"\nFetched/updated : {fetched}   Already-good : {sum(1 for _,a,_ in results if a=='have')}   "
                  f"Skipped : {sum(1 for _,a,_ in results if a=='skip')}   Failed : {len(failed)}")
            if failed:
                print("Failures (left unchanged in DB):")
                for vid, _, detail in failed:
                    print(f"  - {vid}: {detail}")
            print()

    # Code/data licenses that can NEVER apply to video footage ? if one of these is
    # stamped on a remote/broadcast video it is an annotation-vs-video license conflation.
    _CODE_DATA_LICENSES = {LicenseType.MIT, LicenseType.APACHE_2, LicenseType.BSD}

    def reclassify_licenses(self, args):
        """Correct annotation-vs-video license conflation in existing rows.

        A third-party video (one with a remote `video_url` ? i.e. broadcast footage like
        ShuttleSet, whether or not it has since been downloaded to `local_path`) whose
        `license_type` is a code/data license (MIT/Apache-2.0/BSD) is a conflation: that
        license covers the ANNOTATIONS, not the footage. This moves the license to
        `annotation_license`, sets the video `license_type` to Unknown, and recomputes
        `is_commercial=False` so such videos stop being exported as commercially clean.

        Dry-run by default; pass --apply to persist.
        """
        changed = []
        seen = set()
        for status in CurationStatus:
            for video in self.db.get_videos_by_status(status):
                if video.video_id in seen:
                    continue
                seen.add(video.video_id)
                # The broadcast signal is a third-party video_url; a downloaded copy
```

```
                # (local_path set) does not change that the FOOTAGE rights are not ours.
                is_third_party = bool(video.video_url)
                if is_third_party and video.license_type in self._CODE_DATA_LICENSES:
                    changed.append(video)

        if not changed:
            print("No conflated rows found ? every remote video already has a video-footage license.")
            return

        print(f"{'APPLYING' if args.apply else 'DRY-RUN'} ? {len(changed)} conflated row(s):")
        for v in changed:
            print(f"  {v.video_id:40s} {v.license_type.value} (video) -> annotation_license; "
                  f"video license -> Unknown; is_commercial {v.is_commercial} -> False")
            if args.apply:
                v.annotation_license = v.annotation_license or v.license_type
                v.license_type = LicenseType.UNKNOWN
                v.is_commercial = self.validator.check_license_commercial(v.license_type)
                self.db.insert_video(v)

        if args.apply:
            print("\nApplied. Re-run `curate export` to regenerate the manifest without these videos.")
        else:
            print("\nDry-run only. Re-run with --apply to persist, then `curate export`.")

    def export_eval_set(self, args):
        """Export curated annotations as indexer-ready eval/training sets.

        Emits, per qualifying video, a `<out_dir>/<sport>/<video_id>.csv` in the
        indexer's `start,end` ground-truth format (decimal seconds), plus a single
        `<out_dir>/manifest.json` that pairs each CSV with its video file. The
        manifest is the contract the downstream eval/training driver reads: the
        indexer keys videos by *file MD5*, not by our `video_id`, so it must run
        against the exact `local_path`/`video_url` recorded here.

        By default only CURATED + commercially-safe videos are exported, mirroring
        the license gate already enforced elsewhere in the toolchain.
        """
        out_dir = args.out_dir
        commercial_only = not args.include_noncommercial
        statuses = [CurationStatus.CURATED]
        if args.include_staging:
            statuses.append(CurationStatus.STAGING)

        supported = self.config.get("supported_sports", ["badminton"])
        sport_strs = [args.sport] if args.sport else supported

        manifest = []
        skipped_noncomm = 0
        skipped_empty = 0

        for sport_str in sport_strs:
            try:
                sport = SportType(sport_str.lower())
            except ValueError:
                print(f"Warning: skipping unknown sport '{sport_str}'.")
                continue

            videos = []
            for status in statuses:
```

```
                videos.extend(self.db.get_videos_by_status(status, sport))

            for video in videos:
                if commercial_only and not video.is_commercial:
                    skipped_noncomm += 1
                    continue

                intervals = self.db.get_segment_intervals(video.video_id, sport)
                if not intervals:
                    skipped_empty += 1
                    continue

                sport_dir = os.path.join(out_dir, sport.value)
                os.makedirs(sport_dir, exist_ok=True)
                csv_path = os.path.join(sport_dir, f"{video.video_id}.csv")
                with open(csv_path, "w", newline="", encoding="utf-8") as f:
                    writer = csv.writer(f)
                    writer.writerow(["start", "end"])
                    for start, end in intervals:
                        writer.writerow([start, end])

                manifest.append({
                    "video_id": video.video_id,
                    "sport": sport.value,
                    "csv": os.path.relpath(csv_path, out_dir).replace(os.sep, "/"),
                    "local_path": video.local_path,
                    "video_url": video.video_url,
                    "match_name": video.match_name,
                    "fps": video.fps,
                    "resolution": video.resolution,
                    "license_type": video.license_type.value,
                    "is_commercial": video.is_commercial,
                    "status": video.status.value,
                    "num_segments": len(intervals),
                })

        os.makedirs(out_dir, exist_ok=True)
        manifest_doc = {
            "commercial_only": commercial_only,
            "statuses": [s.value for s in statuses],
            "total_videos": len(manifest),
            "total_segments": sum(m["num_segments"] for m in manifest),
            "eval_sets": manifest,
        }
        manifest_path = os.path.join(out_dir, "manifest.json")
        with open(manifest_path, "w", encoding="utf-8") as f:
            json.dump(manifest_doc, f, indent=2)

        print(f"\n=== Exported eval/training set to '{out_dir}' ===")
        if manifest:
            summary = [[m["sport"], m["video_id"], m["num_segments"],
                        "YES" if m["is_commercial"] else "NO",
                        os.path.basename(m["local_path"]) if m["local_path"] else "(no local file)"]
                       for m in manifest]
            print(tabulate(summary, headers=["Sport", "Video ID", "Segments", "Comm?", "Video file"], tablefmt="grid"))
        print(f"\nVideos exported : {len(manifest)}")
        print(f"Total segments  : {manifest_doc['total_segments']}")
        if skipped_noncomm:
            print(f"Skipped (non-commercial) : {skipped_noncomm}  (pass --include-noncommercial to include)")
        if skipped_empty:
            print(f"Skipped (no annotations) : {skipped_empty}")
        print(f"Manifest        : {manifest_path}\n")
```

```
    def convert_cvat(self, args):
        """Convert a vendor CVAT rally export into our canonical per-rally CSV.

        Absorbs the CVAT bounding-box delivery format (rally fields carried as box
        attributes) on the ingestion side: each rally's time is recomputed from
        its frame range at the video's true fps, and content defects (overlaps,
        zero-duration, off-vocabulary endings, coarse sampling) are written to an
        issues report rather than silently "fixed". The emitted CSV imports via
        `import-local`; any blocker issue must be resolved first, since the
        importer's validator rejects overlapping / zero-duration rallies.
        """
        from src.parsers.cvat.rally_cvat import (
            convert, write_canonical_csv, render_issue_report,
        )
        try:
            result = convert(args.input, fps=args.fps,
                             video_path=args.video_path, step=args.step)
        except (FileNotFoundError, ValueError) as e:
            print(f"Error: {e}")
            sys.exit(1)

        if not result.records:
            print(f"Error: the CVAT export parsed to ZERO rallies ({args.input}). "
                  f"Likely the wrong job was exported or the rally_number attribute was "
                  f"renamed (boxes without a matched rally_number are skipped).")
            sys.exit(1)

        write_canonical_csv(result.records, args.out)

        # Observability report next to the canonical CSV. Never raises.
        from src.reporting import emit_delivery_report
        src_name = os.path.splitext(os.path.basename(args.input))[0]
        dur = (result.meta.get("stop_frame") / result.fps) if (result.meta.get("stop_frame") and result.fps) else None
        emit_delivery_report(
            out_dir=os.path.dirname(os.path.abspath(args.out)) or ".", source=src_name,
            records=result.records, issues=result.issues, fps=result.fps, video_path=args.video_path,
            video_duration_s=dur, source_format="cvat", step=result.step,
        )

        report = render_issue_report(result, os.path.basename(args.input))
        if args.issues:
            with open(args.issues, "w", encoding="utf-8") as f:
                f.write(report)
        print(report)
        print(f"Canonical CSV written: {args.out}  ({len(result.records)} rallies)")
        if result.n_blockers:
            print(f"\n  {result.n_blockers} BLOCKER issue(s) ? import-local will reject "
                  f"this file until they are resolved (by the vendor or manual review).")
        else:
            print("\n  No blockers. Next:")
            print(f"    python -m src.cli.curate import-local --sport <sport> "
                  f"--video-path <video> --annotations-path {args.out} --fps {result.fps:g}")

    def validate_delivery(self, args):
        """Run the full validation suite on a rally delivery and emit a report.

        Accepts a CVAT export (--input + --video-path/--fps) or an already-canonical
        CSV (--csv). Writes, into --out-dir:
```

```
      <id>_report.md    human report (scoreboard + issues + rallies to review)
      <id>_report.json  machine summary (for CI / a bulk-ingest loop)
      <id>_review.csv   the annotatable manual-verification sheet
    Exits non-zero if any blocker issue exists, so it can gate a bulk pipeline.
    """
    import json
    from src.parsers.cvat.rally_cvat import (
        convert, load_canonical_csv, detect_issues,
        build_summary, render_validation_md, write_review_csv,
    )
    vid = args.video_id or "delivery"
    fps = None
    step = None
    stop_frame = None
    if args.input:
        try:
            result = convert(args.input, fps=args.fps, video_path=args.video_path)
        except (FileNotFoundError, ValueError) as e:
            print(f"Error: {e}")
            sys.exit(1)
        records, issues, fps = result.records, result.issues, result.fps
        step = result.step
        stop_frame = result.meta.get("stop_frame")
        if not records:
            print(f"Error: the CVAT export parsed to ZERO rallies ({args.input}). "
                  f"The wrong job was exported or the rally_number attribute was renamed. "
                  f"GATE: BLOCKED.")
            sys.exit(2)
    elif args.csv:
        try:
            records = load_canonical_csv(args.csv)
        except FileNotFoundError as e:
            print(f"Error: {e}")
            sys.exit(1)
        if not records:
            print(f"Error: no usable rally rows in {args.csv}")
            sys.exit(1)
        issues = detect_issues(records)
    else:
        print("Error: provide --input <cvat.xml> or --csv <canonical.csv>")
        sys.exit(1)

    os.makedirs(args.out_dir, exist_ok=True)
    summary = build_summary(records, issues, source=vid, fps=fps)
    md = render_validation_md(records, issues, source=vid)
    json_path = os.path.join(args.out_dir, f"{vid}_report.json")
    md_path = os.path.join(args.out_dir, f"{vid}_report.md")
    review_path = os.path.join(args.out_dir, f"{vid}_review.csv")
    with open(json_path, "w", encoding="utf-8") as f:
        json.dump(summary, f, indent=2)
    with open(md_path, "w", encoding="utf-8") as f:
        f.write(md)
    write_review_csv(records, issues, review_path)

    # Observability report (envelope JSON + technical md + customer summary),
    # colocated with the existing artifacts. Never raises.
    from src.reporting import emit_delivery_report
    duration_s = (stop_frame / fps) if (stop_frame and fps) else None
    report_paths = emit_delivery_report(
        out_dir=args.out_dir, source=vid, records=records, issues=issues, fps=fps,
        video_path=args.video_path, video_duration_s=duration_s,
        source_format=("cvat" if args.input else "csv"), step=step,
    )

```

```
            print(md)
            print(f"\nWrote:\n  {md_path}\n  {json_path}\n  {review_path}")
            if report_paths:
                print(f"  {report_paths.get('technical_md')}\n {report_paths.get('customer_md')}\n  {report_paths.get('json')}")
            if summary["n_blockers"]:
                print(f"\n  GATE: BLOCKED ? {summary['n_blockers']} blocker(s) must be resolved.")
                sys.exit(2)
            print("\n  GATE: PASS ? review the flagged rallies, then `import-local`.")

        def curate(self):
            """Interactive loop to review and curate staging video annotations."""
            staging_videos = self.db.get_videos_by_status(CurationStatus.STAGING)
            if not staging_videos:
                print("\nNo staging video annotations to review.\n")
                return

            print(f"\nFound {len(staging_videos)} matches in STAGING state for curation:")
            table_data = []
            for idx, video in enumerate(staging_videos):
                table_data.append([
                    idx + 1,
                    video.video_id,
                    video.match_name,
                    video.sport.value,
                    video.license_type.value,
                    "YES" if video.is_commercial else "NO"
                ])

            print(tabulate(table_data, headers=["#", "Video ID", "Match Name", "Sport", "License", "Commercial Safe?"], tablefmt="grid"))
            print("\nEnter a number (1-{}) to review details, or 'q' to quit:".format(len(staging_videos)))

            choice = input("> ").strip()
            if choice.lower() == 'q':
                return

            try:
                idx = int(choice) - 1
                if idx < 0 or idx >= len(staging_videos):
                    print("Invalid choice.")
                    return
                self._curate_video_interactive(staging_videos[idx])
            except ValueError:
                print("Invalid input.")

        def _curate_video_interactive(self, video: VideoMetadata):
            """Displays details of a specific staging video and prompts for curation action."""

            # Load and count events
            rallies = []
            pts = []
            delivs = []
            if video.sport == SportType.BADMINTON:
                rallies = self.db.get_badminton_rallies(video.video_id)
            elif video.sport == SportType.PICKLEBALL:
                rallies = self.db.get_pickleball_rallies(video.video_id)
            elif video.sport == SportType.TABLE_TENNIS:
                rallies = self.db.get_tabletennis_rallies(video.video_id)
            elif video.sport == SportType.TENNIS:
                pts = self.db.get_tennis_points(video.video_id)
            elif video.sport == SportType.CRICKET:
```

```
                delivs = self.db.get_cricket_deliveries(video.video_id)

            while True:
                print("\n================ REVIEWING VIDEO ================")
                print(f"Match Name   : {video.match_name}")
                print(f"Video ID     : {video.video_id}")
                print(f"Sport        : {video.sport.value}")
                print(f"Video URL    : {video.video_url or 'N/A'}")
                print(f"Local Path   : {video.local_path or 'N/A'}")
                print(f"License      : {video.license_type.value} ({video.license_url or 'no URL'})")
                print(f"Commercial OK: {video.is_commercial}")

                if video.sport in (SportType.BADMINTON, SportType.PICKLEBALL, SportType.TABLE_TENNIS):
                    print(f"Total Rallies: {len(rallies)}")
                    if rallies:
                        print("Sample rallies:")
                        sample_data = [[r.rally_number, f"{r.start_time}s - {r.end_time}s", r.score_after, r.shots_count] for r in rallies[:5]]
                        print(tabulate(sample_data, headers=["Rally #", "Duration", "Score", "Shots"], tablefmt="simple"))
                elif video.sport == SportType.TENNIS:
                    print(f"Total Points : {len(pts)}")
                    if pts:
                        print("Sample points:")
                        sample_data = [[p.point_number, f"{p.start_time}s - {p.end_time}s", p.game_score] for p in pts[:5]]
                        print(tabulate(sample_data, headers=["Point #", "Duration", "Score"], tablefmt="simple"))
                elif video.sport == SportType.CRICKET:
                    print(f"Total Balls  : {len(delivs)}")
                    if delivs:
                        print("Sample deliveries:")
                        sample_data = [[f"{d.over}.{d.ball}", f"{d.start_time}s - {d.end_time}s", d.runs, "Wicket" if d.wicket else ""] for d in delivs[:5]]
                        print(tabulate(sample_data, headers=["Over.Ball", "Duration", "Runs", "Wicket"], tablefmt="simple"))

                print("\nChoose Curation Action:")
                print("  [a] Approve (promote to CURATED)")
                print("  [r] Reject (delete from database)")
                print("  [p] Play a rally/point segment in browser")
                print("  [k] Keep in Staging (do nothing and go back)")

                action = input("> ").strip().lower()
                if action == 'a':
                    self.db.update_status(video.video_id, CurationStatus.CURATED)
                    print(f"Successfully promoted '{video.match_name}' to CURATED.\n")
                    break
                elif action == 'r':
                    self.db.delete_video(video.video_id)
                    print(f"Successfully rejected and deleted '{video.match_name}'.\n")
                    break
                elif action == 'k':
                    print("Curation state unchanged.\n")
                    break
                elif action == 'p':
                    self._play_segment_interactive(video, rallies, pts, delivs)
                else:
                    print("Invalid action selected.")

        def _play_segment_interactive(self, video: VideoMetadata, rallies: List[BadmintonRally], pts: List[TennisPoint], delivs: List[CricketDelivery]):
            import webbrowser
```

```
            # Get start/end based on user input
            start_time = None
            end_time = None
            label = "segment"

            if video.sport in (SportType.BADMINTON, SportType.PICKLEBALL, SportType.TABLE_TENNIS):
                if not rallies:
                    print("No rallies to play.")
                    return
                idx_str = input(f"Enter Rally number to play (1-{len(rallies)}): ").strip()
                try:
                    rally_num = int(idx_str)
                    rally = next((r for r in rallies if r.rally_number == rally_num), None)
                    if rally:
                        start_time = rally.start_time
                        end_time = rally.end_time
                        label = f"Rally {rally_num}"
                except ValueError:
                    pass
            elif video.sport == SportType.TENNIS:
                if not pts:
                    print("No points to play.")
                    return
                idx_str = input(f"Enter Point number to play (1-{len(pts)}): ").strip()
                try:
                    pt_num = int(idx_str)
                    pt = next((p for p in pts if p.point_number == pt_num), None)
                    if pt:
                        start_time = pt.start_time
                        end_time = pt.end_time
                        label = f"Point {pt_num}"
                except ValueError:
                    pass
            elif video.sport == SportType.CRICKET:
                if not delivs:
                    print("No deliveries to play.")
                    return
                over_str = input("Enter Over number: ").strip()
                ball_str = input("Enter Ball number: ").strip()
                try:
                    o_num = int(over_str)
                    b_num = int(ball_str)
                    d = next((x for x in delivs if x.over == o_num and x.ball == b_num), None)
                    if d:
                        start_time = d.start_time
                        end_time = d.end_time
                        label = f"Over {o_num} Ball {b_num}"
                except ValueError:
                    pass

            if start_time is None or end_time is None:
                print("Segment not found or invalid input.")
                return

            print(f"Playing {label} (start: {start_time}s, end: {end_time}s)...")

            # Check video location
            play_url = None
            if video.local_path and os.path.exists(video.local_path):
                abs_path = os.path.abspath(video.local_path)
                # Create file:// URL with media fragment (browsers support #t=start,end fragment)
```

```
            play_url = f"file:///{abs_path.replace(os.sep, '/')}#t={start_time},{end_time}"
        elif video.video_url:
            # YouTube URL: seek to start time
            if "youtube.com" in video.video_url or "youtu.be" in video.video_url:
                t_sec = int(start_time)
                sep = "&" if "?" in video.video_url else "?"
                play_url = f"{video.video_url}{sep}t={t_sec}s"
            else:
                play_url = video.video_url

        if play_url:
            print(f"Opening browser: {play_url}")
            webbrowser.open(play_url)
        else:
            print("Error: No local video file or remote URL available to play.")


def main():
    parser = argparse.ArgumentParser(description="Sports Data Curator CLI")
    subparsers = parser.add_subparsers(dest="command")

    # Status command
    subparsers.add_parser("status", help="Display summary status of sports metadata database")

    # Curate command
    subparsers.add_parser("curate", help="Launch interactive curation tool for staging entries")

    # Import-local command
    import_parser = subparsers.add_parser("import-local", help="Manually import a local video and annotations")
    import_parser.add_argument("--sport", required=True, help="Sport type (e.g. badminton, tennis, cricket)")
    import_parser.add_argument("--video-path", required=True, help="Path to local video file")
    import_parser.add_argument("--annotations-path", required=True, help="Path to local JSON/CSV annotations file")
    import_parser.add_argument("--match-name", help="Name of the match (defaults to video filename)")
    import_parser.add_argument("--video-id", help="Custom alphanumeric video ID")
    import_parser.add_argument("--license", default="Proprietary", help="License type (e.g. MIT, CC-BY-NC, Proprietary)")
    import_parser.add_argument("--fps", type=float, default=30.0, help="Frames per second of the video")
    import_parser.add_argument("--resolution", default="1080p", help="Resolution of the video (e.g., 1080p, 720p)")
    import_parser.add_argument("--force", action="store_true",
                               help="Overwrite an existing video's annotations (default: refuse, to protect golden labels)")
    import_parser.add_argument("--normalize", action="store_true",
                               help="Run normalize-annotations on a CSV input first (drops unusable rows, "
                                    "renumbers fractional/duplicate rally ids 1..N by start time)")

    # Fetch command ? (re)download videos at best resolution, update DB in place
    fetch_parser = subparsers.add_parser(
        "fetch", help="(Re)download videos at best available resolution, updating the DB in place")
    fetch_parser.add_argument("--sport", help="Restrict to a single sport (default: all supported_sports)")
    fetch_parser.add_argument("--video-id", help="Fetch a single video by ID")
    fetch_parser.add_argument("--status", choices=["staging", "curated", "all"],
```

```
default="all",
                                    help="Which curation statuses to fetch (default: all)")
        fetch_parser.add_argument("--min-height", type=int, default=720,
                                    help="Skip videos already at/above this height unless --force (default: 720)")
        fetch_parser.add_argument("--max-height", type=int, default=None,
                                    help="Cap download resolution (e.g. 1080); default: best available")
        fetch_parser.add_argument("--force", action="store_true",
                                    help="Re-download even if a good-enough local file already exists")
        fetch_parser.add_argument("--include-noncommercial", action="store_true",
                                    help="Include videos whose license is not commercially safe (default: commercial only)")
        fetch_parser.add_argument("--cookies-from-browser", dest="cookies_from_browser", metavar="BROWSER",
                                    help="Use a logged-in browser's YouTube cookies (e.g. chrome, edge, firefox, brave). "
                                         "A Premium login gives reliable best-resolution downloads. "
                                         "Chromium browsers must be CLOSED on Windows (cookie DB lock).")
        fetch_parser.add_argument("--cookies", metavar="FILE",
                                    help="Path to a Netscape-format cookies.txt (alternative to --cookies-from-browser)")
        fetch_parser.add_argument("--player-client", dest="player_client", default="default", metavar="CLIENT",
                                    help="YouTube player client yt-dlp uses (default: 'default'). Avoid web/web_safari "
                                         "(SABR-forced -> drops to 144p). 'tv'/'mweb' also serve 1080p if needed.")
        fetch_parser.add_argument("--sleep", type=float, default=4.0, metavar="SECONDS",
                                    help="Pause between videos to avoid YouTube rate-limiting (default: 4)")
        fetch_parser.add_argument("--retries", type=int, default=2, metavar="N",
                                    help="Retries per video when a download fails or returns a low format (default: 2)")
        fetch_parser.add_argument("--retry-backoff", type=float, default=20.0, metavar="SECONDS",
                                    help="Base backoff between retries; grows linearly per attempt (default: 20)")

        # Convert-cvat command ? normalize a vendor CVAT export into our canonical CSV
        cvat_parser = subparsers.add_parser(
            "convert-cvat",
            help="Convert a vendor CVAT rally export (bounding-box XML) into the canonical per-rally CSV")
        cvat_parser.add_argument("--input", required=True,
                                   help="Path to the CVAT XML export (image- or track-mode)")
        cvat_parser.add_argument("--out", required=True,
                                   help="Path to write the canonical per-rally CSV (import-local input)")
        cvat_parser.add_argument("--video-path",
                                   help="Source video; its fps is read via ffprobe (seconds = frame / fps)")
        cvat_parser.add_argument("--fps", type=float,
                                   help="Override fps instead of probing the video")
        cvat_parser.add_argument("--step", type=int,
                                   help="Frame sampling step (default: read from CVAT meta, else inferred)")
        cvat_parser.add_argument("--issues",
                                   help="Path to also write the human-readable issues report")

        # Validate-delivery command ? run the full check suite + emit a structured report
        vd_parser = subparsers.add_parser(
```

```
        "validate-delivery",
        help="Validate a rally delivery (CVAT export or canonical CSV) and emit report + review sheet")
    vd_parser.add_argument("--input", help="CVAT XML export (use with --video-path/--fps)")
    vd_parser.add_argument("--csv", help="An already-canonical per-rally CSV to validate")
    vd_parser.add_argument("--video-path",
                           help="Source video; fps via ffprobe (when using --input)")
    vd_parser.add_argument("--fps", type=float, help="Override fps (when using --input)")
    vd_parser.add_argument("--video-id", help="Names the report files (<id>_report.md, etc.)")
    vd_parser.add_argument("--out-dir", default="data/validation",
                           help="Where to write the report + review sheet")

    # Export command
    export_parser = subparsers.add_parser(
        "export", help="Export curated annotations as indexer-ready eval/training CSVs + manifest")
    export_parser.add_argument("--out-dir", default="data/eval_export",
                               help="Output directory for per-video CSVs and manifest.json")
    export_parser.add_argument("--sport",
                               help="Restrict to a single sport (default: all supported_sports)")
    export_parser.add_argument("--include-staging", action="store_true",
                               help="Also export videos still in staging (default: curated only)")
    export_parser.add_argument("--include-noncommercial", action="store_true",
                               help="Include videos whose license is not commercially safe (default: commercial only)")

    # Reclassify-licenses command (one-shot corrective maintenance)
    reclass_parser = subparsers.add_parser(
        "reclassify-licenses",
        help="Fix annotation-vs-video license conflation (e.g. MIT stamped on broadcast video)")
    reclass_parser.add_argument("--apply", action="store_true",
                                help="Persist changes (default: dry-run preview only)")

    args = parser.parse_args()
    cli = SportsDataCLI()

    if args.command == "status":
        cli.status()
    elif args.command == "reclassify-licenses":
        cli.reclassify_licenses(args)
    elif args.command == "curate":
        cli.curate()
    elif args.command == "import-local":
        cli.import_local(args)
    elif args.command == "convert-cvat":
        cli.convert_cvat(args)
    elif args.command == "validate-delivery":
        cli.validate_delivery(args)
    elif args.command == "export":
        cli.export_eval_set(args)
    elif args.command == "fetch":
        cli.fetch(args)
    else:
        # Default: print help
        parser.print_help()

if __name__ == "__main__":
```

```
1093        main()
1094
```

47. user (2026-06-12T10:28:34.176Z)

```
10-- `bv*+ba/b` selects best video + best audio, falling back to the best
11-  pre-muxed stream. Combined with `-S res,fps,...`, yt-dlp sorts candidate
12-  formats by resolution (then fps) descending and picks the top ? i.e. *always*
13-  the best resolution available, regardless of codec/container quirks.
14-- `--merge-output-format mp4` guarantees an MP4 container, which the downstream
15-  indexer requires.
16:- We never trust the requested resolution; we ffprobe the resulting file and
17-  record the *actual* width/height/fps as the source of truth.
18-"""
19-
20-import os
21-import shutil
22:import subprocess
23-from dataclasses import dataclass
24-from typing import List, Optional, Tuple
25-
26-# Best video+audio, prefer mp4/m4a, fall back to best single stream.
27-_FORMAT = "bv*+ba/b"
28-# Sort candidates so the highest resolution (then fps) wins; prefer mp4/m4a on ties.
--
106-    deno_bin = os.path.join(os.path.expanduser("~"), ".deno", "bin")
107-    if os.path.isdir(deno_bin):
108-        env["PATH"] = deno_bin + os.pathsep + env.get("PATH", "")
109-    return env
110-
111-
112:def probe_resolution_fps(path: str) -> Tuple[Optional[int], Optional[int], Optional[float]]:
113:    """Return (width, height, fps) of a video file via ffprobe, best-effort.
114-
115:    Returns (None, None, None) if ffprobe is unavailable or the file can't be
116:    parsed, so a missing ffprobe degrades gracefully rather than crashing a fetch.
117-    """
118:    if not shutil.which("ffprobe") or not os.path.exists(path):
119-        return (None, None, None)
120-    try:
121:        out = subprocess.run(
122:            ["ffprobe", "-v", "error", "-select_streams", "v:0",
123-             "-show_entries", "stream=width,height,avg_frame_rate",
124-             "-of", "default=noprint_wrappers=1:nokey=1", path],
125:            stdout=subprocess.PIPE, stderr=subprocess.PIPE, text=True, check=True,
126-        ).stdout.split()
127:    except (subprocess.SubprocessError, OSError):
128-        return (None, None, None)
129-
130-    width = int(out[0]) if len(out) > 0 and out[0].isdigit() else None
131-    height = int(out[1]) if len(out) > 1 and out[1].isdigit() else None
132-    fps = None
133-    if len(out) > 2 and "\" in out[2]:
--
156-                        max_size_mb: Optional[float] = None,
157-                        cookies_from_browser: Optional[str] = None,
158-                        cookies_file: Optional[str] = None,
159-                        player_client: Optional[str] = "default") -> DownloadResult:
160-    """Download the best-resolution video for `url` to `dest_path` (MP4).
161-
162:    On success, ffprobes the file and reports actual width/height/fps. If
163-    `max_size_mb` is exceeded the file is removed and a failed result returned,
164-    so callers never register an over-budget download.
165-    """
166-    os.makedirs(os.path.dirname(os.path.abspath(dest_path)), exist_ok=True)
```

```
167-    cmd = build_ytdlp_command(url, dest_path, max_height=max_height,
168-                              cookies_from_browser=cookies_from_browser,
169-                              cookies_file=cookies_file,
170-                              player_client=player_client)
171-    try:
172:        subprocess.run(cmd, stdout=subprocess.PIPE, stderr=subprocess.PIPE,
173-                       text=True, check=True, env=_env_with_deno_on_path())
174-    except FileNotFoundError:
175-        return DownloadResult(success=False, error="yt-dlp not found on PATH")
176:    except subprocess.CalledProcessError as e:
177-        return DownloadResult(success=False, error=(e.stderr or str(e)).strip()[:500])
178-
179-    if not os.path.exists(dest_path):
180-        return DownloadResult(success=False, error="yt-dlp reported success but no file was produced")
181-
182-    size_mb = os.path.getsize(dest_path) / (1024 * 1024)
```

48. user (2026-06-12T10:28:35.031Z)

```
1       video_id,sport,format,match_name,source_filename,drive_link,fps,duration_mmss,expected_rallies,assigned_to,status,notes
2       pilot_badminton_01,badminton,singles,TODO(avidullu): match name,TODO(avidullu): exact source filename,TODO(avidullu): paste Drive link to this video,TODO(avidullu): fps e.g. 30,TODO(avidullu): length e.g. 2:45,10-12,,todo,deliver as pilot_badminton_01.csv
3       pilot_pickleball_01,pickleball,doubles,TODO(avidullu): match name,TODO(avidullu): exact source filename,TODO(avidullu): paste Drive link to this video,TODO(avidullu): fps e.g. 30,TODO(avidullu): length e.g. 2:30,10-12,,todo,deliver as pilot_pickleball_01.csv
4       pilot_tabletennis_01,tabletennis,singles,TODO(avidullu): match name,TODO(avidullu): exact source filename,TODO(avidullu): paste Drive link to this video,TODO(avidullu): fps e.g. 30,TODO(avidullu): length e.g. 2:20,10-12,,todo,deliver as pilot_tabletennis_01.csv
5
```

49. user (2026-06-12T10:28:36.878Z)

```
Name                          Length
----                          ------
test_cli.py                    12556
test_config.py                   779
test_cvat_parser.py            11636
test_cvat_robustness.py         2410
test_db.py                      9997
test_downloader.py              5428
test_import_guard.py            2383
test_licensing_gate.py          4895
test_local_import.py            2025
test_normalize_annotations.py   5304
test_parser.py                  4431
test_parsing.py                  947
test_reporting_faq.py           1675
test_reporting.py               6769
test_schemas.py                 1488
test_validator.py               4974
```

50. assistant (2026-06-12T10:29:31.214Z)

The probe helper, manifest template, and test layout are clear. Let me check the test conventions and pyproject, then write the implementation.

51. user (2026-06-12T10:29:32.132Z)

```
1       """End-to-end coverage for the SportsDataCLI commands.
2
3       Each test runs against a throwaway config + SQLite DB in tmp_path, so the real
```

```
data/ directory is never touched. Network (yt-dlp) and the browser are mocked.
"""
import csv as _csv
import json
from types import SimpleNamespace
from unittest import mock

import pytest
import yaml

from src.cli.curate import SportsDataCLI
from src.core.schemas.base import VideoMetadata, SportType, LicenseType, CurationStatus
from src.core.schemas.badminton import BadmintonRally


@pytest.fixture
def cli(tmp_path):
    cfg = {
        "database_path": str(tmp_path / "test.db"),
        "video_storage": {
            "local_dir": str(tmp_path / "videos"),
            "max_download_size_mb": 100,
            "auto_download": False,
            "max_height": None,
        },
        "commercial_licenses": ["MIT", "Apache-2.0", "BSD", "CC0", "CC-BY", "Public-Domain", "Proprietary"],
        "supported_sports": ["badminton", "tennis", "tabletennis", "pickleball", "cricket"],
    }
    cfg_path = tmp_path / "config.yaml"
    cfg_path.write_text(yaml.safe_dump(cfg), encoding="utf-8")
    return SportsDataCLI(config_path=str(cfg_path))


def _import_args(**kw):
    base = dict(
        sport=None, video_path="nonexistent.mp4", annotations_path=None,
        match_name=None, video_id=None, license="Proprietary", fps=30.0, resolution="1080p",
    )
    base.update(kw)
    return SimpleNamespace(**base)


def _write_csv(path, header, rows):
    with open(path, "w", newline="", encoding="utf-8") as f:
        w = _csv.writer(f)
        w.writerow(header)
        for r in rows:
            w.writerow(r)


# --------------------------------------------------------------------------- #
# import-local: every sport, both JSON and CSV (covers dispatch + parse)       #
# --------------------------------------------------------------------------- #

def test_import_local_badminton_json(cli, tmp_path):
    p = tmp_path / "bad.json"
    p.write_text(json.dumps([
        {"rally_number": 1, "start_time": 10.0, "end_time": 20.0, "shots_count": 5, "ending_reason": "winner"},
        {"rally_number": 2, "start_time": 25.0, "end_time": 30.0, "shots_count": 1, "ending_reason": "service_fault"},
    ]), encoding="utf-8")
```

```
        cli.import_local(_import_args(sport="badminton", annotations_path=str(p), video_id="bad1", match_name="Bad"))
        assert len(cli.db.get_badminton_rallies("bad1")) == 2


    def test_import_local_pickleball_csv(cli, tmp_path):
        p = tmp_path / "pb.csv"
        _write_csv(p, ["rally_number", "start_time", "end_time", "shots_count", "ending_reason"],
                   [[1, 10.0, 20.0, 6, "winner"], [2, 25.0, 27.0, 1, "service_fault"]])
        cli.import_local(_import_args(sport="pickleball", annotations_path=str(p), video_id="pb1", match_name="PB"))
        assert len(cli.db.get_pickleball_rallies("pb1")) == 2


    def test_import_local_tabletennis_json(cli, tmp_path):
        p = tmp_path / "tt.json"
        p.write_text(json.dumps([
            {"rally_number": 1, "start_time": 5.0, "end_time": 9.0, "shots_count": 7, "ending_reason": "winner"},
        ]), encoding="utf-8")
```

52. user (2026-06-12T10:29:32.264Z)

```
[build-system]
requires = ["setuptools>=61.0.0"]
build-backend = "setuptools.build_meta"

[project]
name = "sports-data-collector"
version = "0.1.0"
description = "A collector and curator for sports video datasets with sport-specific schemas and license checks."
readme = "README.md"
requires-python = ">=3.9"
dependencies = [
    "pydantic>=2.0.0",
    "pyyaml>=6.0",
    "tabulate>=0.9.0"
]

[project.optional-dependencies]
# Video acquisition (`fetch` / crawler). Also requires the external binaries
# ffprobe (FFmpeg) and, for YouTube high-res, a JS runtime such as deno ?
# neither is pip-installable; see README "External requirements".
fetch = [
    "yt-dlp>=2024.1.0"
]
dev = [
    "pytest>=7.0.0",
    "pytest-cov>=4.0.0"
]
# Per-video observability reports (SOFT dependency: ingestion degrades to a no-op
# if absent). Shared, permissively-licensed report toolkit. Pin a tag once the
# private repo is published, e.g.:
#   obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
# For local development install it editable from a sibling checkout:
#   pip install -e ../sports-obsreport --no-build-isolation
report = [
    "obsreport>=0.1.3",
]

```

53. user (2026-06-12T10:30:27.172Z)

```
# ---- Python ----
__pycache__/
*.py[cod]
*$py.class
*.egg-info/
.eggs/
build/
dist/
.installed.cfg
*.egg

# ---- Test / coverage ----
.pytest_cache/
.coverage
.coverage.*
htmlcov/
.tox/
.cache/

# ---- Virtual environments ----
.venv/
venv/
env/
ENV/

# ---- Secrets / local config ----
.env
.env.*

# ---- OS / editor cruft ----
.DS_Store
Thumbs.db
.idea/
.vscode/
*.swp

# ---- Project: large media kept outside git (see README) ----
# The video corpus is hundreds of MB+; only metadata (the SQLite DB) and the
# small source annotation CSVs are version-controlled. Re-fetch videos from the
# stored video_url / local_path per the manifest.
data/videos/

# ---- Generated eval/training exports (regenerated via `curate export`) ----
data/eval_export*/

# ---- Generated temp artifacts (crawler scratch space; regenerated each run) ----
data/temp/

# ---- Per-video observability reports (written next to videos / inputs at runtime) ----
*.report.json
*.report.md
*.summary.md

# Local DB backups (e.g. data/sports_metadata.backup-YYYY-MM-DD.db) ? never commit
data/*.backup-*.db

# Model artifact bundles: weights are NEVER committed (private Releases); the
# reviewable surface is each bundle's manifest.json (ADR-008).
artifacts/

```

54. user (2026-06-12T10:30:28.112Z)

```
# Rally Ingestion Pipeline ? End-to-End FAQ

```

How a vendor's rally annotations become training/eval ground truth, and how we
validate each delivery against our conventions. This is the operational companion
to the annotation spec (`docs/annotation/ROORA_BRIEF.md` + `ANNOTATION_GUIDE.md`).

```
  vendor delivery                  OUR side (collector)                 modeling (indexer)
 ??????????????????   convert    ?????????????????  validate  ????????????????
 ? CVAT export    ? ???????????? ? canonical CSV ? ?????????? ? report .md   ?
 ? (boxes + attrs)?              ? (per rally,   ?            ? report .json ?
 ?   OR           ?              ?  decimal secs)?            ? review .csv  ???? human verifies
 ? canonical CSV  ?              ?????????????????            ???????????????? flagged rallies
 ??????????????????                     ?                            ?
                                         ?  import-local (validated)  ?
                                         ?                    golden labels
                                 ?????????????????  export   ????????????????
                                 ? SQLite (rally ? ????????? ? <id>.csv     ? start,end (indexer GT)
                                 ?  tables)      ?           ? manifest.json?
                                 ?????????????????           ????????????????
                                                                     ?
                                                          slice clips / calibrate / train
```

The four commands (in order)

```bash
# 1. Normalise a CVAT export ? our canonical per-rally CSV (skip if the vendor sent a CSV)
python -m src.cli.curate convert-cvat --input job.xml \
    --video-path match.mp4 --out canonical.csv --issues issues.txt

# 2. Validate the delivery ? report + annotatable review sheet (the QA gate)
python -m src.cli.curate validate-delivery --input job.xml \
    --video-path match.mp4 --video-id pilot_badminton_01 --out-dir data/validation
#   ...or validate a CSV directly:
python -m src.cli.curate validate-delivery --csv canonical.csv --video-id pilot_badminton_01

# 3. After human review, import the (corrected) CSV into the DB
python -m src.cli.curate import-local --sport badminton \
    --video-path match.mp4 --annotations-path canonical.csv --fps 59.94

# 4. Export indexer-ready ground truth (start,end CSVs + manifest)
python -m src.cli.curate export --out-dir data/eval_export
```

FAQ

What delivery formats do we accept?
Both. A **CVAT export** (image- or track-mode XML ? rally fields carried as box
attributes) is normalised by `convert-cvat`; an **already-canonical CSV** is taken
as-is. Vendors may keep working in CVAT ? we do **not** require them to hand-author
CSVs.

What is the "canonical" CSV?
One row per rally:
`rally_number, start_mmss, end_mmss, start_time, end_time, shots_count, ending_reason, last_shot_outcome, score_before, score_after, notes`.
`start_time`/`end_time` are **decimal seconds**; `import-local` reads those.

How are timestamps handled? (the fps thing)
We recompute each rally's interval from its **frame range × the true fps**
(`seconds = frame / fps`). A vendor's CVAT time attributes are **ignored** ? a
step-sampled job that assumes 30 fps writes systematically wrong times. fps comes
from `--fps` or ffprobe of `--video-path`. `mm:ss` is also accepted everywhere.

What does validation check?
code	severity	meaning
`overlap`	**blocker**	two rallies' intervals overlap (`end[N] > start[N+1]`)
`zero_duration`	**blocker**	`end <= start` (single-sample / stray box)
`frame_out_of_range`	high	a box frame is past the video's last frame
`shot_density`	medium	long rally with implausibly few shots ? likely an **inflated span** (often hides a merged rally)
`possible_missed_rally`	medium	a long unlabeled gap between two rallies ? check for an **omitted rally**
`ending_reason_vocab`	medium	`ending_reason` outside the agreed set
`outcome_vocab`	low	`last_shot_outcome` outside `{in,net,out,n_a}`

Blocker vs. review ? what's the gate?
`validate-delivery` exits **non-zero if any blocker exists** (so it can gate a bulk
loop / CI). Non-blocker issues don't stop the gate but populate the **review sheet**
for the human pass. `import-local` independently rejects overlaps/zero-duration, so
bad ground truth can't slip into the DB.

How are *missed rallies* caught? (the worst error)
Two complementary checks, because a miss shows up two ways:
1. **Clean omission** (a real hole in the timeline) ? `possible_missed_rally` (gap heuristic).
2. **Merged/inflated** (the missed rally got swallowed into a neighbour's box, leaving
   no gap) ? `shot_density` flags the over-long, under-shot rally, and the human finds
   the second rally inside it.

Neither is a guarantee ? they **surface regions to review**, they don't certify
completeness. (A future enhancement: a motion second-opinion over the raw video.)

What's the review sheet?
`<id>_review.csv` ? one row per rally with `auto_flags` (which checks fired) plus
blank `VERIFIED / true_start_mmss / true_end_mmss / true_ending_reason /
true_last_shot_outcome / reviewer_comment` columns. Open it next to the video
(timestamp overlay on) and fill the flagged rows. This is the human gate that turns
a delivery into trusted golden labels.

Why two ending fields (`ending_reason` + `last_shot_outcome`)?
They're orthogonal: `ending_reason` is *why* the point ended (winner / forced_error /
unforced_error / service_fault / let / other); `last_shot_outcome` is *where* the
last shot went (in / net / out / n_a). A shuttle that flew long is
`forced_error/unforced_error` + `out` ? both kept, no information lost.

How do I onboard a brand-new video?
1. Get fps into the intake manifest (we provide it to the vendor).
2. `convert-cvat` (if CVAT) ? `validate-delivery` ? read the report, fill the review
   sheet for flagged rallies ? correct the CSV ? `import-local` ? `export`.
3. The same suite runs for every video, so quality stays consistent at scale.

How do I pull a single rally out to eyeball it?
The slicer (indexer):
```bash
python -m backend.tools.rally_slicer --video match.mp4 \
    --labels golden_labels.csv --rallies 7,16 --out-dir output/rally_clips
```
Clips are padded to match the highlight compiler (0.5s lead / 1.0s tail).

Does coarse frame sampling hurt boundaries?
Not in practice: ~0.5s sampling slack is absorbed by the highlight compiler's clip

```
padding, so we don't flag sampling density.

### Will a non-commercial license block export?
The video is **kept** in the DB regardless of license, but the default `export` is
**commercial-only** ? a non-commercial source (e.g. `CC-BY-NC-SA`) is *excluded*
from the eval/training set unless you pass `--include-noncommercial`. So if a video
"goes missing" downstream, check its `is_commercial` flag first. (The observability
report raises a `license_noncommercial` warning for exactly this case.)

### What is the observability report? (`*.report.json` / `*.report.md` / `*.summary.md`)
Every `validate-delivery`, `convert-cvat`, and `import-local` run now also writes a
per-video **observability report** (next to the video for `import-local`; into the
`--out-dir` / next to the canonical CSV otherwise):
- **`<id>.report.json`** ? machine-readable: every stage, metric, and finding. Diff
  two runs to see what changed between deliveries.
- **`<id>.report.md`** ? the **technical** report: parse / validation / registration
  stages plus **findings** (one per delivery issue, each citing this FAQ). Read this
  to debug a delivery.
- **`<id>.summary.md`** ? a **delivery summary** for the vendor / data team: the
  rally count and coverage plus the plain-language list of rallies to review ? no
  internal codes or severities.
This is in addition to the existing `<id>_report.md` / `_report.json` / `_review.csv`
(the human review sheet), which are unchanged.

```

55. user (2026-06-12T10:30:30.356Z)

```
ci.yml
name: CI

on:
  push:
    branches: [master]
  pull_request:

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: "3.11"
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -e ".[dev]"
      - name: Run tests
        run: python -m pytest -q
```

56. user (2026-06-12T10:31:38.282Z)

```
Switched to a new branch 'feat/prep-annotation-proxy'
```

57. assistant (2026-06-12T10:32:22.538Z)

```
"""Annotation-proxy generation for vendor handoff (ADR-002).

Vendors annotate in CVAT, where every rally timestamp is later recomputed from
*frame indices* (``seconds = frame / fps`` ? see `src/parsers/cvat/rally_cvat.py`).
A 4K/59.94 source is too heavy to frame-step, which is what pushed the pilot
vendor onto a ~30-frame sampling grid (~0.5 s) and broke the ±0.3 s boundary bar.
The fix is to hand the vendor a light **annotation proxy** of the *same timeline*:
downscaled and scrub-friendly, but identical in frame count, fps, and duration,
```

```
so frame N on the proxy IS frame N on the original and returned labels apply to
the original unchanged.

The ADR-002 proxy contract enforced here:
- video timestamps pass through untouched (``-vsync passthrough``, never ``-r``):
  same frame count, same fps, no trim, no leading padding;
- the proxy is VERIFIED against the source (frame count / fps / duration) and a
  failed proxy must be deleted, never shared ? a silently-wrong proxy poisons
  golden labels;
- provenance is recorded (source + proxy MD5s) so the proxy?original mapping
  survives the handoff (the indexer keys videos by file MD5; downstream
  processing must keep running on the ORIGINAL file).
"""
import csv
import hashlib
import os
import shutil
import subprocess
from dataclasses import dataclass
from typing import Any, Dict, List, Optional


class ProxyError(RuntimeError):
    """A proxy step failed in a way that must stop the handoff (never degrade)."""


#: Columns of the provenance manifest (one row appended per generated proxy).
MANIFEST_FIELDS = [
    "video_id", "source_path", "source_md5", "proxy_path", "proxy_md5",
    "fps", "frame_count", "duration_s", "source_resolution", "proxy_resolution",
    "encoder_settings", "created_at",
]
```

Encode defaults (see ADR-002). Height 1080 not 720 (table-tennis ball visibility); GOP 15 = a keyframe every ~0.25 s at 59.94 fps for snappy back-stepping; CRF 19 keeps the ball/shuttle legible.

```
DEFAULT_HEIGHT = 1080
DEFAULT_CRF = 19
DEFAULT_GOP = 15
DEFAULT_PRESET = "medium"


@dataclass
class ProbeInfo:
    """The frame?time identity of one video stream, as ffprobe reports it."""
    width: Optional[int] = None
    height: Optional[int] = None
    avg_fps: Optional[float] = None     # avg_frame_rate (frames / duration)
    r_fps: Optional[float] = None       # r_frame_rate (the container's base rate)
    nb_frames: Optional[int] = None     # demuxed packet count == frame count
    duration: Optional[float] = None    # video stream duration, else container

    @property
    def resolution(self) -> str:
        if self.width and self.height:
            return f"{self.width}x{self.height}"
        return "unknown"


def parse_rate(value: Optional[str]) -> Optional[float]:
    """Parse an ffprobe rational rate ('60000/1001', '30/1') to float fps."""
    if not value:
        return None
```
```
        return None
```

```python
    try:
        if "/" in value:
            num, den = value.split("/", 1)
            num_f, den_f = float(num), float(den)
            if den_f == 0:
                return None
            return num_f / den_f
        return float(value)
    except ValueError:
        return None


def probe_video(path: str) -> ProbeInfo:
    """Probe the frame?time identity of `path` via ffprobe (loud on failure).

    Uses ``-count_packets`` (demux-only, no decode) so the frame count is exact
    yet cheap even for a 30-minute file. Unlike the fetch path's best-effort
    probe, a failure here RAISES: parity verification is the whole point of the
    proxy flow, so a missing ffprobe must stop it, not degrade it.
    """
    if not shutil.which("ffprobe"):
        raise ProxyError("ffprobe not found on PATH (install FFmpeg)")
    if not os.path.exists(path):
        raise ProxyError(f"video not found: {path}")
    import json as _json
    try:
        out = subprocess.run(
            ["ffprobe", "-v", "error", "-select_streams", "v:0",
             "-count_packets",
             "-show_entries",
             "stream=width,height,avg_frame_rate,r_frame_rate,nb_read_packets,duration"
             ":format=duration",
             "-of", "json", path],
            stdout=subprocess.PIPE, stderr=subprocess.PIPE, text=True, check=True,
        ).stdout
        doc = _json.loads(out)
    except (subprocess.SubprocessError, OSError, ValueError) as e:
        raise ProxyError(f"ffprobe failed on {path}: {e}")

    streams = doc.get("streams") or []
    if not streams:
        raise ProxyError(f"no video stream found in {path}")
    s = streams[0]

    def _f(value) -> Optional[float]:
        try:
            return float(value)
        except (TypeError, ValueError):
            return None

    def _i(value) -> Optional[int]:
        try:
            return int(value)
        except (TypeError, ValueError):
            return None

    # Prefer the video stream's own duration; fall back to the container's
    # (audio encoder priming can stretch the container a few hundredths).
    duration = _f(s.get("duration"))
    if duration is None:
        duration = _f((doc.get("format") or {}).get("duration"))

    return ProbeInfo(
        width=_i(s.get("width")),
        height=_i(s.get("height")),
```

```python
        avg_fps=parse_rate(s.get("avg_frame_rate")),
        r_fps=parse_rate(s.get("r_frame_rate")),
        nb_frames=_i(s.get("nb_read_packets")),
        duration=duration,
    )


def is_vfr(info: ProbeInfo, rel_tol: float = 0.001) -> bool:
    """True when the stream looks variable-frame-rate (or we cannot tell).

    For CFR camera footage r_frame_rate == avg_frame_rate. A mismatch means the
    `frame / fps` time recompute is approximate ? conservative callers should
    require an explicit opt-in before proxying such a source.
    """
    if info.avg_fps is None or info.r_fps is None or info.avg_fps == 0:
        return True
    return abs(info.r_fps - info.avg_fps) / info.avg_fps > rel_tol


def build_proxy_command(src: str, dst: str, height: int = DEFAULT_HEIGHT,
                        crf: int = DEFAULT_CRF, gop: int = DEFAULT_GOP,
                        preset: str = DEFAULT_PRESET) -> List[str]:
    """The ffmpeg invocation that downscales WITHOUT touching the timeline.

    Invariants (ADR-002): no ``-r`` / fps filter (fps must never change), no
    trim, ``-vsync passthrough`` so every source frame maps 1:1 onto a proxy
    frame with its original timestamp. ``min(height, ih)`` never upscales.
    Audio is kept (contact sound is the annotator's best boundary cue).
    """
    return [
        "ffmpeg", "-hide_banner", "-loglevel", "warning", "-stats", "-y",
        "-i", src,
        "-map", "0:v:0", "-map", "0:a?",
        "-vf", f"scale=-2:min({height}\\,ih)",
        "-c:v", "libx264", "-preset", preset, "-crf", str(crf),
        "-g", str(gop), "-pix_fmt", "yuv420p",
        "-vsync", "passthrough",
        "-c:a", "aac", "-b:a", "160k",
        "-movflags", "+faststart",
        dst,
    ]


def run_ffmpeg(cmd: List[str]) -> None:
    """Run an ffmpeg command, streaming its progress to the console."""
    if not shutil.which("ffmpeg"):
        raise ProxyError("ffmpeg not found on PATH (install FFmpeg)")
    try:
        # No capture: -stats progress and any warnings go straight to the user.
        subprocess.run(cmd, check=True)
    except (subprocess.SubprocessError, OSError) as e:
        code = getattr(e, "returncode", "?")
        raise ProxyError(f"ffmpeg failed (exit {code}) ? proxy not produced")


def verify_parity(src: ProbeInfo, proxy: ProbeInfo) -> List[str]:
    """Check the proxy carries the source's exact frame?time identity.

    Returns a list of human-readable failures (empty == parity holds). Frame
    count must match EXACTLY; fps within 0.1%; duration within a small slack
    (AAC priming can pad the container by a few hundredths of a second) that
    still catches any real trim.
    """
    errors = []
```

```
    if src.nb_frames is None or proxy.nb_frames is None:
        errors.append("frame count unavailable on source and/or proxy ? cannot prove parity")
    elif src.nb_frames != proxy.nb_frames:
        errors.append(f"frame count changed: source {src.nb_frames} -> proxy {proxy.nb_frames}")

    if src.avg_fps is None or proxy.avg_fps is None:
        errors.append("fps unavailable on source and/or proxy ? cannot prove parity")
    elif abs(proxy.avg_fps - src.avg_fps) / src.avg_fps > 0.001:
        errors.append(f"fps changed: source {src.avg_fps:.3f} -> proxy {proxy.avg_fps:.3f}")

    if src.duration is None or proxy.duration is None:
        errors.append("duration unavailable on source and/or proxy ? cannot prove parity")
    else:
        tol = max(2.0 / src.avg_fps, 0.2) if src.avg_fps else 0.2
        if abs(proxy.duration - src.duration) > tol:
            errors.append(f"duration changed: source {src.duration:.3f}s -> "
                          f"proxy {proxy.duration:.3f}s (tolerance {tol:.3f}s)")

    return errors


def md5_file(path: str, chunk_size: int = 1024 * 1024) -> str:
    """MD5 of a file's bytes (the identity downstream tooling keys videos by)."""
    digest = hashlib.md5()
    with open(path, "rb") as f:
        for chunk in iter(lambda: f.read(chunk_size), b""):
            digest.update(chunk)
    return digest.hexdigest()


def append_manifest_row(manifest_path: str, row: Dict[str, Any]) -> None:
    """Append one provenance row, writing the header only on first creation."""
    parent = os.path.dirname(os.path.abspath(manifest_path))
    os.makedirs(parent, exist_ok=True)
    is_new = not os.path.exists(manifest_path) or os.path.getsize(manifest_path) == 0
    with open(manifest_path, "a", newline="", encoding="utf-8") as f:
        writer = csv.DictWriter(f, fieldnames=MANIFEST_FIELDS)
        if is_new:
            writer.writeheader()
        writer.writerow({k: row.get(k, "") for k in MANIFEST_FIELDS})
```

58. user (2026-06-12T10:32:22.558Z)

File created successfully at: C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py
(file state is current in your context ? no need to Read it back)

59. assistant (2026-06-12T10:32:27.924Z)

Now wiring the CLI command into `curate.py` ? the method plus the argparse subparser and dispatch:

60. user (2026-06-12T10:32:48.644Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

61. user (2026-06-12T10:33:02.539Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

62. user (2026-06-12T10:33:08.659Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

63. assistant (2026-06-12T10:33:15.092Z)

Now the test suite ? all offline, mirroring `test_cli.py` conventions:

64. assistant (2026-06-12T10:34:01.041Z)

```
"""Tests for the ADR-002 annotation-proxy flow (src/core/proxy.py + the
`prep-annotation-proxy` CLI command).

Everything runs offline: ffmpeg/ffprobe are mocked; only the pure logic
(command construction, parity verification, hashing, manifest append) and the
CLI orchestration (refusals, deletion-on-failure, provenance row) are exercised.
"""
import csv
import os
from types import SimpleNamespace
from unittest import mock

import pytest
import yaml

from src.cli.curate import SportsDataCLI
from src.core.proxy import (
    MANIFEST_FIELDS, ProbeInfo, ProxyError, append_manifest_row,
    build_proxy_command, is_vfr, md5_file, parse_rate, probe_video,
    verify_parity,
)
```

-- #

Pure logic

-- #

```
def test_build_command_preserves_the_timeline():
    cmd = build_proxy_command("in.mp4", "out.mp4")
    # The ADR-002 invariants: no fps change, no trim, timestamps passed through.
    assert "-r" not in cmd
    assert "-ss" not in cmd and "-t" not in cmd and "-to" not in cmd
    assert cmd[cmd.index("-vsync") + 1] == "passthrough"
    # Scrub-friendly + size knobs at their ratified defaults.
    assert cmd[cmd.index("-g") + 1] == "15"
    assert cmd[cmd.index("-crf") + 1] == "19"
    assert cmd[cmd.index("-vf") + 1] == "scale=-2:min(1080\\,ih)"
    assert cmd[cmd.index("-movflags") + 1] == "+faststart"
    # Audio kept (contact sound is the boundary cue), input before output.
    assert "0:a?" in cmd
    assert cmd.index("in.mp4") < cmd.index("out.mp4") == len(cmd) - 1


def test_build_command_custom_knobs():
    cmd = build_proxy_command("a.mp4", "b.mp4", height=720, crf=22, gop=30, preset="fast")
    assert cmd[cmd.index("-vf") + 1] == "scale=-2:min(720\\,ih)"
    assert cmd[cmd.index("-crf") + 1] == "22"
    assert cmd[cmd.index("-g") + 1] == "30"
    assert cmd[cmd.index("-preset") + 1] == "fast"


def test_parse_rate():
    assert parse_rate("30/1") == 30.0
    assert abs(parse_rate("60000/1001") - 59.94006) < 1e-4
```

```python
    assert parse_rate("25") == 25.0
    assert parse_rate("0/0") is None
    assert parse_rate("") is None
    assert parse_rate(None) is None
    assert parse_rate("garbage") is None


def test_is_vfr():
    cfr = ProbeInfo(avg_fps=59.94006, r_fps=59.94006)
    assert not is_vfr(cfr)
    vfr = ProbeInfo(avg_fps=59.2, r_fps=60.0)
    assert is_vfr(vfr)
    # Unknown rates are treated as VFR (conservative: force the explicit flag).
    assert is_vfr(ProbeInfo(avg_fps=None, r_fps=30.0))
    assert is_vfr(ProbeInfo(avg_fps=30.0, r_fps=None))


def _info(frames=5737, fps=59.94006, dur=95.72, w=3840, h=2160):
    return ProbeInfo(width=w, height=h, avg_fps=fps, r_fps=fps,
                     nb_frames=frames, duration=dur)


def test_verify_parity_ok_and_tolerates_audio_priming():
    src = _info()
    # Same frames/fps; container a hair longer (AAC priming) ? still parity.
    proxy = _info(dur=95.76, w=1920, h=1080)
    assert verify_parity(src, proxy) == []


def test_verify_parity_catches_frame_count_change():
    errors = verify_parity(_info(frames=5737), _info(frames=5730, w=1920, h=1080))
    assert any("frame count changed" in e for e in errors)


def test_verify_parity_catches_fps_change():
    errors = verify_parity(_info(fps=59.94006), _info(fps=29.97003, w=1920, h=1080))
    assert any("fps changed" in e for e in errors)


def test_verify_parity_catches_trim():
    errors = verify_parity(_info(dur=95.72), _info(dur=94.20, w=1920, h=1080))
    assert any("duration changed" in e for e in errors)


def test_verify_parity_loud_on_unknown_identity():
    errors = verify_parity(ProbeInfo(), ProbeInfo())
    assert len(errors) == 3  # frames, fps, duration all unprovable


def test_md5_file(tmp_path):
    p = tmp_path / "f.bin"
    p.write_bytes(b"hello world")
    assert md5_file(str(p)) == "5eb63bbbe01eeed093cb22bb8f5acdc3"


def test_append_manifest_header_written_once(tmp_path):
    manifest = tmp_path / "sub" / "proxies.csv"
    append_manifest_row(str(manifest), {"video_id": "v1", "source_md5": "a"})
    append_manifest_row(str(manifest), {"video_id": "v2", "source_md5": "b"})
    with open(manifest, newline="", encoding="utf-8") as f:
        rows = list(csv.reader(f))
    assert rows[0] == MANIFEST_FIELDS
    assert len(rows) == 3
    assert rows[1][0] == "v1" and rows[2][0] == "v2"
```

-- #

probe_video (ffprobe mocked)

-- #

```
_FFPROBE_JSON = """
{
  "streams": [{
    "width": 3840, "height": 2160,
    "r_frame_rate": "60000/1001", "avg_frame_rate": "60000/1001",
    "duration": "95.720000", "nb_read_packets": "5737"
  }],
  "format": {"duration": "95.768000"}
}
"""


def test_probe_video_parses_ffprobe_json(tmp_path):
    p = tmp_path / "v.mp4"
    p.write_bytes(b"x")
    with mock.patch("src.core.proxy.shutil.which", return_value="ffprobe"), \
         mock.patch("src.core.proxy.subprocess.run",
                    return_value=SimpleNamespace(stdout=_FFPROBE_JSON)):
        info = probe_video(str(p))
    assert (info.width, info.height) == (3840, 2160)
    assert info.nb_frames == 5737
    assert abs(info.avg_fps - 59.94006) < 1e-4
    assert info.duration == pytest.approx(95.72)  # stream duration preferred
    assert info.resolution == "3840x2160"


def test_probe_video_requires_ffprobe(tmp_path):
    p = tmp_path / "v.mp4"
    p.write_bytes(b"x")
    with mock.patch("src.core.proxy.shutil.which", return_value=None):
        with pytest.raises(ProxyError, match="ffprobe not found"):
            probe_video(str(p))


def test_probe_video_missing_file():
    with mock.patch("src.core.proxy.shutil.which", return_value="ffprobe"):
        with pytest.raises(ProxyError, match="not found"):
            probe_video("nope.mp4")


def test_probe_video_no_video_stream(tmp_path):
    p = tmp_path / "v.mp4"
    p.write_bytes(b"x")
    with mock.patch("src.core.proxy.shutil.which", return_value="ffprobe"), \
         mock.patch("src.core.proxy.subprocess.run",
                    return_value=SimpleNamespace(stdout='{"streams": []}')):
        with pytest.raises(ProxyError, match="no video stream"):
            probe_video(str(p))
```

-- #

CLI: prep-annotation-proxy

-- #

```
@pytest.fixture
def cli(tmp_path):
    cfg = {
```

```python
        "database_path": str(tmp_path / "test.db"),
        "supported_sports": ["badminton", "pickleball", "tabletennis"],
        "commercial_licenses": ["MIT", "Apache-2.0", "BSD", "Proprietary"],
    }
    cfg_path = tmp_path / "config.yaml"
    cfg_path.write_text(yaml.safe_dump(cfg), encoding="utf-8")
    return SportsDataCLI(config_path=str(cfg_path))


def _prep_args(tmp_path, **kw):
    base = dict(
        video_path=str(tmp_path / "Match One.mp4"),
        out=str(tmp_path / "proxies" / "match_proxy.mp4"),
        video_id=None, height=1080, crf=19, gop=15, preset="medium",
        manifest=str(tmp_path / "proxy_manifest.csv"),
        allow_vfr=False, force=False,
    )
    base.update(kw)
    return SimpleNamespace(**base)


def _cfr(frames=100, w=3840, h=2160):
    return _info(frames=frames, dur=100 / 59.94006, w=w, h=h)


def test_prep_proxy_happy_path(cli, tmp_path):
    args = _prep_args(tmp_path)
    (tmp_path / "Match One.mp4").write_bytes(b"original-bytes")

    def fake_ffmpeg(cmd):
        with open(cmd[-1], "wb") as f:
            f.write(b"proxy-bytes")

    probes = [_cfr(), _cfr(w=1920, h=1080)]  # source probe, then proxy probe
    with mock.patch("src.core.proxy.probe_video", side_effect=probes), \
         mock.patch("src.core.proxy.run_ffmpeg", side_effect=fake_ffmpeg):
        cli.prep_annotation_proxy(args)

    assert os.path.exists(args.out)
    with open(args.manifest, newline="", encoding="utf-8") as f:
        rows = list(csv.DictReader(f))
    assert len(rows) == 1
    row = rows[0]
    # video_id derived like import-local: stem lowercased, spaces -> underscores.
    assert row["video_id"] == "match_one"
    assert row["source_md5"] == md5_file(str(tmp_path / "Match One.mp4"))
    assert row["proxy_md5"] == md5_file(args.out)
    assert row["source_md5"] != row["proxy_md5"]
    assert row["fps"].startswith("59.94")
    assert row["frame_count"] == "100"
    assert row["source_resolution"] == "3840x2160"
    assert row["proxy_resolution"] == "1920x1080"


def test_prep_proxy_parity_failure_deletes_proxy(cli, tmp_path):
    args = _prep_args(tmp_path)
    (tmp_path / "Match One.mp4").write_bytes(b"original-bytes")

    def fake_ffmpeg(cmd):
        with open(cmd[-1], "wb") as f:
            f.write(b"proxy-bytes")

    # Proxy comes back one frame short -> parity check must fail loudly.
    probes = [_cfr(frames=100), _cfr(frames=99, w=1920, h=1080)]
    with mock.patch("src.core.proxy.probe_video", side_effect=probes), \
```

```python
         mock.patch("src.core.proxy.run_ffmpeg", side_effect=fake_ffmpeg):
        with pytest.raises(SystemExit) as exc:
            cli.prep_annotation_proxy(args)
    assert exc.value.code == 1
    assert not os.path.exists(args.out), "a failed proxy must never be left on disk"
    assert not os.path.exists(args.manifest), "no provenance row for a failed proxy"


def test_prep_proxy_refuses_vfr_without_flag(cli, tmp_path):
    args = _prep_args(tmp_path)
    (tmp_path / "Match One.mp4").write_bytes(b"original-bytes")
    vfr_info = ProbeInfo(width=3840, height=2160, avg_fps=59.2, r_fps=60.0,
                         nb_frames=100, duration=1.69)
    with mock.patch("src.core.proxy.probe_video", return_value=vfr_info):
        with pytest.raises(SystemExit) as exc:
            cli.prep_annotation_proxy(args)
    assert exc.value.code == 1
    assert not os.path.exists(args.out)


def test_prep_proxy_allow_vfr_proceeds(cli, tmp_path):
    args = _prep_args(tmp_path, allow_vfr=True)
    (tmp_path / "Match One.mp4").write_bytes(b"original-bytes")
    vfr = ProbeInfo(width=3840, height=2160, avg_fps=59.2, r_fps=60.0,
                    nb_frames=100, duration=100 / 59.2)
    vfr_proxy = ProbeInfo(width=1920, height=1080, avg_fps=59.2, r_fps=60.0,
                          nb_frames=100, duration=100 / 59.2)

    def fake_ffmpeg(cmd):
        with open(cmd[-1], "wb") as f:
            f.write(b"proxy-bytes")

    with mock.patch("src.core.proxy.probe_video", side_effect=[vfr, vfr_proxy]), \
         mock.patch("src.core.proxy.run_ffmpeg", side_effect=fake_ffmpeg):
        cli.prep_annotation_proxy(args)
    assert os.path.exists(args.out)


def test_prep_proxy_refuses_existing_output(cli, tmp_path):
    args = _prep_args(tmp_path)
    (tmp_path / "Match One.mp4").write_bytes(b"original-bytes")
    os.makedirs(os.path.dirname(args.out), exist_ok=True)
    with open(args.out, "wb") as f:
        f.write(b"existing")
    with pytest.raises(SystemExit) as exc:
        cli.prep_annotation_proxy(args)
    assert exc.value.code == 1


def test_prep_proxy_missing_source(cli, tmp_path):
    args = _prep_args(tmp_path, video_path=str(tmp_path / "nope.mp4"))
    with pytest.raises(SystemExit) as exc:
        cli.prep_annotation_proxy(args)
    assert exc.value.code == 1


def test_prep_proxy_out_equals_source_refused(cli, tmp_path):
    src = tmp_path / "Match One.mp4"
    src.write_bytes(b"original-bytes")
    args = _prep_args(tmp_path, out=str(src))
    with pytest.raises(SystemExit) as exc:
        cli.prep_annotation_proxy(args)
    assert exc.value.code == 1
```

65. user (2026-06-12T10:34:01.058Z)

```
File created successfully at: C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py
(file state is current in your context ? no need to Read it back)
```

66. user (2026-06-12T10:34:11.518Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

67. user (2026-06-12T10:34:25.296Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

68. assistant (2026-06-12T10:34:31.251Z)

```
Now running the full collector suite:
```

69. user (2026-06-12T10:34:36.191Z)

```
........................................................................ [ 45%]
........................................................................ [ 90%]
...............                                                          [100%]
159 passed in 2.42s
```

70. user (2026-06-12T10:34:51.542Z)

```
C:\Users\avidu\AppData\Local\Microsoft\WinGet\Packages\Gyan.FFmpeg_Microsoft.Winget.Source_8weky
b3d8bbwe\ffmpeg-8.1.1-full_build\bin\ffmpeg.exe
C:\Users\avidu\AppData\Local\Microsoft\WinGet\Packages\Gyan.FFmpeg_Microsoft.Winget.Source_8weky
b3d8bbwe\ffmpeg-8.1.1-full_build\bin\ffprobe.exe
```

71. user (2026-06-12T10:35:05.328Z)

```
Name             Length
----             ------
Smoke Match.mp4 10387821
```

72. user (2026-06-12T10:35:15.447Z)

```
-vsync is deprecated. Use -fps_mode
frame=  179 fps=0.0 q=-1.0 Lsize=     595KiB time=00:00:02.95 bitrate=1650.5kbits/s speed=6.08x
elapsed=0:00:00.48
Encoding proxy (1920x1080 -> height<=480, fps 59.9401 preserved)...
Hashing source + proxy (provenance)...

=== Annotation proxy ready ===
  Video ID    : smoke_match
  Source      : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\Smoke Match.mp4 (1920x1080, 10 MB,
md5 86280c00c926c319bdc7b61d62c1dd34)
  Proxy       : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\smoke_proxy.mp4 (854x480, 1 MB,
md5 fd6f22e335274d76482ca6860997e26b)
  Parity      : VERIFIED ? 179 frames @ 59.9401 fps on both
  Manifest    : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\proxy_manifest.csv

Next:
  1. Share the PROXY with the vendor (CVAT at frame step 1; first/last box of every rally on the
exact contact/dead frame).
  2. On delivery, convert-cvat may use --video-path with either file (identical fps/frame count
by construction).
  3. import-local / export keep pointing at the ORIGINAL file ? the indexer keys videos by its
MD5.
```

73. user (2026-06-12T10:35:35.426Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

74. user (2026-06-12T10:35:36.565Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

75. user (2026-06-12T10:35:38.884Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

76. user (2026-06-12T10:35:40.067Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

77. user (2026-06-12T10:35:46.499Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

78. assistant (2026-06-12T10:35:51.201Z)

Re-running the suite and the live smoke with the updated flag:

79. user (2026-06-12T10:35:59.065Z)

```
........................................................................ [ 90%]
...............                                                          [100%]
159 passed in 2.48s
=== Annotation proxy ready ===
  Video ID    : smoke_match
  Source      : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\Smoke Match.mp4 (1920x1080, 10 MB,
md5 86280c00c926c319bdc7b61d62c1dd34)
  Proxy       : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\smoke_proxy.mp4 (854x480, 1 MB,
md5 fd6f22e335274d76482ca6860997e26b)
  Parity      : VERIFIED ? 179 frames @ 59.9401 fps on both
  Manifest    : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\proxy_manifest.csv

Next:
  1. Share the PROXY with the vendor (CVAT at frame step 1; first/last box of every rally on the
exact contact/dead frame).
  2. On delivery, convert-cvat may use --video-path with either file (identical fps/frame count
by construction).
  3. import-local / export keep pointing at the ORIGINAL file ? the indexer keys videos by its
MD5.


video_id,source_path,source_md5,proxy_path,proxy_md5,fps,frame_count,duration_s,source_resolutio
n,proxy_resolution,encoder_settings,created_at
smoke_match,C:\Users\avidu\AppData\Local\Temp\proxy_smoke\Smoke Match.mp4,86280c00c926c319bdc7b6
1d62c1dd34,C:\Users\avidu\AppData\Local\Temp\proxy_smoke\smoke_proxy.mp4,fd6f22e335274d76482ca68
60997e26b,59.9401,179,2.986,1920x1080,854x480,libx264 crf=19 g=15 preset=medium height<=480
vsync=passthrough audio=aac160k,2026-06-12T10:35:15Z
smoke_match,C:\Users\avidu\AppData\Local\Temp\proxy_smoke\Smoke Match.mp4,86280c00c926c319bdc7b6
1d62c1dd34,C:\Users\avidu\AppData\Local\Temp\proxy_smoke\smoke_proxy.mp4,fd6f22e335274d76482ca68
60997e26b,59.9401,179,2.986,1920x1080,854x480,libx264 crf=19 g=15 preset=medium height<=480
fps_mode=passthrough audio=aac160k,2026-06-12T10:35:58Z
```

80. user (2026-06-12T10:36:20.894Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

81. user (2026-06-12T10:36:39.643Z)

```
warning: in the working copy of 'src/core/proxy.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'tests/test_proxy.py', LF will be replaced by CRLF the next time
Git touches it
 M .gitignore
M  docs/INGESTION_PIPELINE.md
M  src/cli/curate.py
A  src/core/proxy.py
A  tests/test_proxy.py
[feat/prep-annotation-proxy fd530b5] feat(curate): prep-annotation-proxy - frame-identical
vendor handoff re-encode (ADR-002)
 4 files changed, 724 insertions(+), 3 deletions(-)
 create mode 100644 src/core/proxy.py
 create mode 100644 tests/test_proxy.py
```

82. user (2026-06-12T10:37:14.874Z)

```
Workflow launched in background. Task ID: wyv8524zf
Summary: Adversarial review of the prep-annotation-proxy diff before PR
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_c5f5bca5-d2a
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-sports-data-
collector\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\review-proxy-feature-
wf_c5f5bca5-d2a.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:
"C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-sports-data-
collector\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\review-proxy-feature-
wf_c5f5bca5-d2a.js"} to iterate without resending the script.)
Run ID: wf_c5f5bca5-d2a
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--
Users-avidu-Projects-sports-data-
collector\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\review-proxy-feature-
wf_c5f5bca5-d2a.js", resumeFromRunId: "wf_c5f5bca5-d2a"}) ? completed agents return cached
results.

You will be notified when it completes. Use /workflows to watch live progress.
```

83. user (2026-06-12T10:47:20.718Z)

```
<retrieval_status>timeout</retrieval_status>

<task_id>wyv8524zf</task_id>

<task_type>local_workflow</task_type>

<status>running</status>
```

84. user (2026-06-12T10:47:40.282Z)

```
<retrieval_status>success</retrieval_status>

<task_id>wyv8524zf</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>
[Truncated. Full output: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-
badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\tasks\wyv8524zf.output]
```

dimensions to even; an explicit height expression is used verbatim and no force_divisible_by is
set, so any odd source height below the cap (e.g. 807 < 1080) yields an odd output height. With
-c:v libx264 -pix_fmt yuv420p (proxy.py:172-173), 4:2:0 chroma requires even dims, so ffmpeg
aborts at encoder init ('height not divisible by 2') -> run_ffmpeg raises ProxyError -> CLI
sys.exit(1) (curate.py:701-706). Loud failure, as the claim states. The secondary point also
holds: cleanup of the output file exists only in the parity-verification branch
(curate.py:713-719), not the encode-failure branch, and ffmpeg -y creates the file before
encoder init fails, so an empty/partial proxy is stranded and a re-run then trips the exists-
without---force guard at curate.py:673. The scenario is reachable (arbitrary --video-path;
display-cropped exports/screen captures have odd heights), while mainline >=1080 even-dim camera
footage takes the min->1080 branch ? consistent with the E2E smoke passing. The claim's escaping
analysis is correct (\\, passes Windows list2cmdline verbatim, filtergraph parser unescapes it),
and the proposed fix scale=-2:2*trunc(min(1080\\,ih)/2) is sound ? a 1 px height change does not
affect the temporal frame/fps/duration parity contract. Severity low: edge-case input, loud
failure (no silent corruption, no wrong proxy shipped), residue limited to a stranded empty file
and a cryptic error.",
          "severity": "low"
        }
      },
      {
        "title": "v:0 stream selectors include attached-picture (cover-art) streams in both
probe and encode",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\core\\proxy.py",
        "line": "99 (ffprobe -select_streams v:0) and 170 (-map 0:v:0)",
        "detail": "Lowercase `v` matches video streams INCLUDING attached pics. A source whose
cover-art/thumbnail stream is ordered before the real video (re-wrapped/downloaded files; rare
in raw camera footage) would be probed and encoded as the 1-frame thumbnail. In practice the
avg_fps gate likely refuses such files (attached pics report avg_frame_rate 0/0 -> None) so a
wrong proxy probably never ships, but the resulting 'could not determine fps' error is
misleading and the file is unproxyable with no workaround.",
        "fix": "Use the capital-V specifier, which excludes attached pics: ffprobe
-select_streams V:0 and ffmpeg -map 0:V:0 (both supported on the ffmpeg >= 5.1 baseline this
feature already requires).",
        "verdict": {
          "real": true,
          "reasoning": "Verified directly in the code: proxy.py:99 uses `ffprobe -select_streams
v:0` and proxy.py:170 uses `ffmpeg -map 0:v:0`. Per ffmpeg's stream-specifier semantics,
lowercase `v` matches video streams INCLUDING attached pictures (cover art/thumbnails); capital
`V` excludes them and has existed since FFmpeg 4.1, well under the ffmpeg >= 5.1 baseline this
feature already requires for `-fps_mode passthrough` ? so the proposed fix is valid and zero-
cost. The failure scenario is real but rare: a file with the attached-pic stream ordered before
the real video (re-wrapped/downloaded files; raw camera footage and typical yt-dlp embeds put
thumbnails last). When it fires, probe_video picks the cover art as streams[0]; its
avg_frame_rate of 0/0 makes parse_rate return None (zero denominator), and curate.py's
prep_annotation_proxy then exits with exactly the misleading error the claim predicts ('could
not determine fps ... the frame<->time contract needs it'), with no stream-selection flag as a
workaround. So no wrong proxy ships in the plausible path (the gate is fail-safe), but a valid
video becomes unproxyable with a wrong-cause error. A contrived sub-case (parseable thumbnail
fps + --allow-vfr) could even pass verify_parity with a 1-frame proxy, since both probes look at
the same wrong stream, but that requires stacked unlikely conditions. Net: genuine minor defect
with a trivial correct fix; reviewer's [low] rating and mitigation analysis are accurate.",
          "severity": "low"
        }
      },
      {
        "title": "Failed/partial proxy file is left on disk when ffmpeg or the post-encode probe
fails",
        "severity": "medium",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py",
        "line": "700-706",
        "detail": "The delete-on-failure invariant (module docstring: 'a failed proxy must be
deleted, never shared'; test_prep_proxy_parity_failure_deletes_proxy) is only enforced on the
PARITY branch. In prep_annotation_proxy, the `try: proxylib.run_ffmpeg(cmd); proxy_info =
proxylib.probe_video(out)` block catches ProxyError and exits 1 WITHOUT removing `out`. ffmpeg

runs with -y, so a mid-encode failure (disk full, codec error, Ctrl-C-adjacent kill, old ffmpeg without -fps_mode that dies after output open) leaves a partial .mp4 sitting in data/videos/proxies/ ? exactly the unverified artifact the flow exists to prevent ? and the next run refuses with 'already exists; pass --force', nudging the user toward --force as a habit. Same for a probe failure on a fully-encoded but unverified proxy.",
        "fix": "On ProxyError from run_ffmpeg/probe_video, attempt os.remove(out) with the same delete-or-warn messaging used on the parity-failure branch (extract a small _delete_failed_proxy(out) helper used by both paths). Add a test: run_ffmpeg side_effect that writes the file then raises ProxyError; assert the file is gone.",
        "verdict": {
          "real": true,
          "reasoning": "Confirmed by reading the code. curate.py:701-706 catches ProxyError from run_ffmpeg(cmd) and probe_video(out) and sys.exit(1)s WITHOUT removing `out`; the delete-or-warn logic exists only on the parity-failure branch (curate.py:713-719). proxy.py refutes any internal-cleanup counter-theory: run_ffmpeg (proxy.py:181-190) just raises ProxyError with no cleanup, and build_proxy_command writes directly to the final dst with -y (no temp-file+rename), so a mid-encode failure (disk full, encoder error, external kill) leaves a partial .mp4 in data/videos/proxies/. The next run then hits the existence check at curate.py:673-675 and demands --force, as claimed. Test gap confirmed: only test_prep_proxy_parity_failure_deletes_proxy (tests/test_proxy.py:236) asserts deletion; no test covers run_ffmpeg/probe_video failure. Not intended scope: the module docstring (proxy.py:16, 'a failed proxy must be deleted, never shared'), the existing test's assertion message ('a failed proxy must never be left on disk'), and run_ffmpeg's own error text ('proxy not produced' ? false in the partial-file case) all contradict leaving the artifact. Severity downgraded to low rather than medium: the failure is loud (error printed, exit 1), no manifest row is ever written for the orphan, and a partial mp4 encoded with -movflags +faststart lacks its moov atom (written at finalize) so it is unplayable in CVAT/players and cannot silently poison labels ? the realistic harm is workflow friction plus the --force-habituation nudge. The dangerous variant (fully-encoded playable proxy left unverified after a probe failure) requires ffprobe to fail on a file ffmpeg just wrote, which is rare. One claim detail is wrong: old ffmpeg without -fps_mode rejects the unrecognized option at command-line parse time before opening the output, leaving no file. Proposed fix (shared _delete_failed_proxy helper + a test with a side_effect that writes then raises) is correct and matches the existing parity-branch pattern.",
          "severity": "low"
        }
      },
      {
        "title": "INGESTION_PIPELINE.md onboarding FAQ and top diagram skip the new step 0",
        "severity": "medium",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\INGESTION_PIPELINE.md",
        "line": "7-23, 112-116",
        "detail": "The heading was updated to 'The five commands (in order)' and step 0 added, but (a) the 'How do I onboard a brand-new video?' FAQ ? the operational checklist someone will actually follow for a new video ? still reads 'Get fps into the intake manifest' then jumps straight to convert-cvat, never mentioning prep-annotation-proxy or 'share the PROXY, process the ORIGINAL'; this is the exact workflow ADR-002 calls 'close to a precondition for Phase B', so omitting it here makes the doc actively misleading; (b) the ASCII diagram at the top still begins at 'vendor delivery' with no step-0 box/arrow (PROXY -> vendor), so the diagram and the 'five commands' list disagree about where the pipeline starts.",
        "fix": "Add a first bullet to the onboarding FAQ: 'Run prep-annotation-proxy on the original; share the proxy (and its drive link) with the vendor; fps + video_id go into the intake manifest.' Add a small 'original --prep-annotation-proxy--> proxy --> vendor' leg (or a one-line note) to the diagram.",
        "verdict": {
          "real": true,
          "reasoning": "Both factual sub-claims verify against the actual file and diff.\n\n(a) Onboarding FAQ omission ? CONFIRMED. docs/INGESTION_PIPELINE.md lines 112-116 (\"How do I onboard a brand-new video?\") still read: \"1. Get fps into the intake manifest (we provide it to the vendor). 2. convert-cvat (if CVAT) ? validate-delivery ? ... ? import-local ? export.\" No mention of prep-annotation-proxy or sharing the proxy. The branch diff (git diff master...feat/prep-annotation-proxy -- docs/INGESTION_PIPELINE.md) shows this section was untouched. This creates a genuine internal inconsistency: the same doc's new sampling FAQ (lines 130-133) says \"videos are handed off as annotation proxies (step 0)\", and the ADR-002 quote is real ? docs/DECISIONS.md line 93 literally says a proxy \"is therefore close to a precondition

for Phase B, not an optimization.\" So the named operational checklist for onboarding a new video describes a vendor handoff (step 1 'we provide it to the vendor') that skips the step the rest of the doc calls a precondition; following the FAQ as written repeats the exact pilot failure (coarse sampling grid, every boundary >=0.5s off) documented 20 lines above it.\n\n(b) Diagram omission ? CONFIRMED. The ASCII diagram (lines 7-23) starts at \"vendor delivery / CVAT export\" with no proxy/step-0 leg, while the section immediately below it is now titled \"The five commands (in order)\" beginning with step 0 \"BEFORE handoff\". The diagram and command list disagree about where the pipeline starts. Diff confirms the diagram was not modified.\n\nAttempted refutations and why they fail: (1) \"The diagram is scoped to post-delivery processing\" ? partially defensible, but the doc's own scope expanded (heading changed from four to five commands, step 0 added directly under the diagram), so the diagram is now incomplete relative to the doc's own command list; still, the diagram states nothing false. (2) \"The info is present elsewhere in the same doc\" ? true and significantly mitigating: step 0 is the FIRST item in the prominent commands list near the top, with the exact sentence \"Share the PROXY with the vendor; everything downstream keeps using the ORIGINAL\", plus two dedicated FAQ entries. A reader of this one-page doc is very likely to see it. This mitigation is why I rate severity below the reviewer's medium.\n\nVerdict: real defect ? an incomplete doc update leaving two stale spots ? but docs-only, with the correct workflow prominently and correctly stated in the same file. The proposed fix (one bullet in the FAQ, one diagram leg/note) is appropriate. Severity: low rather than medium, because no code behavior is affected and the authoritative guidance is already the most prominent content in the document; the residual risk is a reader who jumps straight to the bottom FAQ checklist.",
          "severity": "low"
        }
      },
      {
        "title": "out==src guard is case-sensitive; on Windows a case-variant --out can overwrite the source",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py",
        "line": "670",
        "detail": "`os.path.abspath(out) == os.path.abspath(src)` is a case-sensitive string compare. On this repo's primary platform (Windows, case-insensitive filesystem), `--out 'D:\\v\\MATCH.MP4' --video-path 'D:\\v\\match.mp4'` passes the guard, then ffmpeg -y opens the source file as its own output and destroys the original ? the one file the whole contract says is the system of record. Low likelihood (requires an explicit --out aimed at the source), but the guard exists precisely for this fat-finger and is broken on the platform in use.",
        "fix": "Compare os.path.normcase(os.path.realpath(...)) on both sides, and additionally use os.path.samefile(out, src) when `out` exists (also catches hardlinks/subst drives).",
        "verdict": {
          "real": true,
          "reasoning": "VERIFIED REAL, with one mitigating nuance the claim omits. Evidence: (1) C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py:670 is exactly `os.path.abspath(out) == os.path.abspath(src)`; tested on this Windows machine, abspath does NOT normalize case, so 'MATCH.MP4' vs 'match.mp4' in the same directory compare unequal and the guard passes. (2) ffmpeg does not rescue: I reproduced the exact build_proxy_command invocation (src/core/proxy.py:167-178, includes -y) live with input match.mp4 / output MATCH.MP4 ? ffmpeg's own \"Output same as Input\" check is a strcmp and missed the case variant; it exited 0 and the source file was overwritten in place (8448 -> 8385 bytes, replaced by the re-encode). The original is irrecoverably destroyed, and worse, silently: since the re-encode preserves frame count/fps by design, verify_parity passes, the tool prints \"Parity VERIFIED\", and the manifest records source_md5 computed from the already-destroyed file. The parity-failure path (os.remove(out), curate.py:714) would likewise delete the source via the case-variant alias. (3) The proposed fix works as advertised: normcase(realpath()) compares equal and os.path.samefile returns True for the case variant (both verified empirically). Mitigating nuance: the claim's scenario as literally stated is incomplete ? the next guard (curate.py:673, `os.path.exists(out) and not args.force`) DOES fire on the case-insensitive filesystem (verified exists() matches case variants), so destruction additionally requires --force. However that error message itself says \"pass --force to overwrite\", actively nudging a user who fat-fingered --out into the destructive re-run thinking it's a stale proxy ? so the dedicated out==src guard at line 670 remains the broken last line of defense it was written to be. Default --out never collides; trigger requires an explicit case-variant --out aimed at the source plus --force, which keeps likelihood low. Low severity is appropriate: real, platform-relevant (repo's primary platform is Windows), destroys the system-of-record file silently on success, but needs a rare double fat-finger to trigger.",

          "severity": "low"
        }
      },
      {
        "title": "verify_parity raises ZeroDivisionError for avg_fps == 0.0 (reachable via --allow-vfr)",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\core\\proxy.py",
        "line": "210",
        "detail": "parse_rate('0/1') returns 0.0, and the CLI's pre-check at curate.py:683 is `if src_info.avg_fps is None` ? 0.0 passes it. is_vfr() correctly treats avg_fps==0 as VFR, but with --allow-vfr the flow proceeds to verify_parity, where `abs(proxy.avg_fps - src.avg_fps) / src.avg_fps` divides by zero and the command dies with a traceback (after a full, possibly long encode) instead of a clean parity error. The duration branch already guards this (`if src.avg_fps else 0.2`); the fps branch doesn't.",
        "fix": "In verify_parity, treat avg_fps in (None, 0) as 'fps unavailable'; and/or change the CLI guard to `if not src_info.avg_fps`.",
        "verdict": {
          "real": true,
          "reasoning": "Every link verified by reading the code and by direct execution. parse_rate('0/1') returns 0.0 (the den==0 guard at proxy.py:76 only catches zero denominators). The CLI guard at curate.py:682 is `if src_info.avg_fps is None:` ? 0.0 passes it. is_vfr() returns True for avg_fps==0, but its error message tells the user to re-run with --allow-vfr, which bypasses the gate and runs the full encode. verify_parity (proxy.py:208-210) then checks only `is None` before dividing `abs(proxy.avg_fps - src.avg_fps) / src.avg_fps` ? reproduced: ZeroDivisionError raised. The exception escapes the CLI handler (which only catches ProxyError around run_ffmpeg/probe_video), so the command dies with a raw traceback after a possibly long encode, and the unverified proxy file is left on disk because delete-on-failure lives inside the never-reached `if errors:` branch ? a mild violation of the ADR-002 'failed proxy must be deleted' contract. The duration branch three lines below already guards this exact case (`if src.avg_fps else 0.2`), and is_vfr special-cases ==0, refuting any 'avg_fps=0 cannot occur' defense. Severity low as claimed: needs a rare ffprobe result (avg_frame_rate '0/x') plus explicit --allow-vfr opt-in; failure is loud, no manifest row written, no silent label corruption.",
          "severity": "low"
        }
      },
      {
        "title": "Proxy manifest accepts silent duplicate video_ids (re-runs with --force, same-stem/multi-dot sources)",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\core\\proxy.py",
        "line": "233-242 (and curate.py:665-672)",
        "detail": "Help text calls the manifest 'the proxy<->original mapping of record', but append_manifest_row blindly appends: (a) regenerating a proxy with --force adds a second row for the same video_id with a different proxy_md5 ? which row is 'of record' is undefined; (b) filename-derived ids collide easily (the import-local code comment at curate.py:83 says exactly this, and import-local has _guard_import for it) ? e.g. 'Match.A.mp4' and 'Match.B.mp4' both derive stem 'Match'/video_id 'match' AND the same default out path 'Match_proxy.mp4', so with --force the second source quietly replaces the first proxy while the manifest keeps two rows under one key pointing at different source_md5s. Note the multi-dot stem behavior itself exactly mirrors import-local (split('.')[0]) ? the derivation parity asked for in review holds; the gap is only the missing collision guard on the manifest side.",
        "fix": "Before appending, scan the existing manifest for the video_id: if found with a DIFFERENT source_md5, refuse (or require --force with a loud warning); if found with the same source_md5, note that the new row supersedes the old (document last-row-wins in the help text).",
        "verdict": {
          "real": true,
          "reasoning": "Verified in code: append_manifest_row (src/core/proxy.py:233-242) blindly appends with no dedup or collision check, and prep_annotation_proxy (src/cli/curate.py:646-758) never reads the manifest before appending. Derivation at curate.py:665-669 confirms the collision scenario: basename.split('.')[0] makes 'Match.A.mp4' and 'Match.B.mp4' share stem 'Match', video_id 'match', AND the same default out path 'data/videos/proxies/Match_proxy.mp4'. Without --force the second run errors on the existing out

file (curate.py:673-675), but the message ('pass --force to overwrite') does not disclose the existing proxy belongs to a DIFFERENT source, so --force silently clobbers the first source's proxy while the manifest accumulates two rows under one video_id with different source_md5s, the first row's proxy_path now pointing at a file with different content ? a real (operator-mediated) path to converting a vendor delivery against the wrong file/fps, the exact 'silently-wrong proxy poisons golden labels' failure the module docstring forbids. The help text (curate.py:1156-1158) calls the manifest 'the proxy<->original mapping of record' yet no last-row-wins convention is documented anywhere (docs/tests checked). The project's own sibling path proves this collision class is taken seriously: _guard_import (curate.py:63-95, comment at 82-83 'filename-derived video_ids collide easily') refuses such clobbers loudly; the proxy flow lacks the analogous guard. Mitigations keep severity low: each row is self-contained (source_md5 + proxy_md5) so per-row provenance-by-MD5 survives; the dangerous case requires --force plus a stem collision or re-run; no automated in-repo consumer of the manifest exists yet. Real defect, low severity; the proposed guard mirrors the existing _guard_import pattern.",
          "severity": "low"
        }
      },
      {
        "title": "Manifest append after a successful encode can crash with a raw traceback (Windows file lock)",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py",
        "line": "727-741",
        "detail": "The CLI only catches ProxyError; append_manifest_row's open() can raise PermissionError ? very plausible on Windows when data/annotation_proxy_manifest.csv is open in Excel (it's a CSV the user is told to commit/inspect). The expensive encode and both MD5 hashes succeed, then the command dies with an unhandled traceback, no provenance row, and no hint that the proxy itself is fine; a naive re-run re-encodes everything (needs --force) just to write one CSV row.",
        "fix": "Wrap the manifest append in try/except OSError: print the computed row fields (so provenance isn't lost) and a 'close the file and re-append manually / re-run' instruction, exiting non-zero without deleting the verified proxy.",
        "verdict": {
          "real": true,
          "reasoning": "Verified in code: append_manifest_row (src/core/proxy.py:233-242) opens the manifest with a bare open(path, \"a\") and never translates OSError/PermissionError into ProxyError; the call at src/cli/curate.py:727 is outside any try block (the only except ProxyError handlers are at lines 679/704 covering probe/encode), and main() (curate.py:1077-1240) has no top-level handler ? so a Windows file-lock PermissionError (classic when the CSV is open in Excel) surfaces as a raw traceback. The scenario is plausible: docs/INGESTION_PIPELINE.md:143 and the --manifest help text (curate.py:1156-1158) both tell the user to commit/inspect data/annotation_proxy_manifest.csv, and the repo runs on Windows. Consequences match the claim exactly: the encode (702), parity verification (708), and both full-file MD5 hashes (723-724) complete before the append, the computed MD5s are only printed in the success summary AFTER the append (748-749) so provenance is lost on crash, and a naive re-run is refused at curate.py:673-675 (\"pass --force to overwrite\") with --force triggering a full re-encode ? there is no append-only recovery path. It is not intended behavior: the same function catches OSError on the os.remove cleanup path (713-719) and prints friendly errors everywhere else, making the manifest append the one inconsistent step. Severity is low: the verified proxy survives intact, no data corruption or wrong labels ? the cost is a raw-traceback UX, lost provenance output, and a wasteful re-encode in an occasional-but-realistic scenario.",
          "severity": "low"
        }
      },
      {
        "title": "'identical fps ... by construction' message is slightly stronger than what verify_parity proves",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\src\\cli\\curate.py",
        "line": "752-754 (next-steps print), proxy.py:208-219",
        "detail": "The post-run message says convert-cvat may use --video-path with either file because fps/frame count are 'identical by construction'. Frame count is verified EXACTLY, but fps is only verified to 0.1% relative and duration to max(2/fps, 0.2s). A proxy at the tolerance edge, probed for fps by convert-cvat (probe_resolution_fps rounds avg_frame_rate to 3 decimals), shifts recomputed end-of-video times by up to ~0.2s on a 30-minute Phase-B file ? uncomfortably

close to the ±0.3s golden bar the whole feature exists to protect (the duration check is what bounds it to ~0.2s; the fps check alone would allow ~1.7s). In practice -fps_mode passthrough keeps avg_frame_rate bit-identical, so this is a contract-precision nit, not an observed failure.",
        "fix": "Either tighten the fps parity check to exact rational equality of avg_frame_rate strings (compare the raw 'num/den' from ffprobe, falling back to the float tolerance), or soften the message to recommend the ORIGINAL (or --fps from the intake manifest) as convert-cvat's fps source.",
        "verdict": {
          "real": true,
          "reasoning": "Every factual element verifies against the code. The next-steps print (curate.py:755-756) claims \"identical fps/frame count by construction\" and blesses probing EITHER file, but verify_parity (proxy.py:203-219) proves frame count exactly while fps is only checked to 0.1% relative and duration to max(2/fps, 0.2s). convert-cvat with --video-path and no --fps resolves fps via probe_resolution_fps (rally_cvat.py:380-382), which rounds avg_frame_rate to 3 decimals (downloader.py:137); at the verification tolerance edge (duration-bounded ~0.011% ? 0.0066 at 59.94 fps) the proxy would probe to a DIFFERENT 3-decimal fps than the original, shifting end-of-timeline recomputed times by up to ~0.2s on a 30-min file ? the reviewer's arithmetic (fps check alone would allow ~1.7-1.8s; the duration check bounds it to ~0.2s) is correct. Refutation attempts failed: \"by construction\" is defensible only for timestamp passthrough (-fps_mode passthrough, no -r), not for the reported avg_frame_rate that convert-cvat actually consumes ? the very existence of the 0.1%/0.2s tolerances concedes reported fps is not guaranteed bit-stable. Severity is correctly capped at low: passthrough keeps avg_frame_rate identical in practice (E2E smoke passed, VFR is gated), and even the theoretical worst case stays within the ±0.3s golden bar, so this is a message-precision/verification-tightness nit with a cheap fix (exact rational num/den comparison, or recommend the original as fps source), exactly as the reviewer scoped it.",
          "severity": "low"
        }
      },
      {
        "title": "ADR-002 says provenance goes to 'the intake-manifest row'; implementation introduces a separate proxy manifest",
        "severity": "low",
        "file": "C:\\Users\\avidu\\Projects\\sports-data-collector\\docs\\DECISIONS.md",
        "line": "117-119",
        "detail": "ADR-002's provenance bullet reads 'emit/append the intake-manifest row ? video_id, fps, frame count, original MD5 + proxy MD5'. The implementation instead appends to a NEW internal CSV (data/annotation_proxy_manifest.csv) whose columns differ from docs/annotation/INTAKE_MANIFEST_TEMPLATE.csv, and the vendor-facing intake manifest gains no proxy checksum. Probably a deliberate refinement (internal provenance vs vendor sheet), but the ADR text and the shipped behavior now disagree on WHERE the mapping of record lives, and ROORA_BRIEF.md still tells the vendor 'our pipeline identifies a video by its file checksum' with no mention that what they receive is a proxy (ADR lists the brief revision as a Phase-B follow-up, so that part is tracked). Related pre-existing ambiguity made slightly worse: PIPELINE_ROLE.md references the INDEXER repo's 'ADR-002 (this bridge)' unscoped, while this repo now has its own, different ADR-002 that the new code cites.",
        "fix": "One-line amendment to ADR-002 ('provenance lands in data/annotation_proxy_manifest.csv, separate from the vendor-facing intake manifest') and qualify PIPELINE_ROLE.md's ADR references as 'indexer ADR-002/-003/-004'.",
        "verdict": {
          "real": true,
          "reasoning": "Verified against the actual files. (1) docs/DECISIONS.md:117-119 (ADR-002) says provenance is \"emit/append the intake-manifest row ... original MD5 + proxy MD5\"; in this repo \"intake manifest\" unambiguously means the vendor-facing docs/annotation/INTAKE_MANIFEST_TEMPLATE.csv (ADR-002 itself uses the term that way at lines 122 and 127; INGESTION_PIPELINE.md:113 likewise). (2) The implementation instead appends to a new internal CSV ? src/cli/curate.py:1156 defaults --manifest to data/annotation_proxy_manifest.csv, with MANIFEST_FIELDS (src/core/proxy.py:36-40) disjoint from the intake template's columns except video_id/fps ? and the diff (4 files) touches neither DECISIONS.md nor the intake template, so the vendor sheet gains no proxy checksum and the ADR text now disagrees with shipped behavior on where provenance lives. The branch's own FAQ (INGESTION_PIPELINE.md:142-145) documents the new location while citing ADR-002 as the rationale. (3) ROORA_BRIEF.md:53 still says \"our pipeline identifies a video by its file checksum\" (brief revision tracked as a Phase-B follow-up in the ADR, as the claim concedes). (4) PIPELINE_ROLE.md: lines 13-14 are

actually path-scoped to the indexer repo (minor overstatement in the claim), but line 33's bare \"(see ADR-002)\" is genuinely ambiguous now that this repo has its own different ADR-002 cited by the same bare name. Refutation fails: the separate internal manifest is a defensible (likely better) design, but the defect claimed is the ADR/code drift, which factually exists and contradicts ADR-002's \"this ADR records the agreed choice\" role. Docs-only, no functional impact; one-line amendment fixes it ? low severity is correct.",
          "severity": "low"
        }
      }
    ],
    "refuted": [
      {
        "title": "Integration checks verified clean (informational)",
        "why": "No defect ? this is an informational \"checks verified clean\" entry, and every factual assertion in it was independently re-verified as accurate. (1) video_id derivation in prep_annotation_proxy (curate.py ~663-666) exactly mirrors import-local (curate.py:113-114): both basename.split('.')[0] then lower()/space->underscore; test_proxy.py:226 asserts 'Match One.mp4' -> 'match_one'. (2) .gitignore line 41 'data/videos/' covers data/videos/proxies/; data/annotation_proxy_manifest.csv is not matched by any ignore rule and 'commit it' is documented in the --manifest help and FAQ; the artifacts/ .gitignore edit is an uncommitted working-tree change not part of the branch diff (branch touches only the 4 listed files). (3) proxy.py is stdlib-only (csv/hashlib/os/shutil/subprocess/dataclasses/typing + lazy json), compatible with requires-python >=3.9. (4) CI (ubuntu, no ffmpeg, plain pytest) is safe: all ffmpeg/ffprobe touchpoints mocked in test_proxy.py; suite re-run locally reproduces 159 passed in 2.49s; 22 tests collected in test_proxy.py; prep-annotation-proxy --help renders without argparse errors. (5) rally_cvat.py uses video_path only for fps probing (probe_resolution_fps, lines 380-381); export pairs DB-registered local_path files and proxies are never registered, so export/eval manifests are unaffected; the 'run on the ORIGINAL file' contract is consistently restated in the module docstring, CLI output, and FAQ. (6) ADR-002's ratified knobs (1080p not 720, CRF 19 in the 18-20 band, -g 15, audio kept, -fps_mode passthrough, no -r/no trim) match the code defaults (DEFAULT_HEIGHT=1080, DEFAULT_CRF=19, DEFAULT_GOP=15). Only trivial nit: the claim cites the new derivation at lines 665-668; it actually sits at ~663-666 ? immaterial. No action needed; the parent review can rely on these checks without re-auditing."
      }
    ]
  }
}
</output>

85. user (2026-06-12T10:49:00.080Z)

```
10-indexer repo:
11-
12-- `badminton-highlight-indexer/docs/ROADMAP.md` ? the four-phase plan and end-state.
13:- `badminton-highlight-indexer/docs/DECISIONS.md` ? ADR-002 (this bridge), ADR-003
14:  (acquisition policy), ADR-004 (offline learned segmenter).
15-
16-## Why this repo matters
17-
--
30-  table tennis ? share a `(video_id, ordinal, start, end)` shape) and gated on
31-  `is_commercial`, so only licensable data flows
32-  downstream. The annotations are **temporal-only** ? they train/evaluate rally
33:  *segmentation*, not the frame-level shuttle detector (see ADR-002).
34-
35-- **`fetch`** ? (re)downloads videos at **best available resolution**, updating
36-  the DB **in place** so annotations and curation status are never lost. Uses
37-  authenticated cookies for reliable, unthrottled downloads. The why and the
38:  hard-won pitfalls (SABR?144p, Chrome DPAPI, rate-limiting) are in ADR-003.
39-
40-## The one contract that matters: video identity
41-
```

86. assistant (2026-06-12T10:49:06.859Z)

Lines 33 and 38 use bare ADR references that now collide with collector-local numbering. Now implementing all fixes ? `proxy.py` first:

87. user (2026-06-12T10:49:13.057Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

88. user (2026-06-12T10:49:15.766Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

89. user (2026-06-12T10:49:19.529Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

90. user (2026-06-12T10:49:23.763Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

91. user (2026-06-12T10:49:34.598Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

92. user (2026-06-12T10:49:37.016Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\core\proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

93. user (2026-06-12T10:49:49.402Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

94. user (2026-06-12T10:50:14.703Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

95. user (2026-06-12T10:50:28.084Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

96. user (2026-06-12T10:50:35.325Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

97. user (2026-06-12T10:50:37.071Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

98. user (2026-06-12T10:50:47.359Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

99. user (2026-06-12T10:50:51.238Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

100. user (2026-06-12T10:50:58.080Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

101. user (2026-06-12T10:50:54.909Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

102. user (2026-06-12T10:50:56.375Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

103. user (2026-06-12T10:51:07.319Z)

```
113-  **dense keyframes** (`-g 15`) so frame-stepping/back-scrubbing is fast, **audio kept**
114-  (contact sound is the annotator's best boundary cue).
115-- **Verification built in:** assert proxy frame count == original frame count and
116-  durations match (also surfaces VFR sources, where `frame / fps` is already unsafe).
117:- **Provenance:** emit/append the intake-manifest row ? `video_id`, fps, frame count,
118-  **original MD5 + proxy MD5** ? so the proxy?original mapping is durable and downstream
119-  processing provably targets the original.
120-
121-### Rationale (why the collector, why not a loose script)
```

104. user (2026-06-12T10:51:07.814Z)

```
28        `start,end` CSVs + a `manifest.json` pairing each CSV with its video file.
29        Sport-agnostic (all supported sports ? badminton, tennis, cricket, pickleball,
30        table tennis ? share a `(video_id, ordinal, start, end)` shape) and gated on
31        `is_commercial`, so only licensable data flows
32        downstream. The annotations are **temporal-only** ? they train/evaluate rally
33        *segmentation*, not the frame-level shuttle detector (see ADR-002).
34
35      - **`fetch`** ? (re)downloads videos at **best available resolution**, updating
36        the DB **in place** so annotations and curation status are never lost. Uses
37        authenticated cookies for reliable, unthrottled downloads. The why and the
38        hard-won pitfalls (SABR?144p, Chrome DPAPI, rate-limiting) are in ADR-003.
39
40      ## The one contract that matters: video identity
41
```

105. user (2026-06-12T10:51:15.763Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

106. user (2026-06-12T10:51:17.670Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\PIPELINE_ROLE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

107. user (2026-06-12T10:51:19.085Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\PIPELINE_ROLE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

108. user (2026-06-12T10:51:27.447Z)

```
  - **Same fps as the source ? never reduce it.** At a lower fps every CVAT frame index
    changes meaning, and the vendor's 30-frame sampling habit at 30 fps would become a
    **1.0 s** grid ? twice as coarse. Same frame count, no trimming, no leading padding ?
    proxy seconds map 1:1 onto the original.
  - **Downscale 4K ? 1080p** (not 720p: table-tennis ball visibility), CRF ~18?20,
    **dense keyframes** (`-g 15`) so frame-stepping/back-scrubbing is fast, **audio kept**
    (contact sound is the annotator's best boundary cue).
  - **Verification built in:** assert proxy frame count == original frame count and
    durations match (also surfaces VFR sources, where `frame / fps` is already unsafe).
  - **Provenance:** emit/append the intake-manifest row ? `video_id`, fps, frame count,
    **original MD5 + proxy MD5** ? so the proxy?original mapping is durable and downstream
    processing provably targets the original.

  ### Rationale (why the collector, why not a loose script)
  - **This repo owns the vendor flow end-to-end** (`docs/annotation/`, intake manifest,
    `convert-cvat` ? `validate-delivery` ? `import-local`); proxy prep is simply command
```

109. user (2026-06-12T10:51:43.143Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\DECISIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

110. assistant (2026-06-12T10:51:48.192Z)

Now the new and updated tests covering every fix:

111. user (2026-06-12T10:51:56.217Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

112. user (2026-06-12T10:51:59.039Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

113. user (2026-06-12T10:52:06.067Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

114. assistant (2026-06-12T10:52:14.705Z)

Now appending the CLI-level tests for the new failure paths:

115. user (2026-06-12T10:52:39.314Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated successfully. (file state is current in your context ? no need to Read it back)

116. user (2026-06-12T10:52:52.011Z)

```
Exit code 1
tests\test_proxy.py:294: TypeError
=========================== short test summary info ===========================
FAILED tests/test_proxy.py::test_prep_proxy_refuses_vfr_without_flag - TypeEr...
FAILED tests/test_proxy.py::test_prep_proxy_allow_vfr_proceeds - TypeError: P...
2 failed, 167 passed in 2.76s
```

117. assistant (2026-06-12T10:52:57.614Z)

Two tests still construct `ProbeInfo` with the old `duration=` kwarg (now a property). Fixing:

118. user (2026-06-12T10:53:00.468Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

119. user (2026-06-12T10:53:03.363Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

120. user (2026-06-12T10:53:16.541Z)

```
........................................................................ [ 85%]
.........................                                                [100%]
169 passed in 2.71s
  Source      : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\Smoke Match.mp4 (1920x1080, 10 MB,
md5 86280c00c926c319bdc7b61d62c1dd34)
  Proxy       : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\smoke_proxy.mp4 (854x480, 1 MB,
md5 fd6f22e335274d76482ca6860997e26b)
  Parity      : VERIFIED ? 179 frames @ 59.9401 fps on both
  Manifest    : C:\Users\avidu\AppData\Local\Temp\proxy_smoke\proxy_manifest.csv

Next:
  1. Share the PROXY with the vendor (CVAT at frame step 1; first/last box of every rally on the
exact contact/dead frame).
  2. On delivery, run convert-cvat with --video-path pointing at the ORIGINAL (or pass --fps
from the manifest row).
  3. import-local / export keep pointing at the ORIGINAL file ? the indexer keys videos by its
MD5.
```

121. user (2026-06-12T10:53:31.734Z)

```
Exit code 1
Error: --out must differ from --video-path (it would overwrite the original recording).
exit=1
source intact: True
```

122. user (2026-06-12T10:53:51.038Z)

```
warning: in the working copy of 'src/core/proxy.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'tests/test_proxy.py', LF will be replaced by CRLF the next time
Git touches it
 M .gitignore
M  docs/DECISIONS.md
M  docs/INGESTION_PIPELINE.md
M  docs/PIPELINE_ROLE.md
M  src/cli/curate.py
M  src/core/proxy.py
M  tests/test_proxy.py
[feat/prep-annotation-proxy 3c837ef] fix(proxy): harden the proxy flow per adversarial review
(cleanup, same-file, parity)
 6 files changed, 289 insertions(+), 48 deletions(-)
remote:
remote: Create a pull request for 'feat/prep-annotation-proxy' on GitHub by visiting:
remote:      https://github.com/avidullu/sports-data-collector/pull/new/feat/prep-annotation-
proxy
remote:
branch 'feat/prep-annotation-proxy' set up to track 'origin/feat/prep-annotation-proxy'.
To https://github.com/avidullu/sports-data-collector.git
 * [new branch]      feat/prep-annotation-proxy -> feat/prep-annotation-proxy
```

123. user (2026-06-12T10:54:15.241Z)

```
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/22
```

124. user (2026-06-12T10:54:56.484Z)

```
Refreshing checks status every 15 seconds. Press Ctrl+C to quit.

test    pending 0       https://github.com/avidullu/sports-data-
collector/actions/runs/27411256062/job/81012802283
test    pass    20s     https://github.com/avidullu/sports-data-
collector/actions/runs/27411256062/job/81012802283
test    pass    20s     https://github.com/avidullu/sports-data-
collector/actions/runs/27411256062/job/81012802283
```

125. user (2026-06-12T10:55:12.818Z)

```
test: SUCCESS
From https://github.com/avidullu/sports-data-collector
 * branch            master     -> FETCH_HEAD
   4fb2d6f..a068fd1  master     -> origin/master
Updating 4fb2d6f..a068fd1
Fast-forward
 docs/DECISIONS.md          |   7 +-
 docs/INGESTION_PIPELINE.md |  39 +++-
 docs/PIPELINE_ROLE.md      |   4 +-
 src/cli/curate.py          | 202 +++++++++++++++++++
 src/core/proxy.py          | 264 +++++++++++++++++++++++++
 tests/test_proxy.py        | 464 +++++++++++++++++++++++++++++++++++++++++++++
 6 files changed, 971 insertions(+), 9 deletions(-)
 create mode 100644 src/core/proxy.py
 create mode 100644 tests/test_proxy.py
M       .gitignore
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
a068fd1 feat(curate): prep-annotation-proxy ? frame-identical vendor handoff re-encode (ADR-002)
(#22)
4fb2d6f docs: ADR-002 - vendor annotation proxies generated in-repo (curate prep-annotation-
proxy) (#21)
```

master...origin/master

```
 M .gitignore
```

126. assistant (2026-06-12T10:55:20.110Z)

```
Merged. Updating the memory checkpoint, then wrapping up:
```

127. user (2026-06-12T10:55:35.100Z)

```
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
```

128. assistant (2026-06-12T10:55:54.605Z)

```
Done ? **[sports-data-collector#22](https://github.com/avidullu/sports-data-collector/pull/22)
is merged** (squash `a068fd1`, CI green, branch deleted, master synced). ADR-002 is now
implemented end-to-end.
```

What shipped

```
`curate prep-annotation-proxy` plus a new [src/core/proxy.py](C:\Users\avidu\Projects\sports-
data-collector\src\core\proxy.py) engine. One command per video before handing it to Roora:
```

```
python -m src.cli.curate prep-annotation-proxy --video-path match.mp4
```

It downscales to ?1080p with dense keyframes (`-g 15`) and audio kept, while passing the timeline through untouched (`-fps_mode passthrough`, never `-r`) ? then **verifies** the proxy has the source's exact frame count, fps, and duration before declaring it ready. Provenance (source + proxy MD5s, fps, frame count) is appended to `data/annotation_proxy_manifest.csv`, which is meant to be committed. Failed or interrupted encodes are deleted on the spot ? an unverified proxy can never sit under the to-be-shared filename.

How it was verified

- **169 tests green** (+32 new, all offline) and CI passed on the PR.
- **Live end-to-end on your box** (ffmpeg 8.1.1): a synthesized 59.94fps clip went through the real command ? parity verified at 179 frames on both files, 10MB?1MB. The smoke immediately paid for itself by catching ffmpeg 8's `-vsync` deprecation, so the tool now uses the modern `-fps_mode` flag.
- **Adversarial review workflow** (two review lenses, then a refutation agent per finding) confirmed 1 high / 2 medium / 7 low issues, all fixed before merge. The standout: the same-file guard was case-sensitive, and the verifier *live-reproduced* `ffmpeg -y` truncating a source file via a case-variant `--out` + `--force` on NTFS. That's now blocked (re-verified live: refused, source intact). The high was a partial proxy surviving a failed encode; cleanup now covers every failure path including Ctrl-C.

Docs were updated alongside: `INGESTION_PIPELINE.md` now opens with step 0 (proxy ? vendor; everything else runs on the original), and the old "coarse sampling doesn't hurt" FAQ now correctly distinguishes highlight consumption from the ±0.3s golden-label bar.

What's deliberately not in this PR

The two ADR-002 follow-ups remain separate work items: the **Phase-B brief revision** (frame step 1, boundary boxes on exact contact/dead frames ? vendor comms are yours to word) and the **grid-snap QA check** in `validate-delivery` that would auto-catch a delivery like the last one. Say the word on the QA check and I'll build it the same way.

129. user (2026-06-12T11:41:18.185Z)

Please go ahead and build them for use. Additionally, say, I made a directory with subdirectories each (subdirectory) holding a well named video, can you make a cli option that I run the curate and it mirrors the same layout but only with the right videos as Roora requires? Also make a single dedicated doc for HOW to send and receive Roora, so that a human vcan replicate what you do e2e!

130. assistant (2026-06-12T11:42:25.708Z)

Three deliverables: (1) the grid-snap QA check in `validate-delivery` plus the Phase-B brief/guide revision, (2) a batch mode that mirrors your directory layout with proxies, (3) a single end-to-end Roora runbook. Per the one-PR-per-work-item convention I'll ship them as three sequential PRs (no stacking), merging each on green as you've authorized. First, reading the CVAT parser to design the grid-snap check:

131. user (2026-06-12T11:42:26.464Z)

```
"""Adapter: CVAT rally exports -> our canonical per-rally records.

Why this exists
---------------
Our annotation brief (``docs/annotation/ROORA_BRIEF.md``) asks vendors for a
per-rally **CSV** that imports directly via ``curate import-local``. A vendor may
instead do the work in CVAT and hand back the *CVAT* export: per-frame bounding
boxes, with the rally fields (``rally_number``, ``shots_count``,
``ending_reason`` ...) carried as CVAT **object attributes** on each box. The box
```

```
geometry is irrelevant to us ? it is just a carrier the rater attached the rally
metadata to.

This module is the ingestion-side adapter that absorbs that vendor format so the
modeling repo never has to. It does two things, both deliberately on *our* side
so we fix mechanical problems rather than renegotiating them:

1. **Recompute time from the frame range.** A rally's interval is taken from the
   first/last *frame* a box for that rally appears on:
   ``seconds = frame / fps`` at the video's true fps. We **ignore** any
   ``start_time``/``end_time`` attributes the vendor wrote ? a CVAT job that
   samples every ``step`` frames and assumes 30 fps yields systematically wrong
   times (we have seen ``start_time == frame / (step * 30)``, i.e. off by the
   sampling step at the wrong fps). The frame index is reliable; the vendor's
   seconds are not.

2. **Surface ? never silently fix ? content defects.** Overlapping rallies,
   zero-duration rallies, ending_reason values outside the agreed set, and
   too-coarse sampling are reported as :class:`Issue`s. Guessing a corrected
   rally boundary would poison ground truth, so that stays a human/vendor call.

Supported input: the single-file CVAT XML export, in either **image mode**
(``<image><box>...``) or **track/interpolation mode** (``<track><box frame=...>``).
The two PASCAL-VOC / COCO sibling exports encode the same data and can be added
later if a vendor sends only those.
"""
from __future__ import annotations

import os
import re
import zipfile
import xml.etree.ElementTree as ET
from collections import Counter
from dataclasses import dataclass, field
from typing import Any, Dict, List, Optional, Tuple

from src.core.parsing import safe_int
from src.core.downloader import probe_resolution_fps

# The agreed cross-sport ending_reason vocabulary for net-separated sports.
# Kept here as a soft check only (a warning, never a hard failure) so this
# adapter does not depend on the annotation-spec wording.
DEFAULT_ALLOWED_REASONS = {
    "winner",
    "forced_error",
    "unforced_error",
    "service_fault",
    "let",
    "other",
}

# The observational outcome of the last shot ? orthogonal to ending_reason
# (winner can be 'in'; an error can be 'net' or 'out'; service_fault/let -> 'n_a').
DEFAULT_ALLOWED_OUTCOMES = {"in", "net", "out", "n_a"}

# CVAT attribute names we map onto canonical columns. Vendor label schemas
# drift ? suffixed names like "score_before (A-B)", and singular/plural or
# case/separator variants like "shot_count" (the Roora deliveries) for
# "shots_count" ? so both these aliases and the incoming attribute names are
# folded through ``_fold_attr_name`` before matching.
_ATTR_ALIASES = {
    "rally_number": ("rally_number",),
    "shots_count": ("shots_count", "shot_count"),
    "ending_reason": ("ending_reason",),
    "last_shot_outcome": ("last_shot_outcome", "outcome", "last_shot_outcome
```

```
(in/net/out)"),
    "score_before": ("score_before", "score_before (a-b)"),
    "score_after": ("score_after", "score_after (a-b)"),
    "notes": ("notes", "note"),
}


def _fold_attr_name(name: str) -> str:
    """Fold a CVAT attribute name to a tolerant matching key.

    Case ("Shot_Count"), separators ("shot count", "shot-count"), parenthesized
    hints ("score_before (A-B)") and singular/plural ("shot_count" vs
    "shots_count") all fold to the same key, so a vendor's naming drift cannot
    silently drop a column.
    """
    s = re.sub(r"\(.*?\)", " ", (name or "").lower())
    tokens = [t for t in re.split(r"[\s_\-]+", s) if t]
    return "_".join(
        t[:-1] if len(t) > 1 and t.endswith("s") and not t.endswith("ss") else t
        for t in tokens)


_FOLDED_ALIASES = {canon: tuple(dict.fromkeys(_fold_attr_name(a) for a in aliases))
                   for canon, aliases in _ATTR_ALIASES.items()}

CANONICAL_COLUMNS = [
    "rally_number", "start_mmss", "end_mmss", "start_time", "end_time",
    "shots_count", "ending_reason", "last_shot_outcome", "score_before", "score_after",
"notes",
]


@dataclass
class RallyRecord:
    """One rally recovered from the CVAT export, time re-derived from frames."""
    rally_number: int
    start_time: float            # seconds, = frame_min / fps
    end_time: float              # seconds, = frame_max / fps (see padding note)
    shots_count: Optional[int]
    ending_reason: Optional[str]
    last_shot_outcome: Optional[str]
    score_before: Optional[str]
    score_after: Optional[str]
    notes: Optional[str]
    frame_min: int
    frame_max: int
    n_boxes: int
    vendor_start_time: Optional[str] = None   # the (discarded) attribute value
    vendor_end_time: Optional[str] = None

    @property
    def duration(self) -> float:
        return self.end_time - self.start_time


@dataclass
class Issue:
    """A defect found in the delivery. ``blocker`` issues stop a clean import."""
    severity: str   # 'blocker' | 'high' | 'medium' | 'low'
    code: str
    message: str
    rally_number: Optional[int] = None


@dataclass
```

```python
class ConversionResult:
    records: List[RallyRecord]
    issues: List[Issue]
    fps: float
    step: int
    meta: Dict[str, Any] = field(default_factory=dict)

    @property
    def n_blockers(self) -> int:
        return sum(1 for i in self.issues if i.severity == "blocker")


def _mmss(seconds: float) -> str:
    """Format seconds as ``m:ss.mmm`` (matches the brief's mm:ss columns)."""
    m = int(seconds // 60)
    s = seconds - m * 60
    return f"{m}:{s:06.3f}"


def _attr_lookup(attrs: Dict[str, str], canonical: str) -> Optional[str]:
    """Look up a canonical field in a folded-key attrs dict (see _fold_attr_name)."""
    for alias in _FOLDED_ALIASES[canonical]:
        if alias in attrs and attrs[alias] != "":
            return attrs[alias]
    return None


def parse_cvat_boxes(xml_path: str) -> Tuple[Dict[str, Any], List[Dict[str, Any]]]:
    """Parse a CVAT XML export into (meta, boxes).

    ``boxes`` is a flat list of ``{"frame": int, "attrs": {name: value}}`` ? one
    entry per visible box, from either image-mode or track-mode exports. Attr
    names are folded via ``_fold_attr_name`` (lowercase, separators unified,
    parenthesized hints dropped, plural-tolerant), not the raw vendor spelling.
    ``meta`` carries ``step`` (frame_filter) and ``stop_frame`` when present.
    """
    if not os.path.exists(xml_path):
        raise FileNotFoundError(f"CVAT export not found: {xml_path}")

    # Accept the standard CVAT delivery artifact (a .zip containing annotations.xml) as
    # well as a bare .xml. Corrupt/unparseable XML is surfaced as a ValueError (a clean
    # gate failure) rather than an uncaught ParseError traceback.
    try:
        if zipfile.is_zipfile(xml_path):
            with zipfile.ZipFile(xml_path) as zf:
                names = ([n for n in zf.namelist() if n.endswith("annotations.xml")]
                         or [n for n in zf.namelist() if n.endswith(".xml")])
                if not names:
                    raise ValueError(f"CVAT zip contains no annotations.xml: {xml_path}")
                root = ET.fromstring(zf.read(names[0]))
        else:
            root = ET.parse(xml_path).getroot()
    except ET.ParseError as e:
        raise ValueError(f"CVAT XML is corrupt/unparseable ({xml_path}): {e}")
    meta: Dict[str, Any] = {}

    job = root.find(".//job")
    if job is not None:
        stop = job.findtext("stop_frame")
        if stop is not None and stop.strip().isdigit():
            meta["stop_frame"] = int(stop.strip())
        ff = job.findtext("frame_filter") or ""
        if "step=" in ff:
            try:
```

```
                meta["step"] = int(ff.split("step=")[1].split()[0])
            except (ValueError, IndexError):
                pass

    def _attrs(el: ET.Element) -> Dict[str, str]:
        out: Dict[str, str] = {}
        for a in el.findall("attribute"):
            out[_fold_attr_name(a.get("name"))] = (a.text or "").strip()
        return out

    boxes: List[Dict[str, Any]] = []

    # Image mode: <image id=frame> ... <box>...</box>
    for img in root.findall("image"):
        try:
            frame = int(img.get("id"))
        except (TypeError, ValueError):
            continue
        for box in img.findall("box"):
            boxes.append({"frame": frame, "attrs": _attrs(box)})

    # Track mode: <track> ... <box frame=N outside=0>...</box>
    for track in root.findall("track"):
        for box in track.findall("box"):
            if box.get("outside") == "1":
                continue  # a track's terminating keyframe carries no real label
            try:
                frame = int(box.get("frame"))
            except (TypeError, ValueError):
                continue
            boxes.append({"frame": frame, "attrs": _attrs(box)})

    return meta, boxes


def _infer_step(frames: List[int]) -> int:
    """Most common positive gap between consecutive distinct sampled frames."""
    uniq = sorted(set(frames))
    gaps = [b - a for a, b in zip(uniq, uniq[1:]) if b - a > 0]
    if not gaps:
        return 1
    return Counter(gaps).most_common(1)[0][0]


def build_records(boxes: List[Dict[str, Any]], fps: float) -> List[RallyRecord]:
    """Group boxes by rally_number and derive each rally's frame range -> seconds."""
    if fps is None or fps <= 0:
        raise ValueError("fps must be a positive number to convert frames to seconds")

    by_rally: Dict[int, List[Dict[str, Any]]] = {}
    for b in boxes:
        rn = safe_int(_attr_lookup(b["attrs"], "rally_number"), None)
        if rn is None:
            continue
        by_rally.setdefault(rn, []).append(b)

    records: List[RallyRecord] = []
    for rn in sorted(by_rally):
        items = by_rally[rn]
        frames = sorted(b["frame"] for b in items)
        attrs = items[0]["attrs"]
        fmin, fmax = frames[0], frames[-1]
        records.append(RallyRecord(
            rally_number=rn,
            start_time=fmin / fps,
```

```
                end_time=fmax / fps,
                shots_count=safe_int(_attr_lookup(attrs, "shots_count"), None),
                ending_reason=_attr_lookup(attrs, "ending_reason"),
                last_shot_outcome=_attr_lookup(attrs, "last_shot_outcome"),
                score_before=_attr_lookup(attrs, "score_before"),
                score_after=_attr_lookup(attrs, "score_after"),
                notes=_attr_lookup(attrs, "notes"),
                frame_min=fmin,
                frame_max=fmax,
                n_boxes=len(frames),
                vendor_start_time=attrs.get("start_time"),
                vendor_end_time=attrs.get("end_time"),
            ))
        return records


    def detect_issues(records: List[RallyRecord],
                      stop_frame: Optional[int] = None,
                      allowed_reasons: Optional[set] = None,
                      allowed_outcomes: Optional[set] = None,
                      max_gap_s: float = 15.0) -> List[Issue]:
        """Find the defects that block or degrade a clean ground-truth import.

        Note: we deliberately do NOT flag the sampling density (step/fps). A boundary
        slack of ~0.5s is acceptable for this pipeline because the downstream
        highlight compiler pads every clip on output (badminton-highlight-indexer,
        backend/pipeline/stitcher/compiler.py, driven by config.json
        stitching.{begin_padding,end_padding} ? 0.5s/0.5s as shipped), which
        absorbs the sampling uncertainty when highlights are trimmed.
        """
        allowed = allowed_reasons if allowed_reasons is not None else
DEFAULT_ALLOWED_REASONS
        allowed_out = allowed_outcomes if allowed_outcomes is not None else
DEFAULT_ALLOWED_OUTCOMES
        issues: List[Issue] = []

        ordered = sorted(records, key=lambda r: r.rally_number)
        for i, r in enumerate(ordered):
            if r.end_time <= r.start_time:
                issues.append(Issue(
                    "blocker", "zero_duration",
                    f"Rally {r.rally_number}: end <= start "
                    f"(frames {r.frame_min}-{r.frame_max}); single-sample or stray box.",
                    r.rally_number))

            if r.ending_reason is not None and r.ending_reason not in allowed:
                issues.append(Issue(
                    "medium", "ending_reason_vocab",
                    f"Rally {r.rally_number}: ending_reason '{r.ending_reason}' "
                    f"not in the agreed set {sorted(allowed)}.",
                    r.rally_number))

            if r.last_shot_outcome is not None and r.last_shot_outcome not in allowed_out:
                issues.append(Issue(
                    "low", "outcome_vocab",
                    f"Rally {r.rally_number}: last_shot_outcome '{r.last_shot_outcome}' "
                    f"not in {sorted(allowed_out)}.",
                    r.rally_number))

            # Inflated span: a long rally with implausibly few shots usually means a
            # stray/duplicated box stretched the frame range past the real ending.
            if r.shots_count and r.duration >= 5.0:
                sps = r.shots_count / r.duration
                if sps < 0.3:
                    issues.append(Issue(
```

```
                    "medium", "shot_density",
                    f"Rally {r.rally_number}: {r.shots_count} shots over "
                    f"{r.duration:.1f}s ({sps:.2f}/s) is implausibly low ? the span "
                    f"may be inflated by a stray box.",
                    r.rally_number))

        if stop_frame is not None and r.frame_max > stop_frame:
            issues.append(Issue(
                "high", "frame_out_of_range",
                f"Rally {r.rally_number}: frame {r.frame_max} exceeds the video's "
                f"last frame {stop_frame}.",
                r.rally_number))

        if i > 0:
            prev = ordered[i - 1]
            if r.start_time < prev.end_time:
                issues.append(Issue(
                    "blocker", "overlap",
                    f"Rally {prev.rally_number} (ends {prev.end_time:.2f}s) overlaps "
                    f"Rally {r.rally_number} (starts {r.start_time:.2f}s) by "
                    f"{prev.end_time - r.start_time:.2f}s ? rallies must be disjoint.",
                    r.rally_number))
            elif r.start_time - prev.end_time > max_gap_s:
                # A long unlabeled stretch between two rallies is a cheap, label-only
                # second opinion for the worst error (a missed rally). It won't catch a
                # rally MERGED into its neighbour's span (shot_density/inflated catches
                # that), but it does catch an omitted rally that left a visible hole.
                gap = r.start_time - prev.end_time
                issues.append(Issue(
                    "medium", "possible_missed_rally",
                    f"{gap:.1f}s of unlabeled time between Rally {prev.rally_number} "
                    f"(ends {prev.end_time:.1f}s) and Rally {r.rally_number} "
                    f"(starts {r.start_time:.1f}s) ? check for an unlabeled rally in the gap.",
                    r.rally_number))
    return issues


def convert(xml_path: str,
            fps: Optional[float] = None,
            video_path: Optional[str] = None,
            step: Optional[int] = None,
            allowed_reasons: Optional[set] = None) -> ConversionResult:
    """Full pipeline: parse CVAT XML -> records (+ issues) at the true fps.

    ``fps`` resolution order: explicit ``fps`` arg > ffprobe of ``video_path`` >
    the CVAT meta (none) -> error. We never fall back to a guessed 30 fps, since
    that silent guess is exactly the bug this adapter exists to undo.
    """
    meta, boxes = parse_cvat_boxes(xml_path)

    if fps is None and video_path:
        _, _, probed = probe_resolution_fps(video_path)
        fps = probed
    if fps is None or fps <= 0:
        raise ValueError(
            "Could not determine fps. Pass fps=... or a readable video_path so "
            "frames can be converted to seconds (seconds = frame / fps).")

    if step is None:
        step = meta.get("step") or _infer_step([b["frame"] for b in boxes])

    records = build_records(boxes, fps)
    issues = detect_issues(records,
                           stop_frame=meta.get("stop_frame"),
```

```
                           allowed_reasons=allowed_reasons)
    return ConversionResult(records=records, issues=issues, fps=fps, step=step, meta=meta)


def write_canonical_csv(records: List[RallyRecord], out_path: str) -> None:
    """Write records as the canonical per-rally CSV that ``import-local`` reads."""
    import csv
    os.makedirs(os.path.dirname(os.path.abspath(out_path)), exist_ok=True)
    with open(out_path, "w", newline="", encoding="utf-8") as f:
        w = csv.writer(f)
        w.writerow(CANONICAL_COLUMNS)
        for r in sorted(records, key=lambda x: x.rally_number):
            w.writerow([
                r.rally_number,
                _mmss(r.start_time), _mmss(r.end_time),
                f"{r.start_time:.3f}", f"{r.end_time:.3f}",
                r.shots_count if r.shots_count is not None else "",
                r.ending_reason or "",
                r.last_shot_outcome or "",
                r.score_before or "", r.score_after or "", r.notes or "",
            ])


def render_issue_report(result: ConversionResult, source: str) -> str:
    """A human-readable issue report (for review / sending back to the vendor)."""
    lines = [
        f"CVAT rally conversion report ? {source}",
        f"  fps used        : {result.fps:.3f}",
        f"  sampling step   : {result.step} frames (~{result.step / result.fps:.2f}s)",
        f"  rallies parsed  : {len(result.records)}",
        f"  issues          : {len(result.issues)} ({result.n_blockers} blocker)",
        "",
    ]
    if not result.issues:
        lines.append("No issues detected.")
    else:
        for sev in ("blocker", "high", "medium", "low"):
            group = [i for i in result.issues if i.severity == sev]
            if not group:
                continue
            lines.append(f"[{sev.upper()}]")
            for i in group:
                lines.append(f"  - ({i.code}) {i.message}")
            lines.append("")
    return "\n".join(lines)


# --------------------------------------------------------------------------- #
#  Validation reporting ? used by `curate validate-delivery`.                  #
#  These turn a (records, issues) pair into the artifacts a human reviews:     #
#  a machine summary (JSON), a human report (md), and an annotatable sheet.    #
# --------------------------------------------------------------------------- #

def _parse_secs(value: Optional[str]) -> Optional[float]:
    """Decimal seconds or mm:ss / hh:mm:ss -> float; None if blank/unparseable."""
    s = (value or "").strip()
    if not s:
        return None
    if ":" in s:
        try:
            total = 0.0
            for part in s.split(":"):
                total = total * 60 + float(part)
            return total
```

```
        except ValueError:
            return None
    try:
        return float(s)
    except ValueError:
        return None


def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
    """Load a canonical per-rally CSV into RallyRecords, so a *CSV* delivery (not
    just a CVAT export) can be validated. Decimal start/end preferred, mm:ss
    fallback; rows with no usable start/end are skipped. (frame_* are -1 here, as a
    CSV carries no frame info ? only the frame-derived checks become inapplicable.)
    """
    import csv as _csv
    if not os.path.exists(csv_path):
        raise FileNotFoundError(f"CSV not found: {csv_path}")
    records: List[RallyRecord] = []
    with open(csv_path, newline="", encoding="utf-8") as f:
        reader = _csv.DictReader(f)
        reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or [])]
        for raw in reader:
            row = {(k or "").strip().lower(): v for k, v in raw.items()}
            rn = safe_int(row.get("rally_number"), None)
            if rn is None:
                continue
            start = _parse_secs(row.get("start_time"))
            if start is None:
                start = _parse_secs(row.get("start_mmss"))
            end = _parse_secs(row.get("end_time"))
            if end is None:
                end = _parse_secs(row.get("end_mmss"))
            if start is None or end is None:
                continue
            records.append(RallyRecord(
                rally_number=rn, start_time=start, end_time=end,
                shots_count=safe_int(row.get("shots_count"), None),
                ending_reason=(row.get("ending_reason") or "").strip() or None,
                last_shot_outcome=(row.get("last_shot_outcome") or "").strip() or None,
                score_before=(row.get("score_before") or "").strip() or None,
                score_after=(row.get("score_after") or "").strip() or None,
                notes=(row.get("notes") or "").strip() or None,
                frame_min=-1, frame_max=-1, n_boxes=0,
            ))
    return records


def build_summary(records: List[RallyRecord], issues: List[Issue],
                  source: str = "", fps: Optional[float] = None) -> Dict[str, Any]:
    """Machine-readable validation summary (JSON / CI)."""
    by_sev = {s: [i for i in issues if i.severity == s]
              for s in ("blocker", "high", "medium", "low")}
    flagged = sorted({i.rally_number for i in issues if i.rally_number is not None})
    return {
        "source": source,
        "fps": fps,
        "n_rallies": len(records),
        "n_issues": len(issues),
        "counts_by_severity": {s: len(v) for s, v in by_sev.items()},
        "n_blockers": len(by_sev["blocker"]),
        "passed": len(by_sev["blocker"]) == 0,
        "flagged_rallies": flagged,
        "issues": [{"severity": i.severity, "code": i.code,
                    "rally_number": i.rally_number, "message": i.message} for i in
```

```
issues],
    }


def render_validation_md(records: List[RallyRecord], issues: List[Issue], source: str) -> str:
    """Human validation report: scoreboard + issues by severity + rallies to review."""
    n = len(records)
    by_sev = {s: [i for i in issues if i.severity == s]
              for s in ("blocker", "high", "medium", "low")}
    flagged = sorted({i.rally_number for i in issues if i.rally_number is not None})
    clean = n - len(flagged)
    lines = [
        f"# Delivery validation ? {source}",
        "",
        f"- rallies: **{n}**",
        f"- issues: **{len(issues)}**  (blocker {len(by_sev['blocker'])}, "
        f"high {len(by_sev['high'])}, medium {len(by_sev['medium'])}, low {len(by_sev['low'])})",
        f"- structurally-clean rallies: **{clean}/{n}**"
        + (f" ({100 * clean // n}%)" if n else ""),
        f"- gate: **{'PASS' if not by_sev['blocker'] else 'BLOCKED'}**",
        "",
    ]
    if issues:
        for sev in ("blocker", "high", "medium", "low"):
            grp = by_sev[sev]
            if not grp:
                continue
            lines.append(f"## {sev.upper()}")
            lines += [f"- ({i.code}) {i.message}" for i in grp]
            lines.append("")
    else:
        lines.append("No issues detected. ?\n")
    if flagged:
        lines.append(f"## Rallies to review: {flagged}")
    return "\n".join(lines)


def write_review_csv(records: List[RallyRecord], issues: List[Issue], out_path: str) -> None:
    """Auto-generate the annotatable manual-verification sheet (blank `true_*` cols)."""
    import csv
    flags: Dict[int, List[str]] = {}
    for i in issues:
        if i.rally_number is not None:
            flags.setdefault(i.rally_number, []).append(i.code)
    os.makedirs(os.path.dirname(os.path.abspath(out_path)), exist_ok=True)
    cols = ["rally_number", "start_mmss", "end_mmss", "dur_s", "shots_count",
            "ending_reason", "last_shot_outcome", "auto_flags",
            "VERIFIED(Y/N)", "true_start_mmss", "true_end_mmss",
            "true_ending_reason", "true_last_shot_outcome", "reviewer_comment"]
    with open(out_path, "w", newline="", encoding="utf-8") as f:
        w = csv.writer(f)
        w.writerow(cols)
        for r in sorted(records, key=lambda x: x.rally_number):
            w.writerow([
                r.rally_number, _mmss(r.start_time), _mmss(r.end_time),
                f"{r.duration:.1f}", r.shots_count if r.shots_count is not None else "",
                r.ending_reason or "", r.last_shot_outcome or "",
                ";".join(flags.get(r.rally_number, [])),
                "", "", "", "", "", "",
            ])

```

132. user (2026-06-12T11:43:51.922Z)

```
M       .gitignore
Switched to a new branch 'feat/boundary-grid-snap-gate'
```

133. user (2026-06-12T11:43:51.937Z)

```
"""Tests for the CVAT rally adapter (src/parsers/cvat/rally_cvat.py).

The adapter's whole reason for existing is to (a) recompute rally time from the
box FRAME RANGE at the true fps ? ignoring the vendor's wrong start/end_time
attributes ? and (b) report content defects rather than silently fixing them.
These tests pin both behaviours, plus the end-to-end hand-off to import-local.
"""
import csv as _csv

from src.parsers.cvat.rally_cvat import (
    RallyRecord, parse_cvat_boxes, build_records, detect_issues, convert,
    write_canonical_csv, load_canonical_csv, build_summary, render_validation_md,
    write_review_csv,
)
from src.cli.curate import SportsDataCLI


# --- helpers ---------------------------------------------------------------

def _img_box(frame, attrs):
    rows = "".join(f'<attribute name="{k}">{v}</attribute>' for k, v in attrs.items())
    return (f'<image id="{frame}" name="frame_{frame:06d}" width="1920" height="1080">'
            f'<box label="rally" xtl="1" ytl="1" xbr="2" ybr="2">{rows}</box></image>')


def _write_cvat(path, images, step=30, stop_frame=1000):
    body = "".join(images)
    xml = (
        '<?xml version="1.0" encoding="utf-8"?><annotations><version>1.1</version>'
        f'<meta><job><stop_frame>{stop_frame}</stop_frame>'
        f'<frame_filter>step={step}</frame_filter></job></meta>'
        f'{body}</annotations>'
    )
    path.write_text(xml, encoding="utf-8")
    return str(path)


def _rec(rn, start, end, shots=5, reason="winner", fmin=None, fmax=None, n=3,
outcome=None):
    return RallyRecord(
        rally_number=rn, start_time=start, end_time=end, shots_count=shots,
        ending_reason=reason, last_shot_outcome=outcome,
        score_before=None, score_after=None, notes=None,
        frame_min=fmin if fmin is not None else int(start * 60),
        frame_max=fmax if fmax is not None else int(end * 60), n_boxes=n)


# --- the core fix: time from frames, vendor times ignored ------------------

def test_time_recomputed_from_frames_not_attributes(tmp_path):
    # rally 1 spans source frames 60..120; the vendor's start/end attributes are
    # deliberately wrong (the frame/900 bug). At 60 fps the truth is 1.0..2.0s.
    xml = _write_cvat(tmp_path / "j.xml", [
        _img_box(60,  {"rally_number": "1.0", "start_time": "0.2", "end_time": "1.4",
                       "shots_count": "6.0", "ending_reason": "winner",
                       "score_before (A-B)": "0-0", "notes": "clean"}),
        _img_box(90,  {"rally_number": "1.0", "start_time": "0.2", "end_time": "1.4"}),
        _img_box(120, {"rally_number": "1.0", "start_time": "0.2", "end_time": "1.4"}),
```

```
        ])
        res = convert(xml, fps=60.0)
        assert len(res.records) == 1
```

134. user (2026-06-12T11:44:01.913Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

135. user (2026-06-12T11:44:10.853Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

136. user (2026-06-12T11:44:22.348Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

137. user (2026-06-12T11:44:24.973Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

138. user (2026-06-12T11:44:44.107Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md has been updated successfully. (file state is current in your context ? no need to Read it back)

139. user (2026-06-12T11:44:49.649Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md has been updated successfully. (file state is current in your context ? no need to Read it back)

140. user (2026-06-12T11:44:55.817Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md has been updated successfully. (file state is current in your context ? no need to Read it back)

141. user (2026-06-12T11:45:02.390Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md has been updated successfully. (file state is current in your context ? no need to Read it back)

142. assistant (2026-06-12T11:45:08.064Z)

Now the validation-table doc row and the tests for the grid-snap gate:

143. user (2026-06-12T11:45:11.921Z)

The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

144. user (2026-06-12T11:45:31.281Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py has been updated successfully. (file state is current in your context ? no need to Read it back)

145. assistant (2026-06-12T11:45:41.978Z)

Fixing a duplicated line I left in one test:

146. user (2026-06-12T11:45:43.479Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

147. user (2026-06-12T11:45:55.318Z)

```
Exit code 1
tests\test_cvat_parser.py:65: AssertionError
=========================== short test summary info ===========================
FAILED tests/test_cvat_parser.py::test_grid_snap_flags_30_frame_grid_as_blocker
1 failed, 174 passed in 2.82s
```

148. user (2026-06-12T11:46:08.622Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

149. user (2026-06-12T11:46:23.424Z)

```
........................................................................ [ 82%]
...............................                                          [100%]
175 passed in 2.59s
```

150. user (2026-06-12T11:46:37.087Z)

```
<persisted-output>
Output too large (72KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-
Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\tool-
results\ble9g1oqw.txt

Preview (first 2KB):
FullName
--------
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000000.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000030.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000060.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000090.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000120.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000150.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000180.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000210.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000240.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000270.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000300.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000330.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000360.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000390.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000420.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000450.xml
```

```
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000480.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000510.xml
C:\Users\avidu\Projects\Annotation Setup\6 June
2026\badminton\badminton\Annotations\frame_000540.xml
...
</persisted-output>
```

151. user (2026-06-12T11:46:44.143Z)

```
FullName
--------
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\pickleball\pickle ball (1).zip
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\pickleball\pickle ball (2).zip
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\pickleball\pickle ball.zip
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\tabletennis\table tennis (1).zip
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\tabletennis\table tennis (2).zip
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\tabletennis\table tennis.zip
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\badminton\badminton (2)\annotations.xml
```

152. assistant (2026-06-12T11:46:53.738Z)

The pilot zips are there (3 export formats each ? only the CVAT-XML one will parse). Running the new gate over them:

153. user (2026-06-12T11:47:02.192Z)

```
Exit code 1
=== pickle ball (1).zip: GATE CAUGHT boundary_grid_snap (exit 2) ===
- gate: **BLOCKED**
- (boundary_grid_snap) All 12 rally boundaries sit on a 30-frame grid (~0.50s at 59.94 fps) ?
first/last boxes were snapped to sampled frames, so boundaries structurally cannot meet the
±0.3s bar (Brief §4). The first/last box of every rally must be placed on the exact serve-
contact / dead frame (frame-step to it); in-between boxes may stay sparse.
  GATE: BLOCKED ? 1 blocker(s) must be resolved.

pickle ball (2).zip: not the CVAT-XML export (skipped)
pickle ball.zip: not the CVAT-XML export (skipped)
=== table tennis (1).zip: GATE CAUGHT boundary_grid_snap (exit 2) ===
- gate: **BLOCKED**
- (boundary_grid_snap) All 12 rally boundaries sit on a 30-frame grid (~0.50s at 59.94 fps) ?
first/last boxes were snapped to sampled frames, so boundaries structurally cannot meet the
±0.3s bar (Brief §4). The first/last box of every rally must be placed on the exact serve-
contact / dead frame (frame-step to it); in-between boxes may stay sparse.
  GATE: BLOCKED ? 1 blocker(s) must be resolved.

table tennis (2).zip: not the CVAT-XML export (skipped)
table tennis.zip: not the CVAT-XML export (skipped)
```

154. user (2026-06-12T11:47:31.389Z)

```
Exit code 1
[feat/boundary-grid-snap-gate 3d2845b] feat(qa): boundary_grid_snap blocker + Phase-B brief
revision (ADR-002 follow-ups)
 4 files changed, 151 insertions(+), 12 deletions(-)
remote:
remote: Create a pull request for 'feat/boundary-grid-snap-gate' on GitHub by visiting:
remote:      https://github.com/avidullu/sports-data-collector/pull/new/feat/boundary-grid-snap-
gate
remote:
branch 'feat/boundary-grid-snap-gate' set up to track 'origin/feat/boundary-grid-snap-gate'.
To https://github.com/avidullu/sports-data-collector.git
 * [new branch]      feat/boundary-grid-snap-gate -> feat/boundary-grid-snap-gate
```

```
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/23
no checks reported on the 'feat/boundary-grid-snap-gate' branch
```

155. user (2026-06-12T11:48:04.503Z)

```
test    pass    15s     https://github.com/avidullu/sports-data-
collector/actions/runs/27413707946/job/81021024556
test    pass    15s     https://github.com/avidullu/sports-data-
collector/actions/runs/27413707946/job/81021024556
test: SUCCESS
```

156. user (2026-06-12T11:48:19.562Z)

```
From https://github.com/avidullu/sports-data-collector
 * branch            master     -> FETCH_HEAD
   a068fd1..4923e42  master     -> origin/master
Updating a068fd1..4923e42
Fast-forward
 docs/INGESTION_PIPELINE.md     |  1 +
 docs/annotation/ROORA_BRIEF.md | 27 ++++++++++++++---
 src/parsers/cvat/rally_cvat.py | 62 ++++++++++++++++++++++++++++++-----
 tests/test_cvat_parser.py      | 73 +++++++++++++++++++++++++++++++++++++++++
 4 files changed, 151 insertions(+), 12 deletions(-)
M       .gitignore
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
4923e42 feat(qa): boundary_grid_snap blocker + Phase-B brief revision (ADR-002 follow-ups) (#23)
M       .gitignore
Switched to a new branch 'feat/prep-annotation-batch'
```

157. user (2026-06-12T11:49:10.110Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

158. user (2026-06-12T11:49:27.107Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

159. user (2026-06-12T11:49:28.689Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

160. user (2026-06-12T11:49:32.609Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

161. user (2026-06-12T11:49:59.262Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_proxy.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

162. user (2026-06-12T11:50:08.805Z)

```
........................................................................ [ 80%]
....................................                                     [100%]
180 passed in 3.10s
```

163. assistant (2026-06-12T11:50:13.745Z)

180 green. Live batch smoke with real ffmpeg ? a real tree with two synthesized videos in subfolders:

164. user (2026-06-12T11:50:21.108Z)

```
tree built

=== Batch proxy summary ===
+----------+------------------+----------+-------------------------------+
| Folder   | Video            | Status   | Detail                        |
+==========+==================+==========+===============================+
| Court 1  | Morning Game.mp4 | ok       | Court 1\Morning Game_proxy.mp4 |
+----------+------------------+----------+-------------------------------+
| Court 2  | Evening Game.mp4 | ok       | Court 2\Evening Game_proxy.mp4 |
+----------+------------------+----------+-------------------------------+

Proxied : 2   Already-have : 0   Failed : 0
Manifest: C:\Users\avidu\AppData\Local\Temp\batch_smoke\manifest.csv
Share the 'C:\Users\avidu\AppData\Local\Temp\batch_smoke\out' tree with the vendor ? it contains
ONLY proxies; every other stage runs on the originals.



FullName
--------
C:\Users\avidu\AppData\Local\Temp\batch_smoke\out\Court 1\Morning Game_proxy.mp4
C:\Users\avidu\AppData\Local\Temp\batch_smoke\out\Court 2\Evening Game_proxy.mp4
```

165. user (2026-06-12T11:51:25.709Z)

```
[feat/prep-annotation-batch 1ab4390] feat(curate): prep-annotation-batch - mirror a directory
tree into proxies
 3 files changed, 224 insertions(+)
remote:
remote: Create a pull request for 'feat/prep-annotation-batch' on GitHub by visiting:
remote:      https://github.com/avidullu/sports-data-collector/pull/new/feat/prep-annotation-
batch
remote:
branch 'feat/prep-annotation-batch' set up to track 'origin/feat/prep-annotation-batch'.
To https://github.com/avidullu/sports-data-collector.git
 * [new branch]      feat/prep-annotation-batch -> feat/prep-annotation-batch
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/24
test    pass    22s     https://github.com/avidullu/sports-data-
collector/actions/runs/27413857635/job/81021533831
test    pass    22s     https://github.com/avidullu/sports-data-
collector/actions/runs/27413857635/job/81021533831
```

166. user (2026-06-12T11:51:41.962Z)

```
From https://github.com/avidullu/sports-data-collector
 * branch            master     -> FETCH_HEAD
   4923e42..7ad8160  master     -> origin/master
Updating 4923e42..7ad8160
Fast-forward
 docs/INGESTION_PIPELINE.md |   3 ++
 src/cli/curate.py          | 107 ++++++++++++++++++++++++++++++++++++++++++
 tests/test_proxy.py        | 114 +++++++++++++++++++++++++++++++++++++++++++++
 3 files changed, 224 insertions(+)
M       .gitignore
Your branch is up to date with 'origin/master'.
Already on 'master'
```

Already up to date.
7ad8160 feat(curate): prep-annotation-batch - mirror a directory tree into proxies (#24)
M   .gitignore
Switched to a new branch 'docs/roora-runbook'

167. assistant (2026-06-12T11:52:43.634Z)

Roora annotation runbook ? send ? receive ? golden labels, end to end

The **operator's** guide: every command and checklist needed to take a folder of
match videos through Roora and back into trusted golden labels, with nothing in
between left to memory. Vendor-facing docs stay separate: the schema/spec is
[`ROORA_BRIEF.md`](./ROORA_BRIEF.md), the illustrated how-to-annotate is
[`ANNOTATION_GUIDE.md`](./ANNOTATION_GUIDE.md), and the design rationale is
ADR-002 in [`../DECISIONS.md`](../DECISIONS.md).

The one invariant everything below protects:

> The vendor annotates a **proxy** (light, frame-identical re-encode).
> Every other stage ? validation, import, export, the indexer ? runs on the
> **original** file. The proxy manifest (`data/annotation_proxy_manifest.csv`)
> is the mapping of record between the two; commit it.

```
SEND     originals ??prep-annotation-batch??? proxies tree ??Drive??? Roora (CVAT, frame step 1)
RECEIVE  CVAT zip ??validate-delivery??? gate+review ??import-local??? DB ??export??? golden GT
                      (originals from here on)
```

---

0. One-time prerequisites

- This repo cloned, Python ?3.9, `pip install -e ".[dev]"` (tests) ? ffmpeg/ffprobe
  on PATH (`winget install Gyan.FFmpeg` on Windows).
- NDA signed (template [`NDA_TEMPLATE.md`](./NDA_TEMPLATE.md)) **before any footage moves**.
- Roora has the current `ROORA_BRIEF.md` + `ANNOTATION_GUIDE.md` (note which version
  you sent ? ask them to confirm the version in their delivery email).
- A Google Drive folder pair: `<batch>/to_vendor/` (we upload) and `<batch>/from_vendor/`
  (they upload).

---

1. SEND a batch

1.1 Lay out the originals

One folder per match, well-named videos inside (the filename becomes the
`video_id`: lowercased, spaces?underscores, text before the first dot):

```
D:\batches\2026-07-phaseB\
??? Court 1\Mahadevpura Singles.mp4      ? video_id: mahadevpura_singles
??? Court 2\Kushagra Doubles.mp4         ? video_id: kushagra_doubles
??? ...
```

?? Filenames must be **unique across the whole batch** ? duplicate-derived
`video_id`s are refused by the manifest collision guard.

1.2 Generate the proxy tree

```powershell
```

~~~~
python -m src.cli.curate prep-annotation-batch `
    --input-dir "D:\batches\2026-07-phaseB" `
    --out-dir   "D:\batches\2026-07-phaseB_proxies"
```

What this does per video: probes the source, refuses variable-frame-rate files
(`--allow-vfr` to override, after thinking), re-encodes ?1080p with dense
keyframes and audio kept, **verifies** the proxy has the source's exact frame
count/fps/duration (a failed proxy is deleted, never shared), and appends a
provenance row (source + proxy MD5s, fps, frame count) to
`data/annotation_proxy_manifest.csv`. Re-running skips finished proxies, so a
failed batch resumes where it stopped. A single video: `prep-annotation-proxy
--video-path <file>`.

**Checklist before upload:**
- [ ] Batch summary shows `Failed : 0` (a FAIL row names the video and reason).
- [ ] `git add data/annotation_proxy_manifest.csv` + commit ? it is the
      proxy?original mapping of record.
~~~~

1.3 Hand off

~~~~
- [ ] Upload the **`*_proxies` tree only** to `to_vendor/` ? never the originals.
- [ ] Fill the intake manifest
([`INTAKE_MANIFEST_TEMPLATE.csv`](./INTAKE_MANIFEST_TEMPLATE.csv)):
      `video_id` and `fps` come straight from the proxy-manifest rows; add the
      Drive link per video.
- [ ] Email Roora: intake manifest link, brief/guide version, and the two
      non-negotiables in one line ? **CVAT task at frame step 1** on the files
      provided, **first/last box of every rally on the exact contact/dead frame**
      (Brief §7). Ask for the **CVAT for video 1.1 XML** export format back.

---
~~~~

2. RECEIVE a delivery

2.1 Store it

~~~~
Download their export zip(s) to the batch's dated folder (convention:
`C:\Users\avidu\Projects\Annotation Setup\<date>\<sport>\`). The CVAT-XML zip is
the one we ingest ? `validate-delivery` reads it directly (zipped or unzipped);
COCO / Pascal-VOC siblings can be ignored.
~~~~

2.2 Gate it (per video)

~~~~
```powershell
python -m src.cli.curate validate-delivery `
    --input "...\from_vendor\court1.zip" `
    --video-path "D:\batches\2026-07-phaseB\Court 1\Mahadevpura Singles.mp4" `
    --video-id mahadevpura_singles `
    --out-dir data\validation
```

`--video-path` is the **ORIGINAL** (fps via ffprobe; the proxy's fps is identical
by construction ? `--fps <value>` from the proxy manifest works too). Exit code 0
= gate PASS; 2 = BLOCKED.

| outcome | meaning | action |
|---|---|---|
| `boundary_grid_snap` blocker | every boundary sits on a sampling grid (the pickleball/TT pilot
defect) ? rater never frame-stepped to the true boundary | return the whole file for rework;
quote the issue message (it tells them exactly what to fix) |
| `overlap` / `zero_duration` blocker | structurally invalid rows | return for rework (or fix
obvious single-box strays yourself in the canonical CSV) |
| gate PASS + flagged rallies | importable, but specific rallies need eyes | go to 2.3 |
~~~~

| gate PASS, no flags | clean | go to 2.4 |

To send feedback: attach `data\validation\<video_id>_report.md` (human-readable)
? the `.summary.md` observability file is the vendor-friendly variant without
internal codes.

2.3 Human review (the part that makes labels *golden*)

Open `data\validation\<video_id>_review.csv` next to the **original** video in a
frame-steppable player (VLC: `E` = frame step). For each row with `auto_flags`,
check the boundary/fields, fill the `true_*` columns and `VERIFIED(Y/N)`. Spot-check
~20% of unflagged rallies too (Brief §4). Correct the canonical CSV accordingly
(`convert-cvat` writes it; or edit the delivered CSV).

2.4 Import + export (originals, always)

```powershell
python -m src.cli.curate import-local --sport badminton `
    --video-path "D:\batches\2026-07-phaseB\Court 1\Mahadevpura Singles.mp4" `
    --annotations-path data\validation\mahadevpura_singles_canonical.csv `
    --video-id mahadevpura_singles --fps 59.94 --license Proprietary --normalize

python -m src.cli.curate export --out-dir data\eval_export
```

`import-local` refuses to clobber an existing video's annotations (`--force` only
after you understand why it fired). `export` writes the indexer-ready
`start,end` CSVs + `manifest.json` ? the indexer then keys each video by the
original file's MD5, which is why nothing downstream may ever point at a proxy.

---

3. Troubleshooting

symptom	cause	fix
`looks variable-frame-rate` refusal at prep	screen-capture/odd source; `frame/fps` time math is approximate on VFR	prefer a CFR re-export of the source; `--allow-vfr` only if you accept the imprecision
`manifest already maps video_id ... to a DIFFERENT source`	two batch videos share a filename stem	rename the file (or `--video-id`) ? don't reach for `--force`
`could not append to manifest` after a successful encode	the CSV is open in Excel	the proxy is verified and kept; close Excel, append the printed row (or re-run with `--force`)
proxy FAILED parity and was deleted	encode changed the timeline (or genuinely odd source)	re-run; if it persists, that source needs investigation ? do NOT hand-share an unverified file
vendor sent COCO/VOC only	wrong export format	ask for "CVAT for video 1.1" XML; the data is the same job
validate-delivery `ZERO rallies`	wrong job exported or `rally_number` attribute renamed	bounce back with the error text ? it is self-explanatory

4. Why it works this way (30 seconds)

The pilot failed on boundaries because 4K/59.94 originals are too heavy to
frame-step in CVAT, so the vendor sampled every 30th frame and the boundaries
snapped to a ~0.5 s grid ? structurally outside the ±0.3 s bar. The proxy makes
frame-step-1 cheap; the brief now requires exact boundary frames; and
`validate-delivery` auto-blocks any delivery whose boundaries still sit on a
grid. Full decision record: ADR-002 in [`../DECISIONS.md`](../DECISIONS.md);
pipeline FAQ: [`../INGESTION_PIPELINE.md`](../INGESTION_PIPELINE.md).

168. user (2026-06-12T11:52:43.656Z)

File created successfully at: C:\Users\avidu\Projects\sports-data-

```
collector\docs\annotation\ROORA_RUNBOOK.md (file state is current in your context ? no need to
Read it back)
```

169. user (2026-06-12T11:52:54.407Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

170. user (2026-06-12T11:53:42.235Z)

```
....................................                                    [100%]
180 passed in 2.90s
warning: in the working copy of 'docs/annotation/ROORA_RUNBOOK.md', LF will be replaced by CRLF
the next time Git touches it
[docs/roora-runbook ed37a58] docs(annotation): ROORA_RUNBOOK - the operator e2e (send -> receive
-> golden labels)
 2 files changed, 170 insertions(+)
 create mode 100644 docs/annotation/ROORA_RUNBOOK.md
remote:
remote: Create a pull request for 'docs/roora-runbook' on GitHub by visiting:
remote:      https://github.com/avidullu/sports-data-collector/pull/new/docs/roora-runbook
remote:
branch 'docs/roora-runbook' set up to track 'origin/docs/roora-runbook'.
To https://github.com/avidullu/sports-data-collector.git
 * [new branch]      docs/roora-runbook -> docs/roora-runbook
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/25
test    pass    15s     https://github.com/avidullu/sports-data-
collector/actions/runs/27413975852/job/81021923424
test    pass    15s     https://github.com/avidullu/sports-data-
collector/actions/runs/27413975852/job/81021923424
```

171. user (2026-06-12T11:53:56.770Z)

```
From https://github.com/avidullu/sports-data-collector
 * branch            master     -> FETCH_HEAD
   7ad8160..ca61405  master     -> origin/master
Updating 7ad8160..ca61405
Fast-forward
 docs/INGESTION_PIPELINE.md       |   2 +
 docs/annotation/ROORA_RUNBOOK.md | 168 +++++++++++++++++++++++++++++++++++++++
 2 files changed, 170 insertions(+)
 create mode 100644 docs/annotation/ROORA_RUNBOOK.md
M       .gitignore
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
ca61405 docs(annotation): ROORA_RUNBOOK - the operator e2e (send -> receive -> golden labels)
(#25)
7ad8160 feat(curate): prep-annotation-batch - mirror a directory tree into proxies (#24)
4923e42 feat(qa): boundary_grid_snap blocker + Phase-B brief revision (ADR-002 follow-ups) (#23)
a068fd1 feat(curate): prep-annotation-proxy ? frame-identical vendor handoff re-encode (ADR-002)
(#22)
```

master...origin/master

```
 M .gitignore
```

172. user (2026-06-12T11:54:15.004Z)

```
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
```

173. assistant (2026-06-12T11:54:35.720Z)

All three deliverables are built, merged on green CI, and the collector is back on a clean master (`ca61405`, no open PRs).

What shipped

[#23](https://github.com/avidullu/sports-data-collector/pull/23) ? the `boundary_grid_snap` gate + Phase-B brief revision. `validate-delivery` and `convert-cvat` now detect when every rally boundary sits on a common frame-sampling grid coarser than the ±0.3s bar and **block the delivery** with a message telling the vendor exactly what to fix. The proof point: I replayed it against the *actual* June-6 pickleball and table-tennis delivery zips, and both gate-blocked with "All 12 rally boundaries sit on a 30-frame grid (~0.50s at 59.94 fps)" ? the defect that took manual frame-by-frame forensics last time is now caught automatically at intake. The brief's §7 is now a *required* capture method (CVAT at frame step 1, first/last box on the exact contact/dead frame, sparse interior boxes still fine), §4 warns the grid defect is auto-detected, and §2 adds the skip-rallies-in-progress-at-file-edges convention from the pilot review.

[#24](https://github.com/avidullu/sports-data-collector/pull/24) ? `prep-annotation-batch`. Exactly the shape you described: point it at a root directory whose subdirectories each hold a well-named video, and it mirrors the same folder layout into an output tree containing **only the proxies** ? ready to share. Each video runs through the full hardened single-video flow (parity verification, VFR gate, delete-on-failure, provenance manifest), one video's failure doesn't stop the rest, and re-runs skip already-done proxies. Verified live: a two-court tree went through real ffmpeg into a mirrored proxies-only tree with a clean summary table.

```
python -m src.cli.curate prep-annotation-batch --input-dir "D:\batch" --out-dir "D:\batch_proxies"
```

[#25](https://github.com/avidullu/sports-data-collector/pull/25) ? [ROORA_RUNBOOK.md](C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_RUNBOOK.md). The single operator doc: SEND (folder conventions, batch command, commit-the-manifest checklist, intake manifest + handoff email with the two non-negotiables), RECEIVE (gate decision table including what to send back when `boundary_grid_snap` fires, the review-sheet pass, import/export against the originals), plus a troubleshooting table covering every guard the tooling can fire. A human can replicate the whole loop cold from that one file.

Suite is at **180 green** and the full chain is now closed: ADR-002 (decision) ? #22 (proxy tool) ? #23 (enforcement + vendor contract) ? #24 (batch scale) ? #25 (human runbook).

One small flag for your next send: the local guideline docs are still marked DRAFT v2, but you'd mentioned a "v7" possibly living in the email thread ? worth confirming which version Roora gets referenced to before the next batch goes out.

174. user (2026-06-12T12:35:50.771Z)

https://docs.google.com/document/d/1NvAXE0q7t4Z0QddLkEfnta0DU982yDZaqkGNPMe8EY0/edit?tab=t.0 -- is the doc which is a superset of the v7 that I mention. So I think you cana resolve that v7 anomly! It is captured already

175. assistant (2026-06-12T12:36:19.389Z)

Let me pull that doc via the Drive connector to confirm what it covers, then record the resolution durably.

176. user (2026-06-12T12:36:29.529Z)

{"fileContent":"ROORA ANNOTATION GUIDELINES ? RALLY GOLDEN SET\n\nBrief (vendor spec) + Illustrated Guide (rater how-to), in one editable doc.\n\n \n\nPROVENANCE: generated from the source-of-truth repo (sports-data-collector, master ca61405,\n\n2026-06-12). This is the CURRENT finalized version and SUPERSEDES the older Drive copies\n\n(v3 / v5 / v6 / v7). It incorporates the fixes from the 2026-06-10 Roora pilot review that the\n\n\"(v7)\" Drive doc (created

2026-06-04) does NOT have: the last_shot_outcome column,\n\nforced_error/unforced_error in ending_reason, the §7 required boundary-frame method, the §4\n\ngrid-snap auto-reject, and the \"rally already in progress at a file edge gets no row\" exception.\n\n \n\nVERSION-LABEL NOTE: internally the repo tracks this as \"DRAFT v2\"; the email thread with Roora\n\nreferred to \"v7\". Those numbers don't match each other or the Drive copies ? pick ONE label\n\nbefore sending to Roora and use it consistently. (This file is content-current regardless of label.)\n\n \n\nNOTE (plain text): **bold** and TODO(avidullu) markers show as literal asterisks here ? use\n\nFind (Ctrl/Cmd+F) for \"TODO(avidullu)\" to locate the items you must fill before sending.\n\n \n\n\\#\n\n\\#\n\nPART A ? ROORA ANNOTATION BRIEF (vendor-facing spec)\n\n\\#\n\n\\#\n\n \n\nProject: racket-sport rally segmentation golden / regression dataset\n\nSports in scope: Badminton, Pickleball, Table Tennis (each in singles AND doubles)\n\n \n\nThis brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV that\n\nimports directly via \\`python -m src.cli.curate import-local\\`.\n\n
\n\n\\===\n\n0\\. WHAT WE NEED IN ONE
SENTENCE\n\n\\===\n\nFor each provided match video, produce ONE CSV file listing EVERY rally with its start time,\n\nend time, number of shots, and how it ended ? EXCLUDING all dead time.\n\n
\n\n\\===\n\n1\\. DELIVERABLE FORMAT
(CANONICAL)\n\n\\===\n\nOne CSV per video (UTF-8), named exactly \\<video_id\\>.csv (we supply the video_id).\n\nOne row per rally, chronological, header row required.\n\n \n\nCOLUMNS:\n\n rally_number   integer, 1-based, chronological. No gaps, no repeats.\n\n start_mmss   rater-entered time as m:ss.mmm (e.g. 1:12.40) ? serve contact.\n\n end_mmss   rater-entered time as m:ss.mmm ? ball/shuttle dead.\n\n start_time   decimal SECONDS (e.g. 72.40). AUTO-FILLED from start_mmss.\n\n end_time   decimal SECONDS. AUTO-FILLED from end_mmss.\n\n shots_count   integer. Total racket/paddle/bat contacts (serve = shot 1).\n\n ending_reason   one of: winner | forced_error | unforced_error | service_fault | let | other\n\n   (WHY it ended).\n\n last_shot_outcome one of: in | net | out | n_a (WHERE the last shot landed; n_a for\n\n   service_fault / let).\n\n score_before   optional, \"A-B\". Blank if not confidently readable.\n\n score_after   optional, \"A-B\".\n\n notes   optional free text. REQUIRED when ending_reason = other.\n\n \n\nTIMES (mm:ss -\\> decimal seconds): raters type only start_mmss / end_mmss. The template Google\n\nSheet fills start_time / end_time with this formula (shown for start_time, row 2):\n\n =IF(B2=\"\",\"\", IFERROR(VALUE(LEFT(B2,FIND(\":\",B2)-1))*60 + VALUE(MID(B2,FIND(\":\",B2)+1,20)), VALUE(B2)))\n\nOn \"Download as CSV\" the computed decimal seconds are exported. (If your tool already shows\n\ndecimal seconds, type those into start_time / end_time and leave the mm:ss columns blank.)\n\n \n\nEXAMPLE (badminton):\n\n rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot_outcome,score_before,score_after,notes\n\n 1,0:12.50,0:25.00,12.50,25.00,9,winner,in,0-0,1-0,\n\n 2,0:45.00,0:47.30,45.00,47.30,1,service_fault,n_a,1-0,1-1,serve into net\n\n 3,1:01.20,1:18.85,61.20,78.85,14,unforced_error,out,1-1,2-1,final clear sailed long (no pressure)\n\n \n\nCRITICAL FORMAT RULES:\n\n - start_time / end_time that reach us MUST be decimal seconds (the formula guarantees this).\n\n - Times are relative to the START OF THE FILE WE PROVIDE. Do NOT trim, clip, or re-encode the\n\n   video ? our pipeline identifies a video by its file checksum. (The file we share may be an\n\n   ANNOTATION-OPTIMIZED COPY of the original recording ? lighter and faster to frame-step, but\n\n   frame-for-frame identical to it. Treat it exactly like the original: annotate it as-is.)\n\n - end_time \\> start_time for every row.\n\n - No overlaps: end_time\\[N\\] \\<= start_time\\[N+1\\].\n\n - Everything between one rally's end_time and the next rally's start_time is DEAD TIME and gets\n\n   NO row.\n\n
\n\n\\===\n\n2\\. THE THREE NON-NEGOTIABLE RULES (full detail + pictures in the
Guide)\n\n\\===\n\n1\\. IGNORE

ALL DEAD TIME. Only label active play.\n\n2\\. INCLUDE EVERY RALLY ? EXCLUDE NONE. Every served point, including service faults, 1-2 shot\n\n   rallies, and lets/replays. A missed rally is the worst error in this project.\n\n   EXCEPTION: warm-up / knock-up is NOT a rally ? pre-match and between-games warm-up looks like\n\n   rallying but isn't scored play, so don't label it. Tell it apart by: no real serve;\n\n   cooperative (not competitive) hitting; the score not progressing; often several shuttles/balls\n\n   in use (full how-to in the Guide, Part 2). When unsure if the match has started, flag it.\n\n   SECOND EXCEPTION: a rally already in progress when the file starts (or still in progress when\n\n   it ends) gets NO row ? its true start (or end) is outside the footage and must never be\n\n   guessed or extrapolated. Note it in the delivery instead.\n\n3\\. A SHOT = ONE CONTACT with the racket/paddle/bat. Serve = shot 1. Bounces and net-clips are not shots.\n\n
\n\n\\===\n\n3\\. SCOPE & PHASING\n\n\\===\n\nPhase A ? Pilot (this engagement): 3 videos (1 badminton + 1 pickleball + 1 table tennis), each\n\n \\< 3 min with \\~10-12 rallies. This is the paid calibration round; we review jointly and lock the\n\n Guide before scale-up.\n\nPhase B ? Scale (after a successful pilot): \\~12-15 videos PER SPORT (\\~36-45 total), each \\~30 min.\n\n Same format, rules, and template.\n\nDoubles are included across the program. Rally rules are identical to singles; the badminton- and\n\n pickleball-specific doubles notes are in the Guide.\n\n
\n\n\\===\n\n4\\. QUALITY BAR (ACCEPTANCE CRITERIA)\n\n\\===\n\nAutomated validation (no overlaps, positive durations, schema-valid) runs on every file, then a\n\n\\~20% human spot-check. A file is returned for rework (in scope) if it misses:\n\n - Completeness: every rally present; zero missed rallies; zero dead-time rows.\n\n - Boundaries: start_time / end_time within +/- 0.3 s of true serve-contact / landing.\n\n   \\!\\! Boundaries that all sit on a frame-sampling grid (e.g. every 30th frame) STRUCTURALLY FAIL\n\n this bar and are detected automatically ? the whole file is returned for rework. See §7 for\n\n the required method.\n\n - shots_count: exact for rallies \\<= 15 shots; +/- 1 allowed for longer/faster rallies.\n\n - ending_reason: one of the six allowed values, per the Guide's decision tree.\n\n - Format: decimal seconds present, valid columns, non-overlapping, chronological.\n\n
\n\n\\===\n\n5\\. PER-SPORT QUICK REFERENCE (full version + examples in the Guide)\n\n\\===\n\nBadminton (singles & doubles) ? start: racket-shuttle serve contact · end: shuttle floors / fault\n\n · shot: each racket-shuttle contact.\n\nPickleball (singles & doubles) ? start: paddle-ball serve contact · end: double bounce / out / net\n\n / fault · shot: each paddle-ball contact (floor bounces are NOT shots; two-bounce rule is normal play).\n\nTable tennis ? start: racket-ball serve strike (not the toss) · end: failed legal return · shot:\n\n each racket-ball contact (table bounces are NOT shots). Net-cord serve = let.\n\n
\n\n\\===\n\n6\\. ending_reason VALUES (decision tree in the Guide)\n\n\\===\n\nThis vocabulary attributes WHY the point ended; it is shared across all net-separated sports\n\n(badminton, pickleball, table tennis, tennis). Record WHERE the last shot physically landed in the\n\nseparate last_shot_outcome column (in | net | out | n_a) ? e.g. a forced error that flew long is\n\nending_reason=forced_error, last_shot_outcome=out.\n\n \n\n winner   last shot landed IN and the opponent made no legal contact (a clean winner).\n\n forced_error   the losing player erred (netted / out / missed) UNDER PRESSURE ? stretched,\n\n   wrong-footed, or rushed by the opponent's strong shot.\n\n unforced_error the losing player erred on a ROUTINE, UNPRESSURED ball they had time and\n\n position to make.\n\n service_fault   the serve itself failed / was illegal.\n\n let   the point was replayed.\n\n other   anything else / cannot tell. MUST add a note.\n\n
\n\n\\===\n\n7\\. CAPTURE METHOD (REQUIRED FOR BOUNDARY FRAMES)\n\n\\===\n\nUse any tool that shows current time (mm:ss.mmm or seconds, or frame + known fps) and can step one\n\nframe forward/back: a frame-steppable player (VLC frame-step) or a timeline tool (Label Studio /\n\nCVAT). The deliverable is the CSV in the schema above ? or, if you work in CVAT, the CVAT export\n\nas-is (we recompute each rally's time from its box frame range; seconds = frame_number / fps, fps\n\nprovided per video).\n\n \n\nBOUNDARY PLACEMENT IS THE NON-NEGOTIABLE PART:\n\n - Set up the CVAT task at FRAME STEP 1 on the file we provide (it is encoded to make this fast ?\n\n   frame-stepping and back-scrubbing stay snappy).\n\n - The FIRST box of every rally goes on the EXACT serve-contact frame; the LAST box on the EXACT\n\n shuttle/ball-dead frame. Frame-step to find them.\n\n - Boxes IN BETWEEN may stay sparse (e.g.

every 30th frame) ? only the two boundary frames must\n\n   be exact.\n\n - Self-check before submitting: every start/end within +/- 0.3 s of the true frame (Guide).\n\n
\n\n\\==\n\n8\\. LOGISTICS & CONTACTS\n\n\\==\n\n - **TODO(avidullu): paste the Google Drive folder link** where source videos are shared with Roora.\n\n - Per-video fps + video_id: provided in the intake manifest (template INTAKE_MANIFEST_TEMPLATE.csv).\n\n   **TODO(avidullu): fill the manifest (one row per video) and share its link.**\n\n - **TODO(avidullu): paste the Drive folder link** where Roora uploads completed \\<video_id\\>.csv files.\n\n - Blank annotation template Sheet (mm:ss -\\> seconds formula from §1 pre-filled):\n\n   **TODO(avidullu): create it and paste the link.**\n\n - **TODO(avidullu): add the escalation channel** (email / WhatsApp group) for edge-case questions.\n\n - NDA before any footage is shared ? template NDA_TEMPLATE.md.\n\n   **TODO(avidullu): have counsel review, fill the blanks, and have Roora sign.**\n\n - Roora contact on file: info@rooragrp.com (HSR Layout, Bangalore).\n\n
\n\n\\==\n\n9\\. PRICING REQUEST (please
quote)\n\n\\==\n\n 1. Price per rally and per minute of footage.\n\n 2. Pilot price (3 short videos, \\~30-36 rallies) as a paid calibration round.\n\n 3. Indicative scale price for \\~36-45 longer videos.\n\n 4. Turnaround for the pilot and throughput (rallies/day) at scale.\n\n 5. QA process and rework policy.\n\n 6. Whether annotators have racket-sport familiarity (helpful, not required).\n\n
7. Confirmation you can deliver the CSV schema above.\n\n \n\n\\#\n\n\\#\n\nPART B ? ILLUSTRATED ANNOTATION GUIDE (rater how-to)\n\n\\#\n\n\\#\n\n \n\nSports: Badminton, Pickleball, Table Tennis (singles AND doubles).\n\n
\n\nILLUSTRATIONS: boxes marked \\[SCREENSHOT: ...\\] are placeholders.\n\n**TODO(avidullu): replace every \\[SCREENSHOT: ...\\] placeholder with a real frame grab from the\n\npilot videos before sending this guide to Roora.**\n\n
\n\n\\==\n\nPART 1 ? WHAT YOU ARE PRODUCING\n\n\\==\n\nOne CSV per video (from the template Sheet), one row per rally:\n\n rally_number, start_mmss, end_mmss, start_time, end_time, shots_count, ending_reason, last_shot_outcome, score_before, score_after, notes\n\n \n\nYou type: rally_number, start_mmss, end_mmss, shots_count, ending_reason, last_shot_outcome\n\n(+ optional score/notes). The start_time / end_time decimal-second columns FILL THEMSELVES via a\n\nformula already in the template.\n\n \n\nFor every rally you record:\n\n - When it started (serve contact) -\\> start_mmss (m:ss.mmm, e.g. 1:12.40)\n\n - When it ended (shuttle/ball dead) -\\> end_mmss\n\n - How many shots happened -\\> shots_count\n\n - Why it ended -\\> ending_reason · Where the last shot landed -\\> last_shot_outcome\n\n \n\nEverything else (gaps between rallies) you IGNORE.\n\n
\n\n\\==\n\nPART 2 ? THE THREE GOLDEN
RULES\n\n\\==\n\n \n\nRULE 1: IGNORE ALL DEAD TIME\n\nA rally is CONTINUOUS active play. The moment play stops, the rally is over. The time until the\n\nnext serve is DEAD TIME and is never labeled.\n\n \n\n DEAD RALLY 1 DEAD RALLY 2 DEAD\n\n (ignore) |==================| (ignore) |==============| (ignore)\n\n ^ ^ ^ ^\n\n serve shuttle serve shuttle\n\n contact lands contact lands\n\n =start_1 =end_1 =start_2 =end_2\n\n \n\nDead time includes: walking back to serve, picking up the shuttle/ball, wiping sweat, bouncing the\n\nball before serving, the score being announced, replays/slow-mo, crowd or coach shots, changeovers,\n\ntowel/medical timeouts, and PRE-MATCH / BETWEEN-GAMES WARM-UP (knock-up) ? see the Rule 2 caveat below.\n\n \n\nRULE 2: INCLUDE EVERY RALLY ? EVEN THE UGLY ONES\n\nEvery served point gets a row, including:\n\n - Service faults (serve into net/out/illegal) ? usually shots_count = 1, ending_reason = service_fault.\n\n - Very short rallies (serve + one error).\n\n - Lets / replays ? ending_reason = let; the replayed point is its OWN next row.\n\nA missed rally is the worst error in this project ? the model must learn that short and failed\n\nrallies are still

rallies.\n\n \n\n \\!\\! BUT WARM-UP / KNOCK-UP IS NOT A RALLY ? a common trap, because Rule 2
says \"include everything.\"\n\n Pre-match and between-games warm-up LOOKS like rallying but is
not scored play; do not label it.\n\n How to tell it apart:\n\n   - No real serve. Real play
starts with a proper serve from the service position (diagonal,\n\n   behind the service
line). Warm-up has players feeding cooperatively from mid-court with no\n\n   serve -\\> not a
rally.\n\n   - Cooperative, not competitive. Gentle hits straight to each other; nobody is
trying to win the\n\n   point (no smashes-to-win, no scrambling/diving).\n\n   - Score isn't
progressing. No umpire / scoreboard not running / players casually retrieve and\n\n   re-feed;
the score doesn't change. Often several shuttles/balls in use at once.\n\n   - Per sport:
badminton & pickleball ? real play starts with the underhand serve from the service\n\n
court / behind the baseline; table tennis ? real play starts with the tossed, two-bounce
serve\n\n   (knock-up is cooperative rallying without it).\n\n When unsure: find the first
proper serve where the score starts to progress ? labeling begins\n\n there. If you genuinely
can't tell whether the match has started, flag the clip to us ? don't guess.\n\n \n\nRULE 3: A
SHOT = ONE CONTACT WITH THE RACKET / PADDLE / BAT\n\nCount every time the shuttle/ball touches a
racket, paddle, or bat. The serve is shot 1.\n\n - Count: any racket/paddle/bat contact (serve,
return, smash, block...).\n\n - Don't count: floor bounces, table bounces, net clips, the toss
before a serve.\n\n5-shot example: serve(1) -\\> return(2) -\\> hit(3) -\\> hit(4) -\\>
winner(5) -\\> shots_count = 5,\n\nending_reason = winner.\n\n
\n\n\\==\n\nPART 3 ? WHERE
EXACTLY DOES A RALLY START AND
END?\n\n\\==\n\nSTART
(start_mmss): the exact frame the server's racket/paddle/bat CONTACTS the shuttle/ball.
Not\n\nthe toss, not the wind-up ? the contact frame.   \\[SCREENSHOT: serve-contact
frame\\]\n\n \n\nEND (end_mmss): the exact frame the shuttle/ball becomes DEAD:\n\n -
Badminton: shuttle touches the floor, or play stops on a fault.\n\n - Pickleball: the second
floor bounce, or the frame the ball hits the net / lands out.\n\n - Table tennis: the second
bounce on one side, a non-serve net hit, or off the table.\n\n \\[SCREENSHOT: end frame\\]\n\n
\n\nIf occluded/off-screen, pick the closest clear frame and add a note.\n\nBOUNDARY TOLERANCE:
+/- 0.3 s (\\~+/- 9 frames at 30 fps).\n\n
\n\n\\==\n\nPART 4 ?
ending_reason: HOW TO
CHOOSE\n\n\\==\n\nPick
exactly ONE value for how the rally ended. Stop at the first match:\n\n 1. Did the SERVE itself
fail / was illegal (and ended the point)? -\\> service_fault\n\n 2. Was the point REPLAYED
(let / interruption)? -\\> let\n\n 3. Did the final shot land IN and the opponent make NO legal
contact? -\\> winner (a clean winner)\n\n 4. Did the losing player make an error (netted / out
/ missed) UNDER PRESSURE ? stretched,\n\n   wrong-footed, or rushed by a strong shot? -\\>
forced_error\n\n 5. Did the losing player make an error on a ROUTINE, UNPRESSURED ball they
had the time and\n\n   position to make? -\\> unforced_error\n\n 6. None of the above /
cannot tell -\\> other (add a note)\n\n \n\nFORCED vs UNFORCED ? the one judgment call. Decide
using the opponent's PRECEDING shot:\n\n - forced_error = the opponent's shot CAUSED the miss
(pace, placement, deception; the player\n\n   barely reached it or was pulled out of
position).\n\n - unforced_error = a clean, makeable ball the player missed on their own.\n\n
- When genuinely unsure between the two, choose unforced_error and add a brief note.\n\n
\n\nNotes:\n\n - A clean winner means NO legal contact by the opponent. If they got a
racket/paddle/bat on it\n\n   but netted or hit out, that is a forced_error /
unforced_error (their shot), not a winner.\n\n - A serve into the net is service_fault
(rule 1 beats everything).\n\n \n\nTHEN SET last_shot_outcome ? where the last shot
physically went. This is a SEPARATE, observational\n\ncolumn (no judgment): just look at the
last shot.\n\n - in ? landed in (winners; also a put-away the opponent couldn't reach).\n\n -
net ? went into the net.\n\n - out ? landed outside the court / past the lines.\n\n - n_a ?
not applicable: use for service_fault and let.\n\nSo each rally gets both: WHY
(ending_reason) and WHERE (last_shot_outcome) ? e.g. a smash winner =\n\nwinner, in; a
clear forced long = forced_error, out; a routine ball netted = unforced_error, net.\n\n
\n\n\\==\n\nPART 5 ? PER-
SPORT DETAIL\n\n\\==\n\n
\n\nBADMINTON\n\n - Serve: underhand, below the waist, diagonal. Start = racket-shuttle
contact.\n\n - Rally end: shuttle hits the floor, or play stops on a fault.\n\n - Shots: every
racket-shuttle contact; serve = shot 1.\n\n - Service faults (1-shot rally, service_fault):
serve into net, serve outside service court,\n\n   contact above waist, foot fault.\n\n Worked
example:\n\n   \\[SCREENSHOT: badminton serve\\]\n\n   Rally 4: serve 1:28.30, long rally,
smash winner, shuttle lands 1:44.10, 12 contacts.\n\n   4, 1:28.30, 1:44.10, (auto), (auto),
12, winner, in, 3-5, 4-5,\n\n   Rally 5: serve 1:57.00 straight into the net.\n\n   5,
1:57.00, 1:57.90, (auto), (auto), 1, service_fault, n_a, 4-5, 4-6,\n\n \n\n BADMINTON ?

DOUBLES (2 players per side). All rally rules are IDENTICAL to singles:\n\n   - start = serve contact; end = shuttle down / fault; a SHOT = any racket-shuttle contact by\n\n   EITHER player on a side.\n\n   - shots_count = TOTAL contacts across both teams in the rally. Do not track which player hit ?\n\n   just count every contact.\n\n   - Net exchanges (drives, flat pushes) come fast ? frame-step so you neither miss nor\n\n   double-count contacts.\n\n   - Only the designated receiver returns the serve; if the wrong player returns, the rally ends\n\n on a fault ? still a row (service_fault if it's a serve/receiving-position fault, else other + note).\n\n   - Two teammates may NOT both hit the shuttle in succession (double-hit). Two contacts on the\n\n   SAME side back-to-back = a fault ending the rally ? count both contacts, then the rally ends.\n\n \n\nPICKLEBALL\n\n - Serve: underhand, paddle below waist, hit diagonally; must clear the non-volley zone (the\n\n   \"kitchen\"). Start = paddle-ball contact.\n\n - Two-bounce rule: the serve must bounce once in the receiver's court and the return must bounce\n\n   too, before either side may volley. These bounces are NORMAL play, not rally ends, and bounces\n\n   are NOT shots.\n\n - Rally end: ball bounces twice, lands out, hits the net (fails to cross), or a fault (e.g.\n\n   volleying in the kitchen).\n\n - Shots: every paddle-ball contact only; serve = shot 1; floor bounces are not shots.\n\n - Service faults (service_fault): serve into net, out/long, short in the kitchen, foot fault.\n\n Worked example:\n\n   \\[SCREENSHOT: pickleball serve + kitchen line\\]\n\n   Rally 2: serve 0:33.10; serve bounces, return bounces, a few volleys; receiver nets an easy\n\n   one at 0:41.85; 6 paddle contacts.\n\n   2, 0:33.10, 0:41.85, (auto), (auto), 6, unforced_error, net, 1-0, 2-0, easy ball netted\n\n \n\n PICKLEBALL ? DOUBLES (the standard format, 2 per side). All rally rules IDENTICAL to singles:\n\n   - start = serve contact; a SHOT = any paddle-ball contact by EITHER teammate; floor bounces\n\n   still not shots; two-bounce rule still applies.\n\n   - shots_count = TOTAL paddle contacts across both teams; don't track which player.\n\n   - Serving sequence (FYI only ? you do NOT annotate it): both partners serve before the serve\n\n   passes to the opponents, except the very first service turn of the game. The called score has\n\n   three numbers (server score - receiver score - server \\#1/\\#2). Ignore for annotation; just\n\n   mark boundaries / shots / ending.\n\n   - If both teammates contact the ball in succession, that's a fault -\\> rally ends; count the\n\n   contacts, set the ending per the fault.\n\n \n\nTABLE TENNIS\n\n - Serve: ball tossed up from an open palm, then struck so it bounces once on each side. Start =\n\n   the racket-ball strike (not the toss).\n\n - Rally end: a player fails a legal return ? ball double-bounces on their side, goes off the\n\n   table, or is hit into the net on a non-serve shot.\n\n - Shots: every racket-ball contact only; serve = shot 1; table bounces are not shots.\n\n - Let: a serve that clips the net but is otherwise legal -\\> the point is replayed.\n\n   ending_reason = let, and the replay is its own next row.\n\n - Service faults (service_fault): missed/illegal toss, serve into net, serve off the table.\n\n - Speed warning: contacts can be \\< 0.2 s apart ? step frame-by-frame to count shots.\n\n - Doubles (if a clip appears): partners must hit in STRICT ALTERNATION and serve diagonally; a\n\n missed alternation is a fault that ends the rally. Same rule ? count every racket-ball contact\n\n   as a shot; shots_count is the team total.\n\n Worked example:\n\n \\[SCREENSHOT: table-tennis serve contact\\]\n\n   Rally 9: serve 4:10.40; fast 7-contact rally; forehand winner, opponent no contact; dead 4:14.05.\n\n   9, 4:10.40, 4:14.05, (auto), (auto), 7, winner, in, 6-9, 6-10,\n\n   Rally 10: serve 4:22.10 clips the net (otherwise good) -\\> let.\n\n   10, 4:22.10, 4:23.00, (auto), (auto), 1, let, n_a, 6-10, 6-10, net-cord serve, replayed\n\n \n\n\\===\n\nPART 6 ? EDGE CASES & FAQ\n\n\\===\n\n - Two rallies with almost no gap? Still two rows. end_time\\[N\\] \\<= start_time\\[N+1\\], never overlap.\n\n - Interrupted & replayed? Label the interrupted attempt as its own row (ending_reason=let), then\n\n   the replayed point as the next row.\n\n - Serve tossed but caught / re-served (no contact)? No contact = no rally yet. Start at the\n\n   actual serve contact; don't label the aborted toss.\n\n - Opponent reached the ball but netted/hit out? That's a forced_error/unforced_error (their shot),\n\n   not winner ? forced if your shot pressured them, unforced if it was routine.\n\n - Can't read the scoreboard? Leave score_before/score_after blank. Never guess.\n\n - \"Let\" where the rules say play on? If play continued, it's NOT a let ? keep one rally and judge\n\n   the real ending.\n\n - Warm-up / knock-up / practice points? Not real rallies ? do not label them. Start at the first\n\n real, scored serve. See the Rule 2 caveat (Part 2). When unsure whether the match has started,\n\n   flag the clip ? don't guess.\n\n - Doubles? Same rules ? a shot is any one racket/paddle contact by whichever player hits it. See\n\n   the badminton and pickleball DOUBLES notes in Part 5; never attribute shots_count to an\n\n   individual, just total the contacts.\n\n - Server completely whiffs (no contact)? Rare. True zero contact -\\> shots_count = 0,\n\n   ending_reason = service_fault, add a note. A graze = 1 contact = 1 shot.\n\n \n\n\\===\n\nPART 7

? SELF-CHECK BEFORE YOU SUBMIT (per video)\n\n\\==\n\n \\[\\] rally_number is 1,2,3... no gaps or repeats.\n\n \\[\\] Every start_time \\< end_time (check the auto-filled decimal columns populated).\n\n \\[\\] No overlaps: end_time\\[N\\] \\<= start_time\\[N+1\\].\n\n \\[\\] Times entered as m:ss.mmm; decimal start_time/end_time auto-filled and look right.\n\n \\[\\] No dead-time rows (scan the gaps ? dead time, not missing rallies).\n\n \\[\\] Every served point present, including service faults and lets.\n\n \\[\\] shots_count looks right (re-step any rally you rushed; doubles = team total).\n\n \\[\\] ending_reason in {winner, forced_error, unforced_error, service_fault, let, other}; notes\n\n   filled when other.\n\n \\[\\] last_shot_outcome in {in, net, out, n_a} (n_a for service_fault/let).\n\n \\[\\] File downloaded as \\<video_id\\>.csv, header row present, UTF-8.\n\n \\[\\] Did NOT trim / re-encode / rename the source video.\n\n
\n\n\\==\n\nPART 8 ? MINI GLOSSARY\n\n\\==\n\n Rally ? one continuous passage of play from serve contact to the shuttle/ball dead.\n\n Shot ? one contact between the shuttle/ball and a racket/paddle/bat (either teammate in doubles).\n\n Dead time ? any time the shuttle/ball is not in active play. Never labeled.\n\n Service fault ? a failed/illegal serve that ends the point. Still a labeled rally.\n\n Let ? a replayed point. Labeled as its own rally with ending_reason=let.\n\n Winner / forced_error / unforced_error ? the final shot was a clean winner (opponent made no\n\n   legal contact) / the loser's error was forced by a pressuring shot / the loser's error was on a\n\n   routine, unpressured ball.\n\n start_time / end_time ? decimal seconds from the start of the provided file (auto-filled from mm:ss).\n\n "}

177. assistant (2026-06-12T12:37:01.538Z)

```
---
name: roora-guidelines-drive-doc
description: The canonical vendor-facing Roora guidelines Google Doc ? supersedes the
v3/v5/v6/v7 Drive copies; the "v7 anomaly" is RESOLVED
metadata:
  type: reference
---

**Canonical Roora vendor-facing guidelines doc (Brief + Illustrated Guide in one):**
https://docs.google.com/document/d/1NvAXE0q7t4Z0QddLkEfnta0DU982yDZaqkGNPMe8EY0/

- Generated from the source-of-truth repo (sports-data-collector, master `ca61405`,
  2026-06-12); per its own provenance header it **SUPERSEDES the older Drive copies
  (v3/v5/v6/v7)**. The "(v7)" Drive doc (created 2026-06-04) predates the 2026-06-10
  pilot-review fixes ? owner confirmed 2026-06-12 this doc is a superset of v7, so the
  **v7 anomaly noted in the 2026-06-10 pilot-review checkpoint is RESOLVED**.
- Content-current with: `last_shot_outcome` column, forced/unforced `ending_reason`
  taxonomy, §7 required boundary-frame method (frame step 1, exact contact/dead frames),
  §4 grid-snap auto-reject, "rally in progress at a file edge gets no row" exception.
- ? Before sending to Roora: the doc's own VERSION-LABEL NOTE says pick ONE version label
  and use it consistently (repo says "DRAFT v2", email thread said "v7" ? mismatched), and
  it still carries `TODO(avidullu)` markers (Drive folder links, template Sheet, escalation
  channel, NDA, [SCREENSHOT] placeholders in the Guide).
- Operational entry point: collector `docs/annotation/ROORA_RUNBOOK.md` (see
  [[golden-set-vendoring]]).
```

178. user (2026-06-12T12:37:01.550Z)

File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\roora-guidelines-drive-doc.md (file state is current in your context ? no need to Read it back)

179. user (2026-06-12T12:37:07.766Z)

<system-reminder>This memory is 5 days old. Memories are point-in-time observations, not live state ? claims about code behavior or file:line citations may be outdated. Verify against current code before asserting as fact.</system-reminder>

```
# Memory Index

## ? Start here
- [Current state](current-state.md) ? **live micro-checkpoint** (updated every checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.**
- [Session handoff](session-handoff.md) ? current state · next steps · open decisions · ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift, session wrap) ? see working-agreement-checkpoints staleness tripwire.
- [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read `docs/CODE_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.

- [Working agreement: attribution](working-agreement-attribution.md) ? never assert fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The silver-bullet rule.
- [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.
- [Feedback: tail long-job logs](feedback-tail-long-job-logs.md) ? proactively tail progress/summaries from long-running background jobs while they run, not just the final result.

## Project: rally segmentation pipeline
- [Candidate segments & fine-tuning](candidate-segments-finetuning.md) ? why raw detector candidates are persisted (detector-agnostic table); golden-eval-set goal
- [TrackNetV4/WASB via WSL2](tracknet-wasb-wsl2-decision.md) ? self-hosted shuttle trajectory models (not LLM APIs); WASB/TrackNetV3 = MIT, commercial-clear
- [Collector?indexer bridge](collector-indexer-bridge.md) ? 4-phase plan to train/eval a generic temporal rally segmenter; Phases 1?2 done, 3 in progress, 4 next
- [Golden-set vendoring](golden-set-vendoring.md) ? generate golden rally labels via 3rd-party vendors (Bangalore-local) + extensible regression flow; Roora selected for a 3-sport pilot, guideline + collector ingestion built, GS-1/2/4 landed, GS-3 + gold key pending
- [Audio (E1) findings](audio-e1-findings.md) ? quantitative gains/losses of the classical-onset audio cue (recall 100% but discrimination ~2.2×, band-pass null) + reusable lessons for adding the NEXT signal. Audio NOT adopted for fusion; detector kept (PR #35).

## Monetization (hosted API)
- [Monetization plan](monetization-plan.md) ? hosted video?highlights API on a GPU VM; licensing & per-video cost are first-class; user is Bangalore-based
- [Monetization research status](monetization-research-status.md) ? audit done: Ultralytics AGPL swapped out (PR #8); MonoTrack/YOLO-NAS weights avoided; codec=watch; full report in docs/

## Forward-looking
- [Sibling internship repo](sibling-internship-repo.md) ? khelacharya shot-intelligence layer builds ON TOP of our rally segmentation; kept off master at branch level (NOT a separate GitHub repo ? corrected 2026-06-06)
```

180. user (2026-06-12T12:37:18.235Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

181. user (2026-06-12T12:37:28.501Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

182. assistant (2026-06-12T12:37:38.146Z)

Memory updated. Now the one-line runbook closure ? pointing operators at the canonical doc:

183. user (2026-06-12T12:37:40.421Z)

```
M       .gitignore
Switched to a new branch 'docs/runbook-canonical-guideline-doc'
```

184. user (2026-06-12T12:37:53.498Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_RUNBOOK.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

185. user (2026-06-12T12:38:02.919Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_RUNBOOK.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

186. user (2026-06-12T12:39:05.491Z)

```
180 passed in 2.75s
[docs/runbook-canonical-guideline-doc 9208389] docs(runbook): point operators at the canonical
vendor guidelines Google Doc
 1 file changed, 12 insertions(+), 6 deletions(-)
remote:
remote: Create a pull request for 'docs/runbook-canonical-guideline-doc' on GitHub by visiting:
remote:      https://github.com/avidullu/sports-data-collector/pull/new/docs/runbook-canonical-
guideline-doc
remote:
branch 'docs/runbook-canonical-guideline-doc' set up to track 'origin/docs/runbook-canonical-
guideline-doc'.
To https://github.com/avidullu/sports-data-collector.git
 * [new branch]      docs/runbook-canonical-guideline-doc -> docs/runbook-canonical-guideline-
doc
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/26
test    pass    17s     https://github.com/avidullu/sports-data-
collector/actions/runs/27416150806/job/81029363977
```

187. user (2026-06-12T12:39:24.059Z)

```
From https://github.com/avidullu/sports-data-collector
 * branch            master     -> FETCH_HEAD
   ca61405..0a0cff5  master     -> origin/master
Updating ca61405..0a0cff5
Fast-forward
 docs/annotation/ROORA_RUNBOOK.md | 18 ++++++++++++------
 1 file changed, 12 insertions(+), 6 deletions(-)
M       .gitignore
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
0a0cff5 docs(runbook): point operators at the canonical vendor guidelines Google Doc (#26)
```

188. assistant (2026-06-12T12:39:39.921Z)

```
Resolved and recorded. **[#26](https://github.com/avidullu/sports-data-collector/pull/26) is
merged** (`0a0cff5`) ? the collector is back on a clean master with no open PRs.

What I did with the link:

- **Read the doc** via the Drive connector and confirmed it closes the anomaly from the
strongest possible angle: its own provenance header says it was generated from this repo at
master `ca61405` (today's HEAD), explicitly **supersedes the v3/v5/v6/v7 Drive copies**, and
already contains everything that shipped today ? the §7 required boundary-frame method, the §4
grid-snap auto-reject, the file-edge exception, plus the earlier pilot-review fixes
(`last_shot_outcome`, forced/unforced taxonomy) that the stale 2026-06-04 "v7" copy never had.
So the anomaly resolves as: *the "v7" in the email thread referred to an outdated Drive copy;
this doc is the canonical vendor-facing Brief+Guide.*
```

- **Runbook updated** ([#26](https://github.com/avidullu/sports-data-collector/pull/26)): `ROORA_RUNBOOK.md` §0 and the handoff-email checklist now point at the canonical doc instead of the vague "note which version you sent."
- **Memory updated**: a new `roora-guidelines-drive-doc` reference (indexed in MEMORY.md) plus the current-state checkpoint now record the resolution, so no future session re-flags v7.

One heads-up the doc itself raises before it goes to Roora: it still carries `TODO(avidullu)` markers (Drive folder links, the template Sheet, escalation channel, NDA) and `[SCREENSHOT: ?]` placeholders in the Guide part ? and its version-label note asks you to pick **one** label going forward (repo says "DRAFT v2", the thread said "v7"; my suggestion: just call it **v8** in the next email and move on). Those are all yours to fill since they're vendor-comms; everything else in the loop is now tooled, gated, and documented.

189. user (2026-06-12T12:42:46.160Z)

I know I am the loose end in this project ;)

190. assistant (2026-06-12T12:42:59.331Z)

Ha ? to be fair, you're not the loose end, you're the *gate*. Every remaining item is something the tooling deliberately can't do: sign the NDA, paste your own Drive links, pick what to call the doc, and decide which footage Roora gets. That's ownership, not slack.

For the record, the entire human surface area left is about 30 minutes of work, all in one place (the doc's `TODO(avidullu)` markers plus the runbook §0/§1.3 checklists):

1. Fill the Drive/Sheet/escalation links in the guidelines doc, swap the `[SCREENSHOT]` placeholders for real frame grabs.
2. Stamp it "v8" and tell Roora that's the only version that exists now.
3. Run `prep-annotation-batch` on the next batch and hit send.

After that, the machines take it from there ? and `validate-delivery` will be waiting at the door when the annotations come back. ?

191. user (2026-06-12T12:43:31.843Z)

ha perfect! I will get to it and we will do aa test run before we send it to Roora! See you laater!

192. assistant (2026-06-12T12:43:51.331Z)

Before you go ? pinning the agreed next step into the checkpoint so the next session picks up exactly here:

193. user (2026-06-12T12:43:56.462Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

194. assistant (2026-06-12T12:44:03.207Z)

Done ? the checkpoint now records the plan: you fill the doc TODOs and label it v8, then we do a full dry run of the loop on real footage (and we can replay an old delivery through `validate-delivery` as the mock-receive leg) before anything ships to Roora.

Everything's parked clean: both repos on synced master, zero open PRs, suite at 180 green, and the whole vendor loop tooled, gated, and written down. Enjoy the break ? see you at the test run! ?

195. user (2026-06-12T14:20:37.363Z)

There was a pdf tht you helped me generate to give to potential recruits for going to academies across bangalore and get soe feedback. Can you pls point me to it?

196. assistant (2026-06-12T14:21:02.189Z)

Let me hunt for it ? searching the filesystem and past session transcripts in parallel.

197. user (2026-06-12T14:21:08.025Z)

```
FullName
--------

C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Pickleball
and Table Tennis (2026-0?
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\HDFC_Avi_Dullu_31072024_150007641.pdf
C:\Users\avidu\Projects\Annotation Setup\6 June 2026\Feedback\Roora Pilot Review - Badminton
(2026-06-06).pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\unacademy.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\tataneu.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\swiggy.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\makemytrip-2.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\makemytrip-1.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\flipkart.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\dataquest.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\airasia-3.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\airasia-2.pdf
C:\Users\avidu\Projects\Verifiers\muneem\muneem\testdata\promo-emails\airasia-1.pdf
C:\Users\avidu\Projects\Annotation Setup\Illustrated Annotation Guide ? Rally Labeling
(Supplement).pdf
C:\Users\avidu\Projects\Annotation Setup\Roora Annotation Brief ? Rally Golden Set (Format Spec,
Scope, QA).pdf
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\Avi_Dullu_30062025_105046061.pdf
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\Avi_Dullu_30092025_094157457.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for October
2023.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for August 2025.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for January
2026.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for August 2024.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for May 2025.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for June 2025.pdf
C:\Users\avidu\Projects\muneem\testdata\aubank-statements\Statement_2024MTH07_XXXXX859_146.pdf
C:\Users\avidu\Projects\muneem\testdata\aubank-statements\Statement_2026MTH04_XXXXX859_849.pdf
C:\Users\avidu\Projects\muneem\testdata\aubank-statements\Statement_2026MTH01_XXXXX859_721.pdf
C:\Users\avidu\Projects\muneem\testdata\aubank-statements\Statement_2026MTH03_XXXXX859_108.pdf
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\Avi_Dullu_31082025_205409615.pdf
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\Avi_Dullu_30112025_110905081.pdf
C:\Users\avidu\Projects\muneem\testdata\axis-statements\AXIS BANK  Statement for October
2025.pdf
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\Avi_Dullu_30042026_001858675.pdf
C:\Users\avidu\Projects\muneem\testdata\hdfc-statements\Avi_Dullu_28022026_183854522.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\tataneu.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\unacademy.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\swiggy.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\makemytrip-1.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\flipkart.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\airasia-3.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\airasia-2.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\airasia-1.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\makemytrip-2.pdf
C:\Users\avidu\Projects\muneem\testdata\promo-emails\dataquest.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\TestSet\tataneu.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\swiggy.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\unacademy.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\makemytrip.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\flipkart.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\airasia3.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\airasia2.pdf
```

```
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category2\airasia.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category1\makemytrip2.pdf
C:\Users\avidu\Projects\Email-Mining\textExtract\pdfFiles\Category1\dataQuest.pdf
```

198. assistant (2026-06-12T14:21:19.772Z)

No obvious match under `Projects` ? searching past session transcripts and other likely folders:

199. user (2026-06-12T14:21:22.303Z)

This tool requires user approval and is unavailable in unsupervised mode.

200. user (2026-06-12T14:21:24.569Z)

```
FullName
LastWriteTime
--------
-------------
C:\Users\avidu\Downloads\system_design_interview_conversation.pdf 5/30/2026 11:23:57 PM
```

201. assistant (2026-06-12T14:21:46.902Z)

Transcript search isn't available in this mode. Widening the net ? all recent PDFs anywhere in your profile, full `%TEMP%` scan, and Google Drive:

202. user (2026-06-12T14:21:53.185Z)

```
Exit code 1
FullName                                             LastWriteTime
--------                                             -------------
C:\Users\avidu\AppData\Local\Temp\2308.12199.pdf     6/6/2026 11:04:27 PM
C:\Users\avidu\AppData\Local\Temp\apsipa316.pdf      6/6/2026 11:09:50 PM
C:\Users\avidu\AppData\Local\Temp\huang11.pdf        6/6/2026 11:10:03 PM
C:\Users\avidu\AppData\Local\Temp\p2308.pdf          6/6/2026 11:03:54 PM
C:\Users\avidu\AppData\Local\Temp\paper.pdf          6/6/2026 11:03:13 PM
C:\Users\avidu\AppData\Local\Temp\tennis2012.pdf     6/6/2026 11:09:24 PM
C:\Users\avidu\AppData\Local\Temp\tmpg1h_o1_s\cc.pdf 6/10/2026 11:35:34 PM
C:\Users\avidu\Projects\Annotation Setup\6 June 202? 6/10/2026 3:42:59 PM
C:\Users\avidu\Downloads\HDFC_Avi_Dullu_31072024_15? 6/8/2026 11:20:47 AM
C:\Users\avidu\Projects\Annotation Setup\6 June 202? 6/6/2026 3:52:01 PM
C:\Users\avidu\OneDrive\Desktop\Request_for_change_? 6/5/2026 4:51:18 PM
C:\Users\avidu\OneDrive\Desktop\Google-Gemini.pdf    6/4/2026 1:17:21 PM
C:\Users\avidu\Projects\Annotation Setup\Illustrate? 6/2/2026 5:19:41 PM
C:\Users\avidu\Projects\Annotation Setup\Roora Anno? 6/2/2026 5:19:29 PM
C:\Users\avidu\Downloads\system_design_interview_co? 5/30/2026 11:23:57 PM
C:\Users\avidu\OneDrive\Documents\khelacharya_shot_? 5/30/2026 1:57:48 PM
C:\Users\avidu\OneDrive\Documents\khelacharya_inter? 5/30/2026 12:47:07 PM
C:\Users\avidu\OneDrive\Documents\khelacharya_final? 5/30/2026 12:42:22 PM
C:\Users\avidu\OneDrive\Documents\khelacharya_works? 5/30/2026 12:41:40 PM
C:\Users\avidu\Downloads\sigir_colbert.pdf           5/25/2026 1:08:42 PM
C:\Users\avidu\Downloads\Policy_Surrender_Form.pdf   5/20/2026 1:28:32 PM
C:\Users\avidu\Downloads\Sovereign Debt Collateral ? 5/14/2026 11:50:51 AM
```

203. assistant (2026-06-12T14:22:03.954Z)

Found a strong cluster ? four `khelacharya_*` PDFs in OneDrive Documents from May 30. Getting their full names:

204. user (2026-06-12T14:22:06.658Z)

```
Name          : khelacharya_final_flyer.pdf
LastWriteTime : 5/30/2026 12:42:22 PM
KB            : 65
```

Name   : khelacharya_intern_guide.pdf
LastWriteTime : 5/30/2026 12:47:07 PM
KB   : 34

Name   : khelacharya_shot_intelligence_layer.pdf
LastWriteTime : 5/30/2026 1:57:48 PM
KB   : 9

Name   : khelacharya_workspace_guide.pdf
LastWriteTime : 5/30/2026 12:41:40 PM
KB   : 65

205. assistant (2026-06-12T14:22:16.212Z)

Verifying which one is the recruit-facing flyer by reading it:

206. user (2026-06-12T14:22:16.910Z)

PDF file read: C:\Users\avidu\OneDrive\Documents\khelacharya_final_flyer.pdf (65.3KB)

207. user (2026-06-12T14:22:41.962Z)

{"files":[{"canAddChildren":false,"contentSnippet":"# UI Proposal ? khelstura.guru web app (alpha ? beta)\n\nStatus: PROPOSAL ? awaiting owner ratification (no implementation starts without explicit approval, per CLAUDE.md). Date: 2026-06-10 · Inputs: PMF_MARKET_RESEARCH, COMMERCIALIZATION, PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (\\#16), ROADMAP, VIDEO_RECORDING_GUIDELINES, reporting/spec.yaml, current API + static UI. Companion: docs/HOSTING_PLAN.md (domain, topology, costs).\n\n## 1\\. Strategic frame ? why a UI now, and what it secretly is\n\nThe UI is not just product surface; it is the corpus-growth engine. The single highest-leverage item in NEXT_STEPS is growing the golden corpus, and the v1 scope already commits to a human correction loop. Every tester who accepts/rejects/nudges a rally boundary in the review screen produces exactly the labels the Gen-0/M0.x track is starved for (subject to the per-user training opt-in guardrail). 10?25 alpha testers ? a steady stream of labeled, real-world, multi-court-biased footage ? the distribution where the owned model's 0.66 ship target lives.\n\nServing economics per the 2026-06-10 quality table: alpha serves the Gemini path (0.93 on clean footage, \\~$0.05/run) ? no GPU required in the request path. The WASB/owned-model path stays on the owner's box and is not exposed in the alpha UI (segmenter choice hidden).\n\nThe CLI is not replaced. The web app talks to the same FastAPI surface the CLI drives; the CLI remains the power-user/ops tool.\n\n## 2\\. Personas (from PMF research)\n\n - Coach / academy (primary, B2B, India): batches of match recordings, wants rally review, per-player reels for students/parents, per-rally stats. Already records on static cameras.\n - Enthusiast (B2C funnel): records own matches, wants a highlight reel + basic stats, shares with friends. Tripod discipline is the known adoption risk ? the UI must teach recording guidelines before upload, not fail videos after.\n\n## 3\\. ALPHA ? \"quick & dirty\" (goal: 10?25 invited testers, end-to-end, zero hand-holding)\n\nStack: React + Vite + Tailwind SPA, deployed per HOSTING_PLAN (static hosting, $0). API = existing FastAPI on the GPU box via tunnel at api.khelstura.guru. Auth: Cloudflare Access email-allowlist (OTP), free ?50 users, zero auth code ? backend reads the authenticated-email header for per-user scoping (alpha-grade tenancy: header is trustworthy because the tunnel only accepts Access-authenticated traffic).\n\n### Alpha screens (6)\n\n| | | | |\n| :-: | :-: | :-: | :-: |\n| \\# | Screen | What it does | Backed by |\n| A1 | Matches (home) | User's videos, status chip per video (verdict ?/??/?), upload CTA | videos + owner scoping (B4) |\n| A2 | Upload & Process | Drag-drop; inline recording-guidelines checklist (pre-upload education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing | B2 upload endpoint ? B1 job |\n| A3 | Job progress | Stage list (validation ? segmentation ? AI confirm ? report) with live status; verdict banner on completion | B1 GET /api/jobs/{id} poll |\n| A4 | Rally review ? | Video player + rally list; click rally ? seek & play that span; per-rally Accept / Reject / nudge start?end ±0.5 s; bulk \"all good\"; chips: duration, shot count (+confidence) | B6 correction endpoints ? candidate_segments.human_label/notes ? the corpus engine |\n| A5 | Highlights | Select accepted rallies ? name reel ? compile ? download; list of past reels | /api/compile + B3 download endpoint |\n| A6 | Report | Renders the customer-audience report (verdict, notes, FAQ refs); \"nerd view\" toggle shows technical.md | existing reporting artifacts |\n\n### Backend work the alpha actually requires (honest dependency list)\n\n| | | |\n| :-: | :-: | :-: |\n| ID | Item | Notes |\n| B1 | Async job model ? DEFERRED_HARDENING_PLAN \\#16 | The committed

next PLATFORM slice; POST /api/process ? 202 + job_id, jobs table (user_version 1?2),
ThreadPoolExecutor, GET /api/jobs/{id}. Approving this proposal = greenlighting \\#16. A 5?60
min blocking HTTP call cannot sit behind a tunnel UI. |\n| B2 | Chunked upload endpoint | POST
/api/upload accepting ?90 MB parts (free-tier proxy body limit is 100 MB), assemble ? MD5 ?
register under allowed_video_root/\\<user\\>/. Size cap (e.g. 4 GB) + mp4/mov allowlist.
|\n| B3 | Download endpoint | GET /api/highlights/{video_id}/{name} streaming the compiled
reel (today /api/compile returns a server-local path the browser can't reach). |\n| B4 | Per-
user scoping | owner_email column on videos (migration), read from the Access header, filter
every query. Alpha-grade, single DB. |\n| B5 | Retire/replace static UI | The SPA supersedes
backend/api/static (which has a known rallies vs play_segments field bug ? don't fix,
replace). |\n| B6 | Correction-loop endpoints | PATCH /api/segments/{id} ?
accept/reject/boundary-nudge persisted to candidate_segments.human_label/notes (+
provenance: who, when). This is v1-scope \"correction loop\" made real. |\n| B7 | Exposure
hardening | CORS narrowed to the app origin; ...","createdTime":"2026-06-
10T17:20:02.270Z","fileSize":"7690","id":"1fg7BpTR0dcbP_alH9rva_I00eB2myMoBeX_Ih-
ikvRo","mimeType":"application/vnd.google-apps.document","modifiedTime":"2026-06-
10T17:20:04.401Z","owner":"avi.dullu@gmail.com","parentId":"0AJinnK-OT1qhUk9PVA","title":"UI_PRO
POSAL.md","viewUrl":"https://docs.google.com/document/d/1fg7BpTR0dcbP_alH9rva_I00eB2myMoBeX_Ih-i
kvRo/edit?usp=drivesdk&ouid=108278129040148130635","viewedByMeTime":"2026-06-
11T03:24:37.745Z"},{"canAddChildren":false,"contentSnippet":"The China issue\n\n120TH
ANNIVERSARY ISSUE\n\nGenes, chips, qubits, rockets, reactors, surveillance, and sand?the tools
of a rising superpower\n\nVol. 122\n\nJan/Feb\n\n$9.99 USD No. 1\n\n2019\n\n$10.99
CAD\n\nbattelle.org/tech\n\nConnecting vehicles\n\nProtecting pedestrians\n\nMaking cities
smarter\n\nReducing accidents\n\nMaximizing efficiency\n\n02 From the editor\n\nIn November 2018
a Chinese researcher, He Jiankui, announced that he had pro- duced the first ever gene-edited
chil- dren. (MIT Technology Review was first to report that he had embarked on the attempt.) The
story stunned and unnerved the world, not just because a medical taboo had been broken but
because of where it had happened. It seemed to confirm China?s popular image as a country with
growing tech- nological powers and few limits on using them.\n\nSimilar anxieties drove the US-
China trade war that broke out in 2018 and the restrictions several countries imposed on Chinese
telecom giants Huawei and ZTE. In November the US began to con- Gideon Lichfield is editor in
chief of MIT Technology Review.\n\nsider tighter controls on exports of AI and other
technologies.\n\nIt?s fortuitous but fitting, then, that this 120th anniversary issue of MIT
Technology Review is about China?s sta- tus as a rising technology superpower. Two centuries ago
China was the world?s largest economy, with some 15 times the GDP of the US. But wars,
rebellions, and a lack of industrializa- tion made it stagnate. By the time our first issue
appeared?sporting the\n\nwalrus- whiskered president of MIT, James Mason Crafts, on the
cover?the US was slightly ahead. By 1950 its GDP was several times larger than China?s, which
had scarcely budged in real terms in 130 years.\n\nToday, however, the two are once again
roughly on a par, thanks to China?s explo- sive growth since the 1970s. Visiting the country
brings to mind the impressions of European travelers to America a century ago?of a land where
everything is bigger and happens faster, a place bursting with energy and ideas. Our goal in
this issue was to answer the question ?What is China good at?? The common prejudice that China
doesn?t innovate and steals all its intel- lectual property from abroad has been\n\nTechnology
Review?s first issue, in 1899, profiled MIT?s president, James Mason Crafts.\n\noutdated for a
while, but can its com- panies build world-changing products, and can its scientists win Nobels?
Can it meet its goal, laid out in various long- term plans, of achieving supremacy in key areas
of technology? Could its top- down system of government even make it better than the world?s
increasingly fractious democracies at tackling urgent problems like climate change? Or will the
Xi administration?s authoritarianism choke off innovation?\n\nOur writers examine China?s prog-
ress in autonomous and electric vehicles (page 24), microchips (page 28), nuclear power (page
42), high-voltage grids (page 48), space exploration (page 52), quantum computing and communica-
tions (page 56), and gene editing (page 62). We look at how the government has let the tech
industry flourish while co-opting it into the growth of a sur- veillance state (page 8), wielded
censor- ship and squeezed out foreign firms like Google (page 78), and bent scientists to its
ideological will (page 70).\n\nWe profile Shenzhen, dubbed the ?Silicon Valley of hardware?
(page 16), explain how China is extending its geopolitical influence by building the world?s
biggest dredging industry (page 36), and analyze how the rivalry between China and the US may
play out (pages 46 and 96). Finally, we sample the art that has sprung up critiquing the
country?s consumerist culture, internet aesthetics, and authoritarian tendencies (pages 14, 34,
and 60), and end with a dystopian short story specially commis- sioned from one of China?s
leading sci- ence fiction writers (page 88).\n\nI hope this issue leaves you with a picture of
China?s ambitions and its strengths, but also of its limitations. We believe both China and
other countries have more to gain from collaborating than from driving up barriers to trade,
travel, and the free flow of knowledge. I?m interested in hearing your opin- ion, as well as

what stories you think we missed. Write to me at gideon.lichfield@ technologyreview.com and let me know.\n\nWORLD-CLASS EXECUTIVE EDUCATION AT BROWN UNIVERSITY\n\nTransforming Mid-Career Professionals\n\nAn Executive Master?s degree from Brown University will prepare you to lead your organization, transform your field, and build a powerful lifelong professional network.\n\nYou will join a vibrant learning community and apply your new knowledge and skills through a critical challenge project.\n\nThe results can be seen in our alumni who are proven leaders; impacting their organizations and the world.\n\nbrown.edu/professional\n\nIE Brown Executive MBA\n\nExecutive Master in Cybersecurity\n\nExecutive Master in Science and Technology Leadership\n\nExecutive Master of Healthcare Leadership\n\nBrown-34343 Print Ad_81875x105.indd 1 3/29/18...","createdTime":"2025-05-12T11:33:46.590Z","fileExtension":"pdf","fileSize":"16731702","id":"1Hev0ST6548FJwvYIvlWxTpE6nPVm9CE7","mimeType":"application/pdf","modifiedTime":"2019-02-11T02:01:37Z","owner":"avi.dullu@gmail.com","parentId":"1skSEg-_D3_6_PlevXbY4PLpf_JeW0qLi","title":"MIT-Technology-Review-2019-01.pdf","viewUrl":"https://documentcloud.adobe.com/gsuiteintegration/index.html?state=%7B%22ids%22:%5B%221Hev0ST6548FJwvYIvlWxTpE6nPVm9CE7%22%5D,%22action%22:%22open%22,%22userId%22:%22108278129040148130635%22,%22resourceKeys%22:%7B%7D%7D","viewedByMeTime":"2025-12-19T05:28:43.185Z"}]}

208. user (2026-06-12T14:23:08.402Z)

<system-reminder>This memory is 6 days old. Memories are point-in-time observations, not live state ? claims about code behavior or file:line citations may be outdated. Verify against current code before asserting as fact.</system-reminder>

```
---
name: sibling-internship-repo
description: "A separate khelacharya repo builds a shot-intelligence layer ON TOP of this project's rally segmentation ? keep them separate"
metadata:
  node_type: memory
  type: project
  originSessionId: aa329f09-28b7-4acc-b062-98ec7e905abc
---

There is a sibling **working directory** at `C:\Users\avidu\Projects\badminton-highlight-indexer-khelacharya\` (the user works across machines) holding an **internship execution plan** ? `docs/INTERNSHIP_KHELACHARYA_SHOT_INTELLIGENCE_EXECUTION_PLAN.md` (+ a recruiting-flyer generator stub `scripts/generate_khelacharya_flyer.py`).

**IMPORTANT correction (verified 2026-06-06):** this is NOT a separate GitHub repo. That folder is a second local *clone of THIS same repo* (`origin` = `avidullu/badminton-highlight-indexer`); no `?-khelacharya` GitHub repo exists. The separation is currently enforced **at the branch level**: the shot-intelligence scaffolding (`backend/shot_intelligence/` ShotEvent schema, `docs/SHOT_TAXONOMY.md`, the execution plan) lives on branch **`feat/khelacharya-shot-intelligence`** and is deliberately kept OFF `master`. That branch was PR **#11**, now **CLOSED (branch retained, reopenable)** to keep the PR list scoped to the rally-segmentation foundation. Only the flyer stub `scripts/generate_khelacharya_flyer.py` reached `master`. If/when the internship resumes, split the branch into its own GitHub repo.

**The boundary the user explicitly wants kept (2026-05-30):** *this* project (`badminton-highlight-indexer`) is the **foundation** ? reliable **rally segmentation** across sports (badminton, pickleball, table-tennis, tennis). The intern's "KhelAcharya Shot Intelligence Layer" builds **on top**: it takes each segmented rally and maps **each shot within it** to actionable output (shot type, player attribution, outcome). **Rally boundaries are the contract between the two.**

**How to apply:**
- **Keep the repos separate.** Don't bleed shot-level work into this repo; don't pull this repo's churn into the internship plan.
- The intern plan deliberately reuses *our* patterns (`BasePlaySegmenter`-style `BaseShotDetector`, the eval-harness `metrics.py` symmetry at shot granularity, `SportProfile` registry, TrackNet/WASB outputs) and names our `feat/permissive-detector-swap` direction as a principle ? so improvements to our substrate (segmentation quality, permissive licensing, multi-sport profiles) directly enable the intern's layer.
- Document owner: Avi Dullu. This is forward-looking context, not active work in *this* repo.
```

209. user (2026-06-12T14:23:22.238Z)

<tool_use_error>String to replace not found in file.
String: **Document owner: Avi Dullu. This is forward-looking context, not active work in *this* repo.**</tool_use_error>

210. user (2026-06-12T14:23:33.737Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\sibling-internship-repo.md has been updated successfully. (file state is current in your context ? no need to Read it back)

211. assistant (2026-06-12T14:23:41.268Z)

Pointer saved to memory too, so this is a one-lookup answer next time.

TL;DR: `C:\Users\avidu\OneDrive\Documents\khelacharya_final_flyer.pdf` ? the two-page KhelAcharya internship flyer from May 30, with the apply-by-email instructions and the camera-shake application challenge. Its three companions (`intern_guide`, `workspace_guide`, `shot_intelligence_layer`) sit in the same folder.

And the caveat stands: if you were picturing a PDF specifically about *visiting academies for feedback* (the C13 interviewer angle), that one was never made ? the flyer is the only recruit-facing PDF in existence. Say the word and I'll draft it to match the flyer's look.

212. user (2026-06-12T14:24:36.179Z)

right.. I have talked to a bunch of folks and the PMF validation needs to be handled by physically visiting the academies and looking for signals which will establish that Khelsutra is heading in the right direction

213. assistant (2026-06-12T14:25:41.620Z)

That's the C13 thread coming to a head ? and it matches the audit's own conclusion that field contact is the fastest PMF falsifier. Let me ground in the existing PMF/commercialization docs first, then build you the field-validation kit: a signals framework + visit protocol in the repo, and the giveaway one-pager for the people doing the visits.

214. user (2026-06-12T14:25:43.588Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-36-| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic `setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting your approval** before code. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-37-| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md` (#16). |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-38-| | **? IN-FLIGHT (in progress / watch)** | | | | |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-39-| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
docs\COMMERCIALIZATION.md:40:| C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See §6. |
docs\COMMERCIALIZATION.md-41-| C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006:

Roora (Bengaluru) selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN_SET_VENDORING.md`. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-42-[Omitted long context line]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-43-[Omitted long context line]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-44-| C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-200-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-201-## Changelog
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-202-
docs\COMMERCIALIZATION.md-203-- **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
docs\COMMERCIALIZATION.md:204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-205- "beachhead validation" workstream. Wired the summary into §6.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-206-- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
docs\COMMERCIALIZATION.md-207- audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-208- removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-52-   calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
docs\NEXT_STEPS.md-53-4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-54-   badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-55-   by C3 (provenance multiplies per sport).
docs\NEXT_STEPS.md:56:5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-57-   "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
docs\NEXT_STEPS.md-58-   (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
docs\NEXT_STEPS.md-59-   `output\MahadevpuraSingles_highlights.mp4`). **Recruiter hand-out ready: `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md`**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-60-   (script, ground rules, Phase-2 consent rules; owner fills ?/contact placeholders).
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-182-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-183-### 4.1 Entities ? generic core, with the coach model as an optional overlay
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-184-The foundational schema is deliberately **domain-neutral** so we don't overfit to the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-185-coach mindset ? the coach/athlete framing is a **market-fit hypothesis that will iterate**
docs\PLATFORM_ARCHITECTURE.md:186:(see C13 / PMF), not a load-bearing assumption. **Generic core:**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-187-```
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-188-User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md-189-
???< PlaySegment ??< Event
docs\PLATFORM_ARCHITECTURE.md-190-Video \ Highlight ??< Job (processing record: status, compute backend, timings)
--

```
docs\UI_ALPHA_EXECUTION_PLAN.md-50-- [ ] **D3:** deployed smoke E2E (owner-only allowlist); go-live security checklist (HOSTING_PLAN §7) all green
docs\UI_ALPHA_EXECUTION_PLAN.md-51-- [ ] **GATE GD:** checklist signed off
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-52-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-53-### Phase R ? rollout
docs\UI_ALPHA_EXECUTION_PLAN.md:54:- [ ] Wave 1: 5 testers (C13/camera-pilot contacts) + feedback channel (WhatsApp group or Google Form)
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-55-- [ ] 2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL §4)
docs\UI_ALPHA_EXECUTION_PLAN.md-56-- Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2 tester retention
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-57-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-58-## Working agreement
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-96-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-97-1. **Greenlight** alpha scope above (= greenlights DEFERRED_HARDENING_PLAN #16)?
docs\UI_PROPOSAL.md-98-2. **RESOLVED 2026-06-10:** product name = **KhelSutra Guru**; domain = **khelsutra.guru** (GoDaddy receipt, 3-yr registration + domain protection, renews 2029-06).
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-99-3. Per-user **training opt-in default OFF** ? confirm (recommended; conservative per guardrails).
docs\UI_PROPOSAL.md:100:4. Alpha invite list: the C13 interview academies + camera-pilot contacts?
docs\UI_PROPOSAL.md-101-5. SPA stack confirm: React + Vite + Tailwind (proposed) vs anything you prefer.
docs\UI_PROPOSAL.md-102-6. **RESOLVED 2026-06-10:** separate **private repo `khelsutra`** (owner-ratified ? the product's
docs\UI_PROPOSAL.md-103-   "heart and brain"): SPA in `web/`, control plane joins at beta; backend PRs B1?B7 stay in the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-104-   indexer repo. Keeps the indexer's licensing/test discipline untouched.
```

215. user (2026-06-12T14:25:46.535Z)

```
Name                               KB
----                               --
BACKLOG.md                         13
CODE_MAP.md                        66
COMMERCIALIZATION.md               17
DEFERRED_HARDENING_PLAN.md          7
EVALUATION.md                       7
FIELD_RESEARCH_ASSOCIATE_FLYER.md   6
GOLDEN_SET_IMPLEMENTATION.md        6
GOLDEN_SET_PHASE1_VIDEOS.md         8
GOLDEN_SET_VENDORING.md            16
HOSTING_PLAN.md                     8
HOW_RALLY_DETECTION_WORKS.md        4
NEXT_STEPS.md                       7
OWNED_MODEL_IMPLEMENTATION_PLAN.md 15
PLATFORM_ARCHITECTURE.md           47
PMF_MARKET_RESEARCH.md             16
README.md                           2
ROADMAP.md                          8
TRACKNET_WSL_SETUP.md               6
UI_ALPHA_EXECUTION_PLAN.md          6
UI_PROPOSAL.md                      8
VIDEO_RECORDING_GUIDELINES.md       7
```

216. user (2026-06-12T14:26:00.571Z)

```
1       # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3       **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
```

academies are active)
Location: Bengaluru ? in-person visits to badminton academies
Compensation: ? ___ per completed academy visit + travel reimbursement *(discussed individually)*
Reports to: Avi (phone/WhatsApp: ___)

The gig in one paragraph

We're building an AI tool that takes ordinary practice/match video from a badminton academy and automatically finds every rally, builds highlight reels, and generates match reports. Before building further, we need honest, on-the-ground answers from real academies about how they record, what they'd use, and what they'd pay. **Your job: visit 10?15 good academies, have a friendly 15?20 minute conversation with the owner/head coach using the script below, and write down what they actually say.** No selling. No tech knowledge needed. If a conversation goes well, there's a Phase-2 follow-up (coordinating a paid video-recording session) with additional compensation.

Phase 1 ? The conversation (what you actually do)

Who to talk to: the academy owner or head coach ? the person who decides what the academy spends money on.
If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.

Opening (keep it this honest):
> "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice
> video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to
> understand how you work with video today. Could I ask you a few questions? Takes 15 minutes, and we're not selling
> anything today."

The questions (ask in this order; the starred ones matter most):

1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?
2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,
   tournament days, CCTV) ? roughly how much?
3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?
4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, BadPro+, just YouTube/phone.)
   What do you like / what's missing in it?
5. ? **If you could hand over a full session's video and get back every rally cut out + a highlight reel + a simple
   report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per match?** (Let THEM say a number
   first. Note their face, not just the words.)
6. Your hall has multiple courts running at once, right? Does that make video less useful today?
7. Would you let us record one session here (we pay a small fee for the time) to show you what the tool produces on
   YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

After each visit, fill ONE answer sheet (template on the back / separate page): academy name, location, # courts, person spoken to + role + contact, answers 1?7 (their words, not yours), any exact quotes, your gut read (cold / curious / eager), and whether they're a Phase-2 candidate.

Ground rules & gotchas (read twice ? this is the difference between useful and useless)

1. **You are researching, not selling.** If they ask "how much will it cost", say "that's exactly what he's trying to figure out ? what would feel fair to you?" Never quote a price yourself.
2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did you LAST record a session, and what happened to that file?" gets the truth. Prefer the second kind of question.
3. **Don't promise anything.** No features, no dates, no free trials. The only thing you may offer is the Phase-2 idea (a paid recording session + free sample output) and only as "would you be open to it".
4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a reason is worth MORE to us than five vague yeses. You're paid per completed honest visit, not per positive answer.
5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.** Ask what it does well, what it doesn't, and what they pay. Spend extra time here.
6. **Multi-court is our favorite topic.** If they complain that video is useless because "all the courts are in frame and it's chaos" ? capture that complaint in their exact words. That's the problem we solve.
7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ? notes only. Don't photograph the academy or players unless the owner offers.
8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back to record.
9. **Confidentiality:** don't describe the tech beyond the opening line, don't name other academies' answers, and don't share the answer sheets with anyone except Avi.

Phase 2 (only for academies that said yes ? separate briefing & pay)

You coordinate a recording session: we pay the academy a small fee, set up a phone/camera on a tripod (we provide a one-page capture checklist: behind the court, elevated, whole court in frame, landscape, 1080p/60fps, don't move it), record 1?2 sessions, and later return a sample highlight reel + report to the academy for free. **Hard rules:** written consent from the academy AND the adult players being recorded; **sessions with minors are not recorded for our library** ? adult sessions only; the academy gets told plainly what the footage is for (building and demonstrating the tool).

Interested?

Message Avi at ___ with "FIELD RESEARCH" and your availability. First 2 visits are paired (you + Avi or a phone walkthrough) so you're never improvising.

217. user (2026-06-12T14:26:28.964Z)

PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports

> **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year; validate live before relying on them for investment/pricing decisions. This is the market-research-of-record behind the PMF section of the living [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker (§6). Product framing: a closed-source hosted API/SaaS (plus local mode) that ingests long-form, self-recorded, **static-camera, full-court amateur** racket-sports video and auto-detects rallies and compiles highlight reels. Primary sport: badminton; sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.

> **Confidence convention.** Figures from named industry reports, federations, or company pricing pages are tagged with source + year. Inferred items are tagged **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely between vendors; ranges are shown rather than false precision.

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1. Executive summary

The product sits at the intersection of three favorable trends: (a) a fast-growing **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034 depending on vendor); (b) a structural shift to **amateur self-recording** on action cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest segment); and (c) **strong racket-sport participation** ? 220M+ badminton players, 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in 2024, +45.8% YoY).

The central tension is the **static-camera constraint**: the tool works best on fixed, full-court amateur footage and *fails* on broadcast/panning footage. This is both a moat (most competitors target team sports / broadcast) and a filter on the addressable user ? the customer must already set a camera on a tripod behind the court. That behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not the median casual player.

Recommended beachhead (strongest single): **badminton coaching academies and serious club/competitive players in India** ? a large, rapidly formalizing academy ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing willingness to pay for coaching, an underserved tooling gap (incumbents are team-sports/broadcast), and a direct-but-immature competitor field. The product is *native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball players & clubs** (capital-rich, fastest-growing, but SwingVision already strong); **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).

Pricing clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo) and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A software-only BYO-camera model undercuts the hardware players; realistic anchors **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**

Top risks: (1) amateurs won't reliably set up a tripod ? TAM collapses to coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

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2. Market sizing & trends

2.1 Sports analytics / sports-video market
- **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.6?6.0B. *(2025)*
- **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). *(reports 2024?2025)*
- **Video analytics is the leading segment** within sports analytics (Grand View /

Fortune BI) ? the directly relevant sub-market.
- **Interpretation (estimate):** the *amateur/club auto-highlight* slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption
- **2024 size:** ~USD 6.5B (GMInsights) ? ~USD 16.9B by 2034 (~10% CAGR). *(2024?2025)*
- **Sports is the largest end-use segment** of action cameras (GMInsights, 2024).
- **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:** the capture-side behavior the product depends on is already widespread and growing ? *for the segments that record* ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries	BWF est. via BadmintonBites (2023)
Badminton ? China	~80M registered, up to ~100M (largest base)	industry compilations (2023)
Badminton ? India	20M+ recreational by 2023; academy boom + Khelo India	industry compilations (2023)
Badminton ? Indonesia	National sport; ~1M club players	industry (est.)
Badminton ? Denmark	100,000+ registered club members (Europe #1)	industry (2023)
Badminton ? Japan/Korea	Major school-club + workplace sport	industry (2023)
Tennis	**106M players (2024)**, +25.6% over 5 yrs (87M in 2021)	ITF Global Tennis Report (2021, 2024)
Tennis ? Asia	~1/3 of players (33M of 87M)	ITF (2021)
Pickleball ? US	**19.8M players (2024)**, +45.8% YoY, +311% over 3 yrs; ~6.2M core	SFIA / Pickleheads / The Dink (2024)
Pickleball ? US courts	~70,000 courts, +55% places to play YoY	SFIA / Pickleheads (2024)
Pickleball ? global	IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim **(low confidence)**	various (2024?2025)
Table tennis	30M+ players; 227 member associations	ITTF (current)
Cricket	2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders)	Cricket Rise / Nielsen (2024?2025)

- **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR (Market.us/Custom Market Insights, 2024?2025).
- **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

Geographic read: badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket is huge as *fandom* but its amateur self-recording + static-full-court fit is weak (large field, not a fixed-frame court). For *this product's static-camera fit*, badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap item, not a beachhead.

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3. Customer segment evaluation

Segment	Market-size signal	Willingness to pay	Accessibility	Static-camera fit	B2C/B2B	Verdict
Individual amateur/enthusiast	Largest raw TAM (100s of millions)	**Low** ? competes with free; ~USD 5?15/mo ceiling	High via app stores; high CAC/churn	**Mixed** ? most won't set a tripod	B2C	Volume play, weak economics; use as funnel
Coaches & coaching academies	India academy boom; thousands across Asia	**High** ? already sell coaching; USD 300?1,500/yr plausible	Reachable via federations/distributors/social	**Strong** ? already film fixed-angle	B2B/B2B2C	**Best fit** ? buyer, recorder, payer aligned

Clubs & leagues	Denmark 100k+; thousands of clubs Asia/EU	Medium-high; USD 500?3,000/yr	Medium ? committee sales cycle	**Strong**	B2B/B2B2C	Strong secondary; longer cycle
Tournament/event organizers	Many small/regional events	Medium ? per-event/seasonal	Medium ? episodic	**Strong**	B2B	Good per-event upsell; lumpy
Schools & universities	Large (esp. Asia/Japan clubs)	Low-medium ? budget-constrained	Low ? slow procurement	Strong	B2B	Slow; grant/Khelo-India dependent
National/regional federations	Few, high-value	Medium-high; procurement-heavy/political	Low ? long cycles	Strong	B2B	Lighthouse/credibility, not beachhead
Broadcasters / streamers	High value per deal	High	Low ? incumbents entrenched	**Poor ? panning = failure mode**	B2B	**Avoid** ? violates core constraint

Key dynamics: the static-camera constraint is a **B2B filter** ? coaches/academies/ clubs already record from a fixed angle, so the product fits existing behavior (cleanest PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters explicitly violate the constraint ? exclude.

---

4. Competitive & pricing benchmarks

Product	Category	Model	Price	Notes (year)
Veo (Cam 3)	Team-sports auto-filming	Hardware + sub	Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr	15,000+ clubs/80 countries; $115M raised (2025)
Trace	Soccer auto-filming	**Free camera + sub**	USD 180/yr (Basic) ? 300 (Pro)	Camera leased free with sub (2025)
Pixellot (Air)	Auto camera/streaming	Hardware + sub	Camera ~USD 949 + USD 69?167/mo	Institutional/league; 10,000+ clubs (2025)
Spiideo	Auto-filming/analytics	Sub incl. camera	~USD 3,800/yr (1 team) ? 6,100/yr (club)	Institutional budgets (2025)
Hudl	Team video analysis	SaaS tiers	From ~USD 400/yr	Broad team-sports incumbent (2026)
SwingVision	**Tennis/pickleball AI, single iPhone**	Software freemium	**USD 179.99/yr**; free 2 hrs/mo	Closest analog; 200k+ monthly users; $10M+ raised (2025)
BadPro+	**Badminton AI on phone**	**Pay-per-use, no sub**	Per-analysis (? "< one coaching session")	Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025)
Niche pickleball/badminton apps	Various	Mixed	Varies	Fragmented; few true auto-highlight tools (2025)

What buyers actually pay (synthesis): consumer software-only anchors at ~USD 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club, USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ? signaling a real badminton AI market but **unproven subscription WTP** (opportunity + warning). **Positioning:** software-only + BYO-camera undercuts hardware players and sells worldwide via SaaS. Realistic anchors **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline self-improving detection, per-rally stats/tagging, clean licensing) support a **B2B** value story more than a B2C one.

---

5. Recommended beachhead

Ranked by **WTP × participation density × existing recording behavior × static-camera fit × competitive whitespace.**

#1 (strongest) ? Badminton coaching academies & serious players, **India**
20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo India tailwind; academies already record fixed-angle; coaches are buyers who monetize instruction (high relative WTP); incumbents ignore badminton; the only direct badminton AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*. **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not the individual.

#2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts; strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:** **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ monthly users) ? head-to-head differentiation, not greenfield.

#3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan, Korea) + Denmark**
Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea school-club & workplace systems), mature club infra, fixed-court recording. **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

Bottom line: lead with **India badminton academies** (whitespace + behavior + buyer alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C motion**, then expand to **pickleball-US** (out-build SwingVision) and **badminton-Asia/Denmark** (density).

---

6. Risks & open questions (what would invalidate the thesis)

1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies + 50?100 enthusiasts on whether they already record fixed-angle.
2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle, occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real amateur clips; define a "good-capture" spec + auto-reject (ties to `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy INR licensing.
5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports). *Validate:* ship a badminton-quality advantage incumbents can't trivially match (offline self-improving detection, per-rally tagging).
6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/ academy self-serve before institutional deals.
7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5?10 buyer interviews.

Cheapest next steps: (a) 10?15 discovery interviews with Indian badminton academy owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

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7. Sources

- Sports analytics market: Precedence Research; Fortune Business Insights; Grand View Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
  - https://www.precedenceresearch.com/sports-analytics-market
  - https://www.fortunebusinessinsights.com/sports-analytics-market-102217
  - https://www.grandviewresearch.com/industry-analysis/sports-analytics-market
  - https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp
  - https://www.futuremarketinsights.com/reports/sports-analytics-market
- Action camera market: https://www.gminsights.com/industry-analysis/action-camera-market (2024);
  https://www.precedenceresearch.com/action-camera-market (2024);
  https://electroiq.com/stats/action-camera-statistics/ (2024)
- Badminton participation/market: https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/ ;
  https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-

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popular/ ;
      https://en.wikipedia.org/wiki/Badminton ;
      https://www.marketgrowthreports.com/market-reports/badminton-market-110049 ;
      https://www.thesportsschool.com/top-10-badminton-academies-in-india/
    - Tennis: https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-
participation-hits-106-million-in-five-years/ ;
      https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-
gains/
    - Pickleball: https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-
in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-
court-and-facility-demand/ ;
      https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-
nearly-20-million-in-2024/ ;
      https://www.pickleheads.com/guides/pickleball-statistics ;
      https://www.custommarketinsights.com/report/pickleball-market/ ;
https://market.us/report/pickleball-market/
    - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
    - Cricket: https://cricketrise.com/cricket-fans-in-india/ ;
      https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/
    - Competitors & pricing: https://www.veo.com/en-us/pricing ; https://traceup.com/pricing
;
      https://www.pixellot.tv/ ; https://www.hudl.com/pricing ;
https://swing.vision/subscribe ;
      https://www.badproplusapp.com/
```

218. user (2026-06-12T14:26:30.070Z)

```
Found 1 file
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md
```

219. user (2026-06-12T14:27:24.143Z)

```
Switched to a new branch 'docs/pmf-field-signals'
Already up to date.
```

220. assistant (2026-06-12T14:28:24.824Z)

C13 Field Signals ? what academy visits must tell us (and when to change course)

Status: ACTIVE ? the decision layer for the C13 beachhead validation
(owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
visiting academies and reading signals on the ground*).
Companions: [`FIELD_RESEARCH_ASSOCIATE_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
(the recruit hand-out: script + ground rules) ·
[`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture form) ·
[`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).

Why this doc exists: the flyer collects answers; this doc fixes ? **before the
first visit** ? what those answers *mean* and what totals trigger which decision.
Pre-registering the thresholds is the honesty mechanism: it stops us from reading
15 ambiguous conversations as whatever we hoped to hear.

---

1. The hypotheses under test (each maps to script questions)

ID	Hypothesis (from PMF research)	Falsified looks like	Script Q
H1	**Recording behavior exists** ? academies already produce fixed-angle / CCTV / multi-court footage without us	nobody records; "we'd have to start for you"	Q1
H2	**Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)	footage is small, or already gets watched	Q2 ?

H3	**An audience exists** ? coach, player, or parent would actually consume rally clips/reels	shrug; "who has time to watch clips"	Q3
H4	**Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave a gap we fit	"we use X, it's fine, fully solved"	Q4
H5	**Willingness to pay per match** at a price that clears unit economics	"should be free" / <?100 everywhere	Q5 ?
H6	**Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court footage)	multi-court video already useful to them	Q6
H7	**Behavioral commitment** ? they'll let us record on their court (Phase-2 door)	polite no across the board	Q7

Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs roughly ?75?150 to process, so **?300/match has real margin, ?100/match is break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.

2. Per-visit scoring ? GREEN / AMBER / RED anchors

Score each hypothesis on the answer sheet **from what they said and showed, not what you inferred**. When between two grades, take the lower one.

	GREEN (strong confirm)	AMBER (weak/mixed)	RED (disconfirm)
H1 record	shows or names actual setup (tripod/CCTV/phone rig) used ?monthly	records occasionally (tournaments only)	doesn't record; no plans
H2 dead footage	points at real unwatched volume (?~5 hrs, "CCTV runs all day", tournament backlog)	some footage, partially watched	no backlog exists
H3 audience	names WHO would watch and WHEN (e.g. "parents ask for clips", "I review with students Mondays")	"could be useful, maybe"	no consumer identified
H4 whitespace	no tool today, or uses one and volunteers a gap we cover (multi-court, BYO footage, price)	uses a competitor, mildly satisfied	competitor solves it, satisfied, sticky
H5 WTP	volunteers ??300/match unprompted, or proposes a budget framing of their own (per-student/per-month)	lands at ~?100 with hesitation	refuses to name a number / "free"
H6 multi-court	complains about it in their own words before/without prompting	agrees when asked	multi-court footage already works for them
H7 Phase-2	yes + gives a name/number/slot	"maybe, come back"	no

Weighting rule (behavior > words): H7-GREEN and H2-GREEN are *behavioral* signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a polite conversation, not a customer.

Disconfirming-evidence quota: every visit sheet must record at least one verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a sheet has only positives, the visit gets re-done or discounted ? per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

3. Pre-registered decision rules (n = 10?15 academies)

Counts, not percentages ? at this sample size we're reading direction, not estimating proportions. Evaluate after **every 5 completed sheets**; final call at the agreed n. "GREEN" below means GREEN on that hypothesis row.

Verdict	Trigger (pre-registered)	What we then do
PROCEED ? beachhead validated	?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN	Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest
PIVOT A ? "index the backlog" ? "record + index" service	H2 mostly RED (<3 GREEN) but H3+H5 healthy on *fresh* recordings	The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo

PIVOT B ? partner/API motion	H4 RED at ?5 academies (incumbents entrenched and liked)	Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision × PodPlay template from the 2026-06-10 audit)
PIVOT C ? payer is the parent, not the academy	H5 RED from owners but H3 repeatedly names parents/players asking for clips	Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer
FALSIFIED ? beachhead assumption fails	<3 academies record at all (H1) **and** 0 H7-GREEN	The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further

Mixed outcomes that match no row: write up the per-hypothesis tallies and decide explicitly with the owner ? but **the tallies above get computed first**, so the discussion starts from the pre-registered read, not from anecdotes.

Competitor-intel side-channel (always-on): any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update `PMF_MARKET_RESEARCH.md` §4 with field-verified rows (today's table is web-sourced).

4. Mechanics

- One [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) per academy, same day as the visit, in the associate's words + verbatim quotes.
- The owner (or Claude session) transcribes each sheet into one row of the rollup table below (keep it in this doc or a sheet next to it); tallies and the decision-rule read-out update per 5 visits.
- Demo asset if a conversation earns it: the real compiled reel (`output/MahadevpuraSingles_highlights.mp4`) on a phone ? *show, never promise* (flyer ground rule 3 still applies; the associate doesn't carry the demo, the owner does on follow-ups).

Rollup table (one row per completed visit)

#	Academy	Area	Courts	Role spoken to	H1	H2	H3	H4	H5 (?)	H6	H7	Gut	Phase-2?
1													

Tally ? decision read-out (recompute per 5 visits)

Metric	Count	Threshold in play
H2 GREEN (dead footage)	/n	?6 ? PROCEED leg; <3 ? PIVOT A leg
H5 at ?300+	/n	?4 ? PROCEED leg
H7 GREEN (Phase-2 yes + contact)	/n	?3 ? PROCEED leg; 0 ? FALSIFIED leg
H4 RED (incumbent satisfied)	/n	?5 ? PIVOT B
H1 any recording	/n	<3 ? FALSIFIED leg
H5 RED + H3 names parents	/n	recurring ? PIVOT C

5. Guardrails that bind the field work

- **Phase 1 records nothing** ? notes only (flyer rule 7).
- **Phase 2 consent rules are hard:** written consent from the academy AND adult players; **sessions with minors are never recorded for our library** (training-pool guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
- Answer sheets contain personal contact details ? they stay between the associate and the owner; transcribe to the rollup without phone numbers.

221. user (2026-06-12T14:28:24.841Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md (file state is current in your context ? no need to Read it back)

222. assistant (2026-06-12T14:28:46.590Z)

Field Visit Answer Sheet ? one per academy (fill the same day)

> Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
> is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
> Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.

Visit basics

Date / time	
Academy name	
Area (neighbourhood)	
# courts in the hall	
Spoke to (name + role)	
Contact (phone/WhatsApp) ? *only if offered*	
How long the conversation ran	

The seven questions ? their answers (verbatim where possible)

Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?

>

Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?

>

Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?

>

Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)? What's good, what's missing, what does it cost them?

>

Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did THEY say first (?)? What was their face doing?

>

Q6 ? Multiple courts in frame ? does that make video useless today? Their words:

>

Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who to call)

>

Required: the most negative thing they said (verbatim)

>

Best quotes (anything vivid, in their words)

>

Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md §2; when torn, score lower)

Hypothesis	Score
H1 recording behavior	GREEN / AMBER / RED
H2 dead footage	GREEN / AMBER / RED
H3 audience	GREEN / AMBER / RED
H4 whitespace	GREEN / AMBER / RED
H5 willingness to pay ? their ? number: ______	GREEN / AMBER / RED
H6 multi-court pain	GREEN / AMBER / RED
H7 Phase-2 door	GREEN / AMBER / RED

Competitor block (only if they named one)

Product	
What it does well (their words)	
What it misses (their words)	
What they pay	

Your gut read (circle): COLD · POLITE · CURIOUS · EAGER

Phase-2 candidate? (circle): YES ? call: ____________ · MAYBE · NO

Anything else worth knowing (layout, lighting, camera positions you noticed, who really decides)

>

223. user (2026-06-12T14:28:46.601Z)

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indexer\docs\FIELD_VISIT_ANSWER_SHEET.md (file state is current in your context ? no need to
Read it back)
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224. user (2026-06-12T14:29:01.530Z)

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The file C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is
current in your context ? no need to Read it back)
```

225. user (2026-06-12T14:29:06.881Z)

```
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
```

226. user (2026-06-12T14:29:12.274Z)

```
50      3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
51         2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
52         calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
53      4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
54         badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
55         by C3 (provenance multiplies per sport).
56      5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
57         "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
58         (consent at capture; demo served via the Gemini path or human-polish ? the reel
```

pattern exists:
59   `output/MahadevpuraSingles_highlights.mp4`). **Recruiter hand-out ready:
`docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md`**
60   (script, ground rules, Phase-2 consent rules; owner fills ?/contact placeholders).
61   6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
62   deletion in AI Studio); **rename repo off "badminton"** (owner action,
63   in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
64

227. user (2026-06-12T14:29:23.184Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

228. user (2026-06-12T14:30:01.813Z)

Workflow launched in background. Task ID: ww0rvaqf3
Summary: Adversarial critique of the C13 field-validation kit before PR
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\critique-field-signals-kit-wf_774b13f4-334.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\critique-field-signals-kit-wf_774b13f4-334.js"} to iterate without resending the script.)
Run ID: wf_774b13f4-334
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\critique-field-signals-kit-wf_774b13f4-334.js", resumeFromRunId: "wf_774b13f4-334"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

229. user (2026-06-12T14:40:07.359Z)

<retrieval_status>timeout</retrieval_status>

<task_id>ww0rvaqf3</task_id>

<task_type>local_workflow</task_type>

<status>running</status>

230. user (2026-06-12T14:44:52.253Z)

<retrieval_status>success</retrieval_status>

<task_id>ww0rvaqf3</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>
[Truncated. Full output: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\tasks\ww0rvaqf3.output]

te from the 2026-06-10 audit' is unresolvable from the repo",
        "severity": "low",

        "detail": "The PIVOT B row (docs/PMF_FIELD_SIGNALS.md line 70) tells the reader to
follow 'the PB Vision x PodPlay template from the 2026-06-10 audit'. Neither 'PB Vision' nor
'PodPlay' appears anywhere else in the repo ? including docs/PMF_MARKET_RESEARCH.md section 4
(the competitor/pricing table) and docs/COMMERCIALIZATION.md. The reference is real (it is
recorded in session memory, memory/current-state.md lines 388/394: PB Vision $99/yr B2C + $8/hr-
of-video partner API, sold through whoever owns the facility relationship), so it is not
fabricated provenance ? but a repo reader who lands on PIVOT B has no way to discover what the
template is, and the doc that PMF_FIELD_SIGNALS names as 'the hypotheses under test'
(PMF_MARKET_RESEARCH.md) does not contain it.",
        "fix": "Add a one-line footnote in PMF_FIELD_SIGNALS.md describing the template (PB
Vision: $99/yr consumer + $8/hr-of-video partner/API motion through facility owners ? e.g.
PodPlay), and/or add an audit-addendum row to PMF_MARKET_RESEARCH.md section 4 so the citation
resolves in-repo.",
        "why": "Factually verified: 'PB Vision'/'PodPlay' appear exactly once in the repo
(docs/PMF_FIELD_SIGNALS.md:70); PMF_MARKET_RESEARCH.md §4 lists only
Veo/Trace/Pixellot/Spiideo/Hudl/SwingVision/BadPro+; no in-repo artifact of the 2026-06-10
audit's PMF findings exists (BACKLOG.md carries only its engineering items). The provenance is
real but lives outside the repo in session memory (memory/current-state.md:388,394). So the
citation is genuinely unresolvable in-repo, and the kit elsewhere DID inline the same audit's
findings (H4 names Machaxi/VISIST/Game Theory/Stupa, all audit-discovered and absent from the
web-sourced §4 table that PMF_FIELD_SIGNALS:81 itself flags as stale) ? making this the one
dangling audit fact, and the one-line footnote fix correct and cheap. However, severity is
downgraded from medium to low: the PIVOT B row is self-contained where it matters ? the trigger
(H4 RED at ?5) is numeric, and the operative action clause ('explore the engine/API route
through whoever owns the academy relationship') fully specifies the pivot without the citation;
the lost content is only the concrete proof-point/pricing example. The decision-rules audience
is the owner + Claude sessions (the flyer tells the associate that doc 'is ours, not yours'),
both of whom resolve the reference via memory, and NEXT_STEPS.md:11-12 gives an in-repo pointer
to memory current-state.md. PIVOT B also triggers exploration with the owner, not cold
execution. The flaw is real doc-traceability drift (memory is prunable, so a pre-registered
pivot's content could silently vanish), but it cannot distort the 10-15-visit verdict the kit
exists to produce."
      },
      {
        "title": "H5 GREEN anchor ('volunteers >=?300 unprompted') is contaminated by the
script's own anchor ladder",
        "severity": "medium",
        "detail": "PMF_FIELD_SIGNALS.md section 2 grades H5 GREEN only when the academy
'volunteers >=?300/match unprompted'. But the flyer's scripted Q5
(docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md line 40-42) has the associate read out the ladder in the
question itself: 'What feels fair ? ?100, ?300, ?500 per match?', with '(Let THEM say a number
first)' only as a parenthetical afterthought. As scripted, every visit is prompted, making
'unprompted' GREEN nearly unreachable/unscorable ? and it also sits awkwardly next to flyer
ground rule 1 ('Never quote a price yourself'). PROCEED needs >=4 'H5 at ?300+', so this
ambiguity directly feeds the headline decision rule. Related ambiguity: the GREEN anchor's
alternate clause ('proposes a budget framing of their own ? per-student/per-month') produces no
?/match number at all, and the doc never says whether such a visit counts toward the 'H5 at
?300+' tally (which is a number-based metric, not a color-based one).",
        "fix": "Restructure Q5 into two explicit steps in the flyer: (1) ask 'would you pay per
match ? what feels fair?' and wait; (2) only if they will not name a number, offer the
?100/300/500 ladder. Add a checkbox to FIELD_VISIT_ANSWER_SHEET.md Q5: 'number came BEFORE /
AFTER the ladder'. In PMF_FIELD_SIGNALS.md, define 'H5 at ?300+' precisely (recorded ? number >=
300 regardless of prompting; state how budget-framing GREENs convert, e.g. owner computes the
per-match equivalent).",
        "why": "Verified against the actual files; every prong of the claim checks out. (1)
PMF_FIELD_SIGNALS.md §2 line 46 grades H5 GREEN on \"volunteers ??300/match unprompted\", while
FIELD_RESEARCH_ASSOCIATE_FLYER.md lines 40-42 embed the anchor ladder inside the bolded scripted
question itself (\"What feels fair ? ?100, ?300, ?500 per match?\"), with \"(Let THEM say a
number first.)\" only as a parenthetical after the script text. A non-technical associate
instructed to \"ask in this order\" will read the ladder aloud, making every visit prompted ?
the script and the GREEN anchor are self-contradictory, and the signals doc (line 33) shows the
ladder was deliberate, so this is a real internal inconsistency, not a misreading. (2) The
answer sheet's Q5 records \"What number did THEY say first (?)?\" but has no field for whether
the ladder was already read, so \"unprompted\" is unverifiable at transcription/rollup time. (3)
The contamination feeds the headline rule directly: PROCEED (line 68) conjunctively requires

\"?4 H5 at ?300+\", and ?300 is the middle rung of the read-aloud ladder ? classic anchoring
lands compliance answers exactly on the threshold; it also suppresses H5-RED (\"refuses to name
a number\"), biasing against PIVOT C detection. (4) The related ambiguity is real: \"H5 at
?300+\" is a number-based metric never precisely defined, and the GREEN alternate clause (per-
student/per-month budget framing) yields no per-match ? for the sheet's \"their ? number:
______\" field or the rollup's \"H5 (?)\" column; under a number-based reading anchored ?300s
count (contaminated), under a color-based reading GREEN clause 1 is structurally unreachable as
scripted (PROCEED nearly impossible). Attempted refutations fail: the §2 weighting rule (\"H5
answers are talk\") only shapes the gut-read, not the pre-registered tally; \"score lower when
torn\" doesn't fix a number-based tally; the paired first-2-visits onboarding is not a written
protocol fix; and at n=10-15 with a threshold of 4, a couple of anchored answers flip the
verdict ? squarely in scope for a kit whose stated purpose (lines 11-14) is preventing exactly
this motivated-reasoning failure. Minor overstatements (ground rule 1 is contextually about the
product price ? the claim correctly only says \"sits awkwardly\"; GREEN is \"nearly\" not fully
unreachable via the budget-framing clause) don't change the verdict. Severity medium: it
corrupts one of three conjunctive PROCEED legs and the PIVOT C signal, but the double-weighted
behavioral legs (H2, H7) are untouched and the proposed fix is a small pre-visit doc edit (two-
step Q5, before/after-ladder checkbox, precise definition of \"H5 at ?300+\" including budget-
framing conversion) within the kit's own fix-before-first-visit window."
      },
      {
        "title": "PIVOT A and PIVOT C triggers are not actually pre-registered (no numeric
thresholds)",
        "severity": "medium",
        "detail": "PMF_FIELD_SIGNALS.md section 3 claims all decision rules are 'pre-
registered', and PROCEED/PIVOT B/FALSIFIED do carry counts. But PIVOT A's trigger is 'H2 mostly
RED (<3 GREEN) but H3+H5 healthy on fresh recordings' ? 'healthy' is undefined ? and PIVOT C's
trigger is 'H5 RED ... but H3 repeatedly names parents', with the section-4 tally table likewise
saying only 'recurring -> PIVOT C'. Two of the five verdict rows therefore still permit exactly
the motivated reading the doc exists to prevent.",
        "fix": "Quantify both before the first visit, e.g. PIVOT A: '<3 H2-GREEN AND >=6 H3
GREEN/AMBER AND >=4 H5 at ?300+'; PIVOT C: '>=4 visits where H5 is RED and the Q3 answer names
parents/players as the watcher'. Update the tally table rows to match.",
        "why": "Verified against docs/PMF_FIELD_SIGNALS.md on branch docs/pmf-field-signals. The
claim is textually accurate: in the §3 \"Pre-registered decision rules\" table, PROCEED (?6
H2-GREEN, ?4 H5 at ?300+, ?3 H7-GREEN), PIVOT B (?5 H4-RED), and FALSIFIED (<3 H1, 0 H7-GREEN)
carry numeric triggers, but PIVOT A's trigger is \"H2 mostly RED (<3 GREEN) but H3+H5 healthy on
fresh recordings\" with \"healthy\" undefined anywhere in the kit (grep across docs/ confirms
no quantification in the answer sheet, flyer, PMF_MARKET_RESEARCH, NEXT_STEPS, or
COMMERCIALIZATION), and PIVOT C's trigger is \"H5 RED from owners but H3 repeatedly names
parents\" with the §4 tally table saying only \"recurring ? PIVOT C\". It is slightly worse than
claimed: the §4 tally table has no metric row at all for PIVOT A's H3+H5-healthy leg, so that
trigger cannot be computed from the table the doc says is \"computed first\". Refutation
attempts fail: (1) not pedantic ? the doc's stated purpose (lines 12?14) and §3 header are pre-
registration as the honesty mechanism, and NEXT_STEPS.md item 5 advertises \"PROCEED / PIVOT A-C
/ FALSIFIED thresholds fixed before the first visit\", so the kit fails its own bar, and the
three quantified rows prove counts were feasible at n=10?15; (2) not mitigated ? the mixed-
outcomes owner-escalation clause only covers outcomes matching NO row and does not stop a
motivated reader from claiming the vague rows matched; raw data on the answer sheet enables
post-hoc counting, which is exactly what pre-registration is meant to preclude; (3) materially
relevant ? PIVOT A/C are the natural escape hatches when the wedge (H2) or WTP (H5) signal
fails, i.e., exactly where motivated reading would flow. Severity is medium, not high: the
highest-risk verdicts (false PROCEED, FALSIFIED kill-switch) are fully quantified, pivots route
to further validation rather than irreversible build, an owner-in-loop fallback with computed
tallies exists, and the fix is a cheap docs-only edit that can land before the first visit as
the doc itself requires. The proposed fix (numeric thresholds for both rows plus matching tally-
table rows, including adding a tally metric for PIVOT A's H3+H5 leg) is correct and low-cost."
      },
      {
        "title": "Incumbent list mismatch: 'Stupa' is in the H4 hypothesis but missing from the
flyer script and answer sheet",
        "severity": "low",
        "detail": "PMF_FIELD_SIGNALS.md H4 (line 25) lists incumbents '(Machaxi, VISIST, Game
Theory, Stupa, BadPro+)', but the instruments the associate actually uses omit Stupa: flyer Q4
says 'Listen for: Machaxi, VISIST, Game Theory, BadPro+, just YouTube/phone' (line 38), flyer

ground rule 5 says 'Machaxi/VISIST/Game Theory etc.', and FIELD_VISIT_ANSWER_SHEET.md Q4 says 'Machaxi / VISIST / Game Theory / BadPro+ / none'. Stupa is a grounded, audit-verified incumbent ($3.3M, BWF-approved, TT-to-badminton ? memory/current-state.md line 391), so the field instrument should listen for it. Secondary nit: the section-1 table header says 'Hypothesis (from PMF research)', but four of H4's five named incumbents (all but BadPro+) come from the 2026-06-10 audit, not from PMF_MARKET_RESEARCH.md ? whose section 4 table contains none of them (the doc does honestly flag that table as web-sourced/stale in the competitor-intel side-channel note).",
        "fix": "Add Stupa to flyer Q4's listen-for list and to the answer sheet's Q4 line so all three docs name the same incumbent set; optionally annotate H4 that the India incumbent set comes from the 2026-06-10 audit pending PMF_MARKET_RESEARCH section 4 field-verification.",
        "why": "Every factual citation verifies: PMF_FIELD_SIGNALS.md line 25 (H4) names five incumbents including Stupa; the flyer Q4 listen-for list (line 38), flyer ground rule 5 (line 65), and FIELD_VISIT_ANSWER_SHEET.md Q4 (line 33) all carry only four ? Stupa is missing from both field instruments. Stupa is genuinely grounded as an audit-verified incumbent (memory/current-state.md line 391: $3.3M, BWF-approved, TT?badminton). The secondary nit is also accurate (PMF_MARKET_RESEARCH.md §4's table contains only BadPro+ of the five, so the 'from PMF research' header is imprecise for four names ? though PMF_FIELD_SIGNALS line 81 honestly flags that table as web-sourced). Refutation attempts cap the impact but don't dissolve the flaw: Q4 is open-ended, the competitor block and side-channel rule capture ANY named product, and H4's scoring anchors plus the PIVOT B trigger are competitor-agnostic, so no pre-registered verdict can flip on this omission. But the kit's core promise is hypothesis-to-instrument alignment, and the listen-for list's specific function is priming a non-technical associate to recognize unfamiliar brand names ? Stupa being precisely the one with the least Bangalore academy-facing recognizability. A pre-registered five-name incumbent set operationalized as a four-name instrument, with no documented rationale, is a genuine consistency defect with a trivial one-word fix, caught at the right time (pre-merge, before sheets are printed). Severity low, matching the critic's own grading: capture paths and decision mechanics survive intact, so the residual cost is only a modest loss of priming/recognition and doc hygiene."
      }
    ],
    "refuted": [
      {
        "title": "Anchors demand data the 20-minute visit never collects",
        "why": "The claim's factual observations are accurate (no frequency question in flyer Q1/sheet Q1; Q6 is a scripted prompt with no spontaneity flag; H4-RED says \"sticky\"), but its conclusion ? \"in each case the pre-registered boundary is undecidable from the artifact that's supposed to decide it\" ? is refuted for all three cases. (1) H1: the only H1 quantity any decision rule consumes is \"records at all\" (FALSIFIED leg; tally row literally reads \"H1 any recording\"), which Q1 captures directly. The GREEN/AMBER (?monthly vs tournaments-only) boundary feeds NO pre-registered trigger; when frequency evidence doesn't surface organically, the pre-registered tie-break (\"when between two grades, take the lower one\", §2) resolves to AMBER with zero verdict impact. Ground rule 2 additionally arms the associate with the \"when did you LAST record\" recency probe. (2) H6 appears in no decision rule whatsoever (PROCEED/PIVOTs/FALSIFIED consume H1/H2/H3/H4/H5/H7 only) ? it is moat-color, not a verdict input. Spontaneity is also recoverable from the sheet: questions are asked in fixed order, answers are captured verbatim per question, and ground rule 6 instructs capturing spontaneous multi-court complaints in exact words ? a complaint recorded under Q1/Q2/Q4 demonstrably predates the Q6 prompt. The \"leading question\" worry is pre-neutralized by the anchor itself: a prompted yes can never exceed AMBER (\"agrees when asked\"), so the leading wording cannot manufacture GREENs. (3) H4-RED is load-bearing (PIVOT B at ?5), but the decision table's own gloss defines the construct observably as \"incumbents entrenched and liked\", and the sheet's competitor block (what it does well / what it misses / what they pay) plus Q4's cost probe capture the standard one-visit proxies: paying today, satisfied, no gap named. Reading \"sticky\" as requiring a longitudinal retention measurement is the pedantic interpretation; where torn, the tie-break pushes toward RED, which biases toward PIVOT B ? i.e., conservative AGAINST the hoped-for PROCEED, exactly the honesty direction the kit pre-registers. Note also the kit's consistent design pattern: every verdict-consumed quantity has a dedicated capture field (H5's \"their ? number: ______\", H7's \"who to call\", H2's \"roughly how much\"); the fuzzy constructs the critic flags are precisely the non-load-bearing ones. The proposed fixes (frequency line, unprompted checkbox, observable stickiness wording) are cheap polish worth adopting, but their absence does not render any PROCEED/PIVOT/FALSIFIED boundary undecidable for a 10?15-visit directional read."
      },
      {

"title": "Who scores G/A/R is contradictory ? and the associate cannot do it",
"why": "The claimed three-way contradiction does not hold up. Only one document actually assigns the scoring role, and it is unambiguous: the flyer (FIELD_RESEARCH_ASSOCIATE_FLYER.md lines 47-51) enumerates the fields the associate fills ? basics, answers 1-7 verbatim, required negative quote, best quotes, gut read, Phase-2 candidate ? deliberately excluding the H1-H7 score grid, then states \"How the answers get scored and what totals trigger which decision lives in PMF_FIELD_SIGNALS.md ? that part is ours, not yours.\" PMF_FIELD_SIGNALS.md is internally consistent with that: line 11 says \"the flyer collects answers; this doc fixes what those answers mean,\" and section 4 says \"The owner (or Claude session) transcribes each sheet into one row of the rollup.\" The associate is never directed to read PMF_FIELD_SIGNALS.md, so section 2's second-person imperative (\"Score each hypothesis...\") has only one operative reader ? the owner ? making it loose drafting, not a role assignment to the associate. The answer sheet's unfenced \"circle one per row\" grid is the only genuine soft spot, but it is a missing OFFICE-USE label, not a contradiction: the sheet is the single per-visit record feeding the rollup, so owner-applied scores naturally land on it at debrief/transcription. The noise-into-tallies harm chain also fails: (a) the first two visits are paired with the owner, catching any grid confusion immediately; (b) the owner personally transcribes every sheet against the section-2 anchors with the verbatims in hand, so scores are re-derivable regardless of stray circles; (c) the decision-critical inputs are captured mechanically, not judgmentally ? H5's \"volunteers >=Rs300 unprompted\" is operationalized by the sheet field \"What number did THEY say first\" plus the flyer's \"Let THEM say a number first\"; H7-GREEN maps to the sheet's \"Phase-2 candidate? YES ? call: ___\"; H2's ~5hr anchor maps to Q2's \"roughly how much\"; (d) borderline tallies hit the pre-registered \"Mixed outcomes\" row requiring explicit owner discussion. The associate-cannot-use-the-anchors sub-claim is moot under owner scoring and overstated anyway: \"Phase-2 door\" is defined in the flyer's Q7 note, multi-court-as-the-problem-we-solve is ground rule 6, and the Rs100/300/500 ladder is in Q5. Residual: a one-line polish (mark the sheet grid \"OFFICE USE ? leave blank\"; reword section 2 to \"the owner scores\") ? worth doing but pedantic for a 10-15-visit directional read, and the claimed high-severity decision-integrity failure is refuted."
},
{
"title": "Required negative-quote + redo/discount policy + per-visit pay = fabrication incentive",
"why": "The claim's causal chain fails at every link when checked against the actual text. (1) \"Lose pay\" is asserted, not sourced: no document ties compensation to sheet content. The \"re-done or discounted\" sentence (PMF_FIELD_SIGNALS.md §2) is an owner-side data-quality rule whose own justification clause is evidentiary (\"a reasoned 'no' is worth more than five vague yeses\"), i.e. the visit's EVIDENCE gets discounted in the rollup ? and §4 Mechanics puts follow-up visits on the owner, not the associate. The critic's \"undisclosed-or-corrupting\" fork is a false dilemma omitting this natural reading. (2) The \"unsatisfiable quota\" scenario misreads the sheet: FIELD_VISIT_ANSWER_SHEET.md line 52 asks for \"the MOST negative thing they said\" ? a relative minimum that a 15?20 min conversation covering price, competitor gaps, multi-court chaos, and a paid-recording ask virtually always yields honestly. The critic's proposed softening (\"most negative, skeptical, or lukewarm thing\") is essentially what the sheet already says. (3) The kit already states the counter-incentive in the associate's own document ? flyer ground rule 4: \"You're paid per completed honest visit, not per positive answer\" ? and even a worst-case fabricated mild negative quote feeds none of the pre-registered machinery: the PROCEED/PIVOT/FALSIFIED triggers consume only H1?H7 tallies (rollup columns contain no negative-quote field), so the verdict the kit exists to protect is insulated; discounting an all-positive sheet only makes count-based PROCEED thresholds harder, i.e. errs conservative against false-proceed, the exact bias pre-registration targets. (4) \"Completed visit never defined\" is overstated: the flyer functionally defines it (15?20 min owner/head-coach conversation per script + same-day sheet), scripts the junior-coach case, marks compensation \"(discussed individually)\" with owner-filled placeholders, and pairs the first 2 visits to set the bar ? adequate for a 10?15-visit directional engagement with one hands-on owner reviewing sheets over WhatsApp. What survives is a one-word drafting nit (\"discounted\" is ambiguous on a hostile reading; \"(as evidence, not in pay)\" would close it) ? polish, not a fabrication-incentive design flaw, and nowhere near the asserted [high] severity."
},
{
"title": "Demo handling is contradictory: 'show, never promise' an asset the associate doesn't carry",
"why": "The claim fails on all three legs when checked against the actual text. (1) No contradiction in PMF_FIELD_SIGNALS.md §4: the very parenthetical the critic quotes already states the clarification their fix demands ? \"the associate doesn't carry the demo, the owner

does on follow-ups.\" 'Show, never promise' is thus already framed as an owner-on-follow-up rule in the same sentence; §4 is owner-facing mechanics anyway (the flyer tells the associate \"that part is ours, not yours\"). (2) The flyer DOES arm the associate for the \"show me\" moment: ground rule 3 makes the Phase-2 idea (\"a paid recording session + free sample output\", offered only as \"would you be open to it\") the single sanctioned offer, and script Q7 is nearly verbatim the critic's proposed line ? \"Would you let us record one session here ... to show you what the tool produces on YOUR court?\" ? already routed to Q7/Phase-2 capture on the answer sheet and scored as H7. The proposed fix re-creates the existing design. (3) The predicted failure (improvised over-description) is speculative and triple-mitigated: rule 9 bans describing tech beyond the opening line, rule 3 caps what may be offered, and the flyer's first-2-visits-are-paired clause (\"so you're never improvising\") plus rule 1's deflection template train the redirect pattern. The only residue ? adding one verbatim deflection line ? is polish, not a flaw, for a 10?15-visit directional read."
      },
      {
        "title": "Contact collection rules conflict: 'only if offered' vs Q7's required 'who to call'",
        "why": "The cited lines all exist, but they don't conflict in a way an associate must guess through. The 'only if offered' qualifier (FIELD_VISIT_ANSWER_SHEET.md line 16) attaches only to the visit-basics contact field ? the soft-touch rule for a cold research conversation. Q7's 'who to call' (sheet lines 47-48, flyer line 45) is conditional on the academy saying yes/maybe to a paid recording session it was just invited to; agreeing to a follow-up inherently offers the follow-up contact, so noting 'who to call' at that moment is contact-offered-in-context, not a rule violation. The two rules are never simultaneously binding in opposite directions: yes + contact given satisfies both; yes without a contact leaves 'YES ? call: ____' blank, which the kit anticipates by design ? PMF_FIELD_SIGNALS §2 anchors H7-GREEN as 'yes + GIVES a name/number/slot' vs AMBER 'maybe, come back', so a contactless yes gracefully degrades to a weaker score rather than breaking the form. 'Who to call' also need not be a phone number (a name/role suffices), and the flyer's paired-first-2-visits rule calibrates any residual micro-ambiguity in visit 1. The transmission-channel point is out of scope for a two-person, ~15-paper-sheet operation: the guardrail (PMF_FIELD_SIGNALS lines 118-119 + flyer rule 9) constrains audience (associate and Avi only) and the durable artifact (rollup transcribed WITHOUT phone numbers ? the only place data persists into the repo); a WhatsApp photo to Avi's own number does not breach 'stays between the associate and the owner'. The proposed clarifications are harmless polish, but the flaw as stated (an unsatisfiable rules conflict plus a meaningful privacy hole) is overstated and already mitigated by the kit's per-question instruction, the H7 scoring anchor, and the no-numbers-in-rollup rule."
      },
      {
        "title": "'Verbatim' quotes are impossible as instructed: no during-visit note-taking guidance, no language rule, no translated script",
        "why": "Refuted on its central premise and out-of-scope on its residue. (1) The claim's strongest leg ? that quotes must be reconstructed hours later ? misreads the kit: flyer rule 7 bans *recordings* and explicitly sanctions notes as the capture medium (\"you record NOTHING ? notes only\"), and rule 4 is a conversation-time imperative (\"Write down disappointing answers verbatim\"), so contemporaneous jotting is effectively instructed; the same-day rule governs the printed sheet, not the notes. The sheet also hedges itself (\"verbatim where possible\"), so \"impossible as instructed\" is false. (2) The language gap is real as an omission but immaterial to the kit's job: every pre-registered threshold in PMF_FIELD_SIGNALS §3 is computed from G/A/R counts, rupee numbers, and yes/no+contact ? all translation-invariant ? with a conservative score-lower tie-break absorbing noise; verbatim-ness is load-bearing only for the disconfirming-evidence quota, which a translated negative quote still satisfies. (3) The \"anchor-sensitive Q2/Q5\" worry is weakest: Q5's anchors are the numerals ?100/300/500 plus \"let THEM say a number first,\" and Q2's anchor is a footage quantity ? neither drifts in translation. (4) Phrasing-drift across visits presumes multiple interviewers; here one locally-fluent associate, calibrated in the kit's built-in paired first-2-visits with the Bangalore-based owner, asks the same script 10-15 times. A language-of-conversation field and a \"don't translate quotes\" line are cheap polish, but their absence doesn't threaten a directional count-based verdict at n=10-15."
      },
      {
        "title": "Session-cost arithmetic slightly off and the match-vs-session unit is conflated",
        "why": "The surface observations are technically accurate but neither rises to a real flaw in the kit. (1) Arithmetic: ?25-50/hr x 2-3 hr strictly spans ?50-150 and the doc prints

\"roughly ?75-150\" ? but the figure is explicitly hedged (\"roughly\"), is downstream of an
already-rounded $?? conversion, and no conclusion changes at either floor: ?300/match clears
costs and ?100/match stays \"break-even-ish\" whether the session costs ?50 or ?75 to process.
No anchor or threshold moves. (2) Session-vs-match unit: the blur exists in Q5's wording, but it
is defused everywhere it could matter. Treating one session tape as one billed match is the
strictly conservative bound ? every other reading (N matches per tape) has more margin ? so the
only possible error direction is pessimism, the opposite of the motivated-reasoning failure the
pre-registration exists to prevent; no decision rule (?4 H5 at ?300+, PIVOT C) flips on the
interpretation. The kit also deliberately declines to pin the unit: H5 GREEN explicitly credits
respondents who \"propose a budget framing of their own (per-student/per-month)\", and the
answer sheet captures the verbatim Q5 answer plus their ? number, so any respondent-side unit
ambiguity is recoverable at owner-side rollup transcription. For a 10-15-visit directional read
(\"reading direction, not estimating proportions\"), pinning the billed unit is later-stage
pricing work, not this kit's job. The proposed one-line edit is harmless polish, but the claimed
flaw is pedantic and already mitigated by the kit's conservative framing and verbatim-capture
mechanics."
      },
      {
        "title": "FALSIFIED row cites 'PMF section 6 risk 1' for a premise that actually lives
in section 5",
        "why": "Every factual assertion checks out: PMF_FIELD_SIGNALS.md line 72 cites '(PMF §6
risk 1)' for the 'academies already record fixed-angle' premise; that exact phrase lives in
PMF_MARKET_RESEARCH.md §5 beachhead #1 (line 163, echoed in §1 and §3); and §6 risk 1's headline
is amateur tripod adoption whose stated failure mode (B2C TAM collapses to coaches/clubs) treats
academies as the refuge. However, the claim fails on materiality and on the defensibility of the
cite. (1) §6 is the 'what would invalidate the thesis' section ? the natural target for a
FALSIFIED verdict ? and risk 1's validate clause ('observe 3?5 academies ... on whether they
already record fixed-angle') is precisely the observation the FALSIFIED trigger (<3 academies
record at all, H1) operationalizes; the cite reads coherently as 'risk 1's academy-arm
validation failed', not as a wrong pointer. (2) The premise is quoted verbatim in the FALSIFIED
row, so there is no ambiguity about what is being falsified and the §5 source is instantly
greppable. (3) The pre-registration function is untouched: the trigger counts and the consequent
action (revisit beachheads #2/#3, which the critic concedes match §5 exactly) are fixed and
self-contained, so the parenthetical offers no lever for motivated reasoning. The critic's own
wording ('adjacent rather than wrong') concedes the core. This is citation hygiene ? a harmless
one-line polish, not a flaw in the kit ? so per the default for pedantic findings it is
refuted."
      },
      {
        "title": "H5 anchor gap: a named number below ?100 matches neither the RED nor the AMBER
anchor",
        "why": "The literal observation checks out ? a volunteered ?50 matches neither the §2
RED anchor ('refuses to name a number / \"free\"') nor AMBER ('lands at ~?100 with hesitation')
? but the kit already mitigates it in three independent ways, and the claim's materiality
argument is mechanically wrong. (1) The pre-registered tie-break \"when between two grades, take
the lower one\" appears in §2's preamble AND is repeated at the point of use on the answer
sheet's scoring table (\"when torn, score lower\"); a number below AMBER's anchor is exactly the
between-grades case, resolving deterministically to RED ? and the rule can only bias
conservative, never toward PROCEED. (2) §1's falsified column (\"<?100 everywhere\") is in the
same pre-registered doc the scorer reads, fixing the intended interpretation of sub-?100 before
visit 1. (3) Decisively: the claim says the gap matters \"since H5 feeds the PROCEED
threshold,\" but the PROCEED leg is \"?4 H5 at ?300+\", computed from the raw rupee number ?
captured on the sheet (\"their ? number: ______\") and in the rollup column \"H5 (?)\" ? not
from the G/A/R color. ?50 fails ?300+ regardless of grade, so the anchor gap cannot move PROCEED
by a single count. The only grade-consuming H5 rule (PIVOT C) is a qualitative \"recurring\"
pattern read with raw numbers visible in the rollup; a one-cell color ambiguity is below the
noise floor of a doc that explicitly frames itself as \"counts, not percentages? reading
direction\" at n=10?15. The proposed RED-anchor extension is harmless polish, but as a flaw in
the kit this is pedantic and already mitigated."
      }
    ]
  }
}
</output>

231. user (2026-06-12T14:46:06.711Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

232. user (2026-06-12T14:46:17.746Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

233. user (2026-06-12T14:46:20.534Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

234. user (2026-06-12T14:46:22.381Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

235. user (2026-06-12T14:46:33.808Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

236. user (2026-06-12T14:46:36.630Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

237. user (2026-06-12T14:46:40.395Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

238. user (2026-06-12T14:46:47.811Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

239. user (2026-06-12T14:47:00.087Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

240. user (2026-06-12T14:47:13.850Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

241. user (2026-06-12T14:47:25.235Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

242. user (2026-06-12T14:47:30.651Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

243. user (2026-06-12T14:47:47.955Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

244. user (2026-06-12T14:48:39.569Z)

```
========================================================================
 RESULT: PASS - all tests green
========================================================================
warning: in the working copy of 'docs/FIELD_VISIT_ANSWER_SHEET.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'docs/PMF_FIELD_SIGNALS.md', LF will be replaced by CRLF the next time Git touches it
M  docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
A  docs/FIELD_VISIT_ANSWER_SHEET.md
M  docs/NEXT_STEPS.md
A  docs/PMF_FIELD_SIGNALS.md
?? arial.ttf
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
[docs/pmf-field-signals 5548eaf] docs(pmf): C13 field-signals kit - pre-registered decision layer for academy visits
 4 files changed, 263 insertions(+), 9 deletions(-)
 create mode 100644 docs/FIELD_VISIT_ANSWER_SHEET.md
 create mode 100644 docs/PMF_FIELD_SIGNALS.md
remote:
remote: Create a pull request for 'docs/pmf-field-signals' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/pmf-field-signals
remote:
branch 'docs/pmf-field-signals' set up to track 'origin/docs/pmf-field-signals'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/pmf-field-signals -> docs/pmf-field-signals
```

245. user (2026-06-12T14:50:39.599Z)

```
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/104
Full test suite (offline, mocked)	pass	1m25s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27423262453/job/81054174395
Mypy (lenient baseline)	pass	30s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27423262453/job/81054174830
Ruff lint	pass	8s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27423262453/job/81054174839
Syntax sweep + import smoke (no GPU stack)	pass	30s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27423262453/job/81054174412
```

246. user (2026-06-12T14:51:00.334Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

247. assistant (2026-06-12T14:51:18.514Z)

The field-validation machinery for your academy visits is built and waiting for your sign-off on **[#104](https://github.com/avidullu/badminton-highlight-indexer/pull/104)** (docs-only, CI 4/4 green ? left unmerged deliberately, since the thresholds are yours to ratify).

What it adds, on top of the flyer that already existed:

[PMF_FIELD_SIGNALS.md](docs/PMF_FIELD_SIGNALS.md) ? the layer that turns 10?15 conversations into a verdict instead of vibes. Each of the flyer's seven questions now maps to a hypothesis (H1?H7) with GREEN/AMBER/RED anchors, and ? fixed *before the first visit* ? five pre-registered outcomes with numeric triggers: **PROCEED** (camera pilot + UI-alpha Wave-1 seeding), **PIVOT A** (the pain is fresh recordings, not the backlog ? record+index service), **PIVOT B** (incumbents entrenched ? partner/API motion, PB Vision×PodPlay template), **PIVOT C** (parents are the payer, not academies), and **FALSIFIED** (academies don't record ? revisit beachheads #2/#3). Plus a pinned stopping rule so nobody ? including you ? ? can stop at whichever n makes the tallies look best.

[FIELD_VISIT_ANSWER_SHEET.md](docs/FIELD_VISIT_ANSWER_SHEET.md) ? the printable per-visit form the flyer promised but never had: verbatim answers, a *required* most-negative quote, a competitor block, and an office-use score grid (the associate captures, you score).

The adversarial critique (38 agents) earned its cost: it caught that the flyer's Q5 **read the ?100/300/500 ladder aloud** ? and center-bias makes anchored ?300 the modal polite answer, which was *exactly* the PROCEED threshold. The instrument would have manufactured its own pass signal. Q5 is now open-ended-first with the ladder as a recorded fallback, and anchored numbers can never count toward the verdict.

Your runway from here: review/merge #104 ? fill the flyer's ?-per-visit and phone placeholders ? recruit the associate ? first two visits paired with you. Once the placeholders are in, I can render the flyer + answer sheet into print-ready PDFs alongside the khelacharya ones in a minute.

248. user (2026-06-12T14:51:22.378Z)

From a product market perspective, I want to hire a person who would be going to academies and talking to coaches and students to understand how does this project help them and also get some feedback regarding improvements. Full stop. I also think we can have an stretch goal where that person can help record some videos, and we'll run through our pipeline and show them a sample result that is how it will look like for them to use it. First, the UI is... anyway, it's going to be ready by then, and I will have a hosting plan already which can be used to expose a service only to a few people. So I need a good flyer to have a reasonable person to understand this, and I will recruit them.

249. assistant (2026-06-12T14:53:02.496Z)

The spec is now sharper than what's on PR #104: the role is **product**-feedback-first (coaches *and* students, "how does this help you, what should improve"), with the recording-and-show-back as a paid stretch goal ? and the recruiting flyer is the deliverable, aimed at the candidate, not the academy. Since #104 is still open and this amends the same work item, I'll evolve it on that branch (no stacking): a new candidate-facing recruiting flyer, plus the playbook/answer-sheet updates the new framing requires, then an adversarial pass, then push to the same PR.

250. assistant (2026-06-12T14:53:20.042Z)

Field & Product Outreach Associate ? KhelSutra (Sports-Tech, Bengaluru)

Help a Bengaluru sports-AI product meet its first real users.

Duration	~3?4 weeks, part-time (flexible; mostly afternoons/evenings when academies are active)
Location	Bengaluru ? in-person visits to badminton academies
Pay	? ___ per completed academy visit + travel reimbursement; recording sessions (stretch goal below) paid extra ? ? ___ per session
Reports to	Avi (phone/WhatsApp: ___)

The product, in plain words

KhelSutra turns ordinary practice/match video from a badminton academy into ready-to-use pieces: it automatically finds **every rally** in the recording, cuts **highlight reels**, and produces a simple **match report**. It works on video from a phone on a tripod ? no special cameras, no hardware to install. A private web app is opening to a small set of invited academies.

The job, in one paragraph

Visit good badminton academies across Bengaluru and have honest conversations with **coaches AND students**: how do they work with video today, would something like this actually help them, and **what would make it genuinely useful for them** ? including what's missing or wrong with the idea. You work from a provided conversation guide, write down what people *actually say* (the negative answers matter most), and debrief with Avi after each visit. **You are gathering feedback, not selling.** No tech background needed.

The stretch goal (extra pay)

For academies that are interested, you coordinate a simple recording session ? a phone on a tripod, one-page checklist provided. We run that video through KhelSutra, and you go back and show them **their own court, their own players, as highlight reels and a report**: *"this is what it looks like for you."* Then you write down exactly what they say. There is no stronger feedback than that moment.

What you'll actually do

- Visit **10?15 academies** (intro + conversation, ~20?30 minutes each).
- Talk to the **owner/head coach** first, and ? with their permission ? a few **senior or adult students** about whether they'd watch/share their own rally clips and what they'd want from them.
- Fill **one structured answer sheet per visit** (we provide it) ? their words, not yours, including the most negative thing said.
- Show a short sample reel on your phone **only at the point the guide says to** (after the open questions ? it changes people's answers if shown early).
- Debrief with Avi after each visit (WhatsApp/call). **Your first two visits are done together with him**, so you're never improvising.
- *Stretch:* coordinate recording sessions and present the sample results back.

You're a good fit if

- You're comfortable walking into an academy and starting a conversation with a stranger ? and just as comfortable hearing "no, this is useless to us."
- You can capture what people said **accurately** ? quotes, not summaries.
- Kannada / Hindi / English, as the conversation needs (you choose the mix).
- You play or follow a racket sport (helps a lot; not required).
- No technical skills needed. Reliability matters more than anything else.

What you get

- Paid per completed honest visit ? **not** per positive answer. A well-documented "they hated it, here's why" is a fully paid, fully successful visit.
- Ground-floor view of a real sports-AI product finding its market; your field notes directly shape what gets built next.
- Flexible hours, short commitment, clear scope.

Ground rules (the short version ? full version provided on joining)

1. Honesty over positivity. Never oversell, never promise features, prices, or dates.
2. **No recording of any person without written consent** ? and sessions involving **minors are never recorded** for our use, period. In regular visits you record nothing at all: notes only.
3. People's contact details stay between you and Avi.

How to apply

Message Avi at ___ with **"FIELD"**, one line about a sport you follow, and one
line about a time you had to get honest feedback out of someone. First reply gets
a 15-minute call.

251. user (2026-06-12T14:53:20.098Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md (file state is current in your context ? no need to Read it back)

252. assistant (2026-06-12T14:53:31.049Z)

Now reconciling the playbook, answer sheet, and signals mechanics with the new framing:

253. user (2026-06-12T14:53:41.960Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

254. user (2026-06-12T14:53:58.443Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

255. user (2026-06-12T14:54:10.990Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

256. user (2026-06-12T14:54:21.221Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

257. user (2026-06-12T14:54:24.587Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

258. user (2026-06-12T14:54:36.318Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

259. user (2026-06-12T14:54:40.992Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

260. assistant (2026-06-12T14:54:49.297Z)

Two consistency fixes ? the signals doc and NEXT_STEPS still call the playbook "the recruit hand-out":

261. user (2026-06-12T14:54:54.930Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

262. user (2026-06-12T14:55:01.406Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

263. user (2026-06-12T14:55:39.420Z)

Workflow launched in background. Task ID: w5aexakz2
Summary: Adversarial critique of the KhelSutra recruiting flyer + reframed field kit
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_ee69079d-571
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\critique-recruiting-flyer-wf_ee69079d-571.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\critique-recruiting-flyer-wf_ee69079d-571.js"} to iterate without resending the script.)
Run ID: wf_ee69079d-571
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\workflows\scripts\critique-recruiting-flyer-wf_ee69079d-571.js", resumeFromRunId: "wf_ee69079d-571"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

264. user (2026-06-12T15:05:31.561Z)

<retrieval_status>success</retrieval_status>

<task_id>w5aexakz2</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>

{
  "summary": "Adversarial critique of the KhelSutra recruiting flyer + reframed field kit",
  "agentCount": 30,
  "logs": [
    "27 raw finding(s)",
    "12 confirmed of 27"
  ],
  "result": {
    "confirmed": [
      {
        "title": "'Senior or adult students' is ambiguous ? academy 'senior' batches routinely include minors",
        "severity": "low",
        "fix": "Pick the rule (adults 18+ only, per the playbook's skip clause) and say it identically in all three docs: 'adult (18+) students' ? drop the word 'senior', or define it: 'adult students (18+; \"senior batch\" players who are minors don't count)'.",
        "why": "Verified verbatim: flyer lines 43-45 say 'senior or adult students' with no definition and no skip clause; playbook line 61 says 'senior/adult students only' while line 64's 'if everyone available is a minor, skip' implies adults-only; the answer sheet line 64 echoes 'senior/adult only ? skip if only minors available'. The contradiction is real, not pedantic: in Indian academy parlance 'senior batch' routinely includes U-15/U-17/U-19 minors, so under a literal read of line 61+64 an associate may interview a 16-year-old 'senior' whenever at least one adult is also present (the skip only fires when EVERYONE is a minor) ? and that mixed-

batch scenario is the modal evening visit, not an edge case. PMF_FIELD_SIGNALS §5 (the binding
guardrails section) doesn't register the conversation-age rule at all, so nothing elsewhere
resolves it. Visits 3-15 are solo and doc-driven (only the first 2 are paired), so 'owner runs
it personally' doesn't neutralize it. The one-word fix proposed is correct. Severity is lower
than claimed, though: every student conversation is coach-permitted, Phase 1 records nothing, no
student names/contacts are ever collected (the sheet's student block has no identity fields),
student answers feed product feedback only and never the verdict tallies, and the owner debriefs
after every visit ? so the worst realized outcome is a benign, anonymous, coach-permitted chat
that gets caught and corrected within a visit or two. The hard minors guardrail (no recording,
CLAUDE.md training-pool rule) is clean in all four docs. Real consistency defect worth the
trivial fix before printing; mild practical consequence."
      },
      {
        "title": "'First reply gets a 15-minute call' parses two ways",
        "severity": "low",
        "fix": "Reword: 'Next step: a 15-minute phone call with Avi.'",
        "why": "The sentence at docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md:82-83 (\"First reply gets
a 15-minute call\") genuinely parses two ways, and the first-come-first-served reading is the
dominant grammatical parse (subject = \"first reply\"), not a strained one. FCFS cannot be the
intended meaning: the same paragraph asks for two composed application lines, which is
incompatible with a speed race, and calling only the literal first replier would defeat the
hire. No other document disambiguates ? this is the sole candidate-facing artifact (the playbook
at FIELD_RESEARCH_ASSOCIATE_FLYER.md:3 routes recruiting through it), and the misread occurs
before the candidate contacts the owner, so personal recruiting only partially mitigates. The
flyer's own audience framing (non-technical, Kannada/Hindi/English mix) and the owner's spec
(\"a reasonable non-technical person must understand the gig\") put the non-native-reader
concern squarely in scope. Impact is small (a few rushed or self-deselected applicants for a
one-person gig) and the fix is a trivial reword, so severity is low ? but it is a real copy
defect in the call-to-action line, not a pedantic invention."
      },
      {
        "title": "Time promised to the coach (15 min) vs time the flyer/playbook budget (20-30
min) ? the honesty-first script underquotes",
        "severity": "low",
        "fix": "Align the script's ask with the budget: '...Could I ask you a few questions?
About 15-20 minutes, and we're not selling anything today.'",
        "why": "All citations check out: the playbook's opening script
(FIELD_RESEARCH_ASSOCIATE_FLYER.md lines 35-36) promises the coach \"Takes 15 minutes\", while
the same file's summary (line 20) budgets a \"20-30 minute conversation with the owner/head
coach\", the recruiting flyer (PRODUCT_FIELD_ASSOCIATE_FLYER.md line 42) budgets \"~20-30
minutes each\", and ground rule 8 (line 91) says leave at 20 minutes. The best refutation ? that
15 min covers only the coach Q&A while 20-30 covers the full visit (intro + separate 5-min
student block + wrap) ? fails because the playbook's own wording assigns the 20-30 minutes to
the coach conversation itself, the demo and its follow-up sit inside that conversation (after
Q6, before Q7), Q5 scripts deliberate waiting, and rule 5 says spend extra time with competitor
users. So by the kit's own numbers the expected coach-facing time is ~20+, and the 15-minute
quote sits directly under a header reading \"keep it this honest\" in a kit whose core mechanic
is methodological honesty (pre/post-ladder tick, mandatory verbatim negatives, paid-per-honest-
visit). It is a small but real internal inconsistency, not pedantry, and is not mitigated
elsewhere. Severity is low: doorstep quotes are conventionally approximate, the coach controls
the interaction, rule 8 caps the overshoot at ~5 minutes, and the worst outcome is a minor
credibility scuff ? the proposed one-phrase fix (\"about 15-20 minutes\") resolves it at zero
cost."
      },
      {
        "title": "Doc naming drift: the 'playbook' file is still named ..._FLYER.md and
PMF_FIELD_SIGNALS cites 'flyer rule N' meaning the playbook",
        "severity": "low",
        "fix": "Rename FIELD_RESEARCH_ASSOCIATE_FLYER.md to FIELD_VISIT_PLAYBOOK.md; update the
references in PRODUCT_FIELD_ASSOCIATE_FLYER.md (none today ? verify),
FIELD_VISIT_ANSWER_SHEET.md line 4, and PMF_FIELD_SIGNALS.md (companions block, 'flyer rule 4',
'flyer rule 7' -> 'playbook rule ...').",
        "why": "Every factual element verified on disk: docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
line 1 is titled \"Field Visit Playbook\" yet the file keeps the _FLYER.md name;
PMF_FIELD_SIGNALS.md line 73 (\"the flyer's ground rule 4\") and line 161 (\"flyer rule 7\")

resolve only against the playbook's 9 ground rules ? the candidate flyer (PRODUCT_FIELD_ASSOCIATE_FLYER.md) has just 3; FIELD_VISIT_ANSWER_SHEET.md line 4 names the FLYER-named playbook as \"Companion script\". Worse than the critic stated, PMF_FIELD_SIGNALS' own companions block (lines 6-9) explicitly defines PRODUCT_FIELD_ASSOCIATE_FLYER.md as \"the recruiting flyer\", then the body uses \"flyer\" (lines 13, 73, 75, 161) to mean the OTHER file ? the doc contradicts its own legend, a genuine internal inconsistency, not mere style drift. Verified the candidate flyer has zero references to the playbook (rename-safe), and found one extra reference the critic missed: NEXT_STEPS.md line 62 also cites the FLYER-named playbook. Severity stays low because (a) misresolution fails loudly ? the candidate flyer has no rule 4/7, so nobody can silently grab a wrong rule; (b) both citations restate their content inline, so the pre-registered verdict machinery cannot be corrupted; (c) all markdown links use exact filenames and resolve correctly ? only prose \"flyer\" is ambiguous; (d) it is internal-only, never candidate/associate-visible. Not refuted as pedantic: the reframe left a real stale-terminology hazard in the decision-layer doc, nothing elsewhere in the kit handles it (the companions block amplifies the contradiction), and the proposed rename + ~5-line reference touch-up (plus NEXT_STEPS.md:62) is the correct cheap fix."
      },
      {
        "title": "\"A private web app is opening to a small set of invited academies\" overstates reality and pre-promises the campaign's own verdict",
        "severity": "medium",
        "fix": "Soften to future/conditional in both docs: 'we're building a private web app that a few invited academies will be able to try'. In the playbook's Phase-2 section, add a scripted reply for 'can we keep using it?': capture it verbatim as the strongest possible signal, say 'I'll pass that straight to Avi', and promise nothing ? consistent with ground rule 3.",
        "why": "Every factual citation verified on disk. Both associate-facing docs assert as present fact \"a private web app is opening to a small set of / a few invited academies\" (PRODUCT_FIELD_ASSOCIATE_FLYER.md L19-20; FIELD_RESEARCH_ASSOCIATE_FLYER.md L18-19). Repo reality: SPA M1-M4 not started, PR-3/PR-4 open, GATE GL/GD unpassed (UI_ALPHA_EXECUTION_PLAN.md), and both HOSTING_PLAN.md and UI_PROPOSAL.md are \"PROPOSAL ? awaiting owner ratification\". The circularity is genuine: PMF_FIELD_SIGNALS §3 gates academy Wave-1 invites on a PROCEED verdict (Phase R Wave-1 = C13/camera-pilot contacts, which exist only on PROCEED), so the flyer states as fact the outcome this very campaign is pre-registered to decide ? in a kit whose decision doc calls pre-registration \"the honesty mechanism\". The unscripted-moment gap also checks out: the Phase-2 reaction sheet anticipates \"whether they asked to keep using it\" (playbook L103), ground rule 3 forbids promising, ground rule 1 scripts the analogous price question ? but no script exists for the access question, while the associate's own briefing tells them access for academies is already a fact. Refutation attempts cap severity, not existence: the owner's \"UI will be ready by then\" defends the app timeline but not \"invited academies\"; the verdict machinery itself is insulated (academy-facing script never mentions the app, ground rule 9 limits tech talk, H1-H7 anchor to pre-demo answers, demo/Phase-2 data are explicitly non-verdict product feedback, first two visits are owner-paired), so the pre-registered tallies cannot actually be flipped by this sentence. Not pedantic ? it is an internal contradiction between companion docs in the same change, stated in the associate's day-one operating doc, priming a good-faith over-promise at the show-back conversation the playbook itself calls \"the single most valuable artifact\". Fix is a two-sentence soften plus one scripted reply. Real, but medium: framing/consistency flaw with a plausible field consequence, not a verdict-corrupting or consent-level defect."
      },
      {
        "title": "\"Simple match report\" claims a deliverable that doesn't exist ? and it's baked into the H5 willingness-to-pay question",
        "severity": "medium",
        "fix": "Describe what exists: 'a simple summary ? how many rallies, how much actual play time' ? in the flyer, the playbook gig paragraph, Q5, the sheet's Q5 block, and the Phase-2 show-back description. If a coach-facing match report is intended for the show-back, make it an explicit prerequisite in the Phase-2 briefing rather than an implied current feature.",
        "why": "Every cited line checks out on disk: the candidate flyer (line 18) and playbook gig paragraph (lines 17-18) both claim, in present tense, that the product \"produces a simple match report\" / \"generates match reports\"; the phrase \"match report\" appears nowhere else in the repo ? no doc, plan, or code defines such a feature. What actually exists (output\\kushagra_1280.summary.md + backend/reporting/spec.yaml) is a per-video PROCESSING report: highlight count, seconds of play, % of video, pipeline-diagnostic \"things to know,\" and a \"? Your highlights are ready\" verdict ? zero match content (no scores, points, or player

stats). UI screen A6 \"Report\" (UI_PROPOSAL.md:48) would render those same artifacts and is unbuilt (M4 unchecked in UI_ALPHA_EXECUTION_PLAN). The market context makes the connotation gap concrete: the script primes Q4 with Stupa, whose actual product is match-stat reports. The claim's mechanism holds: (a) the \"report\" is baked into the Q5 WTP bundle (\"rally cuts + reel + report\", playbook 46-47, answer sheet line 38), and the demo ? the only corrective artifact ? comes AFTER Q5 and shows only the reel, so every H5 ? number across all 10-15 visits prices a bundle whose third item is overstated; H5-at-?300+ is one of three pre-registered PROCEED legs, and the signals doc polices anchoring on this exact question down to a pre/post-ladder tick-box while leaving the offer description inflated ? a systematic optimistic bias inside the honesty machinery itself; (b) the non-technical associate's only briefing source says \"generates match reports,\" so the obvious coach follow-up (\"what's in the report?\") gets answered from an inflated model, violating the kit's own ground rules 1/3 by construction; (c) the Phase-2 show-back (playbook 102) scripts \"reel + report\" where the deliverable is the processing summary. Refutation attempts partially succeed at capping severity but not at dismissing the flaw: the claim's headline is slightly overstated (a report artifact DOES exist ? it is the \"match\" framing that inflates, not a wholly missing deliverable); the coach-facing Q5 wording is the hedged \"a simple report\"; rally cuts + reel dominate the priced value; stated WTP is explicitly downstream-gated by the pre-agreed fake-door/paid-pilot before any build spend; and the show-back's core asset (their own players' reel) is real, so \"underdeliver\" at Phase-2 is overdramatized. Not pedantic, not handled elsewhere (no kit text tells the associate what the report actually contains), and squarely in scope ? the fix is a cheap rewording in ~5 places of unused docs. Net: real flaw, materially touching a verdict-gating signal, with genuine but partial mitigations ? medium."
      },
      {
        "title": "\"Senior or adult\" students is a minors loophole ? 'senior' batches in Indian academies are routinely under-18",
        "severity": "low",
        "fix": "Replace 'senior/adult' with 'adult (18+)' in all three docs, and add one gating line to the playbook's student section: ask the coach to point out which students are 18 or older; if none, skip and note it (the sheet already supports the skip).",
        "why": "All citations check out on disk: 'senior or adult students' (PRODUCT_FIELD_ASSOCIATE_FLYER.md lines 43-45), 'senior/adult students only' (FIELD_RESEARCH_ASSOCIATE_FLYER.md lines 22 and 61), 'senior/adult only' (FIELD_VISIT_ANSWER_SHEET.md line 64). The disjunction 'senior OR adult' only carries meaning if seniors can be non-adults, and in Indian academy parlance the 'senior batch' routinely contains 15-17-year-olds ? so the natural reading licenses interviewing a minor. The playbook's own skip rule ('if everyone available is a minor, skip', line 64) proves adults-only intent, which the eligibility wording directly undercuts; no doc defines 'minor'/'adult' or says 18+, and there is no instruction to have the coach identify adults. Not handled elsewhere: PMF_FIELD_SIGNALS.md gates recording (§5) and contact collection (§4) but never interview age. Refutation attempts fail on the core point. However, severity is capped at LOW rather than the claimed medium: the binding CLAUDE.md minors guardrail (training pool/recordings) is fully intact in all docs; the worst case the loophole permits is a coach-permitted, unrecorded, notes-only, zero-PII 5-minute conversation whose answers feed product feedback only (never verdict tallies) ? not illegal in India, not a training-data or GDPR event. Exposure is internal inconsistency plus academy/parent-goodwill optics, further bounded by paired first visits and per-visit owner debriefs. The proposed fix (replace with 'adult (18+)' in three files plus one coach-gating line) is correct and proportionate."
      },
      {
        "title": "Associate-facing docs link into the scoring/economics layer, undermining the pre-registration design",
        "severity": "low",
        "fix": "Remove the PMF_FIELD_SIGNALS references from the playbook (line 70 already says 'that part is ours, not yours' ? keep that sentence, drop the link) and from the sheet's header; keep only the one-way signals?playbook/sheet references. Move the OFFICE-USE score grid off the printed sheet entirely ? it lives in the rollup at transcription (PMF_FIELD_SIGNALS.md §4 already supports this).",
        "why": "The factual core checks out: FIELD_RESEARCH_ASSOCIATE_FLYER.md line 70 links the associate's day-one playbook directly to PMF_FIELD_SIGNALS.md, and FIELD_VISIT_ANSWER_SHEET.md line 5 cites \"Scoring anchors (G/A/R): PMF_FIELD_SIGNALS.md §2\" in the header of the form the associate prints and carries ? while PMF_FIELD_SIGNALS.md §2 itself stipulates \"the associate only captures answers.\" Following either pointer hands the interviewer the ?300-pre-ladder success bar, the ?25?50/hr COGS floor, and the PROCEED tallies ? knowledge that adds nothing to

their job and creates expectancy-bias risk on precisely the most behavior-sensitive tally (H5
pre-ladder ?300+, ?4 ? PROCEED), which debriefs cannot police (pause length, transcription
shading, the BEFORE/AFTER tick). That is a genuine self-contradiction in the kit, and the
proposed fix is cheap, correct, and supported by signals §4 (scoring already lives at
transcription/rollup). However, the claim overstates both mechanism and severity. (1)
\"Undermines the pre-registration design\" is wrong: pre-registration binds the SCORER's
interpretation (the doc's own stated purpose) and remains binding regardless of who reads the
thresholds. (2) The kit already mitigates associate steering substantially: ground rule 1
forbids voicing any price; the ladder values and the fact that pre-ladder numbers matter are
unavoidably in the script itself; pay is explicitly decoupled from positive answers; first two
visits are paired with the owner; a disconfirming-quote quota discounts all-positive sheets; and
§2 pre-agrees a fake-door/paid-pilot test before any PROCEED-triggered build spend ? so even a
corrupted H5 tally is backstopped. (3) The \"coach can read the grid in the academy\" point is
weak: the printed grid (sheet lines 78?88) is a label-only OFFICE-USE skeleton with no anchors
or thresholds, reveals nothing beyond what the seven questions ask the coach directly, and the
playbook has the sheet filled after the visit. Net: a real doc-hygiene defect worth the two-line
fix (drop the two associate-facing pointers; optionally move the grid to the rollup), but not a
design-undermining flaw ? severity low, not medium."
      },
      {
        "title": "Opening script promises 15 minutes; every other number in the kit says
20-30+",
        "severity": "low",
        "fix": "Align the opening to 'about 20 minutes' (or explicitly script the 15-minute core
+ 'a couple more minutes if it's interesting to you' extension). Make flyer, gig paragraph,
opening, and rule 8 quote the same number.",
        "why": "All four citations verify exactly on disk: the playbook's scripted opening
(docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md line 35) says \"Takes 15 minutes\" under the heading
\"Opening (keep it this honest):\", while the same doc's gig paragraph (line 20) says \"20?30
minute conversation\", ground rule 8 (line 91) says \"20 minutes, thank them, leave\", and the
recruiting flyer (docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md line 42) says \"~20?30 minutes each\".
Refutation attempts only partially land: (a) \"ask for 15 minutes\" is a normal door-opener
convention and any overrun is consensual ? this caps severity but doesn't erase the
inconsistency; (b) the 20?30 figure includes the student block, but the coach's own mandatory
scripted portion (Q1?Q6 incl. the deliberately slow Q5, the required demo + reaction + follow-
up, Q7) realistically exceeds 15 minutes, and rule 8's own cap (20) already exceeds the ask; (c)
rule 8 is a cap, not a promise ? but a cap above the scripted ask still means the kit plans more
time than its first sentence states. Given the kit's explicit honesty framing (the opening is
literally labeled the honesty benchmark, and the doc is otherwise obsessive about verbatim
precision and pre-registration), the number mismatch is a real, verifiable internal
inconsistency with a trivial fix. The claim's \"2x undersell\" is overstated (1.33x vs the kit's
own 20-min cap; 2x only vs the top of the whole-visit range), and no coach is materially harmed
? hence low, not medium."
      },
      {
        "title": "\"Finds every rally\" asserts perfect recall the product's own correction loop
exists to disclaim",
        "severity": "low",
        "fix": "Drop the absolutism and sell the loop instead: 'it found and cut these rallies
automatically ? and a coach can fix any it gets wrong in a couple of clicks'. That is both
honest and a better demo line, since the correction loop is a differentiator.",
        "why": "Verified verbatim: PRODUCT_FIELD_ASSOCIATE_FLYER.md lines 17-18 (\"automatically
finds **every rally**\") and FIELD_RESEARCH_ASSOCIATE_FLYER.md lines 53-55 (scripted demo line
\"every rally found and cut automatically\", said to coaches at the demo moment, plus the same
phrase in the gig paragraph). The repo's own machinery contradicts the absolute: NEXT_STEPS.md
quality table shows best-path F1 of 0.93 clean / 0.66 multi-court ? and multi-court is the
script's own favorite topic (Q6/H6) ? while UI_PROPOSAL A4/B6 define the Accept/Reject/nudge
correction loop precisely because output needs fixing; NEXT_STEPS even allows the demo reel to
be human-polished, making \"cut automatically\" doubly shaky. The \"marketing shorthand\"
refutation fails on the kit's own terms: this kit is built on engineered honesty (flyer rule 1
\"Never oversell\", playbook rule 3 \"show, never promise\", paid-per-honest-visit,
disconfirming-evidence quota, pre-registered thresholds), so a verbatim scripted overclaim is
internally inconsistent, and Phase-2 show-back on the academy's own multi-court footage (~0.66
F1) will visibly contradict the Phase-1 claim to the same coach ? plus a coach anchored on \"it
finds everything\" is less likely to volunteer accuracy complaints, which is exactly the

improvement feedback the owner's spec asks for. Severity stays low: the verdict machinery is
already insulated (demo reactions are product feedback, not scored signals; H1-H7 anchored pre-
demo), first visits are owner-paired, and the fix is a one-word edit. Note the critic's proposed
replacement line (\"coach can fix any it gets wrong in a couple of clicks\") would itself
promise an unshipped feature (A4 rally review is future UI work, alpha is at PR-2), violating
playbook rule 3 ? the minimal fix is just dropping \"every\" ? but the flawed remedy doesn't
invalidate the finding."
      },
      {
        "title": "PMF_FIELD_SIGNALS still says 'flyer' where it means the playbook ? rule-number
references now resolve to the wrong document",
        "severity": "low",
        "fix": "In PMF_FIELD_SIGNALS.md, replace 'flyer rule 7' with 'playbook rule 7', 'the
flyer's ground rule 4' with 'the playbook's ground rule 4', and 'the flyer collects answers'
with 'the playbook scripts the questions and the sheet collects answers'. Add 'trim the demo
reel to ~60s' to the day-one prep list so the asset assumption is real before visit 1.",
        "why": "Verified against the files on disk: PMF_FIELD_SIGNALS.md's own companion block
(lines 6-9) defines PRODUCT_FIELD_ASSOCIATE_FLYER.md as \"the recruiting flyer\" and
FIELD_RESEARCH_ASSOCIATE_FLYER.md as \"the playbook\", yet lines 161 \"(flyer rule 7)\" and 73
\"the flyer's ground rule 4\" cite rule numbers that exist only in the playbook's 9-item ground-
rules list (its rules 7 and 4 match the cited content exactly), while the recruiting flyer has
only 3 ground rules (recording = rule 2, no rule 4 or 7). Line 13 \"the flyer collects answers\"
is likewise stale (the sheet collects; the playbook scripts). The aside also checks out:
output\\MahadevpuraSingles_highlights.mp4 exists but only as the 125 MB full reel ? no ~60s trim
? while playbook line 113 assumes the trimmed reel is on the associate's phone before visit 1.
Attempted refutation (the playbook's filename contains \"FLYER\", so \"flyer\" could mean it)
concedes rather than dissolves the ambiguity, since the doc's own header binds \"flyer\" to the
recruiting doc. Not pedantic: this is a docs-only branch where cross-reference integrity among
the four companion docs is the deliverable. Severity stays low because the impact is nil
operationally ? both citations restate the rule content inline, the doc's only readers are the
owner/Claude (the associate never sees it), and no field behavior or verdict computation can be
misled; worst case is brief reader confusion. The proposed wording fixes are correct and
trivial."
      },
      {
        "title": "Stretch-goal pay and role boundary inconsistent across the two docs and
underspecified within them",
        "severity": "low",
        "fix": "Align both docs on one structure and price it to the real time: Phase-2 = (a)
recording session, associate present for setup + duration, ?X flat; (b) show-back presentation,
paid as a normal visit; (c) the academy's fee is paid by Avi directly (UPI to the academy), the
associate never carries money. Delete \"(discussed individually)\" from the playbook or move
negotiation to genuinely individual extras only.",
        "why": "A narrow kernel of the claim is real: the flyer
(PRODUCT_FIELD_ASSOCIATE_FLYER.md line 9) promises a rate-card structure for the stretch goal
(\"paid extra ? ? ___ per session\") while the day-one playbook
(FIELD_RESEARCH_ASSOCIATE_FLYER.md line 10) reframes the same pay as \"paid extra (discussed
individually)\" ? a genuine cross-doc mismatch in how Phase-2 compensation is structured, seen
by the same person at recruitment vs onboarding. However, the rest of the claim fails
adversarial checking: (1) the within-Phase-2 underspecification (session duration, show-back-
trip pay) is explicitly handled by the playbook's Phase-2 heading \"separate briefing & pay\"
(line 97) ? the kit deliberately defers those details to an opt-in-triggered briefing, which is
proportionate for a contingent stretch goal in a 3?4 week owner-run gig; and (2) the money-
handling \"trust risk\" misreads the text ? \"we pay the academy a small fee\" (lines 58, 99)
uses \"we\" = Avi/KhelSutra, consistent with \"we provide the checklist\"/\"we process the
footage\" throughout; nothing assigns cash-carrying to the associate. With two of three sub-
claims refuted and the survivor being a one-line fix whose practical harm is minimal (same owner
authors both docs, fills both blanks, recruits personally, pairs the first two visits, and
debriefs after every visit), the claim is real but overgraded at medium; severity is low."
      }
    ],
    "refuted": [
      "Flyer never says who finds/schedules the academies, and 'completed visit' (the pay unit)
is undefined",
      "Stretch goal: 'coordinate a recording session' hides whether the associate operates the

camera for hours",
"Candidate cannot size the gig: no hours-per-week or visits-per-week estimate, no payment timing/method",
"Role title 'Field & Product Outreach Associate' mis-signals a sales/BD job",
"Four template blanks with no fill-before-print checklist ? a blank handed flyer has no pay and no apply path",
"No legitimacy anchor ? an unnamed product with blanks for contact reads scam-adjacent to careful candidates",
"Candidate-facing promises diverge from the day-one playbook: Phase-2 pay structure and paired-visit format",
"Pay is 'per completed visit' but 'completed' is undefined ? junior-coach-only visits are a dispute waiting to happen",
"Phase-2 consent bundles demo use and training-library retention into one 'building the tool' consent",
"Application step screens for speed, not for the skills the role lives on",
"\"Completed visit\" ? the unit of pay ? is undefined, and the office-side definition silently diverges",
"Logistics candidates always ask are missing: start date, cadence, who supplies the academy list, walk-in vs. appointment, transport mode",
"Value proposition omits \"what happens after the gig\" ? even though the plan already contains an honest answer",
"Per-visit effort is framed as the 20-30 minute conversation; the paid unit is actually several hours",
"Thin legitimacy signals for a candidate deciding at home, and one product claim ahead of shipped reality"
]
}
}
</output>

265. user (2026-06-12T15:06:23.690Z)

RM docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md -> docs/FIELD_VISIT_PLAYBOOK.md
?? docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md

266. user (2026-06-12T15:06:35.518Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

267. user (2026-06-12T15:06:37.853Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

268. user (2026-06-12T15:06:40.367Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

269. assistant (2026-06-12T15:06:48.800Z)

Now the playbook (renamed file):

270. user (2026-06-12T15:06:53.108Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

271. user (2026-06-12T15:06:59.164Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

272. user (2026-06-12T15:07:11.443Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

273. user (2026-06-12T15:07:02.970Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

274. user (2026-06-12T15:07:05.308Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

275. user (2026-06-12T15:07:07.944Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

276. assistant (2026-06-12T15:07:20.428Z)

The rename made the new path unread ? reading it, then re-applying:

277. user (2026-06-12T15:07:21.652Z)

```
# Field Visit Playbook ? academy conversations (KhelSutra)

> **Recruiting?** Hand candidates [`PRODUCT_FIELD_ASSOCIATE_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
> ? that's the role description. **This document is the working playbook the hired
> associate gets on day one:** the conversation script, the ground rules, and the
> Phase-2 (recording / show-back) procedure.

**Duration:** ~3?4 weeks, part-time (flexible hours, mostly afternoons/evenings when academies are active)
**Location:** Bengaluru ? in-person visits to badminton academies
**Compensation:** ? ___ per completed academy visit + travel reimbursement; Phase-2 sessions paid extra *(discussed individually)*
**Reports to:** Avi (phone/WhatsApp: ___ )

---

## The gig in one paragraph

KhelSutra takes ordinary practice/match video from a badminton academy and automatically finds every rally, builds
highlight reels, and generates match reports; a private web app is opening to a few invited academies. We need honest,
on-the-ground feedback from real academies ? **how they work with video, whether this actually helps them, and what
would make it genuinely useful**. **Your job: visit 10?15 good academies, have a friendly 20?30 minute conversation
with the owner/head coach (and, with permission, a few senior/adult students) using the script below, and write down
what they actually say.** No selling. No tech knowledge needed. For interested academies there's a Phase-2 follow-up
(a recording session ? then you return and show them THEIR OWN footage as highlights) with additional compensation.

---

## Phase 1 ? The conversation (what you actually do)

**Who to talk to:** the academy owner or head coach ? the person who decides what the academy spends money on.
If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.
```

Opening (keep it this honest):
> "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to understand how you work with video today. Could I ask you a few questions? Takes 15 minutes, and we're not selling anything today."

The questions (ask in this order; the starred ones matter most):

1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?
2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings, tournament days, CCTV) ? roughly how much?
3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?
4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, Stupa, BadPro+, just YouTube/phone.) What do you like / what's missing in it?
5. ? **If you could hand over a full session's video and get back every rally cut out + a highlight reel + a simple report ? would you pay per match for that? What would that be worth to you, per match?** Then **stop talking and wait** ? let THEM name the number. If they genuinely won't after a pause and one "roughly?", you may offer the ladder: "?100, ?300, ?500?" **On the sheet, tick whether their number came BEFORE or AFTER you offered the ladder** ? it changes what the answer means. (Note their face, not just the words.)
6. Your hall has multiple courts running at once, right? Does that make video less useful today?

? NOW (and only now) show the sample reel ? a ~60-second real highlight reel we put on your phone. Say: *"this is what it produces ? every rally found and cut automatically."* Then shut up and watch their face. Write down their FIRST reaction verbatim, and ask one follow-up: **"what would this need to do for it to be useful for YOUR academy?"** (Improvement asks, in their words, go on the sheet. Showing it earlier anchors every answer above ? never lead with it.)

7. Would you let us record one session here (we pay a small fee for the time) and come back showing what the tool produces on YOUR court, your players? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

Students (with the coach's permission ? senior/adult students only, 5 minutes): ask 2?3 of them: (a) *"Would you watch clips of your own rallies if they existed after each session?"* (b) *"Would you share them ? with whom?"* (c) *"What would you want them to show?"* Capture their answers in the sheet's student block. Don't collect students' contact details; if everyone available is a minor, skip the student questions ? note that instead.

After each visit, fill ONE answer sheet ? the template is [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (print one per visit): academy name, location, # courts, person spoken to + role + contact, answers 1?7 (their words, not yours), the most negative thing they said (required), exact quotes, your gut read, and whether they're a Phase-2 candidate. *(How the answers get scored and what totals trigger which decision lives in

[`PMF_FIELD_SIGNALS.md`](PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just capture honestly.)*

Ground rules & gotchas (read twice ? this is the difference between useful and useless)

1. **You are researching, not selling.** If they ask "how much will it cost", say "that's exactly what he's trying to figure out ? what would feel fair to you?" Never quote a price yourself.
2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did you LAST record a session, and what happened to that file?" gets the truth. Prefer the second kind of question.
3. **Don't promise anything.** No features, no dates, no free trials. You may SHOW the standard sample reel (it's real output ? at the scripted moment only, after Q6) and you may offer the Phase-2 idea (a paid recording session + free sample output of their own court), and only as "would you be open to it". Nothing else.
4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a reason is worth MORE to us than five vague yeses. You're paid per completed honest visit, not per positive answer.
5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.** Ask what it does well, what it doesn't, and what they pay. Spend extra time here.
6. **Multi-court is our favorite topic.** If they complain that video is useless because "all the courts are in frame and it's chaos" ? capture that complaint in their exact words. That's the problem we solve.
7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ? notes only. Don't photograph the academy or players unless the owner offers.
8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back to record.
9. **Confidentiality:** don't describe the tech beyond the opening line, don't name other academies' answers, and don't share the answer sheets with anyone except Avi.

Phase 2 (only for academies that said yes ? separate briefing & pay)

You coordinate a recording session: we pay the academy a small fee, set up a phone/camera on a tripod (we provide a one-page capture checklist: behind the court, elevated, whole court in frame, landscape, 1080p/60fps, don't move it), record 1?2 sessions ? then we process the footage through KhelSutra and **you return to present the academy its own sample output** (their court, their players, as a highlight reel + report, free) and fill a short reaction sheet: what they said when they saw it, what they'd change, whether they asked to keep using it. That show-back conversation is the single most valuable artifact this whole engagement produces. **Hard rules:** written consent from the academy AND the adult players being recorded; **sessions with minors are not recorded for our library** ? adult sessions only; the academy gets told plainly what the footage is for (building and demonstrating the tool).

Day one

112   Your first 2 visits are paired (you + Avi, or a live phone walkthrough) so you're never improvising. Before visit 1:
113   read this playbook twice, have the sample reel saved offline on your phone, and carry printed answer sheets
114   ([`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) ? one per academy).
115

278. user (2026-06-12T15:07:29.074Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

279. user (2026-06-12T15:07:34.726Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

280. user (2026-06-12T15:07:36.604Z)

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281. user (2026-06-12T15:07:39.026Z)

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282. user (2026-06-12T15:07:40.840Z)

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283. user (2026-06-12T15:07:45.428Z)

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284. user (2026-06-12T15:07:47.889Z)

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285. user (2026-06-12T15:07:51.684Z)

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286. user (2026-06-12T15:07:53.114Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

287. assistant (2026-06-12T15:08:01.595Z)

Now the answer sheet and signals doc references:

288. user (2026-06-12T15:08:04.726Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

289. user (2026-06-12T15:08:07.122Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

290. user (2026-06-12T15:08:09.457Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

291. user (2026-06-12T15:08:11.366Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_ANSWER_SHEET.md has been updated successfully. (file state is current in your context ? no need to Read it back)

292. user (2026-06-12T15:08:20.660Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

293. user (2026-06-12T15:08:23.471Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

294. user (2026-06-12T15:08:26.298Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

295. user (2026-06-12T15:08:28.764Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

296. user (2026-06-12T15:08:40.394Z)

```
docs\FIELD_VISIT_PLAYBOOK.md:3:> **Recruiting?** Hand candidates
[`PRODUCT_FIELD_ASSOCIATE_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
docs\FIELD_VISIT_PLAYBOOK.md:10:**Compensation:** ? ___ per completed academy visit + travel
reimbursement; Phase-2 sessions paid extra ? ? ___ per session *(rates filled at hire; same
numbers as the recruiting flyer)*
docs\NEXT_STEPS.md:61:   `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner
fills ?/contact placeholders and recruits) +
docs\NEXT_STEPS.md:62:   `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (the hired associate's visit
playbook: coach + student script, demo-after-Q6
docs\PMF_FIELD_SIGNALS.md:6:**Companions:**
[`PRODUCT_FIELD_ASSOCIATE_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
docs\PMF_FIELD_SIGNALS.md:7:(the recruiting flyer candidates see) ·
```

297. user (2026-06-12T15:08:49.340Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

298. assistant (2026-06-12T15:08:56.077Z)

All references reconciled. Committing to the #104 branch and updating the PR:

299. user (2026-06-12T15:09:41.394Z)

```
 RESULT: PASS - all tests green
========================================================================
warning: in the working copy of 'docs/FIELD_VISIT_ANSWER_SHEET.md', LF will be replaced by CRLF
the next time Git touches it
```

```
warning: in the working copy of 'docs/PMF_FIELD_SIGNALS.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md', LF will be replaced by CRLF the next time Git touches it
[docs/pmf-field-signals 985a329] docs(pmf): recruiting flyer + product-feedback reframe (owner spec 2026-06-12)
 6 files changed, 249 insertions(+), 114 deletions(-)
 delete mode 100644 docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
 create mode 100644 docs/FIELD_VISIT_PLAYBOOK.md
 create mode 100644 docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md
To https://github.com/avidullu/badminton-highlight-indexer.git
   5548eaf..985a329  docs/pmf-field-signals -> docs/pmf-field-signals
https://github.com/avidullu/badminton-highlight-indexer/pull/104
```

300. user (2026-06-12T15:11:24.866Z)

```
https://github.com/avidullu/badminton-highlight-indexer/pull/104
Full test suite (offline, mocked)       pass    1m30s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27424460444/job/81058400017
Mypy (lenient baseline) pass    30s     https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27424460444/job/81058400253
Ruff lint       pass    12s     https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27424460444/job/81058400114
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27424460444/job/81058400129
```

301. assistant (2026-06-12T15:11:30.794Z)

CI green on the updated PR. Final memory sync:

302. user (2026-06-12T15:11:41.183Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

303. assistant (2026-06-12T15:11:57.579Z)

Your flyer is ready ?
[PRODUCT_FIELD_ASSOCIATE_FLYER.md](docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md) on [#104](https://github.com/avidullu/badminton-highlight-indexer/pull/104) (updated, CI 4/4 green, awaiting your review).

It's written for exactly the reader you described ? a reasonable non-technical person ? and maps your spec one-to-one:

- **The product in plain words**: finds the rallies, cuts highlight reels, produces a simple summary; phone-on-tripod, no hardware; "we're building a private web app a few invited academies will be able to try."
- **The core job, full stop**: visit academies, talk to coaches *and* adult students, understand how it helps them, capture improvement feedback verbatim ? explicitly "gathering feedback, not selling."
- **The stretch goal, paid separately**: coordinate a recording session ? we run it through the pipeline ? they go back and show the academy *its own court as highlights* and capture that reaction.
- An application step that screens for the actual skill ("one line about a time you had to get honest feedback out of someone").

Because the old doc was doing two jobs badly, it's now split by audience: the flyer recruits; **[FIELD_VISIT_PLAYBOOK.md](docs/FIELD_VISIT_PLAYBOOK.md)** (renamed) is what the hired person gets on day one ? script, the show-the-demo-only-after-the-open-questions rule, the student questions, ground rules, and the Phase-2 show-back procedure.

The adversarial pass (30 agents) mattered again ? the fixes worth knowing about: the draft said the web app "**is opening**" (overstates today's reality *and* pre-promised the verdict this very campaign is meant to decide ? softened to honest future tense); "match report" overclaimed what exists (now "simple summary: rallies found, actual play time"); "finds **every** rally" lost its absolutism; and "senior or adult students" was a real minors loophole in Indian academy parlance ? it's now hard-ruled to **adult 18+ only, coach points them out**, with the recording exclusion for minors untouched.

Before you print and recruit: fill the ?-per-visit / ?-per-session / phone blanks (flyer + playbook header), and I'll need to cut the ~60-second trim of the Mahadevpura reel for the associate's phone ? say the word and I'll do that plus render the flyer to a polished PDF alongside your khelacharya ones.

304. user (2026-06-12T16:31:20.462Z)

First, let us remove all the paid component because I have something else in mind. Do we store the video which they upload? How would they upload a 10GB file? expecting a s3 or gcs bucket is hard. Say they can get it onto a windows desktop, then can we do such that there is no upload? there are open quesitons I want you to also make a mental model it.

305. assistant (2026-06-12T16:33:16.939Z)

Three threads here: strip the paid framing from the kit (you've got other plans), answer the storage/upload questions from what the code actually does today, and build the mental model for the "10GB problem." Taking them in order ? first the pay removal on the #104 branch, then I'll ground the upload facts in the code before writing the model.

306. user (2026-06-12T16:33:20.722Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

307. user (2026-06-12T16:33:22.130Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

308. user (2026-06-12T16:33:24.900Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md has been updated successfully. (file state is current in your context ? no need to Read it back)

309. user (2026-06-12T16:33:35.167Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

310. user (2026-06-12T16:33:37.621Z)

<tool_use_error>String to replace not found in file.
String: knowledge needed. For interested academies there's a Phase-2 follow-up (a recording session ? then you return and
show them THEIR OWN footage as highlights) with additional compensation.
(note: Edit also tried swapping \uXXXX escapes and their characters; neither form matched, so the mismatch is likely elsewhere in old_string. Re-read the file and copy the exact surrounding text.)</tool_use_error>

311. user (2026-06-12T16:33:40.270Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

312. user (2026-06-12T16:33:41.841Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

313. user (2026-06-12T16:33:44.420Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

314. user (2026-06-12T16:33:46.972Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

315. user (2026-06-12T16:33:54.901Z)

```
22:few adult (18+) students) using the script below, and write down what they actually say.** No selling. No tech knowledge needed. For interested academies there's a Phase-2 follow-up
23:(a recording session ? then you return and show them THEIR OWN footage as highlights) with additional compensation.
```

316. user (2026-06-12T16:34:01.476Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\FIELD_VISIT_PLAYBOOK.md has been updated successfully. (file state is current in your context ? no need to Read it back)

317. user (2026-06-12T16:34:12.592Z)

```
docs\PMF_FIELD_SIGNALS.md:29:| H5 | **Willingness to pay per match** at a price that clears unit economics | "should be free" / <?100 everywhere | Q5 ? |
docs\PMF_FIELD_SIGNALS.md:34:hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
docs\PMF_FIELD_SIGNALS.md:35:roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
docs\PMF_FIELD_SIGNALS.md:36:break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
docs\PMF_FIELD_SIGNALS.md:51:| **H5 WTP** | names ??300/match **BEFORE the ladder was offered** (the sheet's tick-box records this), or proposes a budget framing of their own whose per-match equivalent is ??300 (see counting rule below) | names a number only **AFTER** the ladder (anchored ? even ?300/?500), or volunteers ~?100 | refuses to name a number / "should be free" / volunteers <?100 |
docs\PMF_FIELD_SIGNALS.md:55:**The H5 counting rule ("H5 at ?300+"):** a visit counts toward the "?300+" tally
docs\PMF_FIELD_SIGNALS.md:56:only when the ? number (a) is ?300 **and** (b) came *before* the ladder was offered
docs\PMF_FIELD_SIGNALS.md:57:? an anchored ?300 is the modal polite answer to a read-aloud 100/300/500 ladder
docs\PMF_FIELD_SIGNALS.md:59:answers convert at transcription: their monthly ? ÷ the matches/month they cited
docs\PMF_FIELD_SIGNALS.md:60:(if they cited none, ÷4); the computed equivalent counts if ??300 and the
docs\PMF_FIELD_SIGNALS.md:63:paid-pilot test already pre-agreed in PMF §6 risk 2 and the C13 row.
docs\PMF_FIELD_SIGNALS.md:71:verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a
docs\PMF_FIELD_SIGNALS.md:101:| **PROCEED ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ (per the §2 counting rule ? anchored numbers don't count) **and** ?3 H7-GREEN | Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); the **fake-door/paid-pilot WTP test** (PMF §6 risk 2) runs before money is spent on build |
docs\PMF_FIELD_SIGNALS.md:102:| **PIVOT A ? "index the backlog" ? "record + index" service** | <3 H2-GREEN **and** ?6 visits with H3 GREEN-or-AMBER **and** ?3 H5 at ?300+ | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer;
```

```
camera pilot becomes the product wedge, not the demo |
docs\PMF_FIELD_SIGNALS.md:104:| **PIVOT C ? payer is the parent, not the academy** | ?4 visits
where H5 is RED **and** the Q3 answer names parents/players as the watcher | Re-test WTP as B2C
per-student reels (?/reel to parents) with the academy as channel, not customer |
docs\PMF_FIELD_SIGNALS.md:145:| # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4
| H5 (?) | H6 | H7 | Gut | Phase-2? |
docs\PMF_FIELD_SIGNALS.md:154:| H5 at ?300+ (pre-ladder / converted only ? §2 counting rule) |
/n | ?4 ? PROCEED leg; ?3 ? PIVOT A leg |
docs\FIELD_VISIT_ANSWER_SHEET.md:37:**Q5 ? ? Would they pay per match for rally cuts + reel +
summary? What number did
docs\FIELD_VISIT_ANSWER_SHEET.md:38:THEY say (?)? What was their face doing?**
docs\FIELD_VISIT_ANSWER_SHEET.md:41:? AFTER I offered ?100/300/500 · ? no number / refused ·
docs\FIELD_VISIT_ANSWER_SHEET.md:58:**Q7 ? Open to a paid recording session on their court (we
return showing THEIR
docs\FIELD_VISIT_ANSWER_SHEET.md:85:| H5 willingness to pay ? their ? number: ______ | GREEN /
AMBER / RED |
docs\FIELD_VISIT_ANSWER_SHEET.md:96:| What they pay | |
docs\FIELD_VISIT_PLAYBOOK.md:47:   summary ? would you pay per match for that? What would that
be worth to you, per match?** Then **stop talking and
docs\FIELD_VISIT_PLAYBOOK.md:49:   ladder: "?100, ?300, ?500?" **On the sheet, tick whether
their number came BEFORE or AFTER you offered the
docs\FIELD_VISIT_PLAYBOOK.md:58:7. Would you let us record one session here (we pay a small fee
for the time) and come back showing what the tool
docs\FIELD_VISIT_PLAYBOOK.md:82:   real output ? at the scripted moment only, after Q6) and you
may offer the Phase-2 idea (a paid recording session
docs\FIELD_VISIT_PLAYBOOK.md:84:4. **Write down disappointing answers verbatim.** A "no, we'd
never pay for that" with a reason is worth MORE to us
docs\FIELD_VISIT_PLAYBOOK.md:100:You coordinate a recording session: we pay the academy a small
fee, set up a phone/camera on a tripod (we provide a
```

318. user (2026-06-12T16:34:32.474Z)

```
warning: in the working copy of 'docs/PMF_FIELD_SIGNALS.md', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md', LF will be replaced by
CRLF the next time Git touches it
[docs/pmf-field-signals 9c84fea] docs(pmf): remove the associate paid component (owner has
another structure in mind)
 4 files changed, 13 insertions(+), 11 deletions(-)
   985a329..9c84fea  docs/pmf-field-signals -> docs/pmf-field-signals
```

319. user (2026-06-12T16:34:32.598Z)

```
backend\api\static\style.css:11:    --card-bg: rgba(20, 26, 38, 0.6);
backend\api\static\style.css:12:    --card-border: rgba(255, 255, 255, 0.06);
backend\api\static\style.css:13:    --card-hover-border: rgba(46, 213, 115, 0.3);
backend\api\static\style.css:55:    background: rgba(0, 0, 0, 0.2);
backend\api\static\style.css:58:    background: rgba(255, 255, 255, 0.1);
backend\api\static\style.css:62:    background: rgba(255, 255, 255, 0.2);
backend\api\static\style.css:99:    background: rgba(255, 255, 255, 0.04);
backend\api\static\style.css:139:    box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.25);
backend\api\static\style.css:147:    border-color: rgba(255, 255, 255, 0.1);
backend\api\static\style.css:162:    border-bottom: 1px solid rgba(255, 255, 255, 0.05);
backend\api\static\style.css:180:    background: rgba(0, 0, 0, 0.2);
backend\api\static\style.css:215:    box-shadow: 0 8px 20px rgba(46, 213, 115, 0.3);
backend\api\static\style.css:223:    background: rgba(255, 255, 255, 0.06);
backend\api\static\style.css:239:    background: rgba(255, 255, 255, 0.1);
backend\api\static\style.css:240:    border-color: rgba(255, 255, 255, 0.2);
backend\api\static\style.css:251:    background: rgba(255, 255, 255, 0.02);
backend\api\static\style.css:252:    border: 1px solid rgba(255, 255, 255, 0.04);
backend\api\static\style.css:267:    background: rgba(235, 77, 75, 0.02);
backend\api\static\style.css:284:    background: rgba(46, 213, 115, 0.1);
backend\api\static\style.css:289:    background: rgba(235, 77, 75, 0.1);
backend\api\static\style.css:294:    background: rgba(255, 255, 255, 0.05);
backend\api\static\style.css:313:    background: rgba(0, 0, 0, 0.15);
```

```
backend\api\static\style.css:346:    background: rgba(0, 0, 0, 0.15);
backend\api\static\style.css:349:    border: 1px solid rgba(255, 255, 255, 0.04);
backend\api\static\style.css:353:    background: rgba(0, 0, 0, 0.3);
backend\api\static\style.css:378:    background: rgba(255, 255, 255, 0.02);
backend\api\static\style.css:379:    border: 1px solid rgba(255, 255, 255, 0.04);
backend\api\static\style.css:403:    background: rgba(255, 255, 255, 0.04);
backend\api\static\style.css:404:    border-color: rgba(255, 255, 255, 0.08);
backend\api\static\style.css:409:    background: rgba(46, 213, 115, 0.03);

[Showing results with pagination = limit: 30]
```

320. user (2026-06-12T16:34:41.172Z)

```
69:# parallel uploads (2026-06-10 ops learning: serial uploads + backoff). The segmenters
311:# --- Chunked upload (B2, PR-2) ---------------------------------------------------
312:# Browser uploads arrive in sequential parts ? security.max_part_mb (free-tier edge
314:# under security.upload_dir, then submitted to /api/process by the returned path.
316:# aborts partial uploads and the client re-inits. Per-user partitioning of upload_dir
319:_uploads: Dict[str, Dict[str, Any]] = {}
320:_uploads_lock = threading.Lock()
332:def _upload_cfg():
334:    upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
335:    max_bytes = int(getattr(sec, "max_upload_mb", 4096) or 4096) * 1024 * 1024
337:    exts = [e.lower().lstrip(".") for e in (getattr(sec, "allowed_upload_extensions", None)
or ["mp4", "mov"])]
338:    return upload_dir, max_bytes, part_bytes, exts
341:def _abort_upload(upload_id: str) -> None:
342:    with _uploads_lock:
343:        st = _uploads.pop(upload_id, None)
351:@app.post("/api/upload/init")
352:def upload_init(req: UploadInitRequest):
353:    """Start a chunked upload. Returns upload_id; PUT sequential parts next."""
354:    upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
365:                            detail=f"File exceeds the {max_bytes // (1024 * 1024)} MB upload
limit.")
366:    os.makedirs(upload_dir, exist_ok=True)
367:    upload_id = uuid.uuid4().hex
368:    tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")
370:    with _uploads_lock:
371:        _uploads[upload_id] = {"filename": name, "tmp_path": tmp_path, "parts": 0, "bytes":
0}
372:    return {"upload_id": upload_id, "max_part_mb": part_bytes // (1024 * 1024),
373:            "next_part": f"/api/upload/{upload_id}/part/0"}
376:@app.put("/api/upload/{upload_id}/part/{index}")
377:async def upload_part(upload_id: str, index: int, request: Request):
379:    _, max_bytes, part_bytes, _ = _upload_cfg()
380:    with _uploads_lock:
381:        st = _uploads.get(upload_id)
383:        raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
401:        _abort_upload(upload_id)
404:    with _uploads_lock:
407:    return {"upload_id": upload_id, "received_parts": st["parts"], "received_bytes":
st["bytes"]}
410:@app.post("/api/upload/{upload_id}/complete")
411:def upload_complete(upload_id: str, req: UploadCompleteRequest):
413:    upload_dir, _, _, _ = _upload_cfg()
414:    with _uploads_lock:

[Showing results with pagination = limit: 40]
```

321. user (2026-06-12T16:35:47.566Z)

```
Switched to a new branch 'docs/video-locality-model'
```

322. assistant (2026-06-12T16:36:45.946Z)

Video locality ? where the bytes live (mental model + open questions)

Status: THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for the upload/storage problem, with the open questions named. **No implementation is greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).
Companions: `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`, `PLATFORM_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries), collector ADR-002 (the frame-identical proxy contract this doc leans on).

0. The questions that triggered this (and the short answers)

"Do we store the video they upload?" Yes ? today the chunked-upload flow assembles the full file under `security.upload_dir` (default `output/uploads`) on our box, registers it by MD5, and nothing ever deletes it (retention policy is an acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks, frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px re-encoded chunks to Google's Files API.

"How would they upload a 10 GB file?" Today they can't: `/api/upload/init` rejects anything over `security.max_upload_mb` (default **4096 MB**), and even under the cap the file moves as sequential ?90 MB parts through a free-tier edge whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps, ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are miserable everywhere; expecting academies to manage S3/GCS buckets is a non-starter. **The 10 GB problem is real and the current design doesn't solve it.**

"Can we make it so there is NO upload?" Mostly ? because of §1.

1. The central insight: timestamps are the product; the video is dead weight

Walk the pipeline and ask what each stage actually consumes:

- The **Gemini segmenter** never sees the original ? it re-encodes chunks down to **1280 px** (`indexing.ai_chunk_scale_width`) before uploading them to Google.
- The **detectors/featurizer** consume downscaled frames and trajectories.
- The only consumer of the original full-res bytes is the **highlight compiler**, and only to cut clips ? a pure-ffmpeg operation that can run **wherever the original lives**.
- What the customer receives is `{rally intervals, report, reel}`. Intervals + report are **kilobytes**. Given the intervals, the reel can be cut on the machine that already has the original.

So: **any design that moves the 10 GB is moving the wrong thing.** The question is never "how do we upload the video" but "what is the smallest artifact that must cross the network for this tier of service."

2. The locality ladder

From most to least bytes leaving the academy's Windows desktop. Sizes are rough ratios for a 4K source ? measure on a real file before committing to any tier.

	What crosses the network	~Bytes (10 GB source)	What it needs	Status
L3 ? full upload (today's flow)	everything	10 GB (blocked: 4 GB cap)	paid edge tier + resumable uploads + server disk + retention policy	built, capped, wrong for big files
L2 ? proxy upload	a timeline-identical ?1080p re-encode	~1?2.5 GB	a tiny client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5 manifest)	contract proven; client wrapper + server acceptance = new work
L1 ? chunks/candidates upload	only the ?1280px AI chunks (or just candidate windows) the

Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU); server becomes a **GPU-less orchestrator** | the server half exists (it's the gemini segmenter's own flow, relocated) |
L1-direct	chunks go **straight from the client to Gemini** ? zero video bytes through our server	~0 through us	per-user key custody / metering / cost ceilings (OQ4)	idea only
L0 ? fully local	nothing (calls to Google only if the Gemini path is used)	0	the owned on-device model (roadmap north star) or a local GPU + our stack	not alpha-feasible; the long-term answer
L4 ? sneakernet	a hard drive	0 over network	a person (the field associate / owner)	how the pilot actually works today

The honest "no upload" answer: *strictly* no bytes leave only at L0 with the
owned model (and even L0 leaks ?1280px chunks to Google while the Gemini path is
the brain ? say that plainly in consent language, OQ3). The pragmatic near-term
answer is **L2 or L1: don't eliminate the upload, shrink it 5?30×** ? at which
point the existing 90 MB-part flow, the 4 GB cap, and a coffee-break wait are all
suddenly fine.

3. Second-order effects worth noticing

- **L1/L2 make the serving box GPU-less for the Gemini tier** ? ffmpeg + API
  orchestration only. Hosting cost collapses; the GPU box stays for WASB/owned-
  model work. (PLATFORM §3's ComputeBackend split falls out naturally.)
- **Storing the proxy instead of the original** is ~10× less disk AND a smaller
  privacy surface; the review UI (alpha screen A4) actually *prefers* streaming a
  1080p proxy to seeking inside a 10 GB original.
- **The correction loop survives** at every tier ?L1: labels are intervals on a
  timeline, and the proxy timeline IS the original timeline by construction
  (the collector ADR-002 invariants: same fps, same frame count, parity-verified).
- **video_id keying:** the indexer keys by MD5 of the file it processed. At L2
  that's the proxy's MD5; the proxy-manifest pattern (original_md5 ? proxy_md5)
  already exists ? but the product needs ONE registered identity rule (OQ2).

4. Open questions (the decision backlog this doc exists to hold)

#	Question	Why it matters / current lean
OQ1	**Retention policy per artifact class** ? original (only L3 has it), proxy, chunks, frames/trajectories, Gemini remote files, intervals/labels	intervals/labels are the asset (keep forever, tiny); originals/proxies need an explicit keep-while-active + delete-on-request rule; Gemini remote-file GC already in BACKLOG
OQ2	**Which file's MD5 is the video's identity** at L2/L1	lean: the processed file's MD5 (proxy), with original_md5 carried in metadata ? the collector manifest pattern, promoted to a product rule
OQ3	**Consent language for the Gemini path** ? even "local" modes ship 1280px chunks to Google	must be plain in the academy-facing terms; paid-tier no-training posture to be cited
OQ4	**Key custody for L1-direct** ? client calls Gemini with whose key? per-user metering, abuse, cost ceiling per video	unsolved; this is what separates L1 from L1-direct
OQ5	**Windows client packaging** ? portable ffmpeg + a signed tiny app vs a PowerShell script for the pilot	pilot can be a script the associate/owner runs; product needs an installer story (ffmpeg LGPL build = licensing-clean)
OQ6	**What does the review UI play** at each tier	L2 streams the proxy (good); L1 has only chunks ? review degrades ? this quietly argues L2 over L1 as the default
OQ7	**Reel delivery** ? server-compiled from proxy (shareable 1080p) vs intervals-back + client cuts full-res	could be both: proxy reel by default, "premium full-res" cut locally
OQ8	**Real academy uplink numbers** ? the L2-vs-L3 case rests on bandwidth assumptions	C13 visits can note connectivity informally; don't redesign around guesses
OQ9	**Recurring volume** ? an academy recording daily generates 10 GB/day; even L2 accumulates	retention (OQ1) + maybe future delta thinking; not an alpha problem
OQ10	**If L3 must survive** (users who refuse any client-side step)	needs resumable uploads + a paid edge tier + a raised cap; treat as the exception path, not the design center

5. Where this lands (recommendation, not a greenlight)

Default the alpha to L2 (proxy upload): a one-command client prep step built on the *already-verified* proxy contract, uploads that fit comfortably inside today's chunked endpoint and 4 GB cap, a GPU-less serving path, a review UI that streams the proxy, and intervals that map 1:1 onto the customer's original. **L1/L1-direct is the fast-follow** once OQ4 has an answer; **L0 is the owned-model milestone**, not an alpha decision. The field pilot keeps running on L4 ? a person with a hard drive is a fine storage backend for n=3 academies.

323. user (2026-06-12T16:36:45.961Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\VIDEO_LOCALITY_MODEL.md (file state is current in your context ? no need to Read it back)

324. user (2026-06-12T16:37:27.790Z)

The user doesn't want to proceed with this tool use. The tool use was rejected (eg. if it was a file edit, the new_string was NOT written to the file). STOP what you are doing and wait for the user to tell you how to proceed.

325. user (2026-06-12T16:37:27.793Z)

[Request interrupted by user for tool use]

326. user (2026-06-12T16:39:02.616Z)

Let us make sure to use the academy visit for building a stronger product. Keeping the cost low has the advantage that for some time, we can afford the bills as long as we don't burn too much GPU.

327. user (2026-06-12T16:40:04.863Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\VIDEO_LOCALITY_MODEL.md has been updated successfully. (file state is current in your context ? no need to Read it back)

328. user (2026-06-12T16:40:08.533Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\VIDEO_LOCALITY_MODEL.md has been updated successfully. (file state is current in your context ? no need to Read it back)

329. user (2026-06-12T16:42:27.988Z)

```
warning: in the working copy of 'docs/VIDEO_LOCALITY_MODEL.md', LF will be replaced by CRLF the next time Git touches it
[docs/video-locality-model 85abe26] docs(platform): video-locality mental model - the 10GB question (owner-requested)
 1 file changed, 44 insertions(+), 4 deletions(-)
branch 'docs/video-locality-model' set up to track 'origin/docs/video-locality-model'.
To https://github.com/avidullu/badminton-highlight-indexer.git
   1e327d4..85abe26  docs/video-locality-model -> docs/video-locality-model
Warning: 9 uncommitted changes
a pull request for branch "docs/video-locality-model" into branch "master" already exists:
https://github.com/avidullu/badminton-highlight-indexer/pull/106
Full test suite (offline, mocked)       pass    1m38s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27429463010/job/81075740466
Mypy (lenient baseline) pass    28s     https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27429463010/job/81075740116
Ruff lint       pass    11s     https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27429463010/job/81075740096
Syntax sweep + import smoke (no GPU stack)      pass    18s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27429463010/job/81075740085
```

330. user (2026-06-12T16:43:00.100Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

331. assistant (2026-06-12T16:43:21.476Z)

All three asks are done. Here's where things stand:

Paid component ? removed (on [#104](https://github.com/avidullu/badminton-highlight-indexer/pull/104))

The flyer and playbook now say only "Engagement terms: discussed individually with Avi" ? no ?-per-visit, no paid stretch goal, no "paid per honest visit" framing (reworded to "judged on honest capture, not positive answers" so the honesty mechanism survives the pay removal). Two things I deliberately kept, flag if your "something else" covers them: the willingness-to-pay *questions* (that's the research subject, not compensation) and the **academy's** small session fee in Q7/Phase-2 (a different party than your recruit).

Your storage questions ? answered from the code, not memory

Do we store uploads? Yes ? the chunked upload assembles the full file under `security.upload_dir` (default `output/uploads`) and nothing ever deletes it; retention policy is a known gap already in BACKLOG. The Gemini path additionally sends ?1280px re-encoded chunks to Google.

How would a 10GB file upload? It wouldn't ? `/api/upload/init` rejects anything over 4096 MB, and even permitted files crawl up as ?90MB sequential parts (a free-tier edge constraint). At Indian uplink speeds 10GB is 1?4.5 hours even if we lifted the cap. Your instinct is right: this path is wrong for real academy footage, and S3/GCS expectations are a non-starter.

Can there be no upload? Mostly yes ? which is the heart of the new mental model.

The mental model ? [VIDEO_LOCALITY_MODEL.md](docs/VIDEO_LOCALITY_MODEL.md) on [#106](https://g

The central insight: **timestamps are the product; the video is dead weight.** Nothing in the pipeline needs the original bytes except the compiler ? and it only cuts clips, which can happen wherever the original lives. So the question is never "how do we move 10GB" but "what's the smallest artifact that must cross the network." That gives a five-rung ladder: full upload (today, capped) ? **proxy upload** (1?2.5GB, reusing the exact frame-identical contract we just shipped for Roora) ? **chunks-only** (~0.3?1GB, the Gemini segmenter's own flow relocated client-side) ? **client-direct-to-Gemini** (zero video bytes through us ? key custody is the open question) ? **fully local** (the owned-model north star). Your two directives are now load-bearing sections: the **frugality frame** (L2/L1 serving is GPU-less *by construction* ? Gemini costs cents per video-hour, GPU work stays on your 2070S, bills stay personally affordable indefinitely at pilot scale) and the **academy-visit dividend** (Phase-2 consented recordings ARE the multi-court corpus the 0.66 ship target needs; the show-back is a live dress rehearsal of the L2 loop; every friction the associate hits is a pre-paid backlog item). Ten open questions are named (OQ1?OQ10), and the recommendation lean ? alpha defaults to L2 ? is explicitly *not* a greenlight.

Both PRs are CI-green and waiting on you. One housekeeping note: #106's base commit appeared on the branch from outside this session (shared checkout), so if that was you in another window, we're already in sync.

332. user (2026-06-13T16:41:17.615Z)

Hello. Can you sync to head and point me to the exact flyer for the new hire

333. assistant (2026-06-13T16:41:18.752Z)

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.